Windows Commands

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Summarize this article for me

All supported versions of Windows and Windows Server have a set of Win32 console commands built in. This set of documentation describes the Windows Commands you can use to automate tasks by using scripts or scripting tools.

Command-line shells

Windows has two command-line shells: the Command shell and PowerShell. Each shell is a software program that provides direct communication between you and the operating system or application, providing an environment to automate IT operations.

The Command shell was the first shell built into Windows to automate routine tasks, like user account management or nightly backups, with batch (.bat) files. With Windows Script Host, you could run more sophisticated scripts in the Command shell. For more information, see cscript or wscript. You can perform operations more efficiently by using scripts than you can by using the user interface. Scripts accept all commands that are available at the command line.

PowerShell was designed to extend the capabilities of the Command shell to run PowerShell commands called cmdlets. Cmdlets are similar to Windows Commands but provide a more extensible scripting language. You can run both Windows Commands and PowerShell cmdlets in PowerShell, but the Command shell can only run Windows Commands and not PowerShell cmdlets.

For the most robust, up-to-date Windows automation, we recommend using PowerShell instead of Windows Commands or Windows Script Host for Windows automation.

A reference of exit and error codes for Windows Commands can be found in the Debug system error codes articles that may be helpful to understanding errors produced. Windows Commands also include command redirection operators. To learn more of their use, see Using command redirection operators.

> **Note**
>
> You can also download and install PowerShell Core, the open source version of PowerShell.

Command shell file and directory name automatic completion

You can configure the Command shell to automatically complete file and directory names on a computer or user session when a specified control character is pressed. By default this control character is configured to be the **tab** key for both file and directory names, although they can be different. To change this control character, run `regedit.exe` and navigate to either of the following registry keys and entries, depending on whether you wish to change the value for the current user only, or for all users of the computer.

> **⊗ Caution**
>
> Incorrectly editing the registry may severely damage your system. Before making the following changes to the registry, you should back up any valued data on the computer.

registry

```
HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Command Processor\CompletionChar
HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Command Processor\PathCompletionChar
```

Set these values to that of the control character you wish to use. See virtual key codes for a complete list. To disable a particular completion character in the registry, use the value for **space** (0x20) as it isn't a valid control character. The type of value for this registry entry is REG_DWORD, and can also be specified by hexadecimal or decimal value.

You can also enable or disable file and directory name completion per instance of a Command shell by running `cmd.exe` with the parameter and switch `/F:ON` or `/F:OFF`. If name completion is enabled with the `/F:ON` parameter and switch, the two control characters used are `Ctrl-D` for directory name completion and `Ctrl-F` for file name completion. User-specified settings take precedence over computer settings, and command-line options take precedence over registry settings.

Command-line reference A-Z

To find information about a specific command, in the following A-Z menu, select the letter that the command starts with, and then select the command name.

A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z

A

- active
- add
- add alias
- add volume
- adprep
- append
- arp
- assign
- assoc
- at
- atmadm
- attach-vdisk
- attrib
- attributes
  - attributes disk
  - attributes volume
- auditpol
  - auditpol backup
  - auditpol clear
  - auditpol get
  - auditpol list
  - auditpol remove
  - auditpol resourcesacl
  - auditpol restore
  - auditpol set
- autochk
- autoconv
- autofmt
- automount

B

- bcdboot
- bcdedit
- bdehdcfg
  - bdehdcfg driveinfo
  - bdehdcfg newdriveletter
  - bdehdcfg quiet
  - bdehdcfg restart
  - bdehdcfg size

    - bdehdcfg target
- begin backup
- begin restore
- bitsadmin
    - bitsadmin addfile
    - bitsadmin addfileset
    - bitsadmin addfilewithranges
    - bitsadmin cache
        - bitsadmin cache and delete
        - bitsadmin cache and deleteurl
        - bitsadmin cache and getexpirationtime
        - bitsadmin cache and getlimit
        - bitsadmin cache and help
        - bitsadmin cache and info
        - bitsadmin cache and list
        - bitsadmin cache and setexpirationtime
        - bitsadmin cache and setlimit
        - bitsadmin cache and clear
    - bitsadmin cancel
    - bitsadmin complete
    - bitsadmin create
    - bitsadmin examples
    - bitsadmin getaclflags
    - bitsadmin getbytestotal
    - bitsadmin getbytestransferred
    - bitsadmin getclientcertificate
    - bitsadmin getcompletiontime
    - bitsadmin getcreationtime
    - bitsadmin getcustomheaders
    - bitsadmin getdescription
    - bitsadmin getdisplayname
    - bitsadmin geterror
    - bitsadmin geterrorcount
    - bitsadmin getfilestotal
    - bitsadmin getfilestransferred
    - bitsadmin gethelpertokenflags
    - bitsadmin gethelpertokensid
    - bitsadmin gethttpmethod
    - bitsadmin getmaxdownloadtime
    - bitsadmin getminretrydelay
    - bitsadmin getmodificationtime

- bitsadmin getnoprogresstimeout
- bitsadmin getnotifycmdline
- bitsadmin getnotifyflags
- bitsadmin getnotifyinterface
- bitsadmin getowner
- bitsadmin getpeercachingflags
- bitsadmin getpriority
- bitsadmin getproxybypasslist
- bitsadmin getproxylist
- bitsadmin getproxyusage
- bitsadmin getreplydata
- bitsadmin getreplyfilename
- bitsadmin getreplyprogress
- bitsadmin getsecurityflags
- bitsadmin getstate
- bitsadmin gettemporaryname
- bitsadmin gettype
- bitsadmin getvalidationstate
- bitsadmin help
- bitsadmin info
- bitsadmin list
- bitsadmin listfiles
- bitsadmin makecustomheaderswriteonly
- bitsadmin monitor
- bitsadmin nowrap
- bitsadmin peercaching
  - bitsadmin peercaching and getconfigurationflags
  - bitsadmin peercaching and help
  - bitsadmin peercaching and setconfigurationflags
- bitsadmin peers
  - bitsadmin peers and clear
  - bitsadmin peers and discover
  - bitsadmin peers and help
  - bitsadmin peers and list
- bitsadmin rawreturn
- bitsadmin removeclientcertificate
- bitsadmin removecredentials
- bitsadmin replaceremoteprefix
- bitsadmin reset
- bitsadmin resume
- bitsadmin setaclflag

- bitsadmin setclientcertificatebyid
- bitsadmin setclientcertificatebyname
- bitsadmin setcredentials
- bitsadmin setcustomheaders
- bitsadmin setdescription
- bitsadmin setdisplayname
- bitsadmin sethelpertoken
- bitsadmin sethelpertokenflags
- bitsadmin seththttpmethod
- bitsadmin setmaxdownloadtime
- bitsadmin setminretrydelay
- bitsadmin setnoprogresstimeout
- bitsadmin setnotifycmdline
- bitsadmin setnotifyflags
- bitsadmin setpeercachingflags
- bitsadmin setpriority
- bitsadmin setproxysettings
- bitsadmin setreplyfilename
- bitsadmin setsecurityflags
- bitsadmin setvalidationstate
- bitsadmin suspend
- bitsadmin takeownership
- bitsadmin transfer
- bitsadmin util
  - bitsadmin util and enableanalyticchannel
  - bitsadmin util and getieproxy
  - bitsadmin util and help
  - bitsadmin util and repairservice
  - bitsadmin util and setieproxy
  - bitsadmin util and version
- bitsadmin wrap

- bootcfg
  - bootcfg addsw
  - bootcfg copy
  - bootcfg dbg1394
  - bootcfg debug
  - bootcfg default
  - bootcfg delete
  - bootcfg ems
  - bootcfg query
  - bootcfg raw

    - bootcfg rmsw
    - bootcfg timeout
- break

C

- cacls
- call
- cd
- certreq
- certutil
- change
    - change logon
    - change port
    - change user
- chcp
- chdir
- chglogon
- chgport
- chgusr
- chkdsk
- chkntfs
- choice
- cipher
- clean
- cleanmgr
- clip
- cls
- cmd
- cmdkey
- cmstp
- color
- comp
- compact
- compact vdisk
- convert
    - convert basic
    - convert dynamic
    - convert gpt
    - convert mbr

- copy
- create
  - create partition efi
  - create partition extended
  - create partition logical
  - create partition msr
  - create partition primary
  - create volume mirror
  - create volume raid
  - create volume simple
  - create volume stripe
- cscript

D

- date
- dcdiag
- dcgpofix
- dcpromo
- defrag
- del
- delete
  - delete disk
  - delete partition
  - delete shadows
  - delete volume
- detach vdisk
- detail
  - detail disk
  - detail partition
  - detail vdisk
  - detail volume
- dfsdiag
  - dfsdiag testdcs
  - dfsdiag testdfsconfig
  - dfsdiag testdfsintegrity
  - dfsdiag testreferral
  - dfsdiag testsites
- dfsrmig
- diantz

- dir
- diskcomp
- diskcopy
- diskpart
- diskperf
- diskraid
- diskshadow
- dispdiag
- dnscmd
- doskey
- driverquery
- dtrace

E

- echo
- edit
- endlocal
- end restore
- erase
- eventcreate
- Evntcmd
- exec
- exit
- expand
- expand vdisk
- expose
- extend
- extract

F

- fc
- filesystems
- find
- findstr
- finger
- flattemp
- fondue
- for

- forfiles
- format
- freedisk
- fsutil
  - fsutil 8dot3name
  - fsutil behavior
  - fsutil devdrv
  - fsutil dirty
  - fsutil file
  - fsutil fsinfo
  - fsutil hardlink
  - fsutil objectid
  - fsutil quota
  - fsutil repair
  - fsutil reparsepoint
  - fsutil resource
  - fsutil sparse
  - fsutil tiering
  - fsutil transaction
  - fsutil usn
  - fsutil volume
  - fsutil wim
- ftp
  - ftp append
  - ftp ascii
  - ftp bell
  - ftp binary
  - ftp bye
  - ftp cd
  - ftp close
  - ftp debug
  - ftp delete
  - ftp dir
  - ftp disconnect
  - ftp get
  - ftp glob
  - ftp hash
  - ftp lcd
  - ftp literal
  - ftp ls
  - ftp mget

    - ftp mkdir
    - ftp mls
    - ftp mput
    - ftp open
    - ftp prompt
    - ftp put
    - ftp pwd
    - ftp quit
    - ftp quote
    - ftp recv
    - ftp remotehelp
    - ftp rename
    - ftp rmdir
    - ftp send
    - ftp status
    - ftp trace
    - ftp type
    - ftp user
    - ftp verbose
    - ftp mdelete
    - ftp mdir
- ftype
- fveupdate

G

- getmac
- gettype
- goto
- gpfixup
- gpresult
- gpt
- gpupdate
- graftabl

H

- help
- helpctr
- hostname

I

- icacls
- if
- import (shadowdisk)
- import (diskpart)
- inactive
- ipconfig
- ipxroute
- irftp

J

- jetpack

K

- klist
- ksetup
  - ksetup addenctypeattr
  - ksetup addhosttorealmmap
  - ksetup addkdc
  - ksetup addkpasswd
  - ksetup addrealmflags
  - ksetup changepassword
  - ksetup delenctypeattr
  - ksetup delhosttorealmmap
  - ksetup delkdc
  - ksetup delkpasswd
  - ksetup delrealmflags
  - ksetup domain
  - ksetup dumpstate
  - ksetup getenctypeattr
  - ksetup listrealmflags
  - ksetup mapuser
  - ksetup removerealm
  - ksetup server
  - ksetup setcomputerpassword
  - ksetup setenctypeattr
  - ksetup setrealm

- ksetup setrealmflags
- ktmutil
- ktpass

L

- label
- list
  - list providers
  - list shadows
  - list writers
- load metadata
- lodctr
- logman
  - logman create
  - logman create alert
  - logman create api
  - logman create cfg
  - logman create counter
  - logman create trace
  - logman delete
  - logman import and logman export
  - logman query
  - logman start and logman stop
  - logman update
  - logman update alert
  - logman update api
  - logman update cfg
  - logman update counter
  - logman update trace
- logoff
- lpq
- lpr

M

- macfile
- makecab
- manage bde
  - manage bde status

    - manage bde on
    - manage bde off
    - manage bde pause
    - manage bde resume
    - manage bde lock
    - manage bde unlock
    - manage bde autounlock
    - manage bde protectors
    - manage bde tpm
    - manage bde setidentifier
    - manage bde forcerecovery
    - manage bde changepassword
    - manage bde changepin
    - manage bde changekey
    - manage bde keypackage
    - manage bde upgrade
    - manage bde wipefreespace
- mapadmin
- md
- merge vdisk
- mkdir
- mklink
- mmc
- mode
- more
- mount
- mountvol
- move
- mqbkup
- mqsvc
- mqtgsvc
- msdt
- msg
- msiexec
- msinfo32
- mstsc

N

- nbtstat

- netcfg
- netdom
  - netdom add
  - netdom computername
  - netdom join
  - netdom move
  - netdom movent4bdc
  - netdom query
  - netdom remove
  - netdom renamecomputer
  - netdom reset
  - netdom resetpwd
  - netdom trust
  - netdom verify
- net print
- net user
- netsh
  - netsh add
  - netsh advfirewall
  - netsh branchcache
  - netsh bridge
  - netsh delete
  - netsh dhcpclient
  - netsh dnsclient
  - netsh dump
  - netsh exec
  - netsh http
  - netsh interface
  - netsh ipsec
  - netsh lan
  - netsh mbn
  - netsh namespace
  - netsh netio
  - netsh nlm
  - netsh ras
  - netsh rpc
  - netsh set
  - netsh show
  - netsh trace
  - netsh wcn
  - netsh wfp

    - netsh winhttp
    - netsh winsock
    - netsh wlan
- netstat
- nfsadmin
- nfsshare
- nfsstat
- nlbmgr
- nltest
- nslookup
    - nslookup exit Command
    - nslookup finger Command
    - nslookup help
    - nslookup ls
    - nslookup lserver
    - nslookup root
    - nslookup server
    - nslookup set
    - nslookup set all
    - nslookup set class
    - nslookup set d2
    - nslookup set debug
    - nslookup set domain
    - nslookup set port
    - nslookup set querytype
    - nslookup set recurse
    - nslookup set retry
    - nslookup set root
    - nslookup set search
    - nslookup set srchlist
    - nslookup set timeout
    - nslookup set type
    - nslookup set vc
    - nslookup view
- ntbackup
- ntcmdprompt
- ntfrsutl

O

- offline
  - offline disk
  - offline volume
- online
  - online disk
  - online volume
- openfiles

P

- pagefileconfig
- path
- pathping
- pause
- pbadmin
- pentnt
- perfmon
- ping
- pktmon
- pnpunattend
- pnputil
- popd
- powershell
- powershell ise
- print
- prncnfg
- prndrvr
- prnjobs
- prnmngr
- prnport
- prnqctl
- prompt
- pubprn
- pushd
- pushprinterconnections
- pwlauncher
- pwsh

Q

- qappsrv
- qprocess
- query
  - query process
  - query session
  - query termserver
  - query user
- quser
- qwinsta

R

- rd
- rdpsign
- recover
- recover disk group
- refsutil
  - refsutil compression
  - refsutil dedup
  - refsutil fixboot
  - refsutil iometrics
  - refsutil leak
  - refsutil salvage
  - refsutil streamsnapshot
  - refsutil triage
- reg
  - reg add
  - reg compare
  - reg copy
  - reg delete
  - reg export
  - reg import
  - reg load
  - reg query
  - reg restore
  - reg save
  - reg unload
- regini
- regsvr32
- relog

- rem
- remove
- ren
- rename
- repadmin
- repair
  - repair bde
- replace
- rescan
- reset
  - reset session
- retain
- revert
- rexec
- risetup
- rmdir
- robocopy
- route ws2008
- rpcinfo
- rpcping
- rsh
- rundll32
- rundll32 printui
- rwinsta

S

- san
- sc config
- sc create
- sc delete
- sc query
- schtasks
- scwcmd
  - scwcmd analyze
  - scwcmd configure
  - scwcmd register
  - scwcmd rollback
  - scwcmd transform
  - scwcmd view

- secedit
  - secedit analyze
  - secedit configure
  - secedit export
  - secedit generaterollback
  - secedit import
  - secedit validate
- select
  - select disk
  - select partition
  - select vdisk
  - select volume
- serverceipoptin
- servermanagercmd
- serverweroptin
- set environmental variables
- set shadow copy
  - set context
  - set id
  - setlocal
  - set metadata
  - set option
  - set verbose
- setlocal
- setspn
- setx
- sfc
- shadow
- shift
- showmount
- shrink
- shutdown
- simulate restore
- sort
- start
- subcommand set device
- subcommand set drivergroup
- subcommand set drivergroupfilter
- subcommand set driverpackage
- subcommand set image
- subcommand set imagegroup

- subcommand set server
- subcommand set transportserver
- subcommand set multicasttransmission
- subcommand start namespace
- subcommand start server
- subcommand start transportserver
- subcommand stop server
- subcommand stop transportserver
- subst
- sxstrace
- sysmon
- sysocmgr
- systeminfo

T

- takeown
- tapicfg
- taskkill
- tasklist
- tcmsetup
- telnet
  - telnet close
  - telnet display
  - telnet open
  - telnet quit
  - telnet send
  - telnet set
  - telnet status
  - telnet unset
- tftp
- time
- timeout
- title
- tlntadmn
- tpmtool
- tpmvscmgr
- tracerpt
- tracert
- tree

- tscon
- tsdiscon
- tsecimp
- tskill
- tsprof
- type
- typeperf
- tzutil

U

- unexpose
- uniqueid
- unlodctr

V

- ver
- verifier
- verify
- vol
- vssadmin
  - vssadmin delete shadows
  - vssadmin list shadows
  - vssadmin list writers
  - vssadmin resize shadowstorage

W

- waitfor
- wbadmin
  - wbadmin delete catalog
  - wbadmin delete systemstatebackup
  - wbadmin disable backup
  - wbadmin enable backup
  - wbadmin get disks
  - wbadmin get items
  - wbadmin get status
  - wbadmin get versions
  - wbadmin restore catalog

    - wbadmin start backup
    - wbadmin start recovery
    - wbadmin start sysrecovery
    - wbadmin start systemstatebackup
    - wbadmin start systemstaterecovery
    - wbadmin stop job
- wdsutil
- wecutil
- wevtutil
- where
- whoami
- winnt
- winnt32
- winrs
- winsat mem
- winsat mfmedia
- wmic
- writer
- wscript

X

- xcopy

Last updated on 02/24/2026

Command-line syntax key

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

This article describes the notation that command-line syntax blocks use, such as the brackets, braces, and vertical bars that indicate how to enter a command. It isn't a catalog of commands or command options. To find a command and the options it supports, see Windows Commands. In the A-Z menu, select the letter that the command starts with, and then select the command name to view its options.

The following table describes each notation element.

Expand table

Notation	Description
Text without brackets or braces	Items you must type as shown.
`<Text inside angle brackets>`	Placeholder for which you must supply a value.
`[Text inside square brackets]`	Optional items.
`{Text inside braces}`	Set of required items. You must choose one.
Vertical bar (`|`)	Separator for mutually exclusive items. You must choose one.
Ellipsis (`...`)	Items that can be repeated and used multiple times.

Last updated on 09/08/2026

Commands by Server role

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

A server role describes the primary function of the server. Administrators can choose to dedicate an entire server to one role, or install multiple server roles and sub-roles on a single computer. Each role might include additional command-line tools, installed as part of the role. The following topics provide a list of commands associated with each server role.

- Print Command Reference
- Services for Network File System Command Reference
- Remote Desktop Services (Terminal Services) Command Reference
- Windows Server Backup Command Reference

Feedback

Was this page helpful? Yes No

Print command-line tool reference

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Information and links to each of the associated print command-line tools:

Expand table

Command	Description
lpq	Displays the status of a print queue on a computer running Line printer Daemon (LPD).
lpr	Sends a file to a computer or printer sharing device running the Line printer Daemon (LPD) service in preparation for printing.
Net print	Displays information about a specified printer queue, displays information about a specified print job, or controls a specified print job.
print	Sends a text file to a printer.
prncnfg	Configures or displays configuration information about a printer.
prndrvr	Adds, deletes, and lists printer drivers.
prnjobs	Pauses, resumes, cancels, and lists print jobs.
prnmngr	Adds, deletes, and lists printers or printer connections, in addition to setting and displaying the default printer.
prnport	Creates, deletes, and lists standard TCP/IP printer ports, in addition to displaying and changing port configuration.
prnqctl	Prints a test page, pauses or resumes a printer, and clears a printer queue.
pubprn	Publishes a printer to the active directory directory service.
rundll32 printui.dll,printUIEntry	Enables you to automate the installation and configuration of printers using scripts or the command prompt.

Feedback

Was this page helpful? Yes No

Services for Network File System command-line tools

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Services for Network File System (NFS) provides a file sharing solution that lets you transfer files between computers running Windows Server and UNIX operating systems using the NFS protocol.

Information and links to each of the associated NFS command-line tools:

Expand table

Command	Description
mapadmin	Manage User Name Mapping for Microsoft Services for Network File System.
mount	Mount Network File System (NFS) network shares.
nfsadmin	Manage Server for NFS and Client for NFS.
nfsshare	Control Network File System (NFS) shares.
nfsstat	Display or reset counts of calls made to Server for NFS.
rpcinfo	List programs on remote computers.
showmount	Display mounted directories.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

Remote Desktop Services (Terminal Services) command-line tools reference

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Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Learn about the available Remote Desktop Services (Terminal Services) command-line tools, with descriptions and links for more detailed information.

> **Note**
>
> To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

Expand table

Command	Description
change	Changes the Remote Desktop Session Host server settings for sign in, COM port mappings, and install mode.
change logon	Enables or disables logons from client sessions on an Remote Desktop Session Host server, or displays current logon status.
change port	Lists or changes the COM port mappings to be compatible with MS-DOS applications.
change user	Changes the install mode for the Remote Desktop Session Host server.
chglogon	Enables or disables logons from client sessions on an Remote Desktop Session Host server, or displays current logon status.
chgport	Lists or changes the COM port mappings to be compatible with MS-DOS applications.
chgusr	Changes the install mode for the Remote Desktop Session Host server.
flattemp	Enables or disables flat temporary folders.
logoff	Signs out a user from a session on an Remote Desktop Session Host server and deletes the session from the server.
msg	Sends a message to a user on an Remote Desktop Session Host server.

Command	Description
mstsc	Creates connections to Remote Desktop Session Host servers or other remote computers.
qappsrv	Displays a list of all Remote Desktop Session Host servers on the network.
qprocess	Displays information about processes that are running on an Remote Desktop Session Host server.
query	Displays information about processes, sessions, and Remote Desktop Session Host servers.
query process	Displays information about processes that are running on an Remote Desktop Session Host server.
query session	Displays information about sessions on an Remote Desktop Session Host server.
query termserver	Displays a list of all Remote Desktop Session Host servers on the network.
query user	Displays information about user sessions on an Remote Desktop Session Host server.
quser	Displays information about user sessions on an Remote Desktop Session Host server.
qwinsta	Displays information about sessions on an Remote Desktop Session Host server.
rdpsign	Enables you to digitally sign a Remote Desktop Protocol (.rdp) file.
reset session	Enables you to reset (delete) a session on an Remote Desktop Session Host server.
rwinsta	Enables you to reset (delete) a session on an Remote Desktop Session Host server.
shadow	Enables you to remotely control an active session of another user on an Remote Desktop Session Host server.
tscon	Connects to another session on an Remote Desktop Session Host server.
tsdiscon	Disconnects a session from an Remote Desktop Session Host server.
tskill	Ends a process running in a session on an Remote Desktop Session Host server.
tsprof	Copies the Remote Desktop Services user configuration information from one user to another.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

Windows Server Backup Command Reference

Article • 03/03/2021 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The following subcommands for **wbadmin** provide backup and recovery functionality from a command prompt.

To configure a backup schedule, you must be a member of the **Administrators** group. To perform all other tasks with this command, you must be a member of the **Backup Operators** or the **Administrators** group, or you must have been delegated the appropriate permissions.

You must run **wbadmin** from an elevated command prompt. (To open an elevated command prompt, click **Start**, right-click **Command Prompt**, and then click **Run as administrator**.)

Expand table

Subcommand	Description
Wbadmin enable backup	Configures and enables a daily backup schedule.
Wbadmin disable backup	Disables your daily backups.
Wbadmin start backup	Runs a one-time backup. If used with no parameters, uses the settings from the daily backup schedule.
Wbadmin stop job	Stops the currently running backup or recovery operation.
Wbadmin get versions	Lists details of backups recoverable from the local computer or, if another location is specified, from another computer.
Wbadmin get items	Lists the items included in a specific backup.
Wbadmin start recovery	Runs a recovery of the volumes, applications, files, or folders specified.
Wbadmin get status	Shows the status of the currently running backup or recovery operation.
Wbadmin get disks	Lists disks that are currently online.

Subcommand	Description
Wbadmin start systemstaterecovery	Runs a system state recovery.
Wbadmin start systemstatebackup	Runs a system state backup.
Wbadmin delete systemstatebackup	Deletes one or more system state backups.
Wbadmin start sysrecovery	Runs a recovery of the full system (at least all the volumes that contain the operating system's state). This subcommand is only available if you are using the Windows Recovery Environment.
Wbadmin restore catalog	Recovers a backup catalog from a specified storage location in the case where the backup catalog on the local computer has been corrupted.
Wbadmin delete catalog	Deletes the backup catalog on the local computer. Use this command only if the backup catalog on this computer is corrupted and you have no backups stored at another location that you can use to restore the catalog.

Feedback

Was this page helpful? Yes No

Adprep

10/22/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Applies to: Windows Server (All supported versions)

The adprep command extends the Active Directory schema and updates permissions as necessary to prepare a forest and domain for a domain controller that runs Windows Server.

`Adprep.exe` is a command-line tool available in Windows Server. You must run **adprep** from an elevated command prompt. To open an elevated command prompt, select **Start**, right-select **Command Prompt**, and then select **Run as administrator**.

Adprep is available in a 32-bit version and a 64-bit version. The 64-bit version runs by default. If you need to run Adprep on a 32-bit computer, run the 32-bit version (Adprep32.exe).

For more information about running `Adprep.exe` and how to resolve errors that can occur when you run it, see Running Adprep.exe.

For examples of how this command can be used, see Examples.

For more information about running `adprep /forestprep`, see Prepare a Windows 2000 or Windows Server 2003 Forest Schema for a Domain Controller that Runs Windows Server 2008 or Windows Server 2008 R2.

For more information about running `adprep /domainprep /gpprep`, see Prepare a Windows 2000 or Windows Server 2003 Domain for a Domain Controller That Runs Windows Server 2008 or Windows Server 2008 R2.

For more information about running `adprep /rodcprep`, see Prepare a Forest for a Read-Only Domain Controller.

Syntax

adprep {/forestprep | /domainprep | /domainprep /gpprep | /rodcprep | /wssg | /silent }

Parameters

Expand table

Parameter	Description
/forestprep	Prepares a forest for the introduction of a domain controller that runs Windows Server. You run this command only once in the forest. You must run this command on the domain controller that holds the schema operations master role (also known as flexible single master operations or FSMO) for the forest. You must be a member of all the following groups to run this command: • The Enterprise Admins group • The Schema Admins group • The Domain Admins group of the domain that hosts the schema master
/domainprep	Prepares a domain for the introduction of a domain controller that runs Windows Server. You run this command after the **forestprep** command finishes and after the changes replicate to all the domain controllers in the forest. Run this command in each domain where you plan to add a domain controller that runs Windows Server. You must run this command on the domain controller that holds the infrastructure operations master role for the domain. You must be a member of the Domain Admins group to run this command.
/domainprep /gpprep	Performs similar updates to **domainprep**. However, this command also provides updates that are necessary to enable the Resultant Set of Policy (RSOP) Planning Mode functionality. In Active Directory environments that run Microsoft Windows, this command performs updates during off-peak hours. Replicating updates to file system permissions and Active Directory permissions on existing Group Policy objects (GPOs) during off-peak hours minimizes replication traffic. Run this command after the **forestprep** command finishes and after the changes replicate to all domain controllers in the forest. You must run this command on the infrastructure master for the domain. For more information about running this command in Windows Active Directory environments, see Prepare Your Infrastructure for Upgrade.
/rodcprep	This parameter updates permissions on the application directory partitions to enable replication of the partitions to read-only domain controllers (RODCs). This operation runs remotely; it contacts the infrastructure master in each domain to update the permissions. You need to run this command only once in the forest. However, you can rerun this command anytime if it fails to complete successfully because an infrastructure master isn't available. You can run this command on any computer in the forest. You must be a member of the Enterprise Admins group to run this command.
/wssg	Returns an expanded set of exit codes, instead of just 0 (Success) and 1 (Failure).
/silent	Specifies that no standard output is returned from an operation. This parameter can be used only if /wssg is also used.
quit	Returns to the prior menu.

Parameter	Description
Help	Displays Help for this command.
?	Displays Help for this command.

Remarks

- To prepare an existing Windows or Windows Server Active Directory environment for a Windows Server domain controller, run the version of adprep included in the Windows Server installation media.
- You can also perform verification steps before and after you run the **adprep** command to help ensure that the operations complete successfully. For more information, see Steps for Extending the Schema.

Exit Codes

The following table lists exit codes that Adprep can return after an operation completes.

Expand table

Return Code	Description
1	Failure
2	Schema conflict error
3	FSMO role error
4	Connection error
5	Schema upgrade error
6	Unable to modify error
7	Server busy error
8	Permission error
9	Unable to initialize log file error
10	Not a domain controller
11	In nonnative mode
12	Need to run forest update first

Return Code	Description
13	Forest update already done
14	Domain update already done
15	GPO update already done
16	Forest update wait replication

Examples

The following example prepares a forest for a domain controller that runs Windows Server:

```cli
adprep /forestprep
```

The following example prepares a domain for a domain controller that runs Windows Server:

```cli
adprep /domainprep
```

The following example prepares a domain for an RODC:

```cli
adprep /rodcprep
```

See also

Command-Line Syntax Key

append

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Allows programs to open data files in specified directories as if they were in the current directory. If used without parameters, **append** displays the appended directory list.

> **Note**
>
> This command is not supported in Windows 10.

Syntax

```
append [[<drive>:]<path>[;...]] [/x[:on|:off]] [/path:[:on|:off] [/e]
append ;
```

Parameters

Expand table

Parameter	Description
`[\<drive>:]<path>`	Specifies a drive and directory to append.
/x:on	Applies appended directories to file searches and launching applications.
/x:off	Applies appended directories only to requests to open files. The **/x:off** option is the default setting.
/path:on	Applies appended directories to file requests that already specify a path. **/path:on** is the default setting.
/path:off	Turns off the effect of **/path:on**.
/e	Stores a copy of the appended directory list in an environment variable named APPEND. **/e** may be used only the first time you use **append** after starting your system.

Parameter	Description
;	Clears the appended directory list.
/?	Displays help at the command prompt.

Examples

To clear the appended directory list, type:

```
append ;
```

To store a copy of the appended directory to an environment variable named *append*, type:

```
append /e
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

arp

08/15/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `arp` command displays and modifies entries in the Address Resolution Protocol (ARP) cache. The ARP cache contains one or more tables that are used to store IP addresses and their resolved Ethernet or Token Ring physical addresses. There's a separate table for each Ethernet or Token Ring network adapter installed on your computer.

Syntax

```
arp -s [inet_addr] [eth_addr] [if_addr]
arp -d [inet_addr] [if_addr]
arp -a [inet_addr] [-N if_addr] [-v]
```

Parameters

Expand table

Parameter	Description
`-a`	Displays the current ARP entries by querying the protocol data. If `inet_addr` is specified, only the ARP entry for that IP address is shown. If multiple network interfaces use ARP, entries for each interface are displayed.
`-g`	Functions identically to `-a`.
`-v`	Displays ARP entries in verbose mode, including additional details such as invalid entries and entries on the loopback interface.
`inet_addr`	Specifies the IP address to display, delete, or add to the ARP cache. In the context of `-s`, it's the IP you want to associate with a MAC address.
`-N if_addr`	Shows ARP entries for the network interface with the specified IP address (`if_addr`). This parameter is case-sensitive.
`-d`	Deletes the ARP entry for the specified `inet_addr`. Use `*` as a wildcard to delete all entries.
`-s`	Adds a static ARP entry, associating `inet_addr` with the physical address `eth_addr`. Optionally, specify `if_addr` to add the entry to a specific interface. The entry is permanent until the TCP/IP stack is restarted.

Parameter	Description
`eth_addr`	Specifies the physical (MAC) address to associate with the IP address when using `-s`. The format is six hexadecimal bytes separated by hyphens.
`if_addr`	Specifies the IP address of the network interface whose ARP table is to be modified, used optionally with `-s` and `-d`. If omitted, the first applicable interface is used.
`/?`	Displays help at the command prompt.

Remarks

- The IP addresses for `inetaddr` and `ifaceaddr` are expressed in dotted decimal notation (for example, 10.0.0.1).
- The physical MAC address for `etheraddr` consists of six bytes expressed in hexadecimal notation and separated by hyphens (for example, 00-AA-00-4F-2A-9C).
- Entries added with the `-s` parameter are removed if the TCP/IP protocol is stopped and started. To create permanent static arp cache entries, place the appropriate `arp` commands in a batch file and use Scheduled Tasks to run the batch file at startup.

Examples

To display the arp cache tables for all interfaces, type:

```
arp -a
```

To display the arp cache table for the interface where the IP address *10.0.0.99* is assigned, type:

```
arp -a -N 10.0.0.99
```

To add a static arp cache entry that resolves the IP address *10.0.0.80* to the physical address *00-AA-00-4F-2A-9C*, type:

```
arp -s 10.0.0.80 00-AA-00-4F-2A-9C
```

Related links

- Command-Line Syntax Key

assoc

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Displays or modifies file name extension associations. If used without parameters, **assoc** displays a list of all the current file name extension associations.

> **Note**
>
> This command is only supported within cmd.exe and is not available from PowerShell. Though you can use `cmd /c assoc` as a workaround.

Syntax

```
assoc [<.[ext]>[=[<filetype>]]]
```

> **Note**
>
> In the syntax, square brackets `[ ]` indicate optional items and angle brackets `< >` indicate placeholder values that you replace with your own text. Don't type the brackets. For example, to remove the association for the .jar file name extension, type `assoc .jar=`, not `assoc .jar=<filetype>`. If you type the angle brackets, cmd.exe treats `<` and `>` as redirection operators and returns an error. For more information, see Command-Line Syntax Key.

Parameters

Expand table

Parameter	Description
`<.ext>`	Specifies the file name extension.
`<filetype>`	Specifies the file type to associate with the specified file name extension.

Parameter	Description
/?	Displays help at the command prompt.

Remarks

- To make changes in associations, you need administrator privileges.
- To remove the file type association for a file name extension, type the extension and an equal sign with nothing after it, for example `assoc .ext=`. You don't need to add a trailing space after the equal sign.
- To associate files without extension to a file type, use just a dot (see the examples).
- To view current file types that have open command strings defined, use the **ftype** command.
- To redirect the output of **assoc** to a text file, use the `>` redirection operator.

Examples

To view the current file type association for the file name extension .txt, type:

```
assoc .txt
```

To remove the file type association for the file name extension .bak, type:

```
assoc .bak=
```

Type nothing after the equal sign. Don't type placeholder text such as `<filetype>`, and don't add a trailing space.

To remove the file type association for the file name extension .jar, type:

```
assoc .jar=
```

To view the output of **assoc** one screen at a time, type:

```
assoc | more
```

To send the output of **assoc** to the file assoc.txt, type:

```
assoc>assoc.txt
```

Associate **.log** to text files:

```
assoc .log=txtfile
```

Associate files with no extension to text files:

```
assoc .=txtfile
```

Related links

- Command-Line Syntax Key
- ftype command

Last updated on 09/08/2026

at

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Schedules commands and programs to run on a computer at a specified time and date. You can use **at** only when the Schedule service is running. Used without parameters, **at** lists scheduled commands. You must be a member of the local Administrators group to run this command.

Syntax

```
at [\computername] [[id] [/delete] | /delete [/yes]]
at [\computername] <time> [/interactive] [/every:date[,...] |
/next:date[,...]] <command>
```

Parameters

Expand table

Parameter	Description
`\<computername\>`	Specifies a remote computer. If you omit this parameter, **at** schedules the commands and programs on the local computer.
`<id>`	Specifies the identification number assigned to a scheduled command.
/delete	Cancels a scheduled command. If you omit *ID*, all of the scheduled commands on the computer are canceled.
/yes	Answers yes to all queries from the system when you delete scheduled events.
`<time>`	Specifies the time when you want to run the command. time is expressed as Hours:Minutes in 24-hour notation (that is, 00:00 (midnight) through 23:59).
interactive	Allows *command* to interact with the desktop of the user who is logged on at the time *Command* runs.
every:	Runs *command* on every specified day or days of the week or month (for example, every Thursday, or the third day of every month).

Parameter	Description
`<date>`	Specifies the date when you want to run the command. You can specify one or more days of the week (that is, type **M**,**T**,**W**,**Th**,**F**,**S**,**Su**) or one or more days of the month (that is, type 1 through 31). Separate multiple date entries with commas. If you omit *date*, **at** uses the current day of the month.
next:	Runs *command* on the next occurrence of the day (for example, next Thursday).
`<command>`	Specifies the Windows command, program (that is, .exe or .com file), or batch program (that is, .bat or .cmd file) that you want to run. When the command requires a path as an argument, use the absolute path (that is, the entire path beginning with the drive letter). If the command is on a remote computer, specify Universal Naming Convention (UNC) notation for the server and share name, rather than a remote drive letter.
/?	Displays help at the command prompt.

Remarks

- This command doesn't automatically load cmd.exe before running commands. If you're not running an executable (.exe) file, you must explicitly load cmd.exe at the beginning of the command as follows:

```
cmd /c dir > c:\test.out
```

- If using this command without command-line options, scheduled tasks appear in a table formatted similar to the following:

```
Status  ID   Day        time        Command Line
OK      1    Each F     4:30 PM     net send group leads status due
OK      2    Each M     12:00 AM    chkstor > check.file
OK      3    Each F     11:59 PM    backup2.bat
```

- If including an identification number (*ID*) with this command, only information for a single entry appears in a format similar to the following:

```
Task ID: 1
Status: OK
```

```
Schedule: Each  F
Time of Day: 4:30 PM
Command: net send group leads status due
```

- After you schedule a command, especially a command that has command-line options, check that the command syntax is correct by typing **at** without any command-line options. If the information in the **Command Line** column is wrong, delete the command and retype it. If it's still incorrect, retype the command using fewer command-line options.

- Commands scheduled with **at** run as background processes. Output is not displayed on the computer screen. To redirect output to a file, use the redirection symbol `>`. If you redirect output to a file, you need to use the escape symbol `^` before the redirection symbol, whether you are using **at** at the command line or in a batch file. For example, to redirect output to *output.txt*, type:

  ```
  at 14:45 c:\test.bat ^>c:\output.txt
  ```

  The current directory for the executing command is the systemroot folder.

- If you change the system time after you schedule a command to run, synchronize the **at** scheduler with the revised system time by typing **at** without command-line options.

- Scheduled commands are stored in the registry. As a result, you don't lose scheduled tasks if you restart the Schedule service.

- Do not use a redirected drive for scheduled jobs that access the network. The Schedule service might not be able to access the redirected drive, or the redirected drive might not be present if a different user is logged on at the time the scheduled task runs. Instead, use UNC paths for scheduled jobs. For example:

  ```
  at 1:00pm my_backup \\server\share
  ```

  Do not use the following syntax, where **x:** is a connection made by the user:

  ```
  at 1:00pm my_backup x:
  ```

If you schedule an **at** command that uses a drive letter to connect to a shared directory, include an **at** command to disconnect the drive when you are finished using the drive. If the drive is not disconnected, the assigned drive letter won't be available at the command prompt.

- By default, tasks scheduled using this command will stop after 72 hours. You can modify the registry to change this default value.

  To modify the registry

  > ⊗ **Caution**
  >
  > Incorrectly editing the registry may severely damage your system. Before making changes to the registry, you should back up any valued data on the computer.

  1. Start the registry editor (regedit.exe).
  2. Locate and click the following key in the registry:
     `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Schedule`
  3. On the **Edit** menu, click **Add Value**, and then add the following registry values:
     - **Value Name.** atTaskMaxHours
     - **Data type.** reg_DWOrd
     - **Radix.** Decimal
     - **Value Data:** 0. A value of **0** in the **Value Data** field indicates no limit and doesn't not stop. Values from 1 through 99 indicates the number of hours.

- You can use the Scheduled Tasks folder to view or modify the settings of a task that was created by using this command. When you schedule a task using this command, the task is listed in the Scheduled Tasks folder, with a name such as the following:**at3478**. However, if you modify a task through the Scheduled Tasks folder, it's upgraded to a normal scheduled task. The task is no longer visible to the **at** command, and the at account setting no longer applies to it. You must explicitly enter a user account and password for the task.

Examples

To display a list of commands scheduled on the Marketing server, type:

```
at \\marketing
```

To learn more about a command with the identification number 3 on the Corp server, type:

```
at \\corp 3
```

To schedule a net share command to run on the Corp server at 8:00 A.M. and redirect the listing to the Maintenance server, in the Reports shared directory, and the Corp.txt file, type:

```
at \\corp 08:00 cmd /c net share reports=d:\marketing\reports >>
\\maintenance\reports\corp.txt
```

To back up the hard drive of the Marketing server to a tape drive at midnight every five days, create a batch program called Archive.cmd, which contains the backup commands, and then schedule the batch program to run, type:

```
at \\marketing 00:00 /every:5,10,15,20,25,30 archive
```

To cancel all commands scheduled on the current server, clear the **at** schedule information as follows:

```
at /delete
```

To run a command that is not an executable (.exe) file, precede the command with **cmd /c** to load cmd.exe as follows:

```
cmd /c dir > c:\test.out
```

Related links

- Command-Line Syntax Key
- schtasks. Another command-line scheduling tool.

Feedback

Was this page helpful? Yes No

atmadm

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Monitors connections and addresses that are registered by the atM call Manager on an asynchronous transfer mode (atM) network. You can use **atmadm** to display statistics for incoming and outgoing calls on atM adapters. Used without parameters, **atmadm** displays statistics for monitoring the status of active atM connections.

Syntax

```
atmadm [/c][/a][/s]
```

Parameters

Expand table

Parameter	Description
/c	Displays call information for all current connections to the atM network adapter installed on this computer.
/a	Displays the registered atM network service access point (NSAP) address for each adapter installed in this computer.
/s	Displays statistics for monitoring the status of active atM connections.
/?	Displays help at the command prompt.

Remarks

- The **atmadm /c** command produces output similar to the following:

```
Windows atM call Manager Statistics
atM Connections on Interface : [009] Olicom atM PCI 155 Adapter
   Connection   VPI/VCI   remote address/
```

```
                                Media Parameters (rates in bytes/sec)
    In  PMP SVC    0/193    47000580FFE1000000F21A2E180020481A2E180B
                            Tx:UBR,Peak 0,Avg 0,MaxSdu 1516
                            Rx:UBR,Peak 16953936,Avg 16953936,MaxSdu 1516
    Out P-P SVC    0/192    47000580FFE1000000F21A2E180020481A2E180B
                            Tx:UBR,Peak 16953936,Avg 16953936,MaxSdu 1516
                            Rx:UBR,Peak 16953936,Avg 16953936,MaxSdu 1516
    In  PMP SVC    0/191    47000580FFE1000000F21A2E180020481A2E180B
                            Tx:UBR,Peak 0,Avg 0,MaxSdu 1516
                            Rx:UBR,Peak 16953936,Avg 16953936,MaxSdu 1516
    Out P-P SVC    0/190    47000580FFE1000000F21A2E180020481A2E180B
                            Tx:UBR,Peak 16953936,Avg 16953936,MaxSdu 1516
                            Rx:UBR,Peak 16953936,Avg 16953936,MaxSdu 1516
    In  P-P SVC    0/475    47000580FFE1000000F21A2E180000C110081501
                            Tx:UBR,Peak 16953984,Avg 16953984,MaxSdu 9188
                            Rx:UBR,Peak 16953936,Avg 16953936,MaxSdu 9188
    Out PMP SVC    0/194    47000580FFE1000000F21A2E180000C110081501 (0)
                            Tx:UBR,Peak 16953984,Avg 16953984,MaxSdu 9180
                            Rx:UBR,Peak 0,Avg 0,MaxSdu 0
    Out P-P SVC    0/474    4700918100000000613E5BFE010000C110081500
                            Tx:UBR,Peak 16953984,Avg 16953984,MaxSdu 9188
                            Rx:UBR,Peak 16953984,Avg 16953984,MaxSdu 9188
    In  PMP SVC    0/195    47000580FFE1000000F21A2E180000C110081500
                            Tx:UBR,Peak 0,Avg 0,MaxSdu 0
                            Rx:UBR,Peak 16953936,Avg 16953936,MaxSdu 9180
```

The following table contains descriptions of each element in the **atmadm /c** sample output.

Expand table

Type of Data	Screen Display	Description
Connection Information	In/Out	Direction of the call. **In** is to the atM network adapter from another device. **Out** is from the atM network adapter to another device.
PMP	Point-to-multipoint call.	
P-P	Point-to-point call.	
SVC	Connection is on a switched virtual circuit.	
PVC	Connection is on a permanent virtual circuit.	
VPI/VCI Information	VPI/VCI	Virtual path and virtual channel of

Type of Data	Screen Display	Description
		the incoming or outgoing call.
Remote address/Media Parameters	47000580FFE1000000F21A2E180000C110081500	NSAP address of the calling **(In)** or called **(Out)** atM device.
Tx	The **Tx** parameter includes the following three elements: ◦ Default or specified bit-rate type (UBR, CBR, VBR, or ABR) ◦ Default or specified line speed ◦ Specified service data unit (SDU) size.	
Rx	The **Rx** parameter includes the following three elements: ◦ Default or specified bit-rate type (UBR, CBR, VBR, or ABR) ◦ Default or specified line speed ◦ Specified SDU size.	

- The **atmadm /a** command produces output similar to the following:

```
Windows atM call Manager Statistics
atM addresses for Interface : [009] Olicom atM PCI 155 Adapter
47000580FFE1000000F21A2E180000C110081500
```

- The **atmadm /s** command produces output similar to the following:

```
Windows atM call Manager Statistics
atM call Manager statistics for Interface : [009] Olicom atM PCI 155
Adapter
Current active calls                        = 4
Total successful Incoming calls             = 1332
Total successful Outgoing calls             = 1297
Unsuccessful Incoming calls                 = 1
Unsuccessful Outgoing calls                 = 1
calls Closed by remote                      = 1302
calls Closed Locally                        = 1323
Signaling and ILMI Packets Sent            = 33655
Signaling and ILMI Packets Received        = 34989
```

The following table contains descriptions of each element in the **atmadm /s** sample output.

Expand table

Call Manager Statistic	Description
Current active calls	Calls currently active on the atM adapter installed on this computer.
Total successful Incoming calls	Calls successfully received from other devices on this atM network.
Total successful Outgoing calls	Calls successfully completed to other atM devices on this network from this computer.
Unsuccessful Incoming calls	Incoming calls that failed to connect to this computer.
Unsuccessful Outgoing calls	Outgoing calls that failed to connect to another device on the network.
Calls Closed by remote	Calls closed by a remote device on the network.
Calls Closed Locally	Calls closed by this computer.
Signaling and ILMI Packets Sent	Number of integrated local management interface (ILMI) packets sent to the switch to which this computer is attempting to connect.
Signaling and ILMI Packets Received	Number of ILMI packets received from the atM switch.

Examples

To display call information for all current connections to the atM network adapter installed on this computer, type:

```
atmadm /c
```

To display the registered atM network service access point (NSAP) address for each adapter installed in this computer, type:

```
atmadm /a
```

To display statistics for monitoring the status of active atM connections, type:

```
atmadm /s
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

attrib

Article • 09/25/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays, sets, or removes attributes assigned to files or directories. If used without parameters, **attrib** displays attributes of all files in the current directory.

Syntax

```
attrib [{+|-}r] [{+|-}a] [{+|-}s] [{+|-}h] [{+|-}o] [{+|-}i] [{+|-}x]
[{+|-}p] [{+|-}u] [{+|-}b] [<drive>:][<path>][<filename>] [/s [/d] [/l]]
```

Parameters

Expand table

Parameter	Description
`{+|-}r`	Sets (+) or clears (-) the Read-only file attribute.
`{+|-}a`	Sets (+) or clears (-) the Archive file attribute. This attribute set marks files that have changed since the last time they were backed up. The **xcopy** command uses archive attributes.
`{+|-}s`	Sets (+) or clears (-) the System file attribute. If a file uses this attribute set, you must clear the attribute before you can change any other attributes for the file.
`{+|-}h`	Sets (+) or clears (-) the Hidden file attribute. If a file uses this attribute set, you must clear the attribute before you can change any other attributes for the file.
`{+|-}o`	Sets (+) or clears (-) the Offline file attribute.
`{+|-}i`	Sets (+) or clears (-) the Not Content Indexed file attribute.
`{+|-}x`	Sets (+) or clears (-) the Scrub file attribute.
`{+|-}p`	Sets (+) or clears (-) the Pinned file attribute.

Parameter	Description
`{+|-}u`	Sets (+) or clears (-) the Unpinned file attribute.
`{+|-}b`	Sets (+) or clears (-) the SMR Blob file attribute.
`[<drive>:][<path>][<filename>]`	Specifies the location and name of the directory, file, or group of files that you want to view or change attributes. You can use the **?** and * wildcard characters in the *filename* parameter to display or change the attributes for a group of files.
/s	Applies **attrib** and any command-line options to matching files in the current directory and all of its subdirectories.
/d	Applies **attrib** and any command-line options to directories.
/l	Applies **attrib** and any command-line options to the Symbolic Link, rather than the target of the Symbolic Link.
/?	Displays help at the command prompt.

Examples

To display the attributes of a file named News86 that is located in the current directory, type:

```
attrib news86
```

To assign the Read-only attribute to the file named report.txt, type:

```
attrib +r report.txt
```

To remove the Read-only attribute from files in the public directory and its subdirectories on a disk in drive b:, type:

```
attrib -r b:\public\*.* /s
```

To set the Archive attribute for all files on drive a:, and then clear the Archive attribute for files with the .bak extension, type:

```
attrib +a a:*.* & attrib -a a:*.bak
```

Related links

- Command-Line Syntax Key
- xcopy command

Feedback

Was this page helpful? Yes No

auditpol

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays information about and performs functions to manipulate audit policies, including:

- Setting and querying a system audit policy.
- Setting and querying a per-user audit policy.
- Setting and querying auditing options.
- Setting and querying the security descriptor used to delegate access to an audit policy.
- Reporting or backing up an audit policy to a comma-separated value (CSV) text file.
- Loading an audit policy from a CSV text file.
- Configuring global resource SACLs.

Syntax

```
auditpol command [<sub-command><options>]
```

Parameters

Expand table

Sub-command	Description
/get	Displays the current audit policy. For more information, see auditpol get for syntax and options.
/set	Sets the audit policy. For more information, see auditpol set for syntax and options.

Sub-command	Description
/list	Displays selectable policy elements. For more information, see auditpol list for syntax and options.
/backup	Saves the audit policy to a file. For more information, see auditpol backup for syntax and options.
/restore	Restores the audit policy from a file that was previously created by using auditpol /backup. For more information, see auditpol restore for syntax and options.
/clear	Clears the audit policy. For more information, see auditpol clear for syntax and options.
/remove	Removes all per-user audit policy settings and disables all system audit policy settings. For more information, see auditpol remove for syntax and options.
/resourceSACL	Configures global resource system access control lists (SACLs). **Note:** Applies only to Windows 7 and Windows Server 2008 R2. For more information, see auditpol resourceSACL.
/?	Displays help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

auditpol backup

Article • 05/22/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Applies to: Windows Server 2022, Windows Server 2019, Windows Server 2016, Windows Server 2012 R2, Windows Server 2012

The auditpol backup command backs up system audit policy settings, per-user audit policy settings for all users, and all auditing options to a comma-separated value (CSV) text file.

To perform *backup* operations on the *per-user* and *system* policies, you need **Write** or **Full Control** permissions for that object set in the security descriptor. You can also perform *backup* operations if you have the **Manage auditing and security log** (SeSecurityPrivilege) user right. However, this right allows access that isn't necessary to perform the overall *backup* operations.

Syntax

```
auditpol /backup /file:<filename>
```

Parameters

Expand table

Parameter	Description
/file	Specifies the file name of the for the back up.
/?	Displays help at the command prompt.

Examples

You can back up the following settings to a CSV-formatted text file:

- Per-user audit policy settings for all users

- System audit policy settings
- All auditing options

To back up these settings to a file named auditpolicy.csv, type the following command:

```
auditpol /backup /file:C:\auditpolicy.csv
```

> **Note**
>
> If no drive is specified, the current directory is used.

Related links

- Command-Line Syntax Key
- auditpol restore
- auditpol commands

Feedback

Was this page helpful? Yes No

auditpol clear

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Deletes the per-user audit policy for all users, resets (disables) the system audit policy for all subcategories, and sets all the auditing options to disabled.

To perform *clear* operations on the *per-user* and *system* policies, you must have **Write** or **Full Control** permission for that object set in the security descriptor. You can also perform *clear* operations if you have the **Manage auditing and security log** (SeSecurityPrivilege) user right. However, this right allows additional access that is not necessary to perform the overall *clear* operations.

Syntax

```
auditpol /clear [/y]
```

Parameters

Expand table

Parameter	Description
/y	Suppresses the prompt to confirm if all audit policy settings should be cleared.
/?	Displays help at the command prompt.

Examples

To delete the per-user audit policy for all users, reset (disable) the system audit policy for all subcategories, and set all the audit policy settings to disabled, at a confirmation prompt, type:

```
auditpol /clear
```

To delete the per-user audit policy for all users, reset the system audit policy settings for all subcategories, and set all the audit policy settings to disabled, without a confirmation prompt, type:

```
auditpol /clear /y
```

> **Note**
>
> The preceding example is useful when using a script to perform this operation.

Related links

- Command-Line Syntax Key
- auditpol commands

Feedback

Was this page helpful? Yes No

auditpol get

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Retrieves the system policy, per-user policy, auditing options, and audit security descriptor object.

To perform *get* operations on the *per-user* and *system* policies, you must have **Read** permission for that object set in the security descriptor. You can also perform *get* operations if you have the **Manage auditing and security log** (SeSecurityPrivilege) user right. However, this right allows additional access that is not necessary to perform the overall *get* operations.

Syntax

```
auditpol /get
[/user[:<username>|<{sid}>]]
[/category:*|<name>|<{guid}>[,:<name|<{guid}> ]]
[/subcategory:*|<name>|<{guid}>[,:<name|<{guid}> ]]
[/option:<option name>]
[/sd]
[/r]
```

Parameters

Expand table

Parameter	Description
/user	Displays the security principal for whom the per-user audit policy is queried. Either the /category or /subcategory parameter must be specified. The user may be specified as a security identifier (SID) or name. If no user account is specified, then the system audit policy is queried.
/category	One or more audit categories specified by globally unique identifier (GUID) or name. An asterisk (*) may be used to indicate that all audit categories should be queried.
/subcategory	One or more audit subcategories specified by GUID or name.

Parameter	Description
/sd	Retrieves the security descriptor used to delegate access to the audit policy.
/option	Retrieves the existing policy for the CrashOnAuditFail, FullprivilegeAuditing, AuditBaseObjects, or AuditBasedirectories options.
/r	Displays the output in report format, comma-separated value (CSV).
/?	Displays help at the command prompt.

Remarks

All categories and subcategories can be specified by the GUID or name enclosed by quotation marks ("). Users can be specified by SID or name.

Examples

To retrieve the per-user audit policy for the Guest account and display the output for the System, detailed Tracking, and Object Access categories, type:

```
auditpol /get /user:{S-1-5-21-1443922412-3030960370-963420232-51}
/category:System,detailed Tracking,Object Access
```

> **Note**
>
> This command is useful in two scenarios. 1) When monitoring a specific user account for suspicious activity, you can use the `/get` command to retrieve the results in specific categories by using an inclusion policy to enable additional auditing. 2) if audit settings on an account are logging numerous but superfluous events, you can use the `/get` command to filter out extraneous events for that account with an exclusion policy. For a list of all categories, use the `auditpol /list /category` command.

To retrieve the per-user audit policy for a category and a particular subcategory, which reports the inclusive and exclusive settings for that subcategory under the System category for the Guest account, type:

```
auditpol /get /user:guest /category:System /subcategory:{0ccee921a-69ae-
11d9-bed3-505054503030}
```

To display the output in report format and include the computer name, policy target, subcategory, subcategory GUID, inclusion settings, and exclusion settings, type:

```
auditpol /get /user:guest /category:detailed Tracking /r
```

To retrieve the policy for the System category and subcategories, which reports the category and subcategory policy settings for the system audit policy, type:

```
auditpol /get /category:System /subcategory:{0ccee921a-69ae-11d9-bed3-
505054503030}
```

To retrieve the policy for the detailed Tracking category and subcategories in report format and include the computer name, policy target, subcategory, subcategory GUID, inclusion settings, and exclusion settings, type:

```
auditpol /get /category:detailed Tracking /r
```

To retrieve the policy for two categories with the categories specified as GUIDs, which reports all the audit policy settings of all the subcategories under two categories, type:

```
auditpol /get /category:{69979849-797a-11d9-bed3-505054503030},{69997984a-
797a-11d9-bed3-505054503030} subcategory:{0ccee921a-69ae-11d9-bed3-
505054503030}
```

To retrieve the state, either enabled or disabled, of the AuditBaseObjects option, type:

```
auditpol /get /option:AuditBaseObjects
```

Where the available options are AuditBaseObjects, AuditBaseOperations, and FullprivilegeAuditing. To retrieve the state enabled, disabled, or 2 of the

CrashOnAuditFail option, type:

```
auditpol /get /option:CrashOnAuditFail /r
```

Related links

- Command-Line Syntax Key
- auditpol commands

Feedback

Was this page helpful? Yes No

auditpol list

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Lists audit policy categories and subcategories, or lists users for whom a per-user audit policy is defined.

To perform *list* operations on the *per-user* policy, you must have **Read** permission for that object set in the security descriptor. You can also perform *list* operations if you have the **Manage auditing and security log** (SeSecurityPrivilege) user right. However, this right allows additional access that is not necessary to perform the overall *list* operations.

Syntax

```
auditpol /list
[/user|/category|subcategory[:<categoryname>|<{guid}>|*]]
[/v] [/r]
```

Parameters

Expand table

Parameter	Description
/user	Retrieves all users for whom the per-user audit policy has been defined. If used with the /v parameter, the security identifier (SID) of the user is also displayed.
/category	Displays the names of categories understood by the system. If used with the /v parameter, the category globally unique identifier (GUID) is also displayed.
/subcategory	Displays the names of subcategories and their associated GUID.
/v	Displays the GUID with the category or subcategory, or when used with /user, displays the SID of each user.
/r	Displays the output as a report in comma-separated value (CSV) format.
/?	Displays help at the command prompt.

Examples

To list all users who have a defined audit policy, type:

```
auditpol /list /user
```

To list all users who have a defined audit policy and their associated SID, type:

```
auditpol /list /user /v
```

To list all categories and subcategories in report format, type:

```
auditpol /list /subcategory:* /r
```

To list the subcategories of the detailed Tracking and DS Access categories, type:

```
auditpol /list /subcategory:detailed Tracking,DS Access
```

Related links

- Command-Line Syntax Key
- auditpol commands

Feedback

Was this page helpful? Yes No

auditpol remove

Article • 05/22/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Applies to: Windows Server 2022, Windows Server 2019, Windows Server 2016, Windows Server 2012 R2, Windows Server 2012

The auditpol remove command removes the per-user audit policy for a specified account or all accounts.

To perform *remove* operations on the *per-user* policy, you must have **Write** or **Full Control** permissions for that object set in the security descriptor. You can also perform *remove* operations if you have the **Manage auditing and security log** (`SeSecurityPrivilege`) user right. However, this right allows access that isn't necessary to perform the overall *remove* operations.

Syntax

```
auditpol /remove [/user[:<username>|<{SID}>]]
[/allusers]
```

Parameters

Expand table

Parameter	Description
/user	Specifies the security identifier (SID) or user name for the user for whom the per-user audit policy is to be deleted.
/allusers	Removes the per-user audit policy for all users.
/?	Displays help at the command prompt.

Examples

To remove the per-user audit policy for user `mikedan` by name, type:

```
auditpol /remove /user:mikedan
```

To remove the per-user audit policy for user `mikedan` by SID, type:

```
auditpol /remove /user:{S-1-5-21-397123471-12346959}
```

To remove the per-user audit policy for all users, type:

```
auditpol /remove /allusers
```

Related links

- Command-Line Syntax Key
- auditpol commands

Feedback

Was this page helpful? Yes No

auditpol resourceSACL

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Configures global resource system access control lists (SACLs).

To perform *resourceSACL* operations, you must have **Write** or **Full Control** permissions for that object set in the security descriptor. You can also perform *resourceSACL* operations if you have the **Manage auditing and security log** (SeSecurityPrivilege) user right.

Syntax

```
auditpol /resourceSACL
[/set /type:<resource> [/success] [/failure] /user:<user> [/access:<access
flags>]]
[/remove /type:<resource> /user:<user> [/type:<resource>]]
[/clear [/type:<resource>]]
[/view [/user:<user>] [/type:<resource>]]
```

Parameters

Expand table

Parameter	Description
/set	Adds a new entry to or updates an existing entry in the resource SACL for the resource type specified.
/remove	Removes all entries for the given user in the global object access auditing list.
/clear	Removes all entries from the global object access auditing list.
/view	Lists the global object access auditing entries in a resource SACL. The user and resource types are optional.
/?	Displays help at the command prompt.

Arguments

Expand table

Argument	Description
/type	The resource for which object access auditing is being configured. The supported, case-sensitive, argument values are *File* (for directories and files) and *Key* (for registry keys).
/success	Specifies success auditing.
/failure	Specifies failure auditing.
/user	Specifies a user in one of the following forms: • DomainName\Account (such as DOM\Administrators) • StandaloneServer\Group Account (see LookupAccountName function) • {S-1-x-x-x-x} (x is expressed in decimal, and the entire SID must be enclosed in curly braces). For example: {S-1-5-21-5624481-130208933-164394174-1001} **Note:** If the SID form is used, no check is done to verify the existence of this account.
/access	Specifies a permission mask that can be specified through: Generic access rights, including: • GA - GENERIC ALL • GR - GENERIC READ • GW - GENERIC WRITE • GX - GENERIC EXECUTE Access rights for files, including: • FA - FILE ALL ACCESS • FR - FILE GENERIC READ • FW - FILE GENERIC WRITE • FX - FILE GENERIC EXECUTE Access rights for registry keys, including: • KA - KEY ALL ACCESS • KR - KEY READ • KW - KEY WRITE • KX - KEY EXECUTE For example: `/access:FRFW` enables audit events for read and write operations. A hexadecimal value representing the access mask (such as 0x1200a9) This is useful when using resource-specific bit masks that are not part of the security descriptor definition language (SDDL) standard. If omitted, Full access is used.

Examples

To set a global resource SACL to audit successful access attempts by a user on a registry key:

```
auditpol /resourceSACL /set /type:Key /user:MYDOMAIN\myuser /success
```

To set a global resource SACL to audit successful and failed attempts by a user to perform generic read and write functions on files or folders:

```
auditpol /resourceSACL /set /type:File /user:MYDOMAIN\myuser /success
/failure /access:FRFW
```

To remove all global resource SACL entries for files or folders:

```
auditpol /resourceSACL /type:File /clear
```

To remove all global resource SACL entries for a particular user from files or folders:

```
auditpol /resourceSACL /remove /type:File /user:{S-1-5-21-56248481-
1302087933-1644394174-1001}
```

To list the global object access auditing entries set on files or folders:

```
auditpol /resourceSACL /type:File /view
```

To list the global object access auditing entries for a particular user that are set on files or folders:

```
auditpol /resourceSACL /type:File /view /user:MYDOMAIN\myuser
```

Related links

- Command-Line Syntax Key
- auditpol commands

Feedback

Was this page helpful? Yes No

auditpol restore

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Restores system audit policy settings, per-user audit policy settings for all users, and all auditing options from a file that is syntactically consistent with the comma-separated value (CSV) file format used by the /backup option.

To perform *restore* operations on the *per-user* and *system* policies, you must have **Write** or **Full Control** permission for that object set in the security descriptor. You can also perform *restore* operations if you have the **Manage auditing and security log** (SeSecurityPrivilege) user right, which is useful when restoring the security descriptor in the event of an error or malicious attack.

Syntax

```
auditpol /restore /file:<filename>
```

Parameters

Expand table

Parameter	Description
/file	Specifies the file from which the audit policy should be restored. The file must have been created by using the /backup option or must be syntactically consistent with the CSV file format used by the /backup option.
/?	Displays help at the command prompt.

Examples

To restore system audit policy settings, per-user audit policy settings for all users, and all auditing options from a file named auditpolicy.csv that was created by using the /backup command, type:

```
auditpol /restore /file:c:\auditpolicy.csv
```

Related links

- Command-Line Syntax Key
- auditpol backup
- auditpol commands

Feedback

Was this page helpful? Yes No

auditpol set

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the per-user audit policy, system audit policy, or auditing options.

To perform *set* operations on the *per-user* and *system* policies, you must have **Write** or **Full Control** permission for that object set in the security descriptor. You can also perform *set* operations if you have the **Manage auditing and security log** (SeSecurityPrivilege) user right. However, this right allows additional access that is not necessary to perform the overall *set* operations.

Syntax

```
auditpol /set
[/user[:<username>|<{sid}>][/include][/exclude]]
[/category:<name>|<{guid}>[,:<name|<{guid}> ]]
[/success:<enable>|<disable>][/failure:<enable>|<disable>]
[/subcategory:<name>|<{guid}>[,:<name|<{guid}> ]]
[/success:<enable>|<disable>][/failure:<enable>|<disable>]
[/option:<option name> /value: <enable>|<disable>]
```

Parameters

Expand table

Parameter	Description
/user	The security principal for whom the per-user audit policy specified by the category or subcategory is set. Either the category or subcategory option must be specified, as a security identifier (SID) or name.
/include	Specified with /user; indicates that the user's per-user policy will cause an audit to be generated even if it is not specified by the system audit policy. This setting is the default and is automatically applied if neither the /include nor /exclude parameters are explicitly specified.
/exclude	Specified with /user; indicates that the user's per-user policy will cause an audit to be suppressed regardless of the system audit policy. This setting is ignored for

Parameter	Description
	users who are members of the local Administrators group.
/category	One or more audit categories specified by globally unique identifier (GUID) or name. If no user is specified, the system policy is set.
/subcategory	One or more audit subcategories specified by GUID or name. If no user is specified, the system policy is set.
/success	Specifies success auditing. This setting is the default and is automatically applied if neither the /success nor /failure parameters are explicitly specified. This setting must be used with a parameter indicating whether to enable or disable the setting.
/failure	Specifies failure auditing. This setting must be used with a parameter indicating whether to enable or disable the setting.
/option	Sets the audit policy for the CrashOnAuditFail, FullprivilegeAuditing, AuditBaseObjects, or AuditBasedirectories options.
/sd	Sets the security descriptor used to delegate access to the audit policy. The security descriptor must be specified by using the Security Descriptor Definition Language (SDDL). The security descriptor must have a discretionary access control list (DACL).
/?	Displays help at the command prompt.

Examples

To set the per-user audit policy for all subcategories under the detailed Tracking category for the user mikedan so that all the user's successful attempts will be audited, type:

```
auditpol /set /user:mikedan /category:detailed Tracking /include
/success:enable
```

To set the per-user audit policy for categories specified by name and GUID, and subcategories specified by GUID to suppress auditing for any successful or failed attempts, type:

```
auditpol /set /user:mikedan /exclude /category:Object Access,System,
{6997984b-797a-11d9-bed3-505054503030}
```

```
/subcategory:{0ccee9210-69ae-11d9-bed3-505054503030},:{0ccee9211-69ae-11d9-
bed3-505054503030}, /success:enable /failure:enable
```

To set the per-user audit policy for the specified user for all the categories for the suppression of auditing of all but successful attempts, type:

```
auditpol /set /user:mikedan /exclude /category:* /success:enable
```

To set the system audit policy for all subcategories under the detailed Tracking category to include auditing for only successful attempts, type:

```
auditpol /set /category:detailed Tracking /success:enable
```

> **Note**
>
> The failure setting is not altered.

To set the system audit policy for the Object Access and System categories (which is implied because subcategories are listed) and subcategories specified by GUIDs for the suppression of failed attempts and the auditing of successful attempts, type:

```
auditpol /set /subcategory:{0ccee9210-69ae-11d9-bed3-505054503030},
{0ccee9211-69ae-11d9-bed3-505054503030}, /failure:disable /success:enable
```

To set the auditing options to the enabled state for the CrashOnAuditFail option, type:

```
auditpol /set /option:CrashOnAuditFail /value:enable
```

Related links

- Command-Line Syntax Key
- auditpol commands

Feedback

Was this page helpful? Yes No

autochk

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Runs when the computer is started and prior to Windows Server starting to verify the logical integrity of a file system.

Autochk.exe is a version of **chkdsk** that runs only on NTFS disks and only before Windows Server starts. **autochk** cannot be run directly from the command-line. Instead, **autochk** runs in the following situations:

- If you try to run **chkdsk** on the boot volume.
- If **chkdsk** cannot gain exclusive use of the volume.
- If the volume is flagged as dirty.

Remarks

> ⚠ **Warning**
>
> The **autochk** command-line tool cannot be directly run from the command-line. Instead, use the **chkntfs** command-line tool to configure the way you want **autochk** to run at startup.
>
> - You can use **chkntfs** with the **/x** parameter to prevent **autochk** from running on a specific volume or multiple volumes.
> - Use the **chkntfs.exe** command-line tool with the **/t** parameter to change the autochk delay from 0 seconds to up to 3 days (259,200 seconds). However, a long delay means that the computer does not start until the time elapses or until you press a key to cancel **autochk**.

Related links

- Command-Line Syntax Key
- chkdsk command

- chkntfs command

Feedback

Was this page helpful? Yes No

autoconv

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Converts file allocation table (Fat) and Fat32 volumes to the NTFS file system, leaving existing files and directories intact at startup after **autochk** runs. volumes converted to the NTFS file system cannot be converted back to Fat or Fat32.

> **Important**
>
> You can't run **autoconv** from the command-line. This can only run at startup, if set through **convert.exe**.

Related links

- Command-Line Syntax Key
- autochk command
- convert command

Feedback

Was this page helpful? Yes No

autofmt

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Starts the Auto File System Format Utility, which formats a drive or partition when called from the Windows Recovery Console.

> **Important**
>
> You cannot run **autofmt** directly from the command-line.

Related links

- Command-Line Syntax Key
- Windows Recovery Environment (WinRE)
- How to use Windows Recovery Environment (WinRE) to troubleshoot common startup issues

Feedback

Was this page helpful? Yes No

bcdboot

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Enables you to quickly set up a system partition, or to repair the boot environment located on the system partition. The system partition is set up by copying a simple set of Boot Configuration Data (BCD) files to an existing empty partition.

Syntax

```
bcdboot <source> [/l] [/s]
```

Parameters

Expand table

Parameter	Description
source	Specifies the location of the Windows directory to use as the source for copying boot environment files.
/l	Specifies the locale. The default locale is US English.
/s	Specifies the volume letter of the system partition. The default is the system partition identified by the firmware.

Examples

For information about where to find BCDboot and examples of how to use this command, see the BCDboot Command-Line Options topic.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

bcdedit

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Boot Configuration Data (BCD) files provide a store that is used to describe boot applications and boot application settings. The objects and elements in the store effectively replace Boot.ini.

BCDEdit is a command-line tool for managing BCD stores. It can be used for a variety of purposes, including creating new stores, modifying existing stores, adding boot menu parameters, and so on. BCDEdit serves essentially the same purpose as Bootcfg.exe on earlier versions of Windows, but with two major improvements:

- Exposes a wider range of boot parameters than Bootcfg.exe.
- Has improved scripting support.

> ⓘ **Note**
>
> Administrative privileges are required to use BCDEdit to modify BCD.

BCDEdit is the primary tool for editing the boot configuration of Windows Vista and later versions of Windows. It is included with the Windows Vista distribution in the %WINDIR%\System32 folder.

BCDEdit is limited to the standard data types and is designed primarily to perform single common changes to BCD. For more complex operations or nonstandard data types, consider using the BCD Windows Management Instrumentation (WMI) application programming interface (API) to create more powerful and flexible custom tools.

Syntax

```
bcdedit /command [<argument1>] [<argument2>] ...
```

Parameters

General BCDEdit Command-Line Options

Expand table

Option	Description
/?	Displays a list of BCDEdit commands. Running this command without an argument displays a summary of the available commands. To display detailed help for a particular command, run **bcdedit /?** `<command>`, where `<command>` is the name of the command you are searching for more information about. For example, **bcdedit /? createstore** displays detailed help for the Createstore command.

Parameters that Operate on a Store

Expand table

Option	Description
/createstore	Creates a new empty boot configuration data store. The created store is not a system store.
/export	Exports the contents of the system store into a file. This file can be used later to restore the state of the system store. This command is valid only for the system store.
/import	Restores the state of the system store by using a backup data file previously generated by using the **/export** option. This command deletes any existing entries in the system store before the import takes place. This command is valid only for the system store.
/store	This option can be used with most BCDedit commands to specify the store to be used. If this option is not specified, then BCDEdit operates on the system store. Running the **bcdedit /store** command by itself is equivalent to running the **bcdedit /enum active** command.

Parameters that Operate on Entries in a Store

Expand table

Parameter	Description
/copy	Makes a copy of a specified boot entry in the same system store.
/create	Creates a new entry in the boot configuration data store. If a well-known identifier is specified, then the **/application**, **/inherit**, and **/device** parameters cannot be

Parameter	Description
	specified. If an identifier is not specified or not well known, an **/application**, **/inherit**, or **/device** option must be specified.
/delete	Deletes an element from a specified entry.

Parameters that Operate on Entry Options

Expand table

Parameter	Description
/deletevalue	Deletes a specified element from a boot entry.
/set	Sets an entry option value.

Parameters that Control Output

Expand table

Parameter	Description
/enum	Lists entries in a store. The **/enum** option is the default value for BCEdit, so running the **bcdedit** command without parameters is equivalent to running the **bcdedit /enum active** command.
/v	Verbose mode. Usually, any well-known entry identifiers are represented by their friendly shorthand form. Specifying **/v** as a command-line option displays all identifiers in full. Running the **bcdedit /v** command by itself is equivalent to running the **bcdedit /enum active /v** command.

Parameters that Control the Boot Manager

Expand table

Parameter	Description
/bootsequence	Specifies a one-time display order to be used for the next boot. This command is similar to the **/displayorder** option, except that it is used only the next time the computer starts. Afterwards, the computer reverts to the original display order.
/default	Specifies the default entry that the boot manager selects when the timeout expires.

Parameter	Description
/displayorder	Specifies the display order that the boot manager uses when displaying boot parameters to a user.
/timeout	Specifies the time to wait, in seconds, before the boot manager selects the default entry.
/toolsdisplayorder	Specifies the display order for the boot manager to use when displaying the **Tools** menu.

Parameters that Control Emergency Management Services

Expand table

Parameter	Description
/bootems	Enables or disables Emergency Management Services (EMS) for the specified entry.
/ems	Enables or disables EMS for the specified operating system boot entry.
/emssettings	Sets the global EMS settings for the computer. **/emssettings** does not enable or disable EMS for any particular boot entry.

Parameters that Control Debugging

Expand table

Parameter	Description
/bootdebug	Enables or disables the boot debugger for a specified boot entry. Although this command works for any boot entry, it is effective only for boot applications.
/dbgsettings	Specifies or displays the global debugger settings for the system. This command does not enable or disable the kernel debugger; use the **/debug** option for that purpose. To set an individual global debugger setting, use the **bcdedit /set** `<dbgsettings> <type> <value>` command.
/debug	Enables or disables the kernel debugger for a specified boot entry.

Related links

For examples of how to use BCDEdit, see the BCDEdit Options Reference article.

To see the notation used to indicate command-line syntax, see Command-Line Syntax Key.

Feedback

Was this page helpful? Yes No

bdehdcfg

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Prepares a hard drive with the partitions necessary for BitLocker Drive Encryption. Most installations of Windows 7 will not need to use this tool because BitLocker setup includes the ability to prepare and repartition drives as required.

> ⚠ **Warning**
>
> There is a known conflict with the **Deny write access to fixed drives not protected by BitLocker** Group Policy setting located in **Computer Configuration\Administrative Templates\Windows Components\BitLocker Drive Encryption\Fixed Data Drives**.
>
> If bdehdcfg is run on a computer when this policy setting is enabled, you may encounter the following issues:
>
> - If you attempted to shrink the drive and create the system drive, the drive size will be successfully reduced and a raw partition will be created. However, the raw partition will not be formatted. The following error message is displayed: The new active Drive cannot be formatted. You may need to manually prepare your drive for BitLocker.
> - If you attempted to use unallocated space to create the system drive, a raw partition will be created. However, the raw partition will not be formatted. The following error message is displayed: The new active Drive cannot be formatted. You may need to manually prepare your drive for BitLocker.
> - If you attempted to merge an existing drive into the system drive, the tool will fail to copy the required boot file onto the target drive to create the system drive. The following error message is displayed: BitLocker setup failed to copy boot files. You may need to manually prepare your drive for BitLocker.
> - If this policy setting is being enforced, a hard drive cannot be repartitioned because the drive is protected. If you are upgrading computers in your organization from a previous version of Windows and those computers were

configured with a single partition, you should create the required BitLocker system partition before applying the policy setting to the computers.

Syntax

```
bdehdcfg [-driveinfo <drive_letter>] [-target {default|unallocated|
<drive_letter> shrink|<drive_letter> merge}] [-newdriveletter] [-size
<size_in_mb>] [-quiet]
```

Parameters

Expand table

Parameter	Description
bdehdcfg: driveinfo	Displays the drive letter, the total size, the maximum free space, and the partition characteristics of the partitions on the drive specified. Only valid partitions are listed. Unallocated space is not listed if four primary or extended partitions already exist.
bdehdcfg: target	Defines which portion of a drive to use as the system drive and makes the portion active.
bdehdcfg: newdriveletter	Assigns a new drive letter to the portion of a drive used as the system drive.
bdehdcfg: size	Determines the size of the system partition when a new system drive is being created.
bdehdcfg: quiet	Prevents the display of all actions and errors in the command-line interface and directs bdehdcfg to use the Yes answer to any Yes/No prompts that may occur during subsequent drive preparation.
bdehdcfg: restart	Directs the computer to restart after the drive preparation has finished.
/?	Displays Help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

bdehdcfg: driveinfo

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays the drive letter, the total size, the maximum free space, and the partition characteristics. Only valid partitions are listed. Unallocated space is not listed if four primary or extended partitions already exist.

> **Note**
>
> This command is informational only and makes no changes to the drive.

Syntax

```
bdehdcfg -driveinfo <drive_letter>
```

Parameters

Expand table

Parameter	Description
<drive_letter>	Specifies a drive letter followed by a colon.

Example

To display the drive information for the C: drive:

```
bdehdcfg  driveinfo C:
```

Related links

- Command-Line Syntax Key
- bdehdcfg

Feedback

Was this page helpful? Yes No

bdehdcfg: newdriveletter

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Assigns a new drive letter to the portion of a drive used as the system drive. As a best practice, we recommend not assigning a drive letter to your system drive.

Syntax

```
bdehdcfg -target {default|unallocated|<drive_letter> shrink|<drive_letter>
merge} -newdriveletter <drive_letter>
```

Parameters

Expand table

Parameter	Description
`<drive_letter>`	Defines the drive letter that will be assigned to the specified target drive.

Examples

To assign the default drive the drive letter `P`:

```
bdehdcfg -target default -newdriveletter P:
```

Related links

- Command-Line Syntax Key
- bdehdcfg

Feedback

Was this page helpful? Yes No

bdehdcfg: quiet

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Informs the bdehdcfg command-line tool that all actions and errors are not to be displayed in the command-line interface. Any Yes/No (Y/N) prompts displayed during the drive preparation will assume a "Yes" answer. To view any error that occurred during drive preparation, review the system event log under the **Microsoft-Windows-BitLocker-DrivePreparationTool** event provider.

Syntax

```
bdehdcfg -target {default|unallocated|<drive_letter> shrink|<drive_letter>
merge} -quiet
```

Parameters

This command has no additional parameters.

Examples

To use the **quiet** command:

```
bdehdcfg -target default -quiet
```

Related links

- Command-Line Syntax Key
- bdehdcfg

Feedback

Was this page helpful? Yes No

bdehdcfg: restart

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Informs the bdehdcfg command-line tool that the computer should be restarted after the drive preparation has concluded. If other users are logged on to the computer and the **quiet** command is not specified, a prompt appears to confirm that the computer should be restarted.

Syntax

```
bdehdcfg -target {default|unallocated|<drive_letter> shrink|<drive_letter>
merge} -restart
```

Parameters

This command has no additional parameters.

Examples

To use the **restart** command:

```
bdehdcfg -target default -restart
```

Related links

- Command-Line Syntax Key
- bdehdcfg

Feedback

Was this page helpful? Yes No

bdehdcfg: size

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Specifies the size of the system partition when a new system drive is being created. If you do not specify a size, the tool will use the default value of 300 MB. The minimum size of the system drive is 100 MB. If you will store system recovery or other system tools on the system partition, you should increase the size accordingly.

> **Note**
>
> The **size** command cannot be combined with the `target <drive_letter> merge` command.

Syntax

```
bdehdcfg -target {default|unallocated|<drive_letter> shrink} -size
<size_in_mb>
```

Parameters

Expand table

Parameter	Description
`<size_in_mb>`	Indicates the number of megabytes (MB) to use for the new partition.

Examples

To allocate 500 MB to the default system drive:

```
bdehdcfg -target default -size 500
```

Related links

- Command-Line Syntax Key
- bdehdcfg

Feedback

Was this page helpful? Yes No

bdehdcfg: target

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Prepares a partition for use as a system drive by BitLocker and Windows Recovery. By default, this partition is created without a drive letter.

Syntax

```
bdehdcfg -target {default|unallocated|<drive_letter> shrink|<drive_letter>
merge}
```

Parameters

Expand table

Parameter	Description
default	Indicates that the command-line tool will follow the same process as the BitLocker setup wizard.
unallocated	Creates the system partition out of the unallocated space available on the disk.
`<drive_letter>` shrink	Reduces the drive specified by the amount necessary to create an active system partition. To use this command, the drive specified must have at least 5 percent free space.
`<drive_letter>` merge	Uses the drive specified as the active system partition. The operating system drive cannot be a target for merge.

Examples

To designate an existing drive (P) as the system drive:

```
bdehdcfg -target P: merge
```

Related links

- Command-Line Syntax Key
- bdehdcfg

Feedback

Was this page helpful? Yes No

bitsadmin

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Bitsadmin is a command-line tool used to create, download or upload jobs, and to monitor their progress. The bitsadmin tool uses switches to identify the work to perform. You can call `bitsadmin /?` or `bitsadmin /help` to get a list of switches.

Most switches require a `<job>` parameter, which you set to the job's display name, or GUID. A job's display name doesn't have to be unique. The **/create** and **/list** switches return a job's GUID.

By default, you can access information about your own jobs. To access information for another user's jobs, you must have administrator privileges. If the job was created in an elevated state, then you must run **bitsadmin** from an elevated window; otherwise, you'll have read-only access to the job.

Many of the switches correspond to methods in the BITS interfaces. For additional details that may be relevant to using a switch, see the corresponding method.

Use the following switches to create a job, set and retrieve the properties of a job, and monitor the status of a job. For examples that show how to use some of these switches to perform tasks, see bitsadmin examples.

Available switches

- bitsadmin /addfile
- bitsadmin /addfileset
- bitsadmin /addfilewithranges
- bitsadmin /cache
- bitsadmin /cache /delete
- bitsadmin /cache /deleteurl
- bitsadmin /cache /getexpirationtime
- bitsadmin /cache /getlimit
- bitsadmin /cache /help
- bitsadmin /cache /info
- bitsadmin /cache /list
- bitsadmin /cache /setexpirationtime
- bitsadmin /cache /setlimit

- bitsadmin /cache /clear
- bitsadmin /cancel
- bitsadmin /complete
- bitsadmin /create
- bitsadmin /examples
- bitsadmin /getaclflags
- bitsadmin /getbytestotal
- bitsadmin /getbytestransferred
- bitsadmin /getclientcertificate
- bitsadmin /getcompletiontime
- bitsadmin /getcreationtime
- bitsadmin /getcustomheaders
- bitsadmin /getdescription
- bitsadmin /getdisplayname
- bitsadmin /geterror
- bitsadmin /geterrorcount
- bitsadmin /getfilestotal
- bitsadmin /getfilestransferred
- bitsadmin /gethelpertokenflags
- bitsadmin /gethelpertokensid
- bitsadmin /gethttpmethod
- bitsadmin /getmaxdownloadtime
- bitsadmin /getminretrydelay
- bitsadmin /getmodificationtime
- bitsadmin /getnoprogresstimeout
- bitsadmin /getnotifycmdline
- bitsadmin /getnotifyflags
- bitsadmin /getnotifyinterface
- bitsadmin /getowner
- bitsadmin /getpeercachingflags
- bitsadmin /getpriority
- bitsadmin /getproxybypasslist
- bitsadmin /getproxylist
- bitsadmin /getproxyusage
- bitsadmin /getreplydata
- bitsadmin /getreplyfilename
- bitsadmin /getreplyprogress
- bitsadmin /getsecurityflags
- bitsadmin /getstate
- bitsadmin /gettemporaryname
- bitsadmin /gettype

- bitsadmin /getvalidationstate
- bitsadmin /help
- bitsadmin /info
- bitsadmin /list
- bitsadmin /listfiles
- bitsadmin /makecustomheaderswriteonly
- bitsadmin /monitor
- bitsadmin /nowrap
- bitsadmin /peercaching
- bitsadmin /peercaching /getconfigurationflags
- bitsadmin /peercaching /help
- bitsadmin /peercaching /setconfigurationflags
- bitsadmin /peers
- bitsadmin /peers /clear
- bitsadmin /peers /discover
- bitsadmin /peers /help
- bitsadmin /peers /list
- bitsadmin /rawreturn
- bitsadmin /removeclientcertificate
- bitsadmin /removecredentials
- bitsadmin /replaceremoteprefix
- bitsadmin /reset
- bitsadmin /resume
- bitsadmin /setaclflag
- bitsadmin /setclientcertificatebyid
- bitsadmin /setclientcertificatebyname
- bitsadmin /setcredentials
- bitsadmin /setcustomheaders
- bitsadmin /setdescription
- bitsadmin /setdisplayname
- bitsadmin /sethelpertoken
- bitsadmin /sethelpertokenflags
- bitsadmin /sethttpmethod
- bitsadmin /setmaxdownloadtime
- bitsadmin /setminretrydelay
- bitsadmin /setnoprogresstimeout
- bitsadmin /setnotifycmdline
- bitsadmin /setnotifyflags
- bitsadmin /setpeercachingflags
- bitsadmin /setpriority
- bitsadmin /setproxysettings

- bitsadmin /setreplyfilename
- bitsadmin /setsecurityflags
- bitsadmin /setvalidationstate
- bitsadmin /suspend
- bitsadmin /takeownership
- bitsadmin /transfer
- bitsadmin /util
- bitsadmin /util /enableanalyticchannel
- bitsadmin /util /getieproxy
- bitsadmin /util /help
- bitsadmin /util /repairservice
- bitsadmin /util /setieproxy
- bitsadmin /util /version
- bitsadmin /wrap

Feedback

Was this page helpful? Yes No

bitsadmin addfile

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds a file to the specified job.

Syntax

```
bitsadmin /addfile <job> <remoteURL> <localname>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
remoteURL	The URL of the file on the server.
localname	The name of the file on the local computer. *Localname* must contain an absolute path to the file.

Examples

To add a file to the job:

```
bitsadmin /addfile myDownloadJob http://downloadsrv/10mb.zip c:\10mb.zip
```

Repeat this call for each file to add. If multiple jobs use *myDownloadJob* as their name, you must replace *myDownloadJob* with the job's GUID to uniquely identify the job.

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin addfileset

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds one or more files to the specified job.

Syntax

```
bitsadmin /addfileset <job> <textfile>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
textfile	A text file, each line of which contains a remote and a local file name. **Note:** Names must space-delimited. Lines starting with a `#` character are treated as a comment.

Examples

```
bitsadmin /addfileset files.txt
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin addfilewithranges

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds a file to the specified job. BITS downloads the specified ranges from the remote file. This switch is valid only for download jobs.

Syntax

```
bitsadmin /addfilewithranges <job> <remoteURL> <localname> <rangelist>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
remoteURL	URL of the file on the server.
localname	Name of the file on the local computer. Must contain an absolute path to the file.
rangelist	Comma-delimited list of offset:length pairs. Use a colon to separate the offset value from the length value. For example, a value of `0:100,2000:100,5000:eof` tells BITS to transfer 100 bytes from offset 0, 100 bytes from offset 2000, and the remaining bytes from offset 5000 to the end of the file.

Remarks

- The token **eof** is a valid length value within the offset and length pairs in the `<rangelist>`. It instructs the service to read to the end of the specified file.
- The `addfilewithranges` command will fail with error code 0x8020002c, if a zero-length range is specified along with another range using same offset, such as:

`c:\bits>bitsadmin /addfilewithranges j2 http://bitsdc/dload/1k.zip c:\1k.zip 100:0,100:5`

Error message: Unable to add file to job - 0x8020002c. The list of byte ranges contains some overlapping ranges, which are not supported.

Workaround: Don't specify the zero-length range first. For example, use `bitsadmin /addfilewithranges j2 http://bitsdc/dload/1k.zip c:\1k.zip 100:5,100:0`

Examples

To transfer 100 bytes from offset 0, 100 bytes from offset 2000, and the remaining bytes from offset 5000 to the end of the file:

```
bitsadmin /addfilewithranges http://downloadsrv/10mb.zip c:\10mb.zip
0:100,2000:100,5000:eof
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin cache

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Contains a list of the bitsadmin /cache switches.

Contains a list of the bitsadmin /cache switches.

Syntax

```
bitsadmin /cache /help
bitsadmin /cache /clear
bitsadmin /cache /delete
bitsadmin /cache /deleteURL
bitsadmin /cache /list
bitsadmin /cache /info
bitsadmin /cache /getlimit
bitsadmin /cache /setlimit
bitsadmin /cache /getexpirationtime
bitsadmin /cache /setexpirationtime
```

Parameters

Expand table

Parameter	Description
bitsadmin cache and help	Displays the command-line usage for the /cache switches.
bitsadmin cache and clear	Purges the local cache.
bitsadmin cache and delete	Deletes a cache entry.
bitsadmin cache and deleteURL	Deletes all cache entries for the given URL.
bitsadmin cache and list	Lists all cache entries.
bitsadmin cache and info	Dumps a specific cache entry.
bitsadmin cache and getlimit	Retrieves the cache limit.

Parameter	Description
bitsadmin cache and setlimit	Sets the cache limit.
bitsadmin cache and getexpirationtime	Retrieves the cache expiration time.
bitsadmin cache and setexpirationtime	Sets the cache expiration time.

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin cache and delete

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Deletes a specific cache entry.

Syntax

```
bitsadmin /cache /delete recordID
```

Parameters

Expand table

Parameter	Description
recordID	The GUID associated with the cache entry.

Examples

To delete the cache entry with the RecordID of {6511FB02-E195-40A2-B595-E8E2F8F47702}:

```
bitsadmin /cache /delete {6511FB02-E195-40A2-B595-E8E2F8F47702}
```

Related links

- Command-Line Syntax Key
- bitsadmin cache command

Feedback

Was this page helpful? Yes No

bitsadmin cache and deleteURL

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Deletes all cache entries for the given URL.

Syntax

```
bitsadmin /deleteURL URL
```

Parameters

Expand table

Parameter	Description
URL	The Uniform Resource Locator that identifies a remote file.

Examples

To delete all cache entries for `https://www.contoso.com/en/us/default.aspx`:

```
bitsadmin /deleteURL https://www.contoso.com/en/us/default.aspx
```

Related links

- Command-Line Syntax Key
- bitsadmin cache command

Feedback

Was this page helpful? Yes No

bitsadmin cache and getexpirationtime

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the cache expiration time.

Syntax

```
bitsadmin /cache /getexpirationtime
```

Examples

To retrieve the cache expiration time:

```
bitsadmin /cache /getexpirationtime
```

Related links

- Command-Line Syntax Key
- bitsadmin cache command

Feedback

Was this page helpful? Yes No

bitsadmin cache and getlimit

Article • 01/26/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the cache limit.

Syntax

```
bitsadmin /cache /getlimit
```

Examples

To retrieve the cache limit:

```
bitsadmin /cache /getlimit
```

Related links

- Command-Line Syntax Key
- bitsadmin cache command

Feedback

Was this page helpful? Yes No

bitsadmin cache and help

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays the command-line usage for the **cache** switches.

Syntax

```
bitsadmin /cache /help
```

Examples

To show the command-line help for the **cache** switches.

```
bitsadmin /cache /help
```

Related links

- Command-Line Syntax Key
- bitsadmin cache command

Feedback

Was this page helpful? Yes No

bitsadmin cache and info

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Dumps a specific cache entry.

Syntax

```
bitsadmin /cache /info recordID [/verbose]
```

Parameters

Expand table

Paramreter	Description
recordID	The GUID associated with the cache entry.

Examples

To dump the cache entry with the recordID value of {6511FB02-E195-40A2-B595-E8E2F8F47702}:

```
bitsadmin /cache /info {6511FB02-E195-40A2-B595-E8E2F8F47702}
```

Related links

- Command-Line Syntax Key
- bitsadmin cache command

Feedback

Was this page helpful? Yes No

bitsadmin cache and list

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Lists all cache entries.

Syntax

```
bitsadmin /cache /list [/verbose]
```

Examples

To list all cache entries in verbose format.

```
bitsadmin /cache /list /verbose
```

Related links

- Command-Line Syntax Key
- bitsadmin cache command

Feedback

Was this page helpful? Yes No

bitsadmin cache and setexpirationtime

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Sets the cache expiration time.

Syntax

```
bitsadmin /cache /setexpirationtime secs
```

Parameters

Expand table

Parameter	Description
secs	The number of seconds until the cache expires.

Examples

To set the cache to expire in 60 seconds:

```
bitsadmin /cache / setexpirationtime 60
```

Related links

- Command-Line Syntax Key
- bitsadmin cache command

Feedback

Was this page helpful? Yes No

bitsadmin cache and setlimit

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the cache size limit.

Syntax

```
bitsadmin /cache /setlimit percent
```

Parameters

Expand table

Parameter	Description
percent	The cache limit defined as a percentage of the total hard disk space.

Examples

To set the cache size limit to 50%:

```
bitsadmin /cache /setlimit 50
```

Related links

- Command-Line Syntax Key
- bitsadmin cache command

Feedback

Was this page helpful? Yes No

bitsadmin cache and clear

Article • 01/27/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Applies to: Windows Server (All supported versions)

The bitsadmin cache and clear command purges the local cache.

Syntax

```
bitsadmin /cache /clear
```

Examples

To purge the local cache:

```
bitsadmin /cache /clear
```

Related links

- Command-Line Syntax Key
- bitsadmin cache command

Feedback

Was this page helpful? Yes No

bitsadmin cancel

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Removes the job from the transfer queue and deletes all temporary files associated with the job.

Syntax

```
bitsadmin /cancel <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To remove the *myDownloadJob* job from the transfer queue:

```
bitsadmin /cancel myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin complete

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Completes the job. Use this switch after the job moves to the transferred state. Otherwise, only those files that have been successfully transferred will be available.

Syntax

```
bitsadmin /complete <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Example

To complete the *myDownloadJob* job, after it reaches the `TRANSFERRED` state:

```
bitsadmin /complete myDownloadJob
```

If multiple jobs use *myDownloadJob* as their name, you must use the job's GUID to uniquely identify it for completion.

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin create

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Creates a transfer job with the given display name.

> ⓘ Note
>
> The **/Upload** and **/Upload-Reply** parameter types aren't supported by BITS 1.2 and earlier.

Syntax

```
bitsadmin /create [type] displayname
```

Parameters

Expand table

Parameter	Description
type	There are three types of jobs: • **/Download.** Transfers data from a server to a local file. • **/Upload.** Transfers data from a local file to a server. • **/Upload-Reply.** Transfers data from a local file to a server and receives a reply file from the server. This parameter defaults to **/Download** if it's not specified.
displayname	The display name assigned to the newly created job.

Examples

To create a download job named *myDownloadJob*:

```
bitsadmin /create myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin resume command
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin examples

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The following examples show how to use the `bitsadmin` tool to perform the most common tasks.

Transfer a file

To create a job, add files, activate the job in the transfer queue, and to complete the job:

```
bitsadmin /transfer myDownloadJob /download /priority normal
https://downloadsrv/10mb.zip c:\\10mb.zip
```

BITSAdmin continues to show progress information in the MS-DOS window until the transfer completes or an error occurs.

Create a download job

To create a download job named *myDownloadJob*:

```
bitsadmin /create myDownloadJob
```

BITSAdmin returns a GUID that uniquely identifies the job. Use the GUID or job name in subsequent calls. The following text is sample output.

Sample output

```
created job {C775D194-090F-431F-B5FB-8334D00D1CB6}
```

Add files to the download job

To add a file to the job:

```
bitsadmin /addfile myDownloadJob https://downloadsrv/10mb.zip c:\\10mb.zip
```

Repeat this call for each file you want to add. If multiple jobs use *myDownloadJob* as their name, you must use the job's GUID to uniquely identify it for completion.

Activate the download job

After you create a new job, BITS automatically suspends the job. To activate the job in the transfer queue:

```
bitsadmin /resume myDownloadJob
```

If multiple jobs use *myDownloadJob* as their name, you must use the job's GUID to uniquely identify it for completion.

Determine the progress of the download job

The **/info** switch returns the state of the job and the number of files and bytes transferred. When the state is shown as `TRANSFERRED`, it means that BITS has successfully transferred all files in the job. You can also add the **/verbose** argument to get complete details of the job, and **/list** or **/monitor** to get all the jobs in the transfer queue.

To return the state of the job:

```
bitsadmin /info myDownloadJob /verbose
```

If multiple jobs use *myDownloadJob* as their name, you must use the job's GUID to uniquely identify it for completion.

Complete the download job

To complete the job after the state changes to `TRANSFERRED`:

```
bitsadmin /complete myDownloadJob
```

You must run the `/complete` switch before the files in the job become available. If multiple jobs use *myDownloadJob* as their name, you must use the job's GUID to uniquely identify it for completion.

Monitor jobs in the transfer queue using the /list switch

To return the state of the job and the number of files and bytes transferred for all jobs in the transfer queue:

```
bitsadmin /list
```

Sample output

```
{6AF46E48-41D3-453F-B7AF-A694BBC823F7} job1 SUSPENDED 0 / 0 0 / 0
{482FCAF0-74BF-469B-8929-5CCD028C9499} job2 TRANSIENT_ERROR 0 / 1 0 /
UNKNOWN

Listed 2 job(s).
```

Monitor jobs in the transfer queue using the /monitor switch

To return the state of the job and the number of files and bytes transferred for all jobs in the transfer queue, refreshing the data every 5 seconds:

```
bitsadmin /monitor
```

> ⓘ **Note**
>
> To stop the refresh, press CTRL+C.

Sample output

```
MONITORING BACKGROUND COPY MANAGER(5 second refresh)
{6AF46E48-41D3-453F-B7AF-A694BBC823F7} job1 SUSPENDED 0 / 0 0 / 0
{482FCAF0-74BF-469B-8929-5CCD028C9499} job2 TRANSIENT_ERROR 0 / 1 0 /
UNKNOWN
{0B138008-304B-4264-B021-FD04455588FF} job3 TRANSFERRED 1 / 1 100379370 /
100379370
```

Monitor jobs in the transfer queue using the /info switch

To return the state of the job and the number of files and bytes transferred:

```
bitsadmin /info
```

Sample output

```
GUID: {482FCAF0-74BF-469B-8929-5CCD028C9499} DISPLAY: myDownloadJob
TYPE: DOWNLOAD STATE: TRANSIENT_ERROR OWNER: domain\user
PRIORITY: NORMAL FILES: 0 / 1 BYTES: 0 / UNKNOWN
CREATION TIME: 12/17/2002 1:21:17 PM MODIFICATION TIME: 12/17/2002 1:21:30
PM
COMPLETION TIME: UNKNOWN
NOTIFY INTERFACE: UNREGISTERED NOTIFICATION FLAGS: 3
RETRY DELAY: 600 NO PROGRESS TIMEOUT: 1209600 ERROR COUNT: 0
PROXY USAGE: PRECONFIG PROXY LIST: NULL PROXY BYPASS LIST: NULL
ERROR FILE:    https://downloadsrv/10mb.zip -> c:\10mb.zip
ERROR CODE:    0x80072ee7 - The server name or address could not be resolved
ERROR CONTEXT: 0x00000005 - The error occurred while the remote file was
being
processed.
DESCRIPTION:
JOB FILES:
0 / UNKNOWN WORKING https://downloadsrv/10mb.zip -> c:\10mb.zip
NOTIFICATION COMMAND LINE: none
```

Delete jobs from the transfer queue

To remove all jobs from the transfer queue, use the /reset switch:

```
bitsadmin /reset
```

Sample output

```
{DC61A20C-44AB-4768-B175-8000D02545B9} canceled.
{BB6E91F3-6EDA-4BB4-9E01-5C5CBB5411F8} canceled.
2 out of 2 jobs canceled.
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getaclflags

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the access control list (ACL) propagations flags, reflecting whether items are inherited by child objects.

Syntax

```
bitsadmin /getaclflags <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Remarks

Returns one or more of the following flag values:

- **o** - Copy owner information with file.
- **g** - Copy group information with file.
- **d** - Copy discretionary access control list (DACL) information with file.
- **s** - Copy system access control list (SACL) information with file.

Examples

To retrieve the access control list propagation flags for the job named *myDownloadJob*:

```
bitsadmin /getaclflags myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getbytestotal

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the size of the specified job.

Syntax

```
bitsadmin /getbytestotal <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the size of the job named *myDownloadJob*:

```
bitsadmin /getbytestotal myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getbytestransferred

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the number of bytes transferred for the specified job.

Syntax

```
bitsadmin /getbytestransferred <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the number of bytes transferred for the job named *myDownloadJob*:

```
bitsadmin /getbytestransferred myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getclientcertificate

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the client certificate from the job.

Syntax

```
bitsadmin /getclientcertificate <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the client certificate for the job named *myDownloadJob*:

```
bitsadmin /getclientcertificate myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getcompletiontime

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the time that the job finished transferring data.

Syntax

```
bitsadmin /getcompletiontime <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the time that the job named *myDownloadJob* finished transferring data:

```
bitsadmin /getcompletiontime myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getcreationtime

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the creation time for the specified job.

Syntax

```
bitsadmin /getcreationtime <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the creation time for the job named *myDownloadJob*:

```
bitsadmin /getcreationtime myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getcustomheaders

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the custom HTTP headers from the job.

Syntax

```
bitsadmin /getcustomheaders <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To get the custom headers for the job named *myDownloadJob*:

```
bitsadmin /getcustomheaders myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getdescription

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the description of the specified job.

Syntax

```
bitsadmin /getdescription <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the description for the job named *myDownloadJob*:

```
bitsadmin /getdescription myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getdisplayname

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the display name of the specified job.

Syntax

```
bitsadmin /getdisplayname <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the display name for the job named *myDownloadJob*:

```
bitsadmin /getdisplayname myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin geterror

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves detailed error information for the specified job.

Syntax

```
bitsadmin /geterror <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the error information for the job named *myDownloadJob*:

```
bitsadmin /geterror myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin geterrorcount

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves a count of the number of times the specified job generated a transient error.

Syntax

```
bitsadmin /geterrorcount <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve error count information for the job named *myDownloadJob*:

```
bitsadmin /geterrorcount myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getfilestotal

Article • 03/03/2021 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the number of files in the specified job.

Syntax

```
bitsadmin /getfilestotal <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the number of files included in the job named *myDownloadJob*:

```
bitsadmin /getfilestotal myDownloadJob
```

See Also

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getfilestransferred

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the number of files transferred for the specified job.

Syntax

```
bitsadmin /getfilestransferred <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the number of files transferred in the job named *myDownloadJob*:

```
bitsadmin /getfilestransferred myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin gethelpertokenflags

10/22/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Returns the usage flags for a helper token that is associated with a BITS transfer job.

> **Note**
>
> This command isn't supported by BITS 3.0 and earlier.

Syntax

```
bitsadmin /gethelpertokenflags <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Remarks

Possible return values, including:

- **0x0001.** The helper token is used to open the local file of an upload job, to create or rename the temporary file of a download job, or to create or rename the reply file of an upload-reply job.
- **0x0002.** The helper token is used to open the remote file of a Server Message Block (SMB) upload or download job, or in response to an HTTP server or proxy challenge for implicit NTLM or Kerberos credentials. You must call `/SetCredentialsJob TargetScheme NULL NULL` to allow the credentials to be sent over HTTP.

Examples

To retrieve the usage flags for a helper token associated with a BITS transfer job named *myDownloadJob*:

```
bitsadmin /gethelpertokenflags myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

bitsadmin gethelpertokensid

10/22/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Returns the SID of a BITS transfer job's helper token, if one is set.

> **Note**
>
> This command isn't supported by BITS 3.0 and earlier.

Syntax

```
bitsadmin /gethelpertokensid <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the SID of a BITS transfer job named *myDownloadJob*:

```
bitsadmin /gethelpertokensid myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

bitsadmin gethttpmethod

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Gets the HTTP verb to use with the job.

Syntax

```
bitsadmin /gethttpmethod <Job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the HTTP verb to use with the job named *myDownloadJob*:

```
bitsadmin /gethttpmethod myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getmaxdownloadtime

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Retrieves the download timeout in seconds.

Syntax

```
bitsadmin /getmaxdownloadtime <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To get the maximum download time for the job named *myDownloadJob* in seconds:

```
bitsadmin /getmaxdownloadtime myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getminretrydelay

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the length of time, in seconds, that the service will wait after encountering a transient error before trying to transfer the file.

Syntax

```
bitsadmin /getminretrydelay <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the minimum retry delay for the job named *myDownloadJob*:

```
bitsadmin /getminretrydelay myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getmodificationtime

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the last time the job was modified or data was successfully transferred.

Syntax

```
bitsadmin /getmodificationtime <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the last modified time for the job named *myDownloadJob*:

```
bitsadmin /getmodificationtime myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getnoprogresstimeout

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the length of time, in seconds, that the service will try to transfer the file after a transient error occurs.

Syntax

```
bitsadmin /getnoprogresstimeout <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the progress time out value for the job named *myDownloadJob*:

```
bitsadmin /getnoprogresstimeout myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getnotifycmdline

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the command-line command to run after the specified job finishes transferring data.

> **Note**
>
> This command isn't supported by BITS 1.2 and earlier.

Syntax

```
bitsadmin /getnotifycmdline <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the command-line command used by the service when the job named *myDownloadJob* completes.

```
bitsadmin /getnotifycmdline myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getnotifyflags

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the notification flags for the specified job.

Syntax

```
bitsadmin /getnotifyflags <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Remarks

The job can contain one or more of the following notification flags:

Expand table

Flag	Description
0x001	Generate an event when all files in the job have been transferred.
0x002	Generate an event when an error occurs.
0x004	Disable notifications.
0x008	Generate an event when the job is modified or transfer progress is made.

Examples

To retrieve the notify flags for the job named *myDownloadJob*:

```
bitsadmin /getnotifyflags myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getnotifyinterface

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Determines whether another program has registered a COM callback interface (the notify interface) for the specified job.

Syntax

```
bitsadmin /getnotifyinterface <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Output

The output for this command displays either, **Registered** or **Unregistered**.

> **Note**
>
> It's not possible to determine the program that registered the callback interface.

Examples

To retrieve the notify interface for the job named *myDownloadJob*:

```
bitsadmin /getnotifyinterface myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getowner

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays the display name or GUID of the owner of the specified job.

Syntax

```
bitsadmin /getowner <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To display the owner for the job named *myDownloadJob*:

```
bitsadmin /getowner myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getpeercachingflags

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Retrieves flags that determine if the files of the job can be cached and served to peers, and if BITS can download content for the job from peers.

Syntax

```
bitsadmin /getpeercachingflags <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the flags for the job named *myDownloadJob*:

```
bitsadmin /getpeercachingflags myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getpriority

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the priority of the specified job.

Syntax

```
bitsadmin /getpriority <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Output

The returned priority for this command can be:

- **FOREGROUND**
- **HIGH**
- **NORMAL**
- **LOW**
- **UNKNOWN**

Examples

To retrieve the priority for the job named *myDownloadJob*:

```
bitsadmin /getpriority myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getproxybypasslist

Article • 05/22/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Applies to: Applies to: Windows Server (All supported versions)

The `bitsadmin /getproxybypasslist` command retrieves the proxy bypass list for the specified job.

Syntax

```
bitsadmin /getproxybypasslist <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Remarks

The bypass list contains the host names or IP addresses, or both, that aren't routed through a proxy. The list can contain `<local>` to refer to all servers on the same LAN. The list can be semicolon (;) or space-delimited.

Examples

To retrieve the proxy bypass list for the job named *myDownloadJob*:

```
bitsadmin /getproxybypasslist myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getproxylist

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the comma-delimited list of proxy servers to use for the specified job.

Syntax

```
bitsadmin /getproxylist <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the proxy list for the job named *myDownloadJob*:

```
bitsadmin /getproxylist myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getproxyusage

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the proxy usage setting for the specified job.

Syntax

```
bitsadmin /getproxyusage <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Output

The returned proxy usage values can be:

- **Preconfig** - Use the owner's Internet Explorer defaults.
- **No_Proxy** - Don't use a proxy server.
- **Override** - Use an explicit proxy list.
- **Autodetect** - Automatically detect the proxy settings.

Examples

To retrieve the proxy usage for the job named *myDownloadJob*:

```
bitsadmin /getproxyusage myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getreplydata

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the server's upload-reply data in hexadecimal format for the job.

> **Note**
>
> This command isn't supported by BITS 1.2 and earlier.

Syntax

```
bitsadmin /getreplydata <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the upload-reply data for the job named *myDownloadJob*:

```
bitsadmin /getreplydata myDownloadJob
```

Related links

- Command-Line Syntax Key

- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getreplyfilename

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Gets the path of the file that contains the server upload-reply for the job.

> **Note**
>
> This command isn't supported by BITS 1.2 and earlier.

Syntax

```
bitsadmin /getreplyfilename <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the upload-reply filename for the job named *myDownloadJob*:

```
bitsadmin /getreplyfilename myDownloadJob
```

Related links

- Command-Line Syntax Key

- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getreplyprogress

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the size and progress of the server upload-reply.

> **Note**
>
> This command isn't supported by BITS 1.2 and earlier.

Syntax

```
bitsadmin /getreplyprogress <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the upload-reply progress for the job named *myDownloadJob*:

```
bitsadmin /getreplyprogress myDownloadJob
```

Related links

- Command-Line Syntax Key

- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getsecurityflags

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Reports the HTTP security flags for URL redirection and checks performed on the server certificate during the transfer.

Syntax

```
bitsadmin /getsecurityflags <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the security flags from a job named *myDownloadJob*:

```
bitsadmin /getsecurityflags myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getstate

Article • 05/22/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Applies to: Windows Server (All supported versions)

The `bitsadmin /getstate` command retrieves the state of a specified job.

Syntax

```
bitsadmin /getstate <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Output

The returned output values can be:

Expand table

State	Description
`Queued`	The job is waiting to run.
`Connecting`	BITS is contacting the server.
`Transferring`	BITS is transferring data.
`Transferred`	BITS has successfully transferred all files in the job.
`Suspended`	The job is paused.

State	Description
`Error`	A nonrecoverable error occurred; the transfer isn't retried.
`Transient_Error`	A recoverable error occurred; the transfer retries when the minimum retry delay expires.
`Acknowledged`	The job completed.
`Canceled`	The job was canceled.

Examples

To retrieve the state for the job named *myDownloadJob*:

```
bitsadmin /getstate myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin gettemporaryname

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Reports the temporary filename of the given file within the job.

Syntax

```
bitsadmin /gettemporaryname <job> <file_index>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
file_index	Starts from 0.

Examples

To report the temporary filename of file 2 for the job named *myDownloadJob*:

```
bitsadmin /gettemporaryname myDownloadJob 1
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin gettype

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the job type of the specified job.

Syntax

```
bitsadmin /gettype <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Output

The returned output values can be:

Expand table

Type	Description
Download	The job is a download.
Upload	The job is an upload.
Upload-Reply	The job is an upload-reply.
Unknown	The job has an unknown type.

Examples

To retrieve the job type for the job named *myDownloadJob*:

```
bitsadmin /gettype myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin getvalidationstate

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Reports the content validation state of the given file within the job.

Syntax

```
bitsadmin /getvalidationstate <job> <file_index>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
file_index	Starts from 0.

Examples

To retrieve the content validation state of file 2 within the job named *myDownloadJob*:

```
bitsadmin /getvalidationstate myDownloadJob 1
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin help

Article • 01/26/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Applies to: Windows Server (All supported versions)

The bitsadmin help command displays help-related information about the bitsadmin command-line parameters and options.

Syntax

```
bitsadmin /help | /?
```

Examples

To retrieve the command-line help.

```
bitsadmin /help
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin info

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays summary information about the specified job.

Syntax

```
bitsadmin /info <job> [/verbose]
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
/verbose	Optional. Provides detailed information about each job.

Examples

To retrieve information about the job named *myDownloadJob*:

```
bitsadmin /info myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin info

Feedback

Was this page helpful? Yes No

bitsadmin list

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Lists the transfer jobs owned by the current user.

Syntax

```
bitsadmin /list [/allusers][/verbose]
```

Parameters

Expand table

Parameter	Description
/allusers	Optional. Lists jobs for all users. You must have administrator privileges to use this parameter.
/verbose	Optional. Provides detailed information about each job.

Examples

To retrieve information about jobs owned by the current user.

```
bitsadmin /list
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin listfiles

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Lists the files in the specified job.

Syntax

```
bitsadmin /listfiles <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve the list of files for the job named *myDownloadJob*:

```
bitsadmin /listfiles myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin makecustomheaderswriteonly

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Make a job's Custom HTTP Headers write-only.

> **ⓘ Important**
>
> This action can't be undone.

Syntax

```
bitsadmin /makecustomheaderswriteonly <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To make Custom HTTP Headers write-only for the job named *myDownloadJob*:

```
bitsadmin /makecustomheaderswriteonly myDownloadJob
```

Related links

- Command-Line Syntax Key

- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin monitor

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Monitors jobs in the transfer queue that are owned by the current user.

Syntax

```
bitsadmin /monitor [/allusers] [/refresh <seconds>]
```

Parameters

Expand table

Parameter	Description
/allusers	Optional. Monitors jobs for all users. You must have administrator privileges to use this parameter.
/refresh	Optional. Refreshes the data at an interval specified by `<seconds>`. The default refresh interval is five seconds. To stop the refresh, press CTRL+C.

Examples

To monitor the transfer queue for jobs owned by the current user and refreshes the information every 60 seconds.

```
bitsadmin /monitor /refresh 60
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin nowrap

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Truncates any line of output text extending beyond the right-most edge of the command window. By default, all switches, except the **monitor** switch, wrap the output. Specify the **nowrap** switch before other switches.

Syntax

```
bitsadmin /nowrap
```

Examples

To retrieve the state for the job named *myDownloadJob* while not wrapping the output:

```
bitsadmin /nowrap /getstate myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin peercaching

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Lists the /peercaching switches.

Lists the /peercaching switches.

Syntax

```
bitsadmin /peercaching /help
bitsadmin /peercaching /setconfigurationflags
bitsadmin /peercaching /getconfigurationflags
```

Parameters

Expand table

Parameter	Description
bitsadmin peercaching and help	Displays the command-line usage for the /peercaching switches.
bitsadmin peercaching and setconfigurationflags	Sets the configuration flags that determine if the computer can serve content to peers and if it can download content from peers.
bitsadmin peercaching and getconfigurationflags	Gets the configuration flags that determine if the computer serves content to peers and if it can download content from peers.

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin peercaching and getconfigurationflags

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Gets the configuration flags that determine if the computer serves content to peers and if it can download content from peers.

Syntax

```
bitsadmin /peercaching /getconfigurationflags <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To get the configuration flags for the job named *myDownloadJob*:

```
bitsadmin /peercaching /getconfigurationflags myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

- bitsadmin peercaching command

Feedback

Was this page helpful? Yes No

bitsadmin peercaching and help

Article • 05/22/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Applies to: Windows Server (All supported versions)

The `bitsadmin` command displays the command-line usage for the `/peercaching` switch when combined with `/help` switch.

Syntax

The following syntax is used:

```
CLI
bitsadmin /peercaching /help
```

Examples

To display the command-line help for the `/peercaching` switches, run the following command.

```
CLI
bitsadmin /peercaching /help
```

Related links

- Command-Line Syntax Key
- bitsadmin command
- bitsadmin peercaching command

Feedback

Was this page helpful? Yes No

bitsadmin peercaching and setconfigurationflags

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the configuration flags that determine if the computer can serve content to peers and if it can download content from peers.

Syntax

```
bitsadmin /peercaching /setconfigurationflags <job> <value>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
value	An unsigned integer with the following interpretation for the bits in the binary representation: • To allow the job's data to be downloaded from a peer, set the least significant bit. • To allow the job's data to be served to peers, set the second bit from the right.

Examples

To specify the job's data to be downloaded from peers for the job named *myDownloadJob*:

```
bitsadmin /peercaching /setconfigurationflags myDownloadJob 1
```

Related links

- Command-Line Syntax Key
- bitsadmin command
- bitsadmin peercaching command

Feedback

Was this page helpful? Yes No

bitsadmin peers

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Lists the available /peers switches.

Lists the available /peers switches.

```
bitsadmin /peers /help
bitsadmin /peers /discover
bitsadmin /peers /clear
bitsadmin /peers /list
```

Parameters

Expand table

Parameter	Description
bitsadmin peers and help	Displays the command-line usage for the /peers switches.
bitsadmin peers and discover	Discovers peers again.
bitsadmin peers and clear	Clears the peers list.
bitsadmin peers and list	Lists all peers.

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin peers and clear

Article • 05/22/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Applies to Windows Server (All supported versions)

The `bitsadmin /peers /clear` command clears the peer list.

Syntax

```
bitsadmin /peers /clear
```

Examples

To clear the peer list.

```
bitsadmin /peers /clear
```

Related links

- Command-Line Syntax Key
- bitsadmin command
- bitsadmin peers command

Feedback

Was this page helpful? Yes No

bitsadmin peers and discover

Article • 01/20/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Applies to: Windows Server (All supported versions)

The bitsadmin peers and discover command rediscovers peers.

Syntax

```
bitsadmin /peers /discover
```

Examples

To rediscover peers:

```
bitsadmin /peers /discover
```

Related links

- Command-Line Syntax Key
- bitsadmin command
- bitsadmin peers command

Feedback

Was this page helpful? Yes No

bitsadmin peers and help

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays the command-line usage for the **/peers** switches.

Syntax

```
bitsadmin /peers /help
```

Examples

To display the command-line usage for the **/peers** switches:

```
bitsadmin /peers /help
```

Related links

- Command-Line Syntax Key
- bitsadmin command
- bitsadmin peers command

Feedback

Was this page helpful? Yes No

bitsadmin peers and list

Article • 05/22/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Applies to: Windows Server (All supported versions)

The `bitsadmin` command lists all peers when it's combined with the **peers** and **list** switches.

Syntax

CLI

```
bitsadmin /peers /list
```

Examples

To list all peers:

CLI

```
bitsadmin /peers /list
```

Related links

- Command-Line Syntax Key
- bitsadmin command
- bitsadmin peers command

Feedback

Was this page helpful? Yes No

bitsadmin rawreturn

Article • 05/22/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Applies to: Windows Server (All supported versions)

The bitsadmin rawreturn command returns data suitable for parsing. Typically, you use this command with the **/create** and **/get*** switches to receive only the value. You must specify this switch before other switches.

> **Note**
>
> This command strips newline characters and formatting from the output.

Syntax

```
bitsadmin /rawreturn
```

Examples

To retrieve the raw data for the state of the job named *myDownloadJob*:

```
bitsadmin /rawreturn /getstate myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin removeclientcertificate

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Removes the client certificate from the job.

Syntax

```
bitsadmin /removeclientcertificate <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To remove the client certificate from the job named *myDownloadJob*:

```
bitsadmin /removeclientcertificate myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin removecredentials

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Removes credentials from a job.

> **Note**
>
> This command isn't supported by BITS 1.2 and earlier.

Syntax

```
bitsadmin /removecredentials <job> <target> <scheme>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
target	Use either **SERVER** or **PROXY**.
scheme	Use one of the following: • **BASIC.** Authentication scheme where the user name and password are sent in clear-text to the server or proxy. • **DIGEST.** A challenge-response authentication scheme that uses a server-specified data string for the challenge. • **NTLM.** A challenge-response authentication scheme that uses the credentials of the user for authentication in a Windows network environment. • **NEGOTIATE (also known as the Simple and Protected Negotiation protocol).** A challenge-response authentication scheme that negotiates with the server or proxy to determine which scheme to use for authentication. Examples are the Kerberos protocol and NTLM. • **PASSPORT.** A centralized authentication service provided by Microsoft that offers a single logon for member sites.

Examples

To remove credentials from the job named *myDownloadJob*:

```
bitsadmin /removecredentials myDownloadJob SERVER BASIC
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin replaceremoteprefix

Article • 03/03/2021 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Changes the remote URL for all files in the job from *oldprefix* to *newprefix*, as necessary.

Syntax

```
bitsadmin /replaceremoteprefix <job> <oldprefix> <newprefix>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
oldprefix	Existing URL prefix.
newprefix	New URL prefix.

Examples

To change the remote URL for all files in job named *myDownloadJob*, from *http://stageserver* to *http://prodserver*.

```
bitsadmin /replaceremoteprefix myDownloadJob http://stageserver
http://prodserver
```

Additional information

- Command-Line Syntax Key

- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin reset

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Cancels all jobs in the transfer queue owned by the current user. You can't reset jobs created by Local System. Instead, you must be an administrator and use the task scheduler to schedule this command as a task using the Local System credentials.

> **Note**
>
> If you have administrator privileges in BITSAdmin 1.5 and earlier, the /reset switch will cancel all the jobs in the queue. Additionally, the /allusers option isn't supported.

Syntax

```
bitsadmin /reset [/allusers]
```

Parameters

Expand table

Parameter	Description
/allusers	Optional. Cancels all jobs in the queue owned by the current user. You must have administrator privileges to use this parameter.

Examples

To cancel all the jobs in the transfer queue for the current user.

```
bitsadmin /reset
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin resume

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Activates a new or suspended job in the transfer queue. If you resumed your job by mistake, or simply need to suspend your job, you can use the bitsadmin suspend switch to suspend the job.

Syntax

```
bitsadmin /resume <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To resume the job named *myDownloadJob*:

```
bitsadmin /resume myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin suspend command
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin setaclflag

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the access control list (ACL) propagations flags for the job. The flags indicate that you want to maintain the owner and ACL information with the file being downloaded. For example, to maintain the owner and group with the file, set the **flags** parameter to `og`.

Syntax

```
bitsadmin /setaclflag <job> <flags>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
flags	Specify one or more of the values, including: • **o** - Copy owner information with file. • **g** - Copy group information with file. • **d** - Copy discretionary access control list (DACL) information with file. • **s** - Copy system access control list (SACL) information with file.

Examples

To set the access control list propagation flags for the job named *myDownloadJob*, so it maintains the owner and group information with the downloaded files.

```
bitsadmin /setaclflags myDownloadJob og
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin setclientcertificatebyid

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Specifies the identifier of the client certificate to use for client authentication in an HTTPS (SSL) request.

Syntax

```
bitsadmin /setclientcertificatebyid <job> <store_location> <store_name>
<hexadecimal_cert_id>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
store_location	Identifies the location of a system store to use for looking up the certificate, including: • CURRENT_USER • LOCAL_MACHINE • CURRENT_SERVICE • SERVICES • USERS • CURRENT_USER_GROUP_POLICY • LOCAL_MACHINE_GROUP_POLICY • LOCAL_MACHINE_ENTERPRISE.
store_name	The name of the certificate store, including: • CA (Certification Authority certificates) • MY (Personal certificates) • ROOT (Root certificates) • SPC (Software Publisher Certificate).
hexadecimal_cert_id	A hexadecimal number representing the hash of the certificate.

Examples

To specify the identifier of the client certificate to use for client authentication in an HTTPS (SSL) request for the job named *myDownloadJob*:

```
bitsadmin /setclientcertificatebyid myDownloadJob
BG_CERT_STORE_LOCATION_CURRENT_USER MY A106B52356D3FBCD1853A41B619358BD
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin setclientcertificatebyname

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Specifies the subject name of the client certificate to use for client authentication in an HTTPS (SSL) request.

Syntax

```
bitsadmin /setclientcertificatebyname <job> <store_location> <store_name>
<subject_name>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
store_location	Identifies the location of a system store to use for looking up the certificate. Possible values include: • 1 (CURRENT_USER) • 2 (LOCAL_MACHINE) • 3 (CURRENT_SERVICE) • 4 (SERVICES) • 5 (USERS) • 6 (CURRENT_USER_GROUP_POLICY) • 7 (LOCAL_MACHINE_GROUP_POLICY) • 8 (LOCAL_MACHINE_ENTERPRISE)
store_name	The name of the certificate store. Possible values include: • CA (Certification Authority certificates) • MY (Personal certificates) • ROOT (Root certificates) • SPC (Software Publisher Certificate)
subject_name	Name of the certificate.

Examples

To specify the name of the client certificate *myCertificate* to use for client authentication in an HTTPS (SSL) request for the job named *myDownloadJob*:

```
bitsadmin /setclientcertificatebyname myDownloadJob 1 MY myCertificate
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin setcredentials

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds credentials to a job.

> **Note**
>
> This command isn't supported by BITS 1.2 and earlier.

Syntax

```
bitsadmin /setcredentials <job> <target> <scheme> <username> <password>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
target	Use either **SERVER** or **PROXY**.
scheme	Use one of the following: • **BASIC.** Authentication scheme where the user name and password are sent in clear-text to the server or proxy. • **DIGEST.** A challenge-response authentication scheme that uses a server-specified data string for the challenge. • **NTLM.** A challenge-response authentication scheme that uses the credentials of the user for authentication in a Windows network environment. • **NEGOTIATE (also known as the Simple and Protected Negotiation protocol).** A challenge-response authentication scheme that negotiates with the server or proxy to determine which scheme to use for authentication. Examples are the Kerberos protocol and NTLM. • **PASSPORT.** A centralized authentication service provided by Microsoft that offers a single logon for member sites.

Parameter	Description
user_name	The name of the user.
password	The password associated with the provided *Username*.

Examples

To add credentials to the job named *myDownloadJob*:

```
bitsadmin /setcredentials myDownloadJob SERVER BASIC Edward password20
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin setcustomheaders

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Add a custom HTTP header to a GET request sent to an HTTP server. For more information about GET requests, see Method Definitions and Header Field Definitions.

Syntax

```
bitsadmin /setcustomheaders <job> <header1> <header2> <...>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
`<header1> <header2>` and so on	The custom headers for the job.

Examples

To add a custom HTTP header for the job named *myDownloadJob*:

```
bitsadmin /setcustomheaders myDownloadJob accept-encoding:deflate/gzip
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin setdescription

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the description for the specified job.

Syntax

```
bitsadmin /setdescription <job> <description>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
description	Text used to describe the job.

Examples

To retrieve the description for the job named *myDownloadJob*:

```
bitsadmin /setdescription myDownloadJob music_downloads
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin setdisplayname

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the display name for the specified job.

Syntax

```
bitsadmin /setdisplayname <job> <display_name>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
display_name	Text used as the displayed name for the specific job.

Examples

To set the display name for the job to *myDownloadJob*:

```
bitsadmin /setdisplayname myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin sethelpertoken

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the current command prompt's primary token (or an arbitrary local user account's token, if specified) as a BITS transfer job's helper token.

> **Note**
>
> This command isn't supported by BITS 3.0 and earlier.

Syntax

```
bitsadmin /sethelpertoken <job> [<user_name@domain> <password>]
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
`<username@domain>` `<password>`	Optional. The local user account credentials for which token to use.

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin sethelpertokenflags

10/22/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Sets the usage flags for a helper token that is associated with a BITS transfer job.

> **Note**
>
> This command isn't supported by BITS 3.0 and earlier.

Syntax

```
bitsadmin /sethelpertokenflags <job> <flags>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
flags	Possible helper token values, including: • **0x0001.** Used to open the local file of an upload job, to create or rename the temporary file of a download job, or to create or rename the reply file of an upload-reply job. • **0x0002.** Used to open the remote file of a Server Message Block (SMB) upload or download job, or in response to an HTTP server or proxy challenge for implicit NTLM or Kerberos credentials. You must call `/setcredentialsjob targetscheme null null` to send the credentials over HTTP.

Related links

- Command-Line Syntax Key
- bitsadmin command

bitsadmin seththttpmethod

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the HTTP verb to use.

Syntax

```
bitsadmin /seththttpmethod <job> <httpmethod>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
httpmethod	The HTTP verb to use. For information about available verbs, see Method Definitions.

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin setmaxdownloadtime

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the download timeout in seconds.

Syntax

```
bitsadmin /setmaxdownloadtime <job> <timeout>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
timeout	The length for the download timeout, in seconds.

Examples

To set the timeout for the job named *myDownloadJob* to 10 seconds.

```
bitsadmin /setmaxdownloadtime myDownloadJob 10
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin setminretrydelay

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the minimum length of time, in seconds, that BITS waits after encountering a transient error before trying to transfer the file.

Syntax

```
bitsadmin /setminretrydelay <job> <retrydelay>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
retrydelay	Minimum length of time for BITS to wait after an error during transfer, in seconds.

Examples

To set the minimum retry delay to 35 seconds for the job named *myDownloadJob*:

```
bitsadmin /setminretrydelay myDownloadJob 35
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin setnoprogresstimeout

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the length of time, in seconds, that BITS tries to transfer the file after the first transient error occurs.

Syntax

```
bitsadmin /setnoprogresstimeout <job> <timeoutvalue>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
timeoutvalue	The length of time that BITS waits to transfer a file after the first error, in seconds.

Remarks

- The "no progress" timeout interval begins when the job encounters its first transient error.
- The timeout interval stops or resets when a byte of data is successfully transferred.
- If the "no progress" timeout interval exceeds the *timeoutvalue*, then the job is placed in a fatal error state.

Examples

To set the "no progress" timeout value to 20 seconds, for the job named *myDownloadJob*:

```
bitsadmin /setnoprogresstimeout myDownloadJob 20
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin setnotifycmdline

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the command-line command that runs after the job finishes transferring data or after a job enters a specified state.

> **Note**
>
> This command isn't supported by BITS 1.2 and earlier.

Syntax

```
bitsadmin /setnotifycmdline <job> <program_name> [program_parameters]
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
program_name	Name of the command to run when the job completes. You can set this value as NULL, but if you do, *program_parameters* must also be set to NULL.
program_parameters	Parameters that you want to pass to *program_name*. You can set this value as NULL. If *program_parameters* isn't set to NULL, then the first parameter in *program_parameters* must match the *program_name*.

Examples

To run Notepad.exe at the completion of the job named *myDownloadJob*:

```
bitsadmin /setnotifycmdline myDownloadJob c:\winnt\system32\notepad.exe NULL
```

To show the EULA text in Notepad.exe, at the completion of the job named myDownloadJob:

```
bitsadmin /setnotifycmdline myDownloadJob c:\winnt\system32\notepad.exe
notepad c:\eula.txt
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin setnotifyflags

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the event notification flags for the specified job.

Syntax

```
bitsadmin /setnotifyflags <job> <notifyflags>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
notifyflags	Can include one or more of the following notification flags, including: • **1.** Generates an event when all files in the job have been transferred. • **2.** Generates an event when an error occurs. • **3.** Generates an event when all files have completed transfer or when an error occurs. • **4.** Disables notifications.

Examples

To set the notification flags to generate an event when an error occurs, for a job named *myDownloadJob*:

```
bitsadmin /setnotifyflags myDownloadJob 2
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin setpeercachingflags

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets flags that determine if the files of the job can be cached and served to peers and if the job can download content from peers.

Syntax

```
bitsadmin /setpeercachingflags <job> <value>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
value	An unsigned integer, including: • **1.** The job can download content from peers. • **2.** The files of the job can be cached and served to peers.

Examples

To allow the job named *myDownloadJob* to download content from peers:

```
bitsadmin /setpeercachingflags myDownloadJob 1
```

Related links

- Command-Line Syntax Key

- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin setpriority

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the priority of the specified job.

Syntax

```
bitsadmin /setpriority <job> <priority>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
priority	Sets the priority of the job, including: • FOREGROUND • HIGH • NORMAL • LOW

Examples

To set the priority for the job named *myDownloadJob* to normal:

```
bitsadmin /setpriority myDownloadJob NORMAL
```

Related links

- Command-Line Syntax Key

- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin setproxysettings

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the proxy settings for the specified job.

Syntax

```
bitsadmin /setproxysettings <job> <usage> [list] [bypass]
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
usage	Sets the proxy usage, including: • **PRECONFIG.** Use the owner's Internet Explorer defaults. • **NO_PROXY.** Don't use a proxy server. • **OVERRIDE.** Use an explicit proxy list and bypass list. The proxy list and proxy bypass information must follow. • **AUTODETECT.** Automatically detects proxy settings.
list	Used when the *Usage* parameter is set to OVERRIDE. Must contain a comma-delimited list of proxy servers to use.
bypass	Used when the *Usage* parameter is set to OVERRIDE. Must contain a space-delimited list of host names or IP addresses, or both, for which transfers are not to be routed through a proxy. This can be `<local>` to refer to all servers on the same LAN. Values of NULL may be used for an empty proxy bypass list.

Examples

To set the proxy settings using the various usage options for the job named *myDownloadJob*:

```
bitsadmin /setproxysettings myDownloadJob PRECONFIG
```

```
bitsadmin /setproxysettings myDownloadJob NO_PROXY
```

```
bitsadmin /setproxysettings myDownloadJob OVERRIDE proxy1:80
```

```
bitsadmin /setproxysettings myDownloadJob OVERRIDE proxy1,proxy2,proxy3 NULL
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin setreplyfilename

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Specifies the path of the file that contains the server upload-reply.

> **Note**
>
> This command isn't supported by BITS 1.2 and earlier.

Syntax

```
bitsadmin /setreplyfilename <job> <file_path>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
file_path	Location to put the server upload-reply.

Examples

To set the upload-reply filename file path for the job named *myDownloadJob*:

```
bitsadmin /setreplyfilename myDownloadJob c:\upload-reply
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin setsecurityflags

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets security flags for HTTP to determine if BITS should check the certificate revocation list, ignore certain certificate errors, and define the policy to use when a server redirects the HTTP request. The value is an unsigned integer.

Syntax

```
bitsadmin /setsecurityflags <job> <value>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.
value	Can include one or more of the following notification flags, including: • Set the least significant bit to enable CRL Check. • Set the 2nd bit from the right to ignore incorrect common names in the server certificate. • Set the 3rd bit from the right to ignore incorrect dates in the server certificate. • Set the 4th bit from the right to ignore incorrect certification authorities in the server certificate. • Set the 5th bit from the right to ignore incorrect usage of the server certificate. • Set the 9th through the 11th bits from the right to implement your specified redirection policy, including: ◦ **0,0,0.** Redirects are automatically allowed. ◦ **0,0,1.** Remote name in the **IBackgroundCopyFile** interface is updated if a redirect occurs. ◦ **0,1,0.** BITS fails the job if a redirect occurs. • Set the 12th bit from the right to allow redirection from HTTPS to HTTP.

Examples

To set the security flags to enable a CRL check for the job named *myDownloadJob*:

```
bitsadmin /setsecurityflags myDownloadJob 0x0001
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin setvalidationstate

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the content validation state of the given file within the job.

Syntax

```
bitsadmin /setvalidationstate <job> <file_index> <TRUE|FALSE>
```

Parameters

Expand table

Parameter	Description
Job	The job's display name or GUID.
file_index	Starts at 0.
TRUE or FALSE	**TRUE** turns on content validation for the specified file, while **FALSE** turns it off.

Examples

To set the content validation state of file 2 to TRUE for the job named *myDownloadJob*:

```
bitsadmin /setvalidationstate myDownloadJob 2 TRUE
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin suspend

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Suspends the specified job. If you suspended your job by mistake, you can use the bitsadmin resume switch to restart the job.

Syntax

```
bitsadmin /suspend <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Example

To suspend the job named *myDownloadJob*:

```
bitsadmin /suspend myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin resume command
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin takeownership

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Lets a user with administrative privileges take ownership of the specified job.

Syntax

```
bitsadmin /takeownership <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To take ownership of the job named *myDownloadJob*:

```
bitsadmin /takeownership myDownloadJob
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin transfer

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Transfers one or more files. By default, the BITSAdmin service creates a download job that runs at **NORMAL** priority and updates the command window with progress information until the transfer is complete or until a critical error occurs,

The service completes the job if it successfully transfers all the files and cancels the job if a critical error occurs. The service does not create the job if it is unable to add files to the job or if you specify an invalid value for *type* or *job_priority*. To transfer more than one file, specify multiple `<RemoteFileName>-<LocalFileName>` pairs. The pairs must be space-delimited.

> **Note**
>
> The BITSAdmin command continues to run if a transient error occurs. To end the command, press CTRL+C.

Syntax

```
bitsadmin /transfer <name> [<type>] [/priority <job_priority>] [/ACLflags
<flags>] [/DYNAMIC] <remotefilename> <localfilename>
```

Parameters

Expand table

Parameter	Description
name	The name of the job. This command can't be a GUID.
type	Optional. Sets the type of job, including: • **/DOWNLOAD.** The default value. Choose this type for download jobs. • **/UPLOAD.** Choose this type for upload jobs.

Parameter	Description
priority	Optional. Sets the priority of the job, including: • FOREGROUND • HIGH • NORMAL • LOW
ACLflags	Optional. Indicates that you want to maintain the owner and ACL information with the file being downloaded. Specify one or more of the values, including: • **o** - Copy owner information with file. • **g** - Copy group information with file. • **d** - Copy discretionary access control list (DACL) information with file. • **s** - Copy system access control list (SACL) information with file.
/DYNAMIC	Configures the job using **BITS_JOB_PROPERTY_DYNAMIC_CONTENT**, which relaxes the server-side requirements.
remotefilename	The name of the file after it's transferred to the server.
localfilename	The name of the file that resides locally.

Examples

To start a transfer job named *myDownloadJob*:

```
bitsadmin /transfer myDownloadJob http://prodserver/audio.wma
c:\downloads\audio.wma
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin util

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Lists the **/util** switches.

Lists the **util** switches.

Syntax

```
bitsadmin /util /help
bitsadmin /util /getieproxy
bitsadmin /util /repairservice
bitsadmin /util /setieproxy
bitsadmin /util /version
```

Parameters

Expand table

Parameter	Description
bitsadmin util and help	Displays the command-line usage for the /Util switches. You can also specify /?.
bitsadmin util and getieproxy	Retrieves the proxy usage for the given service account.
bitsadmin util and repairservice	Repairs known issues with BITS service.
bitsadmin util and setieproxy	Specifies proxy settings to use when transferring files using a service account.
bitsadmin util and version	Displays the version of the BITS service.

Related links

- Command-Line Syntax Key

- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin util and enableanalyticchannel

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Enables or disables the BITS client analytic channel.

Syntax

```
bitsadmin /util /enableanalyticchannel TRUE|FALSE
```

Expand table

Parameter	Description
TRUE or FALSE	**TRUE** turns on content validation for the specified file, while **FALSE** turns it off.

Examples

To turn the BITS client analytic channel on or off.

```
bitsadmin /util / enableanalyticchannel TRUE
```

Related links

- Command-Line Syntax Key
- bitsadmin util command
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin util and getieproxy

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Retrieves the proxy usage for the given service account. This command shows the value for each proxy usage, not just the proxy usage you specified for the service account. For details about setting the proxy usage for specific service accounts, see the bitsadmin util and setieproxy command.

Syntax

```
bitsadmin /util /getieproxy <account> [/conn <connectionname>]
```

Parameters

Expand table

Parameter	Description
account	Specifies the service account whose proxy settings you want to retrieve. Possible values include: • LOCALSYSTEM • NETWORKSERVICE • LOCALSERVICE.
connectionname	Optional. Used with the **/conn** parameter to specify which modem connection to use. If you don't specify the **/conn** parameter, BITS uses the LAN connection.

Examples

To display the proxy usage for the NETWORK SERVICE account:

```
bitsadmin /util /getieproxy NETWORKSERVICE
```

Related links

- Command-Line Syntax Key
- bitsadmin util command
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin util and help

Article • 05/22/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Applies to: Windows Server (All supported versions)

Display help for the `bitsadmin util` command switch when combined with the **help** switch.

Syntax

```
bitsadmin /util /help
```

Examples

Display the command-line help for the **/util** switches by running the following command.

CLI

```
bitsadmin /util /help
```

Related links

- Command-Line Syntax Key
- bitsadmin util command
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin util and repairservice

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

If BITS fails to start, this switch attempts to resolve errors related to incorrect service configuration and dependencies on Windows services (such as LANManworkstation) and the network directory. This switch also generates output that indicates if the issues that were resolved.

> **Note**
>
> This command isn't supported by BITS 1.5 and earlier.

Syntax

```
bitsadmin /util /repairservice [/force]
```

Parameters

Expand table

Parameter	Description
/force	Optional. Deletes and creates the service again.

> **Note**
>
> If BITS creates the service again, the service description string might be set to English even in a localized system.

Examples

To repair the BITS service configuration:

```
bitsadmin /util /repairservice
```

Related links

- Command-Line Syntax Key
- bitsadmin util command
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin util and setieproxy

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Set the proxy settings to use when transferring files using a service account. You must run this command from an elevated command prompt for it to complete successfully.

> **Note**
>
> This command isn't supported by BITS 1.5 and earlier.

Syntax

```
bitsadmin /util /setieproxy <account> <usage> [/conn <connectionname>]
```

Parameters

Expand table

Parameter	Description
account	Specifies the service account whose proxy settings you want to define. Possible values include: • LOCALSYSTEM • NETWORKSERVICE • LOCALSERVICE.
usage	Specifies the form of proxy detection to use. Possible values include: • **NO_PROXY.** Don't use a proxy server. • **AUTODETECT.** Automatically detect the proxy settings. • **MANUAL_PROXY.** Use a specified proxy list and bypass list. You must specify your lists immediately after the usage tag. For example, `MANUAL_PROXY proxy1,proxy2 NULL`. ◦ **Proxy list.** A comma-delimited list of proxy servers to use. ◦ **Bypass list.** A space-delimited list of host names or IP addresses, or both, for which transfers are not to be routed through a proxy. This can be <local> to refer to all servers on the same LAN. Values of NULL or may be used for an empty proxy bypass list.

Parameter	Description
	• **AUTOSCRIPT.** Same as **AUTODETECT**, except it also runs a script. You must specify the script URL immediately after the usage tag. For example, `AUTOSCRIPT http://server/proxy.js`.
connectionname	Optional. Used with the **/conn** parameter to specify which modem connection to use. If you don't specify the **/conn** parameter, BITS uses the LAN connection.

Remarks

Each successive call using this switch replaces the previously specified usage, but not the parameters of the previously defined usage. For example, if you specify **NO_PROXY**, **AUTODETECT**, and **MANUAL_PROXY** on separate calls, BITS uses the last supplied usage, but keeps the parameters from the previously defined usage.

Examples

To set the proxy usage for the LOCALSYSTEM account:

```
bitsadmin /util /setieproxy localsystem AUTODETECT
```

```
bitsadmin /util /setieproxy localsystem MANUAL_PROXY proxy1,proxy2,proxy3 NULL
```

```
bitsadmin /util /setieproxy localsystem MANUAL_PROXY proxy1:80
```

Related links

- Command-Line Syntax Key
- bitsadmin util command
- bitsadmin command

Last updated on 06/24/2026

bitsadmin util and version

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays the version of BITS service (for example, 2.0).

> **Note**
>
> This command isn't supported by BITS 1.5 and earlier.

Syntax

```
bitsadmin /util /version [/verbose]
```

Parameters

Expand table

Parameter	Description
/verbose	Use this switch to display the file version for each BITS-related DLL and to verify whether the BITS service can start.

Examples

To display the version of the BITS Service.

```
bitsadmin /util /version
```

Related links

- Command-Line Syntax Key

- bitsadmin util command
- bitsadmin command

Feedback

Was this page helpful? Yes No

bitsadmin wrap

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Wraps any line of output text extending beyond the rightmost edge of the command window to the next line. You must specify this switch before any other switches.

By default, all switches except the bitsadmin monitor switch, wrap the output text.

Syntax

```
bitsadmin /wrap <job>
```

Parameters

Expand table

Parameter	Description
job	The job's display name or GUID.

Examples

To retrieve information for the job named *myDownloadJob* and wrap the output text:

```
bitsadmin /wrap /info myDownloadJob /verbose
```

Related links

- Command-Line Syntax Key
- bitsadmin command

Feedback

Was this page helpful? Yes No

bootcfg

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Configures, queries, or changes Boot.ini file settings.

Syntax

```
bootcfg <parameter> [arguments...]
```

Parameters

Expand table

Parameter	Description
bootcfg addsw	Adds operating system load options for a specified operating system entry.
bootcfg copy	Makes a copy of an existing boot entry, to which you can add command-line options.
bootcfg dbg1394	Configures 1394 port debugging for a specified operating system entry.
bootcfg debug	Adds or changes the debug settings for a specified operating system entry.
bootcfg default	Specifies the operating system entry to designate as the default.
bootcfg delete	Deletes an operating system entry in the [operating systems] section of the Boot.ini file.
bootcfg ems	Enables the user to add or change the settings for redirection of the Emergency Management Services console to a remote computer.
bootcfg query	Queries and displays the [boot loader] and [operating systems] section entries from Boot.ini.

Parameter	Description
bootcfg raw	Adds operating system load options specified as a string to an operating system entry in the [operating systems] section of the Boot.ini file.
bootcfg rmsw	Removes operating system load options for a specified operating system entry.
bootcfg timeout	Changes the operating system time-out value.

Feedback

Was this page helpful? Yes No

bootcfg addsw

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Adds operating system load options for a specified operating system entry.

Syntax

```
bootcfg /addsw [/s <computer> [/u <domain>\<user> /p <password>]] [/mm
<maximumram>] [/bv] [/so] [/ng] /id <osentrylinenum>
```

Parameters

Expand table

Term	Definition
`/s <computer>`	Specifies the name or IP address of a remote computer (don't use backslashes). The default is the local computer.
`/u <domain>\<user>`	Runs the command with the account permissions of the user specified by `<user>` or `<domain>\<user>`. The default is the permissions of the current logged on user on the computer issuing the command.
`/p <password>`	Specifies the password of the user account that is specified in the **/u** parameter.
`/mm <maximumram>`	Specifies the maximum amount of RAM, in megabytes, that the operating system can use. The value must be equal to or greater than 32 Megabytes.
/bv	Adds the **/basevideo** option to the specified `<osentrylinenum>`, directing the operating system to use standard VGA mode for the installed video driver.
/so	Adds the **/sos** option to the specified `<osentrylinenum>`, directing the operating system to display device driver names while they are being loaded.
/ng	Adds the **/noguiboot** option to the specified `<osentrylinenum>`, disabling the progress bar that appears before the CTRL+ALT+DEL logon prompt.

Term	Definition
`/id <osentrylinenum>`	Specifies the operating system entry line number in the [operating systems] section of the Boot.ini file to which the operating system load options are added. The first line after the [operating systems] section header is 1.
/?	Displays help at the command prompt.

Examples

To use the **bootcfg /addsw** command:

```
bootcfg /addsw /mm 64 /id 2
bootcfg /addsw /so /id 3
bootcfg /addsw /so /ng /s srvmain /u hiropln /id 2
bootcfg /addsw /ng /id 2
bootcfg /addsw /mm 96 /ng /s srvmain /u maindom\hiropln /p p@ssW23 /id 2
```

Related links

- Command-Line Syntax Key
- bootcfg command

Feedback

Was this page helpful? Yes No

bootcfg copy

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Makes a copy of an existing boot entry, to which you can add command-line options.

Syntax

```
bootcfg /copy [/s <computer> [/u <domain>\<user> /p <password>]] [/d
<description>] [/id <osentrylinenum>]
```

Parameters

Expand table

Parameter	Description
`/s <computer>`	Specifies the name or IP address of a remote computer (don't use backslashes). The default is the local computer.
`/u <domain>\<user>`	Runs the command with the account permissions of the user specified by `<user>` or `<domain>\<user>`. The default is the permissions of the current logged on user on the computer issuing the command.
`/p <password>`	Specifies the password of the user account that is specified in the **/u** parameter.
`/d <description>`	Specifies the description for the new operating system entry.
`/id <osentrylinenum>`	Specifies the operating system entry line number in the [operating systems] section of the Boot.ini file to which the operating system load options are added. The first line after the [operating systems] section header is 1.
/?	Displays help at the command prompt.

Examples

To copy boot entry 1 and enter \ABC Server\ as the description:

```
bootcfg /copy /d \ABC Server\ /id 1
```

Related links

- Command-Line Syntax Key
- bootcfg command

Feedback

Was this page helpful? Yes No

bootcfg dbg1394

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Configures 1394 port debugging for a specified operating system entry.

Syntax

```
bootcfg /dbg1394 {on | off}[/s <computer> [/u <domain>\<user> /p
<password>]] [/ch <channel>] /id <osentrylinenum>
```

Parameters

Expand table

Parameter	Description
`{on | off}`	Specifies the value for 1394 port debugging, including: • **on.** Enables remote debugging support by adding the /dbg1394 option to the specified `<osentrylinenum>`. • **off.** Disables remote debugging support by removing the /dbg1394 option from the specified `<osentrylinenum>`.
`/s <computer>`	Specifies the name or IP address of a remote computer (don't use backslashes). The default is the local computer.
`/u <domain>\<user>`	Runs the command with the account permissions of the user specified by `<user>` or `<domain>\<user>`. The default is the permissions of the current logged on user on the computer issuing the command.
`/p <password>`	Specifies the password of the user account that is specified in the **/u** parameter.
`/ch <channel>`	Specifies the channel to use for debugging. Valid values include integers, between 1 and 64. Don't use this parameter if 1394 port debugging is disabled.
`/id <osentrylinenum>`	Specifies the operating system entry line number in the [operating systems] section of the Boot.ini file to which the operating system load options are

Parameter	Description
	added. The first line after the [operating systems] section header is 1.
/?	Displays help at the command prompt.

Examples

To use the **bootcfg /dbg1394**command:

```
bootcfg /dbg1394 /id 2
bootcfg /dbg1394 on /ch 1 /id 3
bootcfg /dbg1394 edit /ch 8 /id 2
bootcfg /s srvmain /u maindom\hiropln /p p@ssW23 /dbg1394 off /id 2
```

Related links

- Command-Line Syntax Key
- bootcfg command

Feedback

Was this page helpful? Yes No

bootcfg debug

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Adds or changes the debug settings for a specified operating system entry.

> **Note**
>
> If you're attempting to debug port 1394, use the bootcfg dbg1394 command instead.

Syntax

```
bootcfg /debug {on | off | edit}[/s <computer> [/u <domain>\<user> /p
<password>]] [/port {COM1 | COM2 | COM3 | COM4}] [/baud {9600 | 19200 |
38400 | 57600 | 115200}] [/id <osentrylinenum>]
```

Parameters

Expand table

Parameter	Description
`{on | off | edit}`	Specifies the value for port debugging, including: • **on.** Enables remote debugging support by adding the /debug option to the specified `<osentrylinenum>`. • **off.** Disables remote debugging support by removing the /debug option from the specified `<osentrylinenum>`. • **edit.** Allows changes to port and baud rate settings by changing the values associated with the /debug option for the specified `<osentrylinenum>`.
`/s <computer>`	Specifies the name or IP address of a remote computer (don't use backslashes). The default is the local computer.
`/u <domain>\<user>`	Runs the command with the account permissions of the user specified by `<user>` or `<domain>\<user>`. The default is the permissions of the

Parameter	Description
	current logged on user on the computer issuing the command.
`/p <password>`	Specifies the password of the user account that is specified in the **/u** parameter.
`/port {COM1 | COM2 | COM3 | COM4}`	Specifies the COM port to be used for debugging. Don't use this parameter if debugging is disabled.
`/baud {9600 | 19200 | 38400 | 57600 | 115200}`	Specifies the baud rate to be used for debugging. Don't use this parameter if debugging is disabled.
`/id <osentrylinenum>`	Specifies the operating system entry line number in the [operating systems] section of the Boot.ini file to which the operating system load options are added. The first line after the [operating systems] section header is 1.
/?	Displays help at the command prompt.

Examples

To use the **bootcfg /debug** command:

```
bootcfg /debug on /port com1 /id 2
bootcfg /debug edit /port com2 /baud 19200 /id 2
bootcfg /s srvmain /u maindom\hiropln /p p@ssW23 /debug off /id 2
```

Related links

- Command-Line Syntax Key
- bootcfg command

Feedback

Was this page helpful? Yes No

bootcfg default

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Specifies the operating system entry to designate as the default.

Syntax

```
bootcfg /default [/s <computer> [/u <domain>\<user> /p <password>]] [/id
<osentrylinenum>]
```

Parameters

Expand table

Parameter	Description
`/s <computer>`	Specifies the name or IP address of a remote computer (don't use backslashes). The default is the local computer.
`/u <domain>\<user>`	Runs the command with the account permissions of the user specified by `<user>` or `<domain>\<user>`. The default is the permissions of the current logged on user on the computer issuing the command.
`/p <password>`	Specifies the password of the user account that is specified in the **/u** parameter.
`/id <osentrylinenum>`	Specifies the operating system entry line number in the [operating systems] section of the Boot.ini file to which the operating system load options are added. The first line after the [operating systems] section header is 1.
/?	Displays help at the command prompt.

Examples

To use the **bootcfg /default** command:

```
bootcfg /default /id 2
bootcfg /default /s srvmain /u maindom\hiropln /p p@ssW23 /id 2
```

Related links

- Command-Line Syntax Key
- bootcfg command

Feedback

Was this page helpful? Yes No

bootcfg delete

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Deletes an operating system entry in the [operating systems] section of the Boot.ini file.

Syntax

```
bootcfg /delete [/s <computer> [/u <domain>\<user> /p <password>]] [/id
<osentrylinenum>]
```

Parameters

Expand table

Parameter	Description
`/s <computer>`	Specifies the name or IP address of a remote computer (don't use backslashes). The default is the local computer.
`/u <domain>\<user>`	Runs the command with the account permissions of the user specified by `<user>` or `<domain>\<user>`. The default is the permissions of the current logged on user on the computer issuing the command.
`/p <password>`	Specifies the password of the user account that is specified in the **/u** parameter.
`/id <osentrylinenum>`	Specifies the operating system entry line number in the [operating systems] section of the Boot.ini file to which the operating system load options are added. The first line after the [operating systems] section header is 1.
/?	Displays help at the command prompt.

Examples

To use the **bootcfg /delete** command:

```
bootcfg /delete /id 1
bootcfg /delete /s srvmain /u maindom\hiropln /p p@ssW23 /id 3
```

Related links

- Command-Line Syntax Key
- bootcfg command

Feedback

Was this page helpful? Yes No

bootcfg ems

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Enables the user to add or change the settings for redirection of the Emergency Management Services console to a remote computer. Enabling Emergency Management Services, adds a `redirect=Port#` line to the [boot loader] section of the Boot.ini file along with a /redirect option to the specified operating system entry line. The Emergency Management Services feature is enabled only on servers.

Syntax

```
bootcfg /ems {on | off | edit}[/s <computer> [/u <domain>\<user> /p
<password>]] [/port {COM1 | COM2 | COM3 | COM4 | BIOSSET}] [/baud {9600 |
19200 | 38400 | 57600 | 115200}] [/id <osentrylinenum>]
```

Parameters

Expand table

Parameter	Description
`{on | off | edit}`	Specifies the value for Emergency Management Services redirection, including: • **on.** Enables remote output for the specified `<osentrylinenum>`. Also adds a /redirect option to the specified `<osentrylinenum>` and a `redirect=com<X>` setting to the [boot loader] section. The value of `com<X>` is set by the **/port** parameter. • **off.** Disables output to a remote computer. Also removes the /redirect option to the specified `<osentrylinenum>` and the `redirect=com<X>` setting from the [boot loader] section. • **edit.** Allows changes to port settings by changing the `redirect=com<X>` setting in the [boot loader] section. The value of `com<X>` is set by the **/port** parameter.
`/s <computer>`	Specifies the name or IP address of a remote computer (don't use backslashes). The default is the local computer.

Parameter	Description
`/u <domain>\<user>`	Runs the command with the account permissions of the user specified by `<user>` or `<domain>\<user>`. The default is the permissions of the current logged on user on the computer issuing the command.
`/p <password>`	Specifies the password of the user account that is specified in the **/u** parameter.
`/port {COM1 | COM2 | COM3 | COM4 | BIOSSET}`	Specifies the COM port to be used for redirection. The BIOSSET parameter directs Emergency Management Services to get the BIOS settings to determine which port should be used for redirection. Don't use this parameter if remotely administered output is disabled.
`/baud {9600 | 19200 | 38400 | 57600 | 115200}`	Specifies the baud rate to be used for redirection. Don't use this parameter if remotely administered output is disabled.
`/id <osentrylinenum>`	Specifies the operating system entry line number to which the Emergency Management Services option is added in the [operating systems] section of the Boot.ini file. The first line after the [operating systems] section header is 1. This parameter is required when the Emergency Management Services value is set to **on** or **off**.
/?	Displays help at the command prompt.

Examples

To use the **bootcfg /ems** command:

```
bootcfg /ems on /port com1 /baud 19200 /id 2
bootcfg /ems on /port biosset /id 3
bootcfg /s srvmain /ems off /id 2
bootcfg /ems edit /port com2 /baud 115200
bootcfg /s srvmain /u maindom\hiropln /p p@ssW23 /ems off /id 2
```

Related links

- Command-Line Syntax Key
- bootcfg command

Feedback

Was this page helpful? Yes No

bootcfg query

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Queries and displays the [boot loader] and [operating systems] section entries from Boot.ini.

Syntax

```
bootcfg /query [/s <computer> [/u <domain>\<user> /p <password>]]
```

Parameters

Expand table

Parameter	Description
`/s <computer>`	Specifies the name or IP address of a remote computer (don't use backslashes). The default is the local computer.
`/u <domain>\<user>`	Runs the command with the account permissions of the user specified by `<user>` or `<domain>\<user>`. The default is the permissions of the current logged on user on the computer issuing the command.
`/p <password>`	Specifies the password of the user account that is specified in the **/u** parameter.
/?	Displays help at the command prompt.

Sample output

Sample output for the **bootcfg /query** command:

```
Boot Loader Settings
----------
timeout: 30
```

```
default: multi(0)disk(0)rdisk(0)partition(1)\WINDOWS
Boot Entries
------
Boot entry ID:   1
Friendly Name:
path: multi(0)disk(0)rdisk(0)partition(1)\WINDOWS
OS Load Options: /fastdetect /debug /debugport=com1:
```

- The **Boot Loader Settings** area shows each entry in the [boot loader] section of Boot.ini.
- The **Boot Entries** area shows more details for each operating system entry in the [operating systems] section of the Boot.ini

Examples

To use the **bootcfg /query** command:

```
bootcfg /query
bootcfg /query /s srvmain /u maindom\hiropln /p p@ssW23
bootcfg /query /u hiropln /p p@ssW23
```

Related links

- Command-Line Syntax Key
- bootcfg command

Feedback

Was this page helpful? Yes No

bootcfg raw

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Adds operating system load options specified as a string to an operating system entry in the [operating systems] section of the Boot.ini file. This command overwrites any existing operating system entry options.

Syntax

```
bootcfg /raw [/s <computer> [/u <domain>\<user> /p <password>]]
<osloadoptionsstring> [/id <osentrylinenum>] [/a]
```

Parameters

Expand table

Parameter	Description
`/s <computer>`	Specifies the name or IP address of a remote computer (don't use backslashes). The default is the local computer.
`/u <domain>\<user>`	Runs the command with the account permissions of the user specified by `<user>` or `<domain>\<user>`. The default is the permissions of the current logged on user on the computer issuing the command.
`/p <password>`	Specifies the password of the user account that is specified in the **/u** parameter.
`<osloadoptionsstring>`	Specifies the operating system load options to add to the operating system entry. These load options replace any existing load options associated with the operating system entry. There is no validation against the `<osloadoptions>` parameter.
`/id <osentrylinenum>`	Specifies the operating system entry line number in the [operating systems] section of the Boot.ini file to which the operating system load options are added. The first line after the [operating systems] section header is 1.

Parameter	Description
/a	Specifies which operating system options should be appended to any existing operating system options.
/?	Displays help at the command prompt.

Examples

This text should contain valid OS Load Options such as **/debug**, **/fastdetect**, **/nodebug**, **/baudrate**, **/crashdebug**, and **/sos**.

To add **/debug /fastdetect** to the end of the first operating system entry, replacing any previous operating system entry options:

```
bootcfg /raw /debug /fastdetect /id 1
```

To use the **bootcfg /raw** command:

```
bootcfg /raw /debug /sos /id 2
bootcfg /raw /s srvmain /u maindom\hiropln /p p@ssW23 /crashdebug  /id 2
```

Related links

- Command-Line Syntax Key
- bootcfg command

Feedback

Was this page helpful? Yes No

bootcfg rmsw

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Removes operating system load options for a specified operating system entry.

Syntax

```
bootcfg /rmsw [/s <computer> [/u <domain>\<user> /p <password>]] [/mm] [/bv] [/so] [/ng] /id <osentrylinenum>
```

Parameters

Expand table

Parameter	Description
`/s <computer>`	Specifies the name or IP address of a remote computer (don't use backslashes). The default is the local computer.
`/u <domain>\<user>`	Runs the command with the account permissions of the user specified by `<user>` or `<domain>\<user>`. The default is the permissions of the current logged on user on the computer issuing the command.
`/p <password>`	Specifies the password of the user account that is specified in the **/u** parameter.
/mm	Removes the /maxmem option and its associated maximum memory value from the specified `<osentrylinenum>`. The /maxmem option specifies the maximum amount of RAM that the operating system can use.
/bv	Removes the /basevideo option from the specified `<osentrylinenum>`. The /basevideo option directs the operating system to use standard VGA mode for the installed video driver.
/so	Removes the /sos option from the specified `<osentrylinenum>`. The /sos option directs the operating system to display device driver names while they are being loaded.

Parameter	Description
/ng	Removes the /noguiboot option from the specified `<osentrylinenum>`. The /noguiboot option disables the progress bar that appears before the CTRL+ALT+DEL logon prompt.
`/id <osentrylinenum>`	Specifies the operating system entry line number in the [operating systems] section of the Boot.ini file to which the operating system load options are added. The first line after the [operating systems] section header is 1.
/?	Displays help at the command prompt.

Examples

To use the **bootcfg /rmsw** command:

```
bootcfg /rmsw /mm 64 /id 2
bootcfg /rmsw /so /id 3
bootcfg /rmsw /so /ng /s srvmain /u hiropln /id 2
bootcfg /rmsw /ng /id 2
bootcfg /rmsw /mm 96 /ng /s srvmain /u maindom\hiropln /p p@ssW23 /id 2
```

Related links

- Command-Line Syntax Key
- bootcfg command

Feedback

Was this page helpful? Yes No

bootcfg timeout

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Changes the operating system time-out value.

Syntax

```
bootcfg /timeout <timeoutvalue> [/s <computer> [/u <domain>\<user> /p
<password>]]
```

Parameters

Expand table

Parameter	Description
`/timeout <timeoutvalue>`	Specifies the timeout value in the [boot loader] section. The `<timeoutvalue>` is the number of seconds the user has to select an operating system from the boot loader screen before NTLDR loads the default. The valid range for `<timeoutvalue>` is 0-999. If the value is 0, NTLDR immediately starts the default operating system without displaying the boot loader screen.
`/s <computer>`	Specifies the name or IP address of a remote computer (don't use backslashes). The default is the local computer.
`/u <domain>\<user>`	Runs the command with the account permissions of the user specified by `<user>` or `<domain>\<user>`. The default is the permissions of the current logged on user on the computer issuing the command.
`/p <password>`	Specifies the password of the user account that is specified in the **/u** parameter.
/?	Displays help at the command prompt.

Examples

To use the **bootcfg /timeout** command:

```
bootcfg /timeout 30
bootcfg /s srvmain /u maindom\hiropln /p p@ssW23 /timeout 50
```

Related links

- Command-Line Syntax Key
- bootcfg command

Feedback

Was this page helpful? Yes No

break

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

> **ⓘ Important**
>
> This command is no longer in use. It is included only to preserve compatibility with existing MS-DOS files, but it has no effect at the command line because the functionality is automatic.

Sets or clears extended CTRL+C checking on MS-DOS systems. If used without parameters, **break** displays the existing setting value.

If command extensions are enabled and running on the Windows platform, inserting the **break** command into a batch file enters a hard-coded breakpoint if being debugged by a debugger.

Syntax

```
break=[on|off]
```

> **ⓘ Note**
>
> Because the break command has no effect, it is often used to create empty files or delete the content of an existing file. For example:

```
rem -- cleans the content of the file --
break>log
```

Related links

- Command-Line Syntax Key

- break command

Feedback

Was this page helpful? Yes No

cacls

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

> **ⓘ Important**
>
> This command has been deprecated. Please use **icacls** instead.

Displays or modifies discretionary access control lists (DACL) on specified files.

Syntax

```
cacls <filename> [/t] [/m] [/l] [/s[:sddl]] [/e] [/c] [/g user:<perm>] [/r user [...]] [/p user:<perm> [...]] [/d user [...]]
```

Parameters

Expand table

Parameter	Description
`<filename>`	Required. Displays ACLs of specified files.
/t	Changes ACLs of specified files in the current directory and all subdirectories.
/m	Changes ACLs of volumes mounted to a directory.
/l	Works on the Symbolic Link itself instead of the target.
/s:sddl	Replaces the ACLs with those specified in the SDDL string. This parameter is not valid for use with the **/e**, **/g**, **/r**, **/p**, or **/d** parameters.
/e	Edit an ACL instead of replacing it.
/c	Continue after access denied errors.
`/g user:<perm>`	Grants specified user access rights, including these valid values for permission: • **n** - None

Parameter	Description
	• **r** - Read • **w** - Write • **c** - Change (write) • **f** - Full control
/r user [...]	Revoke specified user's access rights. Only valid when used with the **/e** parameter.
`[/p user:<perm> [...]`	Replace specified user's access rights, including these valid values for permission: • **n** - None • **r** - Read • **w** - Write • **c** - Change (write) • **f** - Full control
[/d user [...]	Deny specified user access.
/?	Displays help at the command prompt.

Sample output

Expand table

Output	Access control entry (ACE) applies to
OI	Object inherit. This folder and files.
CI	Container inherit. This folder and subfolders.
IO	Inherit only. The ACE does not apply to the current file/directory.
No output message	This folder only.
(OI)(CI)	This folder, subfolders, and files.
(OI)(CI)(IO)	Subfolders and files only.
(CI)(IO)	Subfolders only.
(OI)(IO)	Files only.

Remarks

- You can use wildcards (**?** and *) to specify multiple files.

- You can specify more than one user.

Related links

- Command-Line Syntax Key
- icacls

Feedback

Was this page helpful? Yes No

call

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Calls one batch program from another without stopping the parent batch program. The **call** command accepts labels as the target of the call.

> **Note**
>
> Call has no effect at the command prompt when it is used outside of a script or batch file.

Syntax

```
call [drive:][path]<filename> [<batchparameters>]]
call [:<label> [<arguments>]]
```

Parameters

Expand table

Parameter	Description
`[<drive>:][<path>]<filename>`	Specifies the location and name of the batch program that you want to call. The `<filename>` parameter is required, and it must have a .bat or .cmd extension.
`<batchparameters>`	Specifies any command-line information required by the batch program.
`:<label>`	Specifies the label that you want a batch program control to jump to.
`<arguments>`	Specifies the command-line information to be passed to the new instance of the batch program, beginning at `:<label>`.
/?	Displays help at the command prompt.

Batch parameters

The batch script argument references (**%0**, **%1**, ...) are listed in the following tables.

Using the %* value in a batch script refers to all the arguments (for example, **%1**, **%2**, **%3**...).

You can use the following optional syntaxes as substitutions for batch parameters (**%n**):

Expand table

Batch Parameter	Description
%~1	Expands **%1** and removes surrounding quotation marks.
%~f1	Expands **%1** to a fully qualified path.
%~d1	Expands **%1** to a drive letter only.
%~p1	Expands **%1** to a path only.
%~n1	Expands **%1** to a file name only.
%~x1	Expands **%1** to a file name extension only.
%~s1	Expands **%1** to a fully qualified path that contains short names only.
%~a1	Expands **%1** to the file attributes.
%~t1	Expands **%1** to the date and time of file.
%~z1	Expands **%1** to the size of the file.
%~$PATH:1	Searches the directories listed in the PATH environment variable, and expands **%1** to the fully qualified name of the first directory found. If the environment variable name is not defined or the file is not found by the search, then this modifier expands to the empty string.

The following table shows how you can combine modifiers with the batch parameters for compound results:

Expand table

Batch Parameter with Modifier	Description
%~dp1	Expands **%1** to a drive letter and path only.

Batch Parameter with Modifier	Description
%~nx1	Expands **%1** to a file name and extension only.
%~dp$PATH:1	Searches the directories listed in the PATH environment variable for **%1**, and then expands to the drive letter and path of the first directory found.
%~ftza1	Expands **%1** to display output similar to the **dir** command.

In the above examples, **%1** and PATH can be replaced by other valid values. The %~ syntax is terminated by a valid argument number. The %~ modifiers cannot be used with %*.

Remarks

- Using batch parameters:

  Batch parameters can contain any information that you can pass to a batch program, including command-line options, file names, the batch parameters **%0** through **%9**, and variables (for example, **%baud%**).

- Using the `<label>` parameter:

  By using **call** with the `<label>` parameter, you create a new batch file context and pass control to the statement after the specified label. The first time the end of the batch file is encountered (that is, after jumping to the label), control returns to the statement after the **call** statement. The second time the end of the batch file is encountered, the batch script is exited.

- Using pipes and redirection symbols:

  Do not use pipes `(|)` or redirection symbols (`<` or `>`) with **call**.

- Making a recursive call

  You can create a batch program that calls itself. However, you must provide an exit condition. Otherwise, the parent and child batch programs can loop endlessly.

- Working with command extensions

  If command extensions are enabled, **call** accepts `<label>` as the target of the call. The correct syntax is `call :<label> <arguments>`.

Examples

To run the checknew.bat program from another batch program, type the following command in the parent batch program:

```
call checknew
```

If the parent batch program accepts two batch parameters and you want it to pass those parameters to checknew.bat, type the following command in the parent batch program:

```
call checknew %1 %2
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

cd

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays the name of the current directory or changes the current directory. If used with only a drive letter (for example, `cd C:`), **cd** displays the names of the current directory in the specified drive. If used without parameters, **cd** displays the current drive and directory.

> **Note**
>
> This command is the same as the chdir command.

Syntax

```
cd [/d] [<drive>:][<path>]
cd [..]
chdir [/d] [<drive>:][<path>]
chdir [..]
```

Parameters

Expand table

Parameter	Description
/d	Changes the current drive as well as the current directory for a drive.
`<drive>:`	Specifies the drive to display or change (if different from the current drive).
`<path>`	Specifies the path to the directory that you want to display or change.
[..]	Specifies that you want to change to the parent folder.
/?	Displays help at the command prompt.

Remarks

If command extensions are enabled, the following conditions apply to the **cd** command:

- The current directory string is converted to use the same case as the names on the disk. For example, `cd c:\temp` would set the current directory to C:\Temp if that is the case on the disk.
- Spaces aren't treated as delimiters, so `<path>` can contain spaces without enclosing quotation marks. For example:

  ```
  cd username\programs\start menu
  ```

  is the same as:

  ```
  cd "username\programs\start menu"
  ```

  If extensions are disabled, the quotation marks are required.
- To disable command extensions, type:

  ```
  cmd /e:off
  ```

Examples

To return to the root directory, the top of the directory hierarchy for a drive:

```
cd\
```

To change the default directory on a drive that is different from the one you are on:

```
cd [<drive>:[<directory>]]
```

To verify the change to the directory, type:

```
cd [<drive>:]
```

Related links

- Command-Line Syntax Key
- chdir command

Feedback

Was this page helpful? Yes No

certreq

09/02/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The certreq command can be used to request certificates from a certification authority (CA), to retrieve a response to a previous request from a CA, to create a new request from an .inf file, to accept and install a response to a request, to construct a cross-certification or qualified subordination request from an existing CA certificate or request, and to sign a cross-certification or qualified subordination request.

> **ⓘ Important**
>
> Earlier versions of the certreq command might not provide all of the options described here. To see the options supported based on specific versions of certreq, run the command-line help option, `certreq -v -?`.
>
> The certreq command doesn't support creating a new certificate request based on a Key Attestation template when in a CEP/CES environment.

> **⚠ Warning**
>
> The content for this topic is based on the default settings for Windows Server; for example, setting the key length to 2048, selecting Microsoft Software Key Storage Provider as the CSP, and using Secure Hash Algorithm 1 (SHA1). Evaluate these selections against the requirements of your company's security policy.

Syntax

```
certreq [-submit] [options] [requestfilein [certfileout [certchainfileout [fullresponsefileOut]]]]
certreq -retrieve [options] requestid [certfileout [certchainfileout [fullresponsefileOut]]]
certreq -new [options] [policyfilein [requestfileout]]
certreq -accept [options] [certchainfilein | fullresponsefilein | certfilein]
certreq -sign [options] [requestfilein [requestfileout]]
certreq -enroll [options] templatename
certreq -enroll -cert certId [options] renew [reusekeys]
```

Parameters

Expand table

Parameter	Description
-submit	Submits a request to a certificate authority.
-retrieve `<requestid>`	Retrieves a response to a previous request from a certificate authority.
-new	Creates a new request from an .inf file.
-accept	Accepts and installs a response to a certificate request.
-policy	Sets the policy for a request.
-sign	Signs a cross-certification or qualified subordination request.
-enroll	Enrolls for or renews a certificate.
-?	Displays a list of certreq syntax, options, and descriptions.

Parameter	Description
`<parameter>` -?	Displays help for the parameter specified.
-v -?	Displays a verbose list of the certreq syntax, options, and descriptions.

Examples

certreq -submit

To submit a basic certificate request:

```
certreq -submit certrequest.req certnew.cer certnew.pfx
```

Remarks

- This is the default certreq.exe parameter. If no option is specified at the command-line prompt, certreq.exe attempts to submit a certificate request to a certificate authority. You must specify a certificate request file when using the **-submit** option. If this parameter is omitted, a common **File Open** window appears, letting you select the appropriate certificate request file.
- To request a certificate by specifying the SAN attribute, see the *How to use the certreq.exe utility to create and submit a certificate request* section of Microsoft Knowledge Base article 931351 How to add a Subject Alternative Name to a secure LDAP certificate.

certreq -retrieve

To retrieve certificate ID 20 and to create a certificate file (.cer), named *MyCertificate*:

```
certreq -retrieve 20 MyCertificate.cer
```

Remarks

- Use certreq -retrieve *requestid* to retrieve the certificate after the certificate authority has issued it. The *requestid* PKC can be a decimal or hex with 0x prefix and it can be a certificate serial number with no 0x prefix. You can also use it to retrieve any certificate that has ever been issued by the certificate authority, including revoked or expired certificates, without considering if the certificate's request was ever in the pending state.
- If you submit a request to the certificate authority, the policy module of the certificate authority might leave the request in a pending state and return the *requestid* to the certreq caller for display. Eventually, the certificate authority's administrator issues the certificate or deny the request.

certreq -new

To create a new request:

```
[newrequest]
; At least one value must be set in this section
```

```
subject = CN=W2K8-BO-DC.contoso2.com
```

The following are some of the possible sections that may be added to the INF file:

[newrequest]

This area of the INF file is mandatory for any new certificate request templates, and must include at least one parameter with a value.

Expand table

Key[1]	Description	Value[2]	Example
Subject	Several apps rely on the subject information in a certificate. We recommend specifying a value for this key. If the subject isn't set here, we recommend you include a subject name as part of the subject alternative name certificate extension.	Relative Distinguished Name string values	Subject = CN=computer1.contoso.com Subject=CN=John Smith,CN=Users,DC=Contoso,DC=com
Exportable	If set to TRUE, the private key can be exported with the certificate. To ensure a high level of security, private keys shouldn't be exportable; however, in some cases, it might be required if several computers or users must share the same private key.	`true | false`	`Exportable = TRUE`. CNG keys can distinguish between this and plaintext exportable. CAPI1 keys can't.
ExportableEncrypted	Specifies whether the private key should be set to be exportable.	`true | false`	`ExportableEncrypted = true` **Tip:** Not all public key sizes and algorithms work with all hash algorithms. The specified CSP must also support the specified hash algorithm. To see the list of supported hash algorithms, you can run the command: `certutil -oid 1 | findstr pwszCNGAlgid | findstr /v CryptOIDInfo`
HashAlgorithm	Hash Algorithm to be used for this request.	`Sha256, sha384, sha512, sha1, md5, md4, md2`	`HashAlgorithm = sha1`. To see the list of supported hash algorithms use: certutil -oid 1 | findstr pwszCNGAlgid | findstr /v CryptOIDInfo
KeyAlgorithm	The algorithm that's used by the service provider to generate a public and private key pair.	`RSA, DH, DSA, ECDH_P256, ECDH_P521, ECDSA_P256, ECDSA_P384, ECDSA_P521`	`KeyAlgorithm = RSA`
KeyContainer	We don't recommend setting	Random string value	`KeyContainer = {C347BD28-7F69-4090-AA16-BC58CF4D749C}`

Key[1]	Description	Value[2]	Example
	this parameter for new requests where new key material is generated. The key container is automatically generated and maintained by the system. For requests where the existing key material should be used, this value can be set to the key-container name of the existing key. Use the `certutil -key` command to display the list of available key containers for the machine context. Use the `certutil -key -user` command for the current user's context.	**Tip:** Use double quotes around any INF key value that has blanks or special characters to avoid potential INF parsing issues.	
KeyLength	Defines the length of the public and private key. The key length has an impact on the security level of the certificate. Greater key length usually provides a higher security level; however, some applications may have limitations regarding the key length.	Any valid key length that is supported by the cryptographic service provider.	`KeyLength = 2048`
KeySpec	Determines if the key can be used for signatures, for Exchange (encryption), or for both.	`AT_NONE, AT_SIGNATURE, AT_KEYEXCHANGE`	`KeySpec = AT_KEYEXCHANGE`
KeyUsage	Defines what the certificate key should be used for.	• `CERT_DIGITAL_SIGNATURE_KEY_USAGE -- 80 (128)` • `CERT_NON_REPUDIATION_KEY_USAGE -- 40 (64)` • `CERT_KEY_ENCIPHERMENT_KEY_USAGE -- 20 (32)` • `CERT_DATA_ENCIPHERMENT_KEY_USAGE -- 10 (16)` • `CERT_KEY_AGREEMENT_KEY_USAGE -- 8` • `CERT_KEY_CERT_SIGN_KEY_USAGE -- 4` • `CERT_OFFLINE_CRL_SIGN_KEY_USAGE -- 2` • `CERT_CRL_SIGN_KEY_USAGE -- 2` • `CERT_ENCIPHER_ONLY_KEY_USAGE -- 1` • `CERT_DECIPHER_ONLY_KEY_USAGE -- 8000 (32768)`	`KeyUsage = CERT_DIGITAL_SIGNATURE_KEY_USAGE | CERT_KEY_ENCIPHERMENT_KEY_USAGE` **Tip:** Multiple values use a pipe (|) symbol separator. Ensure that you use double-quotes when using multiple values to avoid INF parsing issues. The values shown are hexadecimal (decimal) values for each bit definition. Older syntax can also be used: a single hexadecimal value with multiple bits set, instead of the symbolic representation. For example, `KeyUsage = 0xa0`.

Key[1]	Description	Value[2]	Example
KeyUsageProperty	Retrieves a value that identifies the specific purpose for which a private key can be used.	• `NCRYPT_ALLOW_DECRYPT_FLAG -- 1` • `NCRYPT_ALLOW_SIGNING_FLAG -- 2` • `NCRYPT_ALLOW_KEY_AGREEMENT_FLAG -- 4` • `NCRYPT_ALLOW_ALL_USAGES -- ffffff (16777215)`	`KeyUsageProperty = NCRYPT_ALLOW_DECRYPT_FLAG | NCRYPT_ALLOW_SIGNING_FLAG`
MachineKeySet	This key is important when you need to create certificates that are owned by the machine and not a user. The key material that is generated is maintained in the security context of the security principal (user or computer account) that has created the request. When an administrator creates a certificate request on behalf of a computer, the key material must be created in the machine's security context and not the administrator's security context. Otherwise, the machine couldn't access its private key since it would be in the administrator's security context.	`true | false`. The default is false.	`MachineKeySet = true`
NotBefore	Specifies a date or date and time before which the request can't be issued. `NotBefore` can be used with `ValidityPeriod` and `ValidityPeriodUnits`.	Date or date and time	`NotBefore = 7/24/2012 10:31 AM` **Tip:** `NotBefore` and `NotAfter` are for R`equestType=cert` only. Date parsing attempts to be locale-sensitive. Using month names will disambiguate and should work in every locale.
NotAfter	Specifies a date or date and time after which the request can't be issued. `NotAfter` can't be used with `ValidityPeriod` or `ValidityPeriodUnits`.	Date or date and time	`NotAfter = 9/23/2014 10:31 AM` **Tip:** `NotBefore` and `NotAfter` are for `RequestType=cert` only. Date parsing attempts to be locale-sensitive. Using month names will disambiguate and should work in every locale.
PrivateKeyArchive	The PrivateKeyArchive setting works only if the corresponding RequestType is set to CMC because only the Certificate Management	`true | false`	`PrivateKeyArchive = true`

Key[1]	Description	Value[2]	Example
	Messages over CMS (CMC) request format allows for securely transferring the requester's private key to the CA for key archival.		
EncryptionAlgorithm	The encryption algorithm to use.	Possible options vary, depending on the operating system version and the set of installed cryptographic providers. To see the list of available algorithms, run the command: `certutil -oid 2 | findstr pwszCNGAlgid`. The specified CSP used must also support the specified symmetric encryption algorithm and length.	`EncryptionAlgorithm = 3des`
EncryptionLength	Length of encryption algorithm to use.	Any length allowed by the specified EncryptionAlgorithm.	`EncryptionLength = 128`
ProviderName	The provider name is the display name of the CSP.	If you don't know the provider name of the CSP you're using, run `certutil -csplist` from a command line. The command displays the names of all CSPs that are available on the local system	`ProviderName = Microsoft RSA SChannel Cryptographic Provider`
ProviderType	The provider type is used to select specific providers based on specific algorithm capability such as RSA Full.	If you don't know the provider type of the CSP you're using, run `certutil -csplist` from a command-line prompt. The command displays the provider type of all CSPs that are available on the local system.	`ProviderType = 1`
RenewalCert	If you need to renew a certificate that exists on the system where the certificate request is generated, you must specify its certificate hash as the value for this key.	The certificate hash of any certificate that is available at the computer where the certificate request is created. If you don't know the certificate hash, use the Certificates MMC Snap-In and look at the certificate that should be renewed. Open the certificate properties and see the `Thumbprint` attribute of the certificate. Certificate renewal requires either a `PKCS#7` or a `CMC` request format.	`RenewalCert = 4EDF274BD2919C6E9EC6A522F0F3B153E9B1582D`
RequesterName	Makes the request to enroll on behalf of another user request. The request must also be signed with an Enrollment Agent certificate, or the CA rejects the request. Use the `-cert` option to specify the enrollment agent certificate. The requester name can be specified for certificate requests if the `RequestType` is set to `PKCS#7` or `CMC`. If the `RequestType` is set to `PKCS#10`, this key is ignored. The `Requestername` can only be set as part of	`Domain\User`	`Requestername = Contoso\BSmith`

Key[1]	Description	Value[2]	Example
	the request. You can't manipulate the `Requestername` in a pending request.		
RequestType	Determines the standard that is used to generate and send the certificate request.	• `PKCS10 -- 1` • `PKCS7 -- 2` • `CMC -- 3` • `Cert -- 4` • `SCEP -- fd00 (64768)` **Tip:** This option indicates a self-signed or self-issued certificate. It doesn't generate a request, but rather a new certificate and then installs the certificate. Self-signed is the default. Specify a signing cert by using the -cert option to create a self-issued certificate that isn't self-signed.	`RequestType = CMC`
SecurityDescriptor	Contains the security information associated with securable objects. For most securable objects, you can specify an object's security descriptor in the function call that creates the object.Strings based on security descriptor definition language. **Tip:** This is relevant only for machine context non-smart card keys.	`SecurityDescriptor = D:P(A;;GA;;;SY)(A;;GA;;;BA)`	
AlternateSignatureAlgorithm	Specifies and retrieves a Boolean value that indicates whether the signature algorithm object identifier (OID) for a PKCS#10 request or certificate signature is discrete or combined.	`true | false`	`AlternateSignatureAlgorithm = false` For an RSA signature, `false` indicates a `Pkcs1 v1.5`, while `true` indicates a `v2.1` signature.
Silent	By default, this option allows the CSP access to the interactive user desktop and request information such as a smart card PIN from the user. If this key is set to TRUE, the CSP must not interact with the desktop and will be blocked from displaying any user interface to the user.	`true | false`	`Silent = true`

Key[1]	Description	Value[2]	Example
SMIME	If this parameter is set to TRUE, an extension with the object identifier value 1.2.840.113549.1.9.15 is added to the request. The number of object identifiers depends on the operating system version installed and CSP capability, which refers to symmetric encryption algorithms that may be used by Secure Multipurpose Internet Mail Extensions (S/MIME) applications such as Outlook.	`true | false`	`SMIME = true`
UseExistingKeySet	This parameter is used to specify that an existing key pair should be used in building a certificate request. If this key is set to TRUE, you must also specify a value for the RenewalCert key or the KeyContainer name. You must not set the Exportable key because you can't change the properties of an existing key. In this case, no key material is generated when the certificate request is built.	`true | false`	`UseExistingKeySet = true`
KeyProtection	Specifies a value that indicates how a private key is protected before use.	• `XCN_NCRYPT_UI_NO_PROTCTION_FLAG -- 0` • `XCN_NCRYPT_UI_PROTECT_KEY_FLAG -- 1` • `XCN_NCRYPT_UI_FORCE_HIGH_PROTECTION_FLAG -- 2`	`KeyProtection = NCRYPT_UI_FORCE_HIGH_PROTECTION_FLAG`
SuppressDefaults	Specifies a Boolean value that indicates whether the default extensions and attributes are included in the request. The defaults are represented by their object identifiers (OIDs).	`true | false`	`SuppressDefaults = true`
FriendlyName	A friendly name for the new certificate.	Text	`FriendlyName = Server1`

Key[1]	Description	Value[2]	Example
ValidityPeriodUnits	Specifies many units that are to be used with ValidityPeriod. Note: This is used only when the `request type=cert`.	Numeric	`ValidityPeriodUnits = 3`
ValidityPeriod	ValidityPeriod must be a US English plural time period. Note: This is used only when the request type=cert.	`Years | Months | Weeks | Days | Hours | Minutes | Seconds`	`ValidityPeriod = Years`

[1]Parameter to the left of the equal sign (=)

[2]Parameter to the right of the equal sign (=)

[Extensions]

This section is optional.

Expand table

Extension OID	Definition	Example
2.5.29.17		2.5.29.17 = {text}
continue		`continue = UPN=User@Domain.com&`
continue		`continue = EMail=User@Domain.com&`
continue		`continue = DNS=host.domain.com&`
continue		`continue = DirectoryName=CN=Name,DC=Domain,DC=com&`
continue		`continue = URL=<http://host.domain.com/default.html&>`
continue		`continue = IPAddress=10.0.0.1&`
continue		`continue = RegisteredId=1.2.3.4.5&`
continue		`continue = 1.2.3.4.6.1={utf8}String&`
continue		`continue = 1.2.3.4.6.2={octet}AAECAwQFBgc=&`
continue		`continue = 1.2.3.4.6.2={octet}{hex}00 01 02 03 04 05 06 07&`
continue		`continue = 1.2.3.4.6.3={asn}BAgAAQIDBAUGBw==&`
continue		`continue = 1.2.3.4.6.3={hex}04 08 00 01 02 03 04 05 06 07`
2.5.29.37		`2.5.29.37={text}`
continue		`continue = 1.3.6.1.5.5.7`
continue		`continue = 1.3.6.1.5.5.7.3.1`
2.5.29.19		`{text}ca=0pathlength=3`
Critical		`Critical=2.5.29.19`
KeySpec		• `AT_NONE -- 0` • `AT_SIGNATURE -- 2`

Extension OID	Definition	Example
		• `AT_KEYEXCHANGE -- 1`
RequestType		• `PKCS10 -- 1` • `PKCS7 -- 2` • `CMC -- 3` • `Cert -- 4` • `SCEP -- fd00 (64768)`
KeyUsage		• `CERT_DIGITAL_SIGNATURE_KEY_USAGE -- 80 (128)` • `CERT_NON_REPUDIATION_KEY_USAGE -- 40 (64)` • `CERT_KEY_ENCIPHERMENT_KEY_USAGE -- 20 (32)` • `CERT_DATA_ENCIPHERMENT_KEY_USAGE -- 10 (16)` • `CERT_KEY_AGREEMENT_KEY_USAGE -- 8` • `CERT_KEY_CERT_SIGN_KEY_USAGE -- 4` • `CERT_OFFLINE_CRL_SIGN_KEY_USAGE -- 2` • `CERT_CRL_SIGN_KEY_USAGE -- 2` • `CERT_ENCIPHER_ONLY_KEY_USAGE -- 1` • `CERT_DECIPHER_ONLY_KEY_USAGE -- 8000 (32768)`
KeyUsageProperty		• `NCRYPT_ALLOW_DECRYPT_FLAG -- 1` • `NCRYPT_ALLOW_SIGNING_FLAG -- 2` • `NCRYPT_ALLOW_KEY_AGREEMENT_FLAG -- 4` • `NCRYPT_ALLOW_ALL_USAGES -- ffffff (16777215)`
KeyProtection		• `NCRYPT_UI_NO_PROTECTION_FLAG -- 0` • `NCRYPT_UI_PROTECT_KEY_FLAG -- 1` • `NCRYPT_UI_FORCE_HIGH_PROTECTION_FLAG -- 2`
SubjectNameFlags	template	• `CT_FLAG_SUBJECT_REQUIRE_COMMON_NAME -- 40000000 (1073741824)` • `CT_FLAG_SUBJECT_REQUIRE_DIRECTORY_PATH -- 80000000 (2147483648)` • `CT_FLAG_SUBJECT_REQUIRE_DNS_AS_CN -- 10000000 (268435456)` • `CT_FLAG_SUBJECT_REQUIRE_EMAIL -- 20000000 (536870912)` • `CT_FLAG_OLD_CERT_SUPPLIES_SUBJECT_AND_ALT_NAME -- 8` • `CT_FLAG_SUBJECT_ALT_REQUIRE_DIRECTORY_GUID -- 1000000 (16777216)` • `CT_FLAG_SUBJECT_ALT_REQUIRE_DNS -- 8000000 (134217728)` • `CT_FLAG_SUBJECT_ALT_REQUIRE_DOMAIN_DNS -- 400000 (4194304)` • `CT_FLAG_SUBJECT_ALT_REQUIRE_EMAIL -- 4000000 (67108864)` • `CT_FLAG_SUBJECT_ALT_REQUIRE_SPN -- 800000 (8388608)` • `CT_FLAG_SUBJECT_ALT_REQUIRE_UPN -- 2000000 (33554432)`
X500NameFlags		• `CERT_NAME_STR_NONE -- 0` • `CERT_OID_NAME_STR -- 2` • `CERT_X500_NAME_STR -- 3` • `CERT_NAME_STR_SEMICOLON_FLAG -- 40000000 (1073741824)` • `CERT_NAME_STR_NO_PLUS_FLAG -- 20000000 (536870912)` • `CERT_NAME_STR_NO_QUOTING_FLAG -- 10000000 (268435456)` • `CERT_NAME_STR_CRLF_FLAG -- 8000000 (134217728)` • `CERT_NAME_STR_COMMA_FLAG -- 4000000 (67108864)` • `CERT_NAME_STR_REVERSE_FLAG -- 2000000 (33554432)` • `CERT_NAME_STR_FORWARD_FLAG -- 1000000 (16777216)` • `CERT_NAME_STR_DISABLE_IE4_UTF8_FLAG -- 10000 (65536)` • `CERT_NAME_STR_ENABLE_T61_UNICODE_FLAG -- 20000 (131072)` • `CERT_NAME_STR_ENABLE_UTF8_UNICODE_FLAG -- 40000 (262144)` • `CERT_NAME_STR_FORCE_UTF8_DIR_STR_FLAG -- 80000 (524288)` • `CERT_NAME_STR_DISABLE_UTF8_DIR_STR_FLAG -- 100000 (1048576)` • `CERT_NAME_STR_ENABLE_PUNYCODE_FLAG -- 200000 (2097152)`

> ⓘ Note
>
> `SubjectNameFlags` allows the INF file to specify which **Subject** and **SubjectAltName** extension fields should be auto-populated by certreq based on the current user or current machine properties: DNS name, UPN, and so on. Using the literal template means the template name flags are used instead. This allows a single INF file to be used in multiple contexts to generate requests with context-specific subject information.
>
> `X500NameFlags` specifies the flags to be passed directly to `CertStrToName` API when the `Subject INF keys` value is converted to an ASN.1 encoded Distinguished **Name**.

Example

To create a policy file (.inf) in Notepad and save it as *requestconfig.inf*:

```
[NewRequest]
Subject = CN=<FQDN of computer you are creating the certificate>
Exportable = TRUE
KeyLength = 2048
KeySpec = 1
KeyUsage = 0xf0
MachineKeySet = TRUE
[RequestAttributes]
CertificateTemplate=WebServer
[Extensions]
OID = 1.3.6.1.5.5.7.3.1
OID = 1.3.6.1.5.5.7.3.2
```

On the computer for which you're requesting a certificate:

```
certreq -new requestconfig.inf certrequest.req
```

To use the [Strings] section syntax for OIDs and other difficult to interpret data. The new {text} syntax example for EKU extension, which uses a comma separated list of OIDs:

```
[Version]
Signature=$Windows NT$

[Strings]
szOID_ENHANCED_KEY_USAGE = 2.5.29.37
szOID_PKIX_KP_SERVER_AUTH = 1.3.6.1.5.5.7.3.1
szOID_PKIX_KP_CLIENT_AUTH = 1.3.6.1.5.5.7.3.2

[NewRequest]
Subject = CN=TestSelfSignedCert
Requesttype = Cert

[Extensions]
%szOID_ENHANCED_KEY_USAGE%={text}%szOID_PKIX_KP_SERVER_AUTH%,
_continue_ = %szOID_PKIX_KP_CLIENT_AUTH%
```

To specify a Subject Alternate Name (SAN), add it in the [Extensions] section of your INF. For example:

```
[Version]
Signature=$Windows NT$
```

```
[Strings]
szOID_ENHANCED_KEY_USAGE = 2.5.29.37
szOID_PKIX_KP_SERVER_AUTH = 1.3.6.1.5.5.7.3.1
szOID_PKIX_KP_CLIENT_AUTH = 1.3.6.1.5.5.7.3.2

[NewRequest]
Subject = CN=TestSelfSignedCert
Requesttype = Cert

[Extensions]
2.5.29.17 = "{text}"
_continue_ = "DNS=example.com"
_continue_ = "DNS=www.example.com"
_continue_ = "IP Address=192.168.1.1"
```

In this example, `2.5.29.17` is the OID defining the SAN. To specify multiple SANs, the `_continue_` extension OID is used, which lets you separate and specify each SAN extension.

certreq -accept

The `-accept` parameter links the previously generated private key with the issued certificate and removes the pending certificate request from the system where the certificate is requested (if there's a matching request).

To manually accept a certificate:

```
certreq -accept certnew.cer
```

> ⚠ **Warning**
>
> Using the `-accept` parameter with the `-user` and `-machine` options indicates whether the installing certificate should be installed in **user** or **machine** context. If there's an outstanding request in either context that matches the public key being installed, then these options aren't needed. If there is no outstanding request, then one of these must be specified.

certreq -policy

The policy.inf file is a configuration file that defines the constraints applied to a CA certification, when a qualified subordination is defined.

To build a cross certificate request:

```
certreq -policy certsrv.req policy.inf newcertsrv.req
```

Using `certreq -policy` without any other parameter opens a dialog window, allowing you to select the requested file (.req, .cmc, .txt, .der, .cer or .crt). After you select the requested file and click **Open**, another dialog window opens, allowing you to select the policy.inf file.

Examples

Find an example of the policy.inf file in the CAPolicy.inf Syntax.

certreq -sign

To create a new certificate request, sign it, and to submit it:

```
certreq -new policyfile.inf myrequest.req
certreq -sign myrequest.req myrequest.req
certreq -submit myrequest_sign.req myrequest_cert.cer
```

Remarks

- Using `certreq -sign` without any other parameter it opens a dialog window so you can select the requested file (req, cmc, txt, der, cer or crt).
- Signing the qualified subordination request may require **Enterprise Administrator** credentials. This is a best practice for issuing signing certificates for qualified subordination.
- The certificate used to sign the qualified subordination request uses the qualified subordination template. Enterprise Administrators have to sign the request or grant user permissions to the individuals signing the certificate.
- You might be required to have additional personnel sign the CMC request after you. This depends on the assurance level associated with the qualified subordination.
- If the parent CA of the qualified subordinate CA you're installing is offline, you must obtain the CA certificate for the qualified subordinate CA from the offline parent. If the parent CA is online, specify the CA certificate for the qualified subordinate CA during the **Certificate Services Installation** wizard.

certreq -enroll

You can use this comment to enroll or renew your certificates.

Examples

To enroll a certificate, using the *WebServer* template, and by selecting the policy server using U/I:

```
certreq -enroll -machine -policyserver * WebServer
```

To renew a certificate using a serial number:

```
certreq -enroll -machine -cert 61 2d 3c fe 00 00 00 00 00 05 renew
```

You can only renew valid certificates. Expired certificates can't be renewed and must be replaced with a new certificate.

Options

Expand table

Options	Description
-any	`Force ICertRequest::Submit` to determine encoding type.
-attrib `<attributestring>`	Specifies the **Name** and **Value** string pairs, separated by a colon. Separate **Name** and **Value** string pairs using `\n` (for example, Name1:value1\nName2:value2).

Options	Description
-binary	Formats output files as binary instead of base64-encoded.
-policyserver `<policyserver>`	ldap: `<path>` Insert the URI or unique ID for a computer running the Certificate Enrollment Policy web service. To specify that you would like to use a request file by browsing, just use a minus (-) sign for `<policyserver>`.
-config `<ConfigString>`	Processes the operation by using the CA specified in the configuration string, which is **CAHostName\CAName**. For an https:\\ connection, specify the enrollment server URI. For the local machine store CA, use a minus (-) sign.
-anonymous	Use anonymous credentials for Certificate Enrollment web services.
-kerberos	Use Kerberos (domain) credentials for Certificate Enrollment web services.
-clientcertificate `<ClientCertId>`	You can replace the `<ClientCertId>` with a certificate thumbprint, CN, EKU, template, email, UPN, or the new `name=value` syntax.
-username `<username>`	Used with Certificate Enrollment web services. You can substitute `<username>` with the SAM name or **domain\user** value. This option is for use with the `-p` option.
-p `<password>`	Used with Certificate Enrollment web services. Substitute `<password>` with the actual user's password. This option is for use with the `-username` option.
-user	Configures the `-user` context for a new certificate request or specifies the context for a certificate acceptance. This is the default context, if none is specified in the INF or template.
-machine	Configures a new certificate request or specifies the context for an a certificate acceptance for the machine context. For new requests it must be consistent with the MachineKeyset INF key and the template context. If this option isn't specified and the template doesn't set a context, then the default is the user context.
-crl	Includes certificate revocation lists (CRLs) in the output to the base64-encoded PKCS #7 file specified by `certchainfileout` or to the base64-encoded file specified by `requestfileout`.
-rpc	Instructs Active Directory Certificate Services (AD CS) to use a remote procedure call (RPC) server connection instead of Distributed COM.
-adminforcemachine	Use the Key Service or impersonation to submit the request from Local System context. Requires that the user invoking this option be a member of Local Administrators.
-renewonbehalfof	Submit a renewal on behalf of the subject identified in the signing certificate. This sets CR_IN_ROBO when calling ICertRequest::Submit method
-f	Force existing files to be overwritten. This also bypasses caching templates and policy.
-q	Use silent mode; suppress all interactive prompts.
-unicode	Writes Unicode output when standard output is redirected or piped to another command, which helps when invoked from Windows PowerShell scripts.
-unicodetext	Sends Unicode output when writing base64 text encoded data blobs to files.

Formats

Expand table

Formats	Description
requestfilein	Base64-encoded or binary input file name: PKCS #10 certificate request, CMS certificate request, PKCS #7 certificate renewal request, X.509 certificate to be cross-certified, or KeyGen tag format certificate request.
requestfileout	Base64-encoded output file name.
certfileout	Base64-encoded X-509 file name.
PKCS10fileout	For use with the `certreq -policy` parameter only. Base64-encoded PKCS10 output file name.

Formats	Description
certchainfileout	Base64-encoded PKCS #7 file name.
fullresponsefileout	Base64-encoded full response file name.
policyfilein	For use with the `certreq -policy` parameter only. INF file containing a textual representation of extensions used to qualify a request.

Additional Resources

The following articles contain examples of certreq usage:

- How to add a subject alternative name to a secure LDAP certificate
- Test Lab Guide: Deploying an AD CS Two-Tier PKI Hierarchy
- Appendix 3: Certreq.exe Syntax
- How to create a web server SSL certificate manually
- Certificate Enrollment for System Center Operations Manager Agent
- Active Directory Certificate Services Overview
- How to enable LDAP over SSL with a third-party certification authority

certutil

Article • 05/01/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

> **⊗ Caution**
>
> `Certutil` isn't recommended to be used in any production code and doesn't provide any guarantees of live site support or application compatibilities. It's a tool utilized by developers and IT administrators to view certificate content information on devices.

Certutil.exe is a command-line program installed as part of Certificate Services. You can use certutil.exe to display certification authority (CA) configuration information, configure Certificate Services, and back up and restore CA components. The program also verifies certificates, key pairs, and certificate chains.

If `certutil` is run on a certification authority without other parameters, it displays the current certification authority configuration. If `certutil` is run on a non-certification authority without other parameters, the command defaults to running the `certutil -dump` command. Not all versions of certutil provide all of the parameters and options that this document describes. You can see the choices that your version of certutil provides by running `certutil -?` or `certutil <parameter> -?`.

> **Tip**
>
> To see complete help for all certutil verbs and options, including ones that are hidden from the `-?` argument, run `certutil -v -uSAGE`. The `uSAGE` switch is case-sensitive.

Parameters

-dump

Dumps the configuration information or files.

Windows Command Prompt

```
certutil [options] [-dump]
certutil [options] [-dump] File
```

Options:

```
Windows Command Prompt
[-f] [-user] [-Silent] [-split] [-p Password] [-t Timeout]
```

-dumpPFX

Dumps the PFX structure.

```
Windows Command Prompt
certutil [options] [-dumpPFX] File
```

Options:

```
Windows Command Prompt
[-f] [-Silent] [-split] [-p Password] [-csp Provider]
```

-asn

Parses and displays the contents of a file using Abstract Syntax Notation (ASN.1) syntax. File types include .CER, .DER and PKCS #7 formatted files.

```
Windows Command Prompt
certutil [options] -asn File [type]
```

- `[type]`: numeric CRYPT_STRING_* decoding type

-decodehex

Decodes a hexadecimal-encoded file.

```
Windows Command Prompt
certutil [options] -decodehex InFile OutFile [type]
```

- `[type]`: numeric CRYPT_STRING_* decoding type

Options:

Windows Command Prompt

```
[-f]
```

-encodehex

Encodes a file in hexadecimal.

Windows Command Prompt

```
certutil [options] -encodehex InFile OutFile [type]
```

- `[type]`: numeric CRYPT_STRING_* encoding type

Options:

Windows Command Prompt

```
[-f] [-nocr] [-nocrlf] [-UnicodeText]
```

-decode

Decodes a Base64-encoded file.

Windows Command Prompt

```
certutil [options] -decode InFile OutFile
```

Options:

Windows Command Prompt

```
[-f]
```

-encode

Encodes a file to Base64.

Windows Command Prompt

```
certutil [options] -encode InFile OutFile
```

Options:

```
Windows Command Prompt

[-f] [-unicodetext]
```

-deny

Denies a pending request.

```
Windows Command Prompt

certutil [options] -deny RequestId
```

Options:

```
Windows Command Prompt

[-config Machine\CAName]
```

-resubmit

Resubmits a pending request.

```
Windows Command Prompt

certutil [options] -resubmit RequestId
```

Options:

```
Windows Command Prompt

[-config Machine\CAName]
```

-setattributes

Sets attributes for a pending certificate request.

```
Windows Command Prompt

certutil [options] -setattributes RequestId AttributeString
```

Where:

- **RequestId** is the numeric Request ID for the pending request.
- **AttributeString** is the request attribute name and value pairs.

Options:

```
Windows Command Prompt

[-config Machine\CAName]
```

Remarks

- Names and values must be colon-separated, while multiple names and value pairs must be newline-separated. For example: `CertificateTemplate:User\nEMail:User@Domain.com` where the `\n` sequence is converted to a newline separator.

-setextension

Set an extension for a pending certificate request.

```
Windows Command Prompt

certutil [options] -setextension RequestId ExtensionName Flags {Long | Date |
String | @InFile}
```

Where:

- **requestID** is the numeric Request ID for the pending request.
- **ExtensionName** is the ObjectId string for the extension.
- **Flags** sets the priority of the extension. `0` is recommended, while `1` sets the extension to critical, `2` disables the extension, and `3` does both.

Options:

```
Windows Command Prompt

[-config Machine\CAName]
```

Remarks

- If the last parameter is numeric, it's taken as a **Long**.

- If the last parameter can be parsed as a date, it's taken as a **Date**.
- If the last parameter starts with `\@`, the rest of the token is taken as the filename with binary data or an ASCII-text hex dump.
- If the last parameter is anything else, it's taken as a String.

-revoke

Revokes a certificate.

Windows Command Prompt

```
certutil [options] -revoke SerialNumber [Reason]
```

Where:

- **SerialNumber** is a comma-separated list of certificate serial numbers to revoke.
- **Reason** is the numeric or symbolic representation of the revocation reason, including:
  - **0. CRL_REASON_UNSPECIFIED** - Unspecified (default)
  - **1. CRL_REASON_KEY_COMPROMISE** - Key compromise
  - **2. CRL_REASON_CA_COMPROMISE** - Certificate Authority compromise
  - **3. CRL_REASON_AFFILIATION_CHANGED** - Affiliation changed
  - **4. CRL_REASON_SUPERSEDED** - Superseded
  - **5. CRL_REASON_CESSATION_OF_OPERATION** - Cessation of operation
  - **6. CRL_REASON_CERTIFICATE_HOLD** - Certificate hold
  - **8. CRL_REASON_REMOVE_FROM_CRL** - Remove from CRL
  - **9: CRL_REASON_PRIVILEGE_WITHDRAWN** - Privilege withdrawn
  - **10: CRL_REASON_AA_COMPROMISE** - AA compromise
  - **-1. Unrevoke** - Unrevokes

Options:

Windows Command Prompt

```
[-config Machine\CAName]
```

-isvalid

Displays the disposition of the current certificate.

Windows Command Prompt

```
certutil [options] -isvalid SerialNumber | CertHash
```

Options:

Windows Command Prompt

```
[-config Machine\CAName]
```

-getconfig

Gets the default configuration string.

Windows Command Prompt

```
certutil [options] -getconfig
```

Options:

Windows Command Prompt

```
[-idispatch] [-config Machine\CAName]
```

-getconfig2

Gets the default configuration string via ICertGetConfig.

Windows Command Prompt

```
certutil [options] -getconfig2
```

Options:

Windows Command Prompt

```
[-idispatch]
```

-getconfig3

Gets configuration via ICertConfig.

Windows Command Prompt

```
certutil [options] -getconfig3
```

Options:

Windows Command Prompt

```
[-idispatch]
```

-ping

Attempts to contact the Active Directory Certificate Services Request interface.

Windows Command Prompt

```
certutil [options] -ping [MaxSecondsToWait | CAMachineList]
```

Where:

- **CAMachineList** is a comma-separated list of CA machine names. For a single machine, use a terminating comma. This option also displays the site cost for each CA machine.

Options:

Windows Command Prompt

```
[-config Machine\CAName] [-Anonymous] [-Kerberos] [-ClientCertificate
ClientCertId] [-UserName UserName] [-p Password]
```

-pingadmin

Attempts to contact the Active Directory Certificate Services Admin interface.

Windows Command Prompt

```
certutil [options] -pingadmin
```

Options:

Windows Command Prompt

```
[-config Machine\CAName]
```

-CAInfo

Displays information about the certification authority.

Windows Command Prompt

```
certutil [options] -CAInfo [InfoName [Index | ErrorCode]]
```

Where:

- **InfoName** indicates the CA property to display, based on the following infoname argument syntax:
  - * - Displays all properties
  - **ads** - Advanced Server
  - **aia [Index]** - AIA URLs
  - **cdp [Index]** - CDP URLs
  - **cert [Index]** - CA cert
  - **certchain [Index]** - CA cert chain
  - **certcount** - CA cert count
  - **certcrlchain [Index]** - CA cert chain with CRLs
  - **certstate [Index]** - CA cert
  - **certstatuscode [Index]** - CA cert verify status
  - **certversion [Index]** - CA cert version
  - **CRL [Index]** - Base CRL
  - **crlstate [Index]** - CRL
  - **crlstatus [Index]** - CRL Publish Status
  - **cross- [Index]** - Backward cross cert
  - **cross+ [Index]** - Forward cross cert
  - **crossstate- [Index]** - Backward cross cert
  - **crossstate+ [Index]** - Forward cross cert
  - **deltacrl [Index]** - Delta CRL
  - **deltacrlstatus [Index]** - Delta CRL Publish Status
  - **dns** - DNS Name
  - **dsname** - Sanitized CA short name (DS name)
  - **error1 ErrorCode** - Error message text
  - **error2 ErrorCode** - Error message text and error code
  - **exit [Index]** - Exit module description
  - **exitcount** - Exit module count
  - **file** - File version
  - **info** - CA info
  - **kra [Index]** - KRA cert

    - **kracount** - KRA cert count
    - **krastate [Index]** - KRA cert
    - **kraused** - KRA cert used count
    - **localename** - CA locale name
    - **name** - CA name
    - **ocsp [Index]** - OCSP URLs
    - **parent** - Parent CA
    - **policy** - Policy module description
    - **product** - Product version
    - **propidmax** - Maximum CA PropId
    - **role** - Role Separation
    - **sanitizedname** - Sanitized CA name
    - **sharedfolder** - Shared folder
    - **subjecttemplateoids** - Subject Template OIDs
    - **templates** - Templates
    - **type** - CA type
    - **xchg [Index]** - CA exchange cert
    - **xchgchain [Index]** - CA exchange cert chain
    - **xchgcount** - CA exchange cert count
    - **xchgcrlchain [Index]** - CA exchange cert chain with CRLs
- **index** is the optional zero-based property index.
- **errorcode** is the numeric error code.

Options:

Windows Command Prompt

```
[-f] [-split] [-config Machine\CAName]
```

-CAPropInfo

Displays CA Property Type information.

Windows Command Prompt

```
certutil [options] -CAInfo [InfoName [Index | ErrorCode]]
```

Options:

Windows Command Prompt

```
[-idispatch] [-v1] [-admin] [-config Machine\CAName]
```

-ca.cert

Retrieves the certificate for the certification authority.

Windows Command Prompt

```
certutil [options] -ca.cert OutCACertFile [Index]
```

Where:

- **OutCACertFile** is the output file.
- **Index** is the CA certificate renewal index (defaults to most recent).

Options:

Windows Command Prompt

```
[-f] [-split] [-config Machine\CAName]
```

-ca.chain

Retrieves the certificate chain for the certification authority.

Windows Command Prompt

```
certutil [options] -ca.chain OutCACertChainFile [Index]
```

Where:

- **OutCACertChainFile** is the output file.
- **Index** is the CA certificate renewal index (defaults to most recent).

Options:

Windows Command Prompt

```
[-f] [-split] [-config Machine\CAName]
```

-GetCRL

Gets a certificate revocation list (CRL).

Windows Command Prompt

```
certutil [options] -GetCRL OutFile [Index] [delta]
```

Where:

- **Index** is the CRL index or key index (defaults to CRL for most recent key).
- **delta** is the delta CRL (default is base CRL).

Options:

Windows Command Prompt

```
[-f] [-split] [-config Machine\CAName]
```

-CRL

Publishes new certificate revocation lists (CRLs) or delta CRLs.

Windows Command Prompt

```
certutil [options] -CRL [dd:hh | republish] [delta]
```

Where:

- **dd:hh** is the new CRL validity period in days and hours.
- **republish** republishes the most recent CRLs.
- **delta** publishes the delta CRLs only (default is base and delta CRLs).

Options:

Windows Command Prompt

```
[-split] [-config Machine\CAName]
```

-shutdown

Shuts down the Active Directory Certificate Services.

Windows Command Prompt

```
certutil [options] -shutdown
```

Options:

Windows Command Prompt

```
[-config Machine\CAName]
```

-installCert

Installs a certification authority certificate.

Windows Command Prompt

```
certutil [options] -installCert [CACertFile]
```

Options:

Windows Command Prompt

```
[-f] [-silent] [-config Machine\CAName]
```

-renewCert

Renews a certification authority certificate.

Windows Command Prompt

```
certutil [options] -renewCert [ReuseKeys] [Machine\ParentCAName]
```

Options:

Windows Command Prompt

```
[-f] [-silent] [-config Machine\CAName]
```

- Use `-f` to ignore an outstanding renewal request, and to generate a new request.

-schema

Dumps the schema for the certificate.

Windows Command Prompt

```
certutil [options] -schema [Ext | Attrib | CRL]
```

Where:

- The command defaults to the Request and Certificate table.
- **Ext** is the extension table.
- **Attribute** is the attribute table.
- **CRL** is the CRL table.

Options:

Windows Command Prompt

```
[-split] [-config Machine\CAName]
```

-view

Dumps the certificate view.

Windows Command Prompt

```
certutil [options] -view [Queue | Log | LogFail | Revoked | Ext | Attrib | CRL]
[csv]
```

Where:

- **Queue** dumps a specific request queue.
- **Log** dumps the issued or revoked certificates, plus any failed requests.
- **LogFail** dumps the failed requests.
- **Revoked** dumps the revoked certificates.
- **Ext** dumps the extension table.
- **Attrib** dumps the attribute table.
- **CRL** dumps the CRL table.
- **csv** provides the output using comma-separated values.

Options:

Windows Command Prompt

```
[-silent] [-split] [-config Machine\CAName] [-restrict RestrictionList] [-out
ColumnList]
```

Remarks

- To display the **StatusCode** column for all entries, type `-out StatusCode`
- To display all columns for the last entry, type: `-restrict RequestId==$`
- To display the **RequestId** and **Disposition** for three requests, type: `-restrict requestID>=37,requestID<40 -out requestID,disposition`
- To display Row IDs **Row IDs** and **CRL numbers** for all Base CRLs, type: `-restrict crlminbase=0 -out crlrowID,crlnumber crl`
- To display Base CRL number 3, type: `-v -restrict crlminbase=0,crlnumber=3 -out crlrawcrl crl`
- To display the entire CRL table, type: `CRL`
- Use `Date[+|-dd:hh]` for date restrictions.
- Use `now+dd:hh` for a date relative to the current time.
- Templates contain Extended Key Usages (EKUs), which are object identifiers (OIDs) that describe how the certificate is used. Certificates don't always include template common names or display names, but they always contain the template EKUs. You can extract the EKUs for a specific certificate template from Active Directory and then restrict views based on that extension.

-db

Dumps the raw database.

Windows Command Prompt

```
certutil [options] -db
```

Options:

Windows Command Prompt

```
[-config Machine\CAName] [-restrict RestrictionList] [-out ColumnList]
```

-deleterow

Deletes a row from the server database.

Windows Command Prompt

```
certutil [options] -deleterow RowId | Date [Request | Cert | Ext | Attrib | CRL]
```

Where:

- **Request** deletes the failed and pending requests, based on submission date.
- **Cert** deletes the expired and revoked certificates, based on expiration date.
- **Ext** deletes the extension table.
- **Attrib** deletes the attribute table.
- **CRL** deletes the CRL table.

Options:

Windows Command Prompt

```
[-f] [-config Machine\CAName]
```

Examples

- To delete failed and pending requests submitted by January 22, 2001, type: `1/22/2001 request`
- To delete all certificates that expired by January 22, 2001, type: `1/22/2001 cert`
- To delete the certificate row, attributes, and extensions for RequestID 37, type: `37`
- To delete CRLs that expired by January 22, 2001, type: `1/22/2001 crl`

> ⓘ **Note**
>
> **Date** expects the format `mm/dd/yyyy` rather than `dd/mm/yyyy`, for example `1/22/2001` rather than `22/1/2001` for January 22, 2001. If your server isn't configured with US regional settings, using the **Date** argument might produce unexpected results.

-backup

Backs up the Active Directory Certificate Services.

Windows Command Prompt

```
certutil [options] -backup BackupDirectory [Incremental] [KeepLog]
```

Where:

- **BackupDirectory** is the directory to store the backed up data.
- **Incremental** performs an incremental backup only (default is full backup).
- **KeepLog** preserves the database log files (default is to truncate log files).

Options:

```
Windows Command Prompt

[-f] [-config Machine\CAName] [-p Password] [-ProtectTo SAMNameAndSIDList]
```

-backupDB

Backs up the Active Directory Certificate Services database.

```
Windows Command Prompt

certutil [options] -backupdb BackupDirectory [Incremental] [KeepLog]
```

Where:

- **BackupDirectory** is the directory to store the backed up database files.
- **Incremental** performs an incremental backup only (default is full backup).
- **KeepLog** preserves the database log files (default is to truncate log files).

Options:

```
Windows Command Prompt

[-f] [-config Machine\CAName]
```

-backupkey

Backs up the Active Directory Certificate Services certificate and private key.

```
Windows Command Prompt

certutil [options] -backupkey BackupDirectory
```

Where:

- **BackupDirectory** is the directory to store the backed up PFX file.

Options:

```
Windows Command Prompt

[-f] [-config Machine\CAName] [-p password] [-ProtectTo SAMNameAndSIDList] [-t
```

```
Timeout]
```

-restore

Restores the Active Directory Certificate Services.

Windows Command Prompt

```
certutil [options] -restore BackupDirectory
```

Where:

- **BackupDirectory** is the directory containing the data to be restored.

Options:

Windows Command Prompt

```
[-f] [-config Machine\CAName] [-p password]
```

-restoredb

Restores the Active Directory Certificate Services database.

Windows Command Prompt

```
certutil [options] -restoredb BackupDirectory
```

Where:

- **BackupDirectory** is the directory containing the database files to be restored.

Options:

Windows Command Prompt

```
[-f] [-config Machine\CAName]
```

-restorekey

Restores the Active Directory Certificate Services certificate and private key.

```
```

Windows Command Prompt

```
certutil [options] -restorekey BackupDirectory | PFXFile
```

Where:

- **BackupDirectory** is the directory containing PFX file to be restored.
- **PFXFile** is the PFX file to be restored.

Options:

Windows Command Prompt

```
[-f] [-config Machine\CAName] [-p password]
```

-exportPFX

Exports the certificates and private keys. For more information, see the `-store` parameter in this article.

Windows Command Prompt

```
certutil [options] -exportPFX [CertificateStoreName] CertId PFXFile [Modifiers]
```

Where:

- **CertificateStoreName** is the name of the certificate store.
- **CertId** is the certificate or CRL match token.
- **PFXFile** is the PFX file to be exported.
- **Modifiers** are the comma-separated list, which can include one or more of the following:
  - **CryptoAlgorithm=** specifies the cryptographic algorithm to use for encrypting the PFX file, such as `TripleDES-Sha1` or `Aes256-Sha256`.
  - **EncryptCert** - Encrypts the private key associated with the certificate with a password.
  - **ExportParameters** -Exports the private key parameters in addition to the certificate and private key.
  - **ExtendedProperties** - Includes all extended properties associated with the certificate in the output file.
  - **NoEncryptCert** - Exports the private key without encrypting it.
  - **NoChain** - Doesn't import the certificate chain.
  - **NoRoot** - Doesn't import the root certificate.

-importPFX

Imports the certificates and private keys. For more information, see the `-store` parameter in this article.

Windows Command Prompt

```
certutil [options] -importPFX [CertificateStoreName] PFXFile [Modifiers]
```

Where:

- **CertificateStoreName** is the name of the certificate store.
- **PFXFile** is the PFX file to be imported.
- **Modifiers** are the comma-separated list, which can include one or more of the following:
  - **AT_KEYEXCHANGE** - Changes the keyspec to key exchange.
  - **AT_SIGNATURE** - Changes the keyspec to signature.
  - **ExportEncrypted** - Exports the private key associated with the certificate with password encryption.
  - **FriendlyName=** - Specifies a friendly name for the imported certificate.
  - **KeyDescription=** - Specifies a description for the private key associated with the imported certificate.
  - **KeyFriendlyName=** - Specifies a friendly name for the private key associated with the imported certificate.
  - **NoCert** - Doesn't import the certificate.
  - **NoChain** - Doesn't import the certificate chain.
  - **NoExport** - Makes the private key non-exportable.
  - **NoProtect** - Doesn't password protect keys by using a password.
  - **NoRoot** - Doesn't import the root certificate.
  - **Pkcs8** - Uses PKCS8 format for the private key in the PFX file.
  - **Protect** - Protects keys by using a password.
  - **ProtectHigh** - Specifies that a high-security password must be associated with the private key.
  - **VSM** - Stores the private key associated with the imported certificate in the Virtual Smart Card (VSC) container.

Options:

Windows Command Prompt

```
[-f] [-Enterprise] [-user] [-GroupPolicy] [-Silent] [-p Password] [-csp Provider]
```

Remarks

- Defaults to personal machine store.

-dynamicfilelist

Displays a dynamic file list.

Windows Command Prompt

```
certutil [options] -dynamicfilelist
```

Options:

Windows Command Prompt

```
[-config Machine\CAName]
```

-databaselocations

Displays database locations.

Windows Command Prompt

```
certutil [options] -databaselocations
```

Options:

Windows Command Prompt

```
[-config Machine\CAName]
```

-hashfile

Generates and displays a cryptographic hash over a file.

Windows Command Prompt

```
certutil [options] -hashfile InFile [HashAlgorithm]
```

-store

Dumps the certificate store.

Windows Command Prompt

```
certutil [options] -store [CertificateStoreName [CertId [OutputFile]]]
```

Where:

- **CertificateStoreName** is the certificate store name. For example:
  - `My, CA (default), Root,`
  - `ldap:///CN=Certification Authorities,CN=Public Key Services,CN=Services,CN=Configuration,DC=cpandl,DC=com?cACertificate?one?objectClass=certificationAuthority (View Root Certificates)`
  - `ldap:///CN=CAName,CN=Certification Authorities,CN=Public Key Services,CN=Services,CN=Configuration,DC=cpandl,DC=com?cACertificate?base?objectClass=certificationAuthority (Modify Root Certificates)`
  - `ldap:///CN=CAName,CN=MachineName,CN=CDP,CN=Public Key Services,CN=Services,CN=Configuration,DC=cpandl,DC=com?certificateRevocationList?base?objectClass=cRLDistributionPoint (View CRLs)`
  - `ldap:///CN=NTAuthCertificates,CN=Public Key Services,CN=Services,CN=Configuration,DC=cpandl,DC=com?cACertificate?base?objectClass=certificationAuthority (Enterprise CA Certificates)`
  - `ldap: (AD computer object certificates)`
  - `-user ldap: (AD user object certificates)`
- **CertId** is the certificate or CRL match token. This ID can be a:
  - Serial number
  - SHA-1 certificate
  - CRL, CTL or public key hash
  - Numeric cert index (0, 1, and so on)
  - Numeric CRL index (.0, .1, and so on)
  - Numeric CTL index (..0, ..1, and so on)
  - Public key
  - Signature or extension ObjectId
  - Certificate subject Common Name
  - E-mail address
  - UPN or DNS name
  - Key container name or CSP name
  - Template name or ObjectId
  - EKU or Application Policies ObjectId
  - CRL issuer Common Name.

Many of these identifiers might result in multiple matches.

- **OutputFile** is the file used to save the matching certificates.

Options:

```
Windows Command Prompt

[-f] [-Enterprise] [-user] [-GroupPolicy] [-Silent] [-split] [-dc DCName]
```

- The `-user` option accesses a user store instead of a machine store.
- The `-enterprise` option accesses a machine enterprise store.
- The `-service` option accesses a machine service store.
- The `-grouppolicy` option accesses a machine group policy store.

For example:

- `-enterprise NTAuth`
- `-enterprise Root 37`
- `-user My 26e0aaaf000000000004`
- `CA .11`

> ⓘ **Note**
>
> Performance issues are observed when using the `-store` parameter given these two aspects:
>
> 1. When the number of certificates in the store exceeds 10.
> 2. When a **CertId** is specified, it's used to match all the listed types for every certificate. For example, if a **serial number** is provided, it will also attempt to match all other listed types.
>
> If you're concerned about performance issues, PowerShell commands are recommended where it will only match the specified certificate type.

-enumstore

Enumerates the certificate stores.

```
Windows Command Prompt

certutil [options] -enumstore [\\MachineName]
```

Where:

- **MachineName** is the remote machine name.

Options:

```
Windows Command Prompt
[-enterprise] [-user] [-grouppolicy]
```

-addstore

Adds a certificate to the store. For more information, see the `-store` parameter in this article.

```
Windows Command Prompt
certutil [options] -addstore CertificateStoreName InFile
```

Where:

- **CertificateStoreName** is the certificate store name.
- **InFile** is the certificate or CRL file you want to add to the store.

Options:

```
Windows Command Prompt
[-f] [-Enterprise] [-user] [-GroupPolicy] [-dc DCName]
```

-delstore

Deletes a certificate from the store. For more information, see the `-store` parameter in this article.

```
Windows Command Prompt
certutil [options] -delstore CertificateStoreName certID
```

Where:

- **CertificateStoreName** is the certificate store name.
- **CertId** is the certificate or CRL match token.

Options:

```
Windows Command Prompt

[-f] [-Enterprise] [-user] [-GroupPolicy] [-Silent] [-dc DCName]
```

-verifystore

Verifies a certificate in the store. For more information, see the `-store` parameter in this article.

```
Windows Command Prompt

certutil [options] -verifystore CertificateStoreName [CertId]
```

Where:

- **CertificateStoreName** is the certificate store name.
- **CertId** is the certificate or CRL match token.

Options:

```
Windows Command Prompt

[-Enterprise] [-user] [-GroupPolicy] [-Silent] [-split] [-dc DCName] [-t Timeout]
```

-repairstore

Repairs a key association or update certificate properties or the key security descriptor. For more information, see the `-store` parameter in this article.

```
Windows Command Prompt

certutil [options] -repairstore CertificateStoreName CertIdList [PropertyInfFile |
SDDLSecurityDescriptor]
```

Where:

- **CertificateStoreName** is the certificate store name.
- **CertIdList** is the comma-separated list of certificate or CRL match tokens. For more information, see the `-store` CertId description in this article.
- **PropertyInfFile** is the INF file containing external properties, including:

Windows Command Prompt

```
[Properties]
    19 = Empty ; Add archived property, OR:
    19 =       ; Remove archived property

    11 = {text}Friendly Name ; Add friendly name property

    127 = {hex} ; Add custom hexadecimal property
        _continue_ = 00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d 0e 0f
        _continue_ = 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d 1e 1f

    2 = {text} ; Add Key Provider Information property
      _continue_ = Container=Container Name&
      _continue_ = Provider=Microsoft Strong Cryptographic Provider&
      _continue_ = ProviderType=1&
      _continue_ = Flags=0&
      _continue_ = KeySpec=2

    9 = {text} ; Add Enhanced Key Usage property
      _continue_ = 1.3.6.1.5.5.7.3.2,
      _continue_ = 1.3.6.1.5.5.7.3.1,
```

Options:

Windows Command Prompt

```
[-f] [-Enterprise] [-user] [-GroupPolicy] [-Silent] [-split] [-csp Provider]
```

-viewstore

Dumps the certificates store. For more information, see the `-store` parameter in this article.

Windows Command Prompt

```
certutil [options] -viewstore [CertificateStoreName [CertId [OutputFile]]]
```

Where:

- **CertificateStoreName** is the certificate store name. For example:
  - `My, CA (default), Root,`
  - `ldap:///CN=Certification Authorities,CN=Public Key Services,CN=Services,CN=Configuration,DC=cpandl,DC=com?cACertificate?one?objectClass=certificationAuthority (View Root Certificates)`
  - `ldap:///CN=CAName,CN=Certification Authorities,CN=Public Key Services,CN=Services,CN=Configuration,DC=cpandl,DC=com?cACertificate?base?`

`objectClass=certificationAuthority (Modify Root Certificates)`
    - `ldap:///CN=CAName,CN=MachineName,CN=CDP,CN=Public Key Services,CN=Services,CN=Configuration,DC=cpandl,DC=com?certificateRevocationList?base?objectClass=cRLDistributionPoint (View CRLs)`
    - `ldap:///CN=NTAuthCertificates,CN=Public Key Services,CN=Services,CN=Configuration,DC=cpandl,DC=com?cACertificate?base?objectClass=certificationAuthority (Enterprise CA Certificates)`
    - `ldap: (AD computer object certificates)`
    - `-user ldap: (AD user object certificates)`

- **CertId** is the certificate or CRL match token. This can be a:
    - Serial number
    - SHA-1 certificate
    - CRL, CTL, or public key hash
    - Numeric cert index (0, 1, and so on)
    - Numeric CRL index (.0, .1, and so on)
    - Numeric CTL index (..0, ..1, and so on)
    - Public key
    - Signature or extension ObjectId
    - Certificate subject Common Name
    - E-mail address
    - UPN or DNS name
    - Key container name or CSP name
    - Template name or ObjectId
    - EKU or Application Policies ObjectId
    - CRL issuer Common Name.

Many of these may result in multiple matches.

- **OutputFile** is the file used to save the matching certificates.

Options:

Windows Command Prompt

```
[-f] [-Enterprise] [-user] [-GroupPolicy] [-dc DCName]
```

- The `-user` option accesses a user store instead of a machine store.
- The `-enterprise` option accesses a machine enterprise store.
- The `-service` option accesses a machine service store.
- The `-grouppolicy` option accesses a machine group policy store.

For example:

- `-enterprise NTAuth`
- `-enterprise Root 37`
- `-user My 26e0aaaf000000000004`
- `CA .11`

-viewdelstore

Deletes a certificate from the store.

Windows Command Prompt

```
certutil [options] -viewdelstore [CertificateStoreName [CertId [OutputFile]]]
```

Where:

- **CertificateStoreName** is the certificate store name. For example:
  - `My, CA (default), Root,`
  - `ldap:///CN=Certification Authorities,CN=Public Key Services,CN=Services,CN=Configuration,DC=cpandl,DC=com?cACertificate?one?objectClass=certificationAuthority (View Root Certificates)`
  - `ldap:///CN=CAName,CN=Certification Authorities,CN=Public Key Services,CN=Services,CN=Configuration,DC=cpandl,DC=com?cACertificate?base?objectClass=certificationAuthority (Modify Root Certificates)`
  - `ldap:///CN=CAName,CN=MachineName,CN=CDP,CN=Public Key Services,CN=Services,CN=Configuration,DC=cpandl,DC=com?certificateRevocationList?base?objectClass=cRLDistributionPoint (View CRLs)`
  - `ldap:///CN=NTAuthCertificates,CN=Public Key Services,CN=Services,CN=Configuration,DC=cpandl,DC=com?cACertificate?base?objectClass=certificationAuthority (Enterprise CA Certificates)`
  - `ldap: (AD computer object certificates)`
  - `-user ldap: (AD user object certificates)`
- **CertId** is the certificate or CRL match token. This can be a:
  - Serial number
  - SHA-1 certificate
  - CRL, CTL, or public key hash
  - Numeric cert index (0, 1, and so on)
  - Numeric CRL index (.0, .1, and so on)

- Numeric CTL index (..0, ..1, and so on)
- Public key
- Signature or extension ObjectId
- Certificate subject Common Name
- E-mail address
- UPN or DNS name
- Key container name or CSP name
- Template name or ObjectId
- EKU or Application Policies ObjectId
- CRL issuer Common Name.

Many of these might result in multiple matches.

- **OutputFile** is the file used to save the matching certificates.

Options:

Windows Command Prompt

```
[-f] [-Enterprise] [-user] [-GroupPolicy] [-dc DCName]
```

- The `-user` option accesses a user store instead of a machine store.
- The `-enterprise` option accesses a machine enterprise store.
- The `-service` option accesses a machine service store.
- The `-grouppolicy` option accesses a machine group policy store.

For example:

- `-enterprise NTAuth`
- `-enterprise Root 37`
- `-user My 26e0aaaf000000000004`
- `CA .11`

-UI

Invokes the certutil interface.

Windows Command Prompt

```
certutil [options] -UI File [import]
```

-TPMInfo

Displays Trusted Platform Module Information.

Windows Command Prompt

```
certutil [options] -TPMInfo
```

Options:

Windows Command Prompt

```
[-f] [-Silent] [-split]
```

-attest

Specifies that the certificate request file should be attested.

Windows Command Prompt

```
certutil [options] -attest RequestFile
```

Options:

Windows Command Prompt

```
[-user] [-Silent] [-split]
```

-getcert

Selects a certificate from a selection UI.

Windows Command Prompt

```
certutil [options] [ObjectId | ERA | KRA [CommonName]]
```

Options:

Windows Command Prompt

```
[-Silent] [-split]
```

-ds

Displays directory service (DS) distinguished names (DNs).

Windows Command Prompt

```
certutil [options] -ds [CommonName]
```

Options:

Windows Command Prompt

```
[-f] [-user] [-split] [-dc DCName]
```

-dsDel

Deletes DS DNs.

Windows Command Prompt

```
certutil [options] -dsDel [CommonName]
```

Options:

Windows Command Prompt

```
[-user] [-split] [-dc DCName]
```

-dsPublish

Publishes a certificate or certificate revocation list (CRL) to Active Directory.

Windows Command Prompt

```
certutil [options] -dspublish CertFile [NTAuthCA | RootCA | SubCA | CrossCA | KRA
| User | Machine]
certutil [options] -dspublish CRLfile [DSCDPContainer [DSCDPCN]]
```

Where:

- **CertFile** is the name of the certificate file to publish.
- **NTAuthCA** publishes the certificate to the DS Enterprise store.
- **RootCA** publishes the certificate to the DS Trusted Root store.
- **SubCA** publishes the CA certificate to the DS CA object.

- **CrossCA** publishes the cross-certificate to the DS CA object.
- **KRA** publishes the certificate to the DS Key Recovery Agent object.
- **User** publishes the certificate to the User DS object.
- **Machine** publishes the certificate to the Machine DS object.
- **CRLfile** is the name of the CRL file to publish.
- **DSCDPContainer** is the DS CDP container CN, usually the CA machine name.
- **DSCDPCN** is the DS CDP object CN based on the sanitized CA short name and key index.

Options:

Windows Command Prompt

```
[-f] [-user] [-dc DCName]
```

- Use `-f` to create a new DS object.

-dsCert

Displays DS certificates.

Windows Command Prompt

```
certutil [options] -dsCert [FullDSDN] | [CertId [OutFile]]
```

Options:

Windows Command Prompt

```
[-Enterprise] [-user] [-config Machine\CAName] [-dc DCName]
```

-dsCRL

Displays DS CRLs.

Windows Command Prompt

```
certutil [options] -dsCRL [FullDSDN] | [CRLIndex [OutFile]]
```

Options:

Windows Command Prompt

```
[-idispatch] [-Enterprise] [-user] [-config Machine\CAName] [-dc DCName]
```

-dsDeltaCRL

Displays DS delta CRLs.

Windows Command Prompt

```
certutil [options] -dsDeltaCRL [FullDSDN] | [CRLIndex [OutFile]]
```

Options:

Windows Command Prompt

```
[-Enterprise] [-user] [-config Machine\CAName] [-dc DCName]
```

-dsTemplate

Displays DS template attributes.

Windows Command Prompt

```
certutil [options] -dsTemplate [Template]
```

Options:

Windows Command Prompt

```
[Silent] [-dc DCName]
```

-dsAddTemplate

Adds DS templates.

Windows Command Prompt

```
certutil [options] -dsAddTemplate TemplateInfFile
```

Options:

Windows Command Prompt

```
[-dc DCName]
```

-ADTemplate

Displays Active Directory templates.

Windows Command Prompt

```
certutil [options] -ADTemplate [Template]
```

Options:

Windows Command Prompt

```
[-f] [-user] [-ut] [-mt] [-dc DCName]
```

-Template

Displays the certificate enrollment policy templates.

Options:

Windows Command Prompt

```
certutil [options] -Template [Template]
```

Options:

Windows Command Prompt

```
[-f] [-user] [-Silent] [-PolicyServer URLOrId] [-Anonymous] [-Kerberos] [-
ClientCertificate ClientCertId] [-UserName UserName] [-p Password]
```

-TemplateCAs

Displays the certification authorities (CAs) for a certificate template.

Windows Command Prompt

```
certutil [options] -TemplateCAs Template
```

Options:

```
Windows Command Prompt

[-f] [-user] [-dc DCName]
```

-CATemplates

Displays templates for the Certificate Authority.

```
Windows Command Prompt

certutil [options] -CATemplates [Template]
```

Options:

```
Windows Command Prompt

[-f] [-user] [-ut] [-mt] [-config Machine\CAName] [-dc DCName]
```

-SetCATemplates

Sets the certificate templates that the Certificate Authority can issue.

```
Windows Command Prompt

certutil [options] -SetCATemplates [+ | -] TemplateList
```

Where:

- The `+` sign adds certificate templates to the CA's available template list.
- The `-` sign removes certificate templates from the CA's available template list.

-SetCASites

Manages site names, including setting, verifying, and deleting Certificate Authority site names.

```
Windows Command Prompt

certutil [options] -SetCASites [set] [SiteName]
certutil [options] -SetCASites verify [SiteName]
certutil [options] -SetCASites delete
```

Where:

- **SiteName** is allowed only when targeting a single Certificate Authority.

Options:

```
Windows Command Prompt

[-f] [-config Machine\CAName] [-dc DCName]
```

Remarks

- The `-config` option targets a single Certificate Authority (default is all CAs).
- The `-f` option can be used to override validation errors for the specified **SiteName** or to delete all CA site names.

> **Note**
>
> For more information about configuring CAs for Active Directory Domain Services (AD DS) site awareness, see AD DS Site Awareness for AD CS and PKI clients.

-enrollmentServerURL

Displays, adds, or deletes enrollment server URLs associated with a CA.

```
Windows Command Prompt

certutil [options] -enrollmentServerURL [URL AuthenticationType [Priority]
[Modifiers]]
certutil [options] -enrollmentserverURL URL delete
```

Where:

- **AuthenticationType** specifies one of the following client authentication methods while adding a URL:
  - **Kerberos** - Use Kerberos SSL credentials.
  - **UserName** - Use a named account for SSL credentials.
  - **ClientCertificate** - Use X.509 Certificate SSL credentials.
  - **Anonymous** - Use anonymous SSL credentials.
- **delete** deletes the specified URL associated with the CA.
- **Priority** defaults to `1` if not specified when adding a URL.
- **Modifiers** is a comma-separated list, which includes one or more of the following:

- **AllowRenewalsOnly** only renewal requests can be submitted to this CA via this URL.
- **AllowKeyBasedRenewal** allows use of a certificate that has no associated account in the AD. This applies only with **ClientCertificate** and **AllowRenewalsOnly** mode.

Options:

```
Windows Command Prompt

[-config Machine\CAName] [-dc DCName]
```

-ADCA

Displays the Active Directory Certificate Authorities.

```
Windows Command Prompt

certutil [options] -ADCA [CAName]
```

Options:

```
Windows Command Prompt

[-f] [-split] [-dc DCName]
```

-CA

Displays the enrollment policy Certificate Authorities.

```
Windows Command Prompt

certutil [options] -CA [CAName | TemplateName]
```

Options:

```
Windows Command Prompt

[-f] [-user] [-Silent] [-split] [-PolicyServer URLOrId] [-Anonymous] [-Kerberos]
[-ClientCertificate ClientCertId] [-UserName UserName] [-p Password]
```

-Policy

Displays the enrollment policy.

Windows Command Prompt

```
certutil [options] -Policy
```

Options:

Windows Command Prompt

```
[-f] [-user] [-Silent] [-split] [-PolicyServer URLOrId] [-Anonymous] [-Kerberos]
[-ClientCertificate ClientCertId] [-UserName UserName] [-p Password]
```

-PolicyCache

Displays or deletes enrollment policy cache entries.

Windows Command Prompt

```
certutil [options] -PolicyCache [delete]
```

Where:

- **delete** deletes the policy server cache entries.
- **-f** deletes all cache entries

Options:

Windows Command Prompt

```
[-f] [-user] [-policyserver URLorID]
```

-CredStore

Displays, adds, or deletes Credential Store entries.

Windows Command Prompt

```
certutil [options] -CredStore [URL]
certutil [options] -CredStore URL add
certutil [options] -CredStore URL delete
```

Where:

- **URL** is the target URL. You can also use `*` to match all entries or `https://machine*` to match a URL prefix.
- **add** adds a credential store entry. Using this option also requires the use of SSL credentials.
- **delete** deletes credential store entries.
- **-f** overwrites a single entry or deletes multiple entries.

Options:

```
Windows Command Prompt

[-f] [-user] [-Silent] [-Anonymous] [-Kerberos] [-ClientCertificate ClientCertId]
[-UserName UserName] [-p Password]
```

-InstallDefaultTemplates

Installs the default certificate templates.

```
Windows Command Prompt

certutil [options] -InstallDefaultTemplates
```

Options:

```
Windows Command Prompt

[-dc DCName]
```

-URL

Verifies certificate or CRL URLs.

```
Windows Command Prompt

certutil [options] -URL InFile | URL
```

Options:

```
Windows Command Prompt

[-f] [-split]
```

-URLCache

Displays or deletes URL cache entries.

Windows Command Prompt

```
certutil [options] -URLcache [URL | CRL | * [delete]]
```

Where:

- **URL** is the cached URL.
- **CRL** runs on all cached CRL URLs only.
- * operates on all cached URLs.
- **delete** deletes relevant URLs from the current user's local cache.
- **-f** forces fetching a specific URL and updating the cache.

Options:

Windows Command Prompt

```
[-f] [-split]
```

-pulse

Pulses an autoenrollment event or NGC task.

Windows Command Prompt

```
certutil [options] -pulse [TaskName [SRKThumbprint]]
```

Where:

- **TaskName** is the task to trigger.
  - **Pregen** is the NGC Key pregen task.
  - **AIKEnroll** is the NGC AIK certificate enrollment task. (Defaults to the autoenrollment event).
- **SRKThumbprint** is the thumbprint of the Storage Root Key
- **Modifiers**:
  - Pregen
  - PregenDelay
  - AIKEnroll
  - CryptoPolicy

- NgcPregenKey
- DIMSRoam

Options:

```
Windows Command Prompt

[-user]
```

-MachineInfo

Displays information about the Active Directory machine object.

```
Windows Command Prompt

certutil [options] -MachineInfo DomainName\MachineName$
```

-DCInfo

Displays information about the domain controller. The default displays DC certificates without verification.

```
Windows Command Prompt

certutil [options] -DCInfo [Domain] [Verify | DeleteBad | DeleteAll]
```

- **Modifiers**:
  - Verify
  - DeleteBad
  - DeleteAll

Options:

```
Windows Command Prompt

[-f] [-user] [-urlfetch] [-dc DCName] [-t Timeout]
```

> **Tip**
>
> The ability to specify an Active Directory Domain Services (AD DS) domain **[Domain]** and to specify a domain controller (**-dc**) was added in Windows Server 2012. To successfully

run the command, you must use an account that is a member of **Domain Admins** or **Enterprise Admins**. The behavior modifications of this command are as follows:

- If a domain isn't specified and a specific domain controller isn't specified, this option returns a list of domain controllers to process from the default domain controller.
- If a domain isn't specified, but a domain controller is specified, a report of the certificates on the specified domain controller is generated.
- If a domain is specified, but a domain controller isn't specified, a list of domain controllers is generated along with reports on the certificates for each domain controller in the list.
- If the domain and domain controller are specified, a list of domain controllers is generated from the targeted domain controller. A report of the certificates for each domain controller in the list is also generated.

For example, assume there's a domain named CPANDL with a domain controller named CPANDL-DC1. You can run the following command to a retrieve a list of domain controllers and their certificates from CPANDL-DC1: `certutil -dc cpandl-dc1 -DCInfo cpandl`.

-EntInfo

Displays information about an enterprise Certificate Authority.

Windows Command Prompt

```
certutil [options] -EntInfo DomainName\MachineName$
```

Options:

Windows Command Prompt

```
[-f] [-user]
```

-TCAInfo

Displays information about the Certificate Authority.

Windows Command Prompt

```
certutil [options] -TCAInfo [DomainDN | -]
```

Options:

Windows Command Prompt

```
[-f] [-Enterprise] [-user] [-urlfetch] [-dc DCName] [-t Timeout]
```

-SCInfo

Displays information about the smart card.

Windows Command Prompt

```
certutil [options] -scinfo [ReaderName [CRYPT_DELETEKEYSET]]
```

Where:

- **CRYPT_DELETEKEYSET** deletes all keys on the smart card.

Options:

Windows Command Prompt

```
[-Silent] [-split] [-urlfetch] [-t Timeout]
```

-SCRoots

Manages smart card root certificates.

Windows Command Prompt

```
certutil [options] -SCRoots update [+][InputRootFile] [ReaderName]
certutil [options] -SCRoots save @OutputRootFile [ReaderName]
certutil [options] -SCRoots view [InputRootFile | ReaderName]
certutil [options] -SCRoots delete [ReaderName]
```

Options:

Windows Command Prompt

```
[-f] [-split] [-p Password]
```

-key

Lists the keys stored in a key container.

Windows Command Prompt

```
certutil [options] -key [KeyContainerName | -]
```

Where:

- **KeyContainerName** is the key container name for the key to verify. This option defaults to machine keys. To switch to user keys, use `-user`.
- Using the `-` sign refers to using the default key container.

Options:

Windows Command Prompt

```
[-user] [-Silent] [-split] [-csp Provider] [-Location AlternateStorageLocation]
```

-delkey

Deletes the named key container.

Windows Command Prompt

```
certutil [options] -delkey KeyContainerName
```

Options:

Windows Command Prompt

```
[-user] [-Silent] [-split] [-csp Provider] [-Location AlternateStorageLocation]
```

-DeleteHelloContainer

Deletes the Windows Hello container, removing all associated credentials that are stored on the device, including any WebAuthn and FIDO credentials.

Users need to sign out after using this option for it to complete.

Windows Command Prompt

```
certutil [options] -DeleteHelloContainer
```

-verifykeys

Verifies a public or private key set.

Windows Command Prompt

```
certutil [options] -verifykeys [KeyContainerName CACertFile]
```

Where:

- **KeyContainerName** is the key container name for the key to verify. This option defaults to machine keys. To switch to user keys, use `-user`.
- **CACertFile** signs or encrypts certificate files.

Options:

Windows Command Prompt

```
[-f] [-user] [-Silent] [-config Machine\CAName]
```

Remarks

- If no arguments are specified, each signing CA certificate is verified against its private key.
- This operation can only be performed against a local CA or local keys.

-verify

Verifies a certificate, certificate revocation list (CRL), or certificate chain.

Windows Command Prompt

```
certutil [options] -verify CertFile [ApplicationPolicyList | -
[IssuancePolicyList]] [Modifiers]
certutil [options] -verify CertFile [CACertFile [CrossedCACertFile]]
certutil [options] -verify CRLFile CACertFile [IssuedCertFile]
certutil [options] -verify CRLFile CACertFile [DeltaCRLFile]
```

Where:

- **CertFile** is the name of the certificate to verify.

- **ApplicationPolicyList** is the optional comma-separated list of required Application Policy ObjectIds.
- **IssuancePolicyList** is the optional comma-separated list of required Issuance Policy ObjectIds.
- **CACertFile** is the optional issuing CA certificate to verify against.
- **CrossedCACertFile** is the optional certificate cross-certified by **CertFile**.
- **CRLFile** is the CRL file used to verify the **CACertFile**.
- **IssuedCertFile** is the optional issued certificate covered by the CRLfile.
- **DeltaCRLFile** is the optional delta CRL file.
- **Modifiers**:
  - Strong - Strong signature verification
  - MSRoot - Must chain to a Microsoft root
  - MSTestRoot - Must chain to a Microsoft test root
  - AppRoot - Must chain to a Microsoft application root
  - EV - Enforce Extended Validation Policy

Options:

Windows Command Prompt

```
[-f] [-Enterprise] [-user] [-Silent] [-split] [-urlfetch] [-t Timeout] [-sslpolicy
ServerName]
```

Remarks

- Using **ApplicationPolicyList** restricts chain building to only chains valid for the specified Application Policies.
- Using **IssuancePolicyList** restricts chain building to only chains valid for the specified Issuance Policies.
- Using **CACertFile** verifies the fields in the file against **CertFile** or **CRLfile**.
- If **CACertFile** isn't specified, the full chain is built and verified against **CertFile**.
- If **CACertFile** and **CrossedCACertFile** are both specified, the fields in both files are verified against **CertFile**.
- Using **IssuedCertFile** verifies the fields in the file against **CRLfile**.
- Using **DeltaCRLFile** verifies the fields in the file against **CertFile**.

-verifyCTL

Verifies the AuthRoot or Disallowed Certificates CTL.

Windows Command Prompt

```
certutil [options] -verifyCTL CTLobject [CertDir] [CertFile]
```

Where:

- **CTLObject** identifies the CTL to verify, including:
  - **AuthRootWU** reads the AuthRoot CAB and matching certificates from the URL cache. Use `-f` to download from Windows Update instead.
  - **DisallowedWU** reads the Disallowed Certificates CAB and disallowed certificate store file from the URL cache. Use `-f` to download from Windows Update instead.
    - **PinRulesWU** reads the PinRules CAB from the URL cache. Use `-f` to download from Windows Update instead.
  - **AuthRoot** reads the registry-cached AuthRoot CTL. Use with `-f` and an untrusted **CertFile** to force the registry cached **AuthRoot** and Disallowed Certificate CTLs to update.
  - **Disallowed** reads the registry-cached Disallowed Certificates CTL. Use with `-f` and an untrusted **CertFile** to force the registry cached **AuthRoot** and Disallowed Certificate CTLs to update.
    - **PinRules** reads the registry cached PinRules CTL. Using `-f` has the same behavior as with **PinRulesWU**.
  - **CTLFileName** specifies the file or http path to the CTL or CAB file.

- **CertDir** specifies the folder containing certificates matching the CTL entries. Defaults to the same folder or website as the **CTLobject**. Using an http folder path requires a path separator at the end. If you don't specify **AuthRoot** or **Disallowed**, multiple locations are searched for matching certificates, including local certificate stores, crypt32.dll resources and the local URL cache. Use `-f` to download from Windows Update, as needed.

- **CertFile** specifies the certificate(s) to verify. Certificates are matched against CTL entries, displaying the results. This option suppresses most of the default output.

Options:

Windows Command Prompt

```
[-f] [-user] [-split]
```

-syncWithWU

Syncs certificates with Windows Update.

Windows Command Prompt

```
certutil [options] -syncWithWU DestinationDir
```

Where:

- **DestinationDir** is the specified directory.
- **f** forces an overwrite.
- **Unicode** writes redirected output in Unicode.
- **gmt** displays times as GMT.
- **seconds** displays times with seconds and milliseconds.
- **v** is a verbose operation.
- **PIN** is the Smart Card PIN.
- **WELL_KNOWN_SID_TYPE** is a numeric SID:
  - 22 - Local System
  - 23 - Local Service
  - 24 - Network Service

Remarks

The following files are downloaded by using the automatic update mechanism:

- *authrootstl.cab* contains the CTLs of non-Microsoft root certificates.
- *disallowedcertstl.cab* contains the CTLs of untrusted certificates.
- *disallowedcert.sst* contains the serialized certificate store, including the untrusted certificates.
- *thumbprint.crt* contains the non-Microsoft root certificates.

For example, `certutil -syncWithWU \\server1\PKI\CTLs`.

- If you use a nonexistent local path or folder as the destination folder, you see the error: `The system can't find the file specified. 0x80070002 (WIN32: 2 ERROR_FILE_NOT_FOUND)`
- If you use a nonexistent or unavailable network location as the destination folder, you see the error: `The network name can't be found. 0x80070043 (WIN32: 67 ERROR_BAD_NET_NAME)`
- If your server can't connect over TCP port 80 to Microsoft Automatic Update servers, you receive the following error: `A connection with the server couldn't be established 0x80072efd (INet: 12029 ERROR_INTERNET_CANNOT_CONNECT)`
- If your server is unable to reach the Microsoft Automatic Update servers with the DNS name `ctldl.windowsupdate.com`, you receive the following error: `The server name or address couldn't be resolved 0x80072ee7 (INet: 12007 ERROR_INTERNET_NAME_NOT_RESOLVED).`

- If you don't use the `-f` switch, and any of the CTL files already exist in the directory, you receive a file exists error: `certutil: -syncWithWU command FAILED: 0x800700b7 (WIN32/HTTP: 183 ERROR_ALREADY_EXISTS) Certutil: Can't create a file when that file already exists.`
- If there's a change in the trusted root certificates, you see: `Warning! Encountered the following no longer trusted roots: <folder path>\<thumbprint>.crt. Use "-f" option to force the delete of the above ".crt" files. Was "authrootstl.cab" updated? If yes, consider deferring the delete until all clients have been updated.`

Options:

Windows Command Prompt

```
[-f] [-Unicode] [-gmt] [-seconds] [-v] [-privatekey] [-pin PIN] [-sid
WELL_KNOWN_SID_TYPE]
```

-generateSSTFromWU

Generates a store file that is synced with Windows Update.

Windows Command Prompt

```
certutil [options] -generateSSTFromWU SSTFile
```

Where:

- **SSTFile** is the `.sst` file to be generated that contains the Third Party Roots downloaded from Windows Update.

Options:

Windows Command Prompt

```
[-f] [-split]
```

-generatePinRulesCTL

Generates a Certificate Trust List (CTL) file that contains a list of pinning rules.

Windows Command Prompt

```
certutil [options] -generatePinRulesCTL XMLFile CTLFile [SSTFile
[QueryFilesPrefix]]
```

Where:

- **XMLFile** is the input XML file to be parsed.
- **CTLFile** is the output CTL file to be generated.
- **SSTFile** is the optional `.sst` file to be created that contains all of the certificates used for pinning.
- **QueryFilesPrefix** are optional *Domains.csv* and *Keys.csv* files to be created for database query.
  - The **QueryFilesPrefix** string is prepended to each created file.
  - The **Domains.csv** file contains rule name, domain rows.
  - The **Keys.csv** file contains rule name, key SHA256 thumbprint rows.

Options:

Windows Command Prompt

```
[-f]
```

-downloadOcsp

Downloads the OCSP responses and writes to the directory.

Windows Command Prompt

```
certutil [options] -downloadOcsp CertificateDir OcspDir [ThreadCount] [Modifiers]
```

Where:

- **CertificateDir** is the directory of a certificate, store and PFX files.
- **OcspDir** is the directory to write OCSP responses.
- **ThreadCount** is the optional maximum number of threads for concurrent downloading. Default is **10**.
- **Modifiers** are comma separated list of one or more of the following:
  - **DownloadOnce** - Downloads once and exits.
  - **ReadOcsp** - Reads from OcspDir instead of writing.

-generateHpkpHeader

Generates the HPKP header using certificates in a specified file or directory.

Windows Command Prompt

```
certutil [options] -generateHpkpHeader CertFileOrDir MaxAge [ReportUri]
[Modifiers]
```

Where:

- **CertFileOrDir** is the file or directory of certificates, which is the source of pin-sha256.
- **MaxAge** is the max-age value in seconds.
- **ReportUri** is the optional report-uri.
- **Modifiers** are comma separated list of one or more of the following:
  - **includeSubDomains** - Appends the includeSubDomains.

-flushCache

Flushes the specified caches in selected process, such as, lsass.exe.

Windows Command Prompt

```
certutil [options] -flushCache ProcessId CacheMask [Modifiers]
```

Where:

- **ProcessId** is the numeric ID of a process to flush. Set to **0** to flush all processes where flush is enabled.
- **CacheMask** is the bit mask of caches to be flushed either numeric or the following bits:
  - **0**: ShowOnly
  - **0x01**: CERT_WNF_FLUSH_CACHE_REVOCATION
  - **0x02**: CERT_WNF_FLUSH_CACHE_OFFLINE_URL
  - **0x04**: CERT_WNF_FLUSH_CACHE_MACHINE_CHAIN_ENGINE
  - **0x08**: CERT_WNF_FLUSH_CACHE_USER_CHAIN_ENGINES
  - **0x10**: CERT_WNF_FLUSH_CACHE_SERIAL_CHAIN_CERTS
  - **0x20**: CERT_WNF_FLUSH_CACHE_SSL_TIME_CERTS
  - **0x40**: CERT_WNF_FLUSH_CACHE_OCSP_STAPLING
- **Modifiers** are comma separated list of one or more of the following:
  - **Show** - Shows the caches being flushed. Certutil must be explicitly terminated.

-addEccCurve

Adds an ECC Curve.

Windows Command Prompt

```
certutil [options] -addEccCurve [CurveClass:]CurveName CurveParameters [CurveOID]
[CurveType]
```

Where:

- **CurveClass** is the ECC Curve Class type:
  - WEIERSTRASS (Default)
  - MONTGOMERY
  - TWISTED_EDWARDS
- **CurveName** is the ECC Curve name.
- **CurveParameters** are one of the following:
  - A certificate filename containing ASN encoded parameters.
  - A file containing ASN encoded parameters.
- **CurveOID** is the ECC Curve OID and is one of the following:
  - A certificate filename containing an ASN encoded OID.
  - An explicit ECC Curve OID.
- **CurveType** is the Schannel ECC NamedCurve point (numeric).

Options:

Windows Command Prompt

```
[-f]
```

-deleteEccCurve

Deletes the ECC Curve.

Windows Command Prompt

```
certutil [options] -deleteEccCurve CurveName | CurveOID
```

Where:

- **CurveName** is the ECC Curve name.
- **CurveOID** is the ECC Curve OID.

Options:

```
Windows Command Prompt
[-f]
```

-displayEccCurve

Displays the ECC Curve.

```
Windows Command Prompt
certutil [options] -displayEccCurve [CurveName | CurveOID]
```

Where:

- **CurveName** is the ECC Curve name.
- **CurveOID** is the ECC Curve OID.

Options:

```
Windows Command Prompt
[-f]
```

-csplist

Lists the cryptographic service providers (CSPs) installed on this machine for cryptographic operations.

```
Windows Command Prompt
certutil [options] -csplist [Algorithm]
```

Options:

```
Windows Command Prompt
[-user] [-Silent] [-csp Provider]
```

-csptest

Tests the CSPs installed on this machine.

Windows Command Prompt

```
certutil [options] -csptest [Algorithm]
```

Options:

Windows Command Prompt

```
[-user] [-Silent] [-csp Provider]
```

-CNGConfig

Displays CNG cryptographic configuration on this machine.

Windows Command Prompt

```
certutil [options] -CNGConfig
```

Options:

Windows Command Prompt

```
[-Silent]
```

-sign

Re-signs a certificate revocation list (CRL) or certificate.

Windows Command Prompt

```
certutil [options] -sign InFileList | SerialNumber | CRL OutFileList [StartDate [+
| -dd:hh] + | -dd:hh] [+SerialNumberList | -SerialNumberList | -ObjectIdList |
@ExtensionFile]
certutil [options] -sign InFileList | SerialNumber | CRL OutFileList
[#HashAlgorithm] [+AlternateSignatureAlgorithm | -AlternateSignatureAlgorithm]
certutil [options] -sign InFileList OutFileList [Subject:CN=...] [Issuer:hex data]
```

Where:

- **InFileList** is the comma-separated list of certificate or CRL files to modify and re-sign.

- **SerialNumber** is the serial number of the certificate to create. The validity period and other options can't be present.
- **CRL** creates an empty CRL. The validity period and other options can't be present.
- **OutFileList** is the comma-separated list of modified certificate or CRL output files. The number of files must match infilelist.
- **StartDate+dd:hh** is the new validity period for the certificate or CRL files, including:
  - optional date plus
  - optional days and hours validity period If multiple fields are used, use a (+) or (-) separator. Use `now[+dd:hh]` to start at the current time. Use `now-dd:hh+dd:hh` to start at a fixed offset from the current time and a fixed validity period. Use `never` to have no expiration date (for CRLs only).
- **SerialNumberList** is the comma-separated serial number list of the files to add or remove.
- **ObjectIdList** is the comma-separated extension ObjectId list of the files to remove.
- **@ExtensionFile** is the INF file that contains the extensions to update or remove. For example:

  Windows Command Prompt

  ```
  [Extensions]
      2.5.29.31 = ; Remove CRL Distribution Points extension
      2.5.29.15 = {hex} ; Update Key Usage extension
      _continue_=03 02 01 86
  ```

- **HashAlgorithm** is the name of the hash algorithm. This must only be the text preceded by the `#` sign.
- **AlternateSignatureAlgorithm** is the alternate signature algorithm specifier.

Options:

Windows Command Prompt

```
[-nullsign] [-f] [-user] [-Silent] [-Cert CertId] [-csp Provider]
```

Remarks

- Using the minus sign (-) removes serial numbers and extensions.
- Using the plus sign (+) adds serial numbers to a CRL.

- You can use a list to remove both serial numbers and **ObjectIds** from a CRL at the same time.
- Using the minus sign before **AlternateSignatureAlgorithm** allows you to use the legacy signature format.
- Using the plus sign allows you to use the alternate signature format.
- If you don't specify **AlternateSignatureAlgorithm**, the signature format in the certificate or CRL is used.

-vroot

Creates or deletes web virtual roots and file shares.

Windows Command Prompt

```
certutil [options] -vroot [delete]
```

-vocsproot

Creates or deletes web virtual roots for an OCSP web proxy.

Windows Command Prompt

```
certutil [options] -vocsproot [delete]
```

-addEnrollmentServer

Adds an Enrollment Server application and application pool if necessary for the specified Certificate Authority. This command doesn't install binaries or packages.

Windows Command Prompt

```
certutil [options] -addEnrollmentServer Kerberos | UserName | ClientCertificate
[AllowRenewalsOnly] [AllowKeyBasedRenewal]
```

Where:

- **addEnrollmentServer** requires you to use an authentication method for the client connection to the Certificate Enrollment Server, including:
  - **Kerberos** uses Kerberos SSL credentials.
  - **UserName** uses named account for SSL credentials.
  - **ClientCertificate** uses X.509 Certificate SSL credentials.

- **Modifiers**:
  - **AllowRenewalsOnly** allows only renewal request submissions to the Certificate Authority through the URL.
  - **AllowKeyBasedRenewal** allows use of a certificate with no associated account in Active Directory. This applies when used with **ClientCertificate** and **AllowRenewalsOnly** mode.

Options:

```
Windows Command Prompt

[-config Machine\CAName]
```

-deleteEnrollmentServer

Deletes an Enrollment Server application and application pool if necessary for the specified Certificate Authority. This command doesn't install binaries or packages.

```
Windows Command Prompt

certutil [options] -deleteEnrollmentServer Kerberos | UserName | ClientCertificate
```

Where:

- **deleteEnrollmentServer** requires you to use an authentication method for the client connection to the Certificate Enrollment Server, including:
  - **Kerberos** uses Kerberos SSL credentials.
  - **UserName** uses named account for SSL credentials.
  - **ClientCertificate** uses X.509 Certificate SSL credentials.

Options:

```
Windows Command Prompt

[-config Machine\CAName]
```

-addPolicyServer

Add a Policy Server application and application pool, if necessary. This command doesn't install binaries or packages.

```
Windows Command Prompt
```

```
certutil [options] -addPolicyServer Kerberos | UserName | ClientCertificate
[KeyBasedRenewal]
```

Where:

- **addPolicyServer** requires you to use an authentication method for the client connection to the Certificate Policy Server, including:
  - **Kerberos** uses Kerberos SSL credentials.
  - **UserName** uses named account for SSL credentials.
  - **ClientCertificate** uses X.509 Certificate SSL credentials.
- **KeyBasedRenewal** allows use of policies returned to the client containing keybasedrenewal templates. This option applies only for **UserName** and **ClientCertificate** authentication.

-deletePolicyServer

Deletes a Policy Server application and application pool, if necessary. This command doesn't remove binaries or packages.

Windows Command Prompt

```
certutil [options] -deletePolicyServer Kerberos | UserName | ClientCertificate
[KeyBasedRenewal]
```

Where:

- **deletePolicyServer** requires you to use an authentication method for the client connection to the Certificate Policy Server, including:
  - **Kerberos** uses Kerberos SSL credentials.
  - **UserName** uses named account for SSL credentials.
  - **ClientCertificate** uses X.509 Certificate SSL credentials.
- **KeyBasedRenewal** allows use of a KeyBasedRenewal policy server.

-Class

Displays COM registry information.

Windows Command Prompt

```
certutil [options] -Class [ClassId | ProgId | DllName | *]
```

Options:

Windows Command Prompt

```
[-f]
```

-7f

Checks certificate for 0x7f length encodings.

Windows Command Prompt

```
certutil [options] -7f CertFile
```

-oid

Displays the object identifier or sets a display name.

Windows Command Prompt

```
certutil [options] -oid ObjectId [DisplayName | delete [LanguageId [type]]]
certutil [options] -oid GroupId
certutil [options] -oid AlgId | AlgorithmName [GroupId]
```

Where:

- **ObjectId** is the ID to be displayed or to add to the display name.
- **GroupId** is the GroupID number (decimal) that ObjectIds enumerate.
- **AlgId** is the hexadecimal ID that objectID looks up.
- **AlgorithmName** is the algorithm name that objectID looks up.
- **DisplayName** displays the name to store in DS.
- **Delete** deletes the display name.
- **LanguageId** is the language ID value (defaults to current: 1033).
- **Type** is the type of DS object to create, including:
  - `1` - Template (default)
  - `2` - Issuance Policy
  - `3` - Application Policy
- `-f` creates a DS object.

Options:

Windows Command Prompt

```
[-f]
```

-error

Displays the message text associated with an error code.

Windows Command Prompt

```
certutil [options] -error ErrorCode
```

-getsmtpinfo

Gets Simple Mail Transfer Protocol (SMTP) information.

Windows Command Prompt

```
certutil [options] -getsmtpinfo
```

-setsmtpinfo

Sets SMTP information.

Windows Command Prompt

```
certutil [options] -setsmtpinfo LogonName
```

Options:

Windows Command Prompt

```
[-config Machine\CAName] [-p Password]
```

-getreg

Displays a registry value.

Windows Command Prompt

```
certutil [options] -getreg [{ca | restore | policy | exit | template | enroll |
chain | PolicyServers}\[ProgId\]] [RegistryValueName]
```

Where:

- **ca** uses a Certificate Authority's registry key.
- **restore** uses Certificate Authority's restore registry key.
- **policy** uses the policy module's registry key.
- **exit** uses the first exit module's registry key.
- **template** uses the template registry key (use `-user` for user templates).
- **enroll** uses the enrollment registry key (use `-user` for user context).
- **chain** uses the chain configuration registry key.
- **PolicyServers** uses the Policy Servers registry key.
- **ProgId** uses the policy or exit module's ProgID (registry subkey name).
- **RegistryValueName** uses the registry value name (use `Name*` to prefix match).
- **value** uses the new numeric, string, or date registry value or filename. If a numeric value starts with `+` or `-`, the bits specified in the new value are set or cleared in the existing registry value.

Options:

Windows Command Prompt

```
[-f] [-Enterprise] [-user] [-GroupPolicy] [-config Machine\CAName]
```

Remarks

- If a string value starts with `+` or `-`, and the existing value is a `REG_MULTI_SZ` value, the string is added to or removed from the existing registry value. To force creation of a `REG_MULTI_SZ` value, add `\n` to the end of the string value.
- If the value starts with `\@`, the rest of the value is the name of the file containing the hexadecimal text representation of a binary value.
- If it doesn't refer to a valid file, it's instead parsed as `[Date][+|-][dd:hh]` which is an optional date plus or minus optional days and hours.
- If both are specified, use a plus sign (+) or minus sign (-) separator. Use `now+dd:hh` for a date relative to the current time.
- Use `i64` as a suffix to create a REG_QWORD value.
- Use `chain\chaincacheresyncfiletime @now` to effectively flush cached CRLs.
- Registry aliases:
  - Config
  - CA
  - Policy - PolicyModules
  - Exit - ExitModules

- Restore - RestoreInProgress
- Template - Software\Microsoft\Cryptography\CertificateTemplateCache
- Enroll - Software\Microsoft\Cryptography\AutoEnrollment (Software\Policies\Microsoft\Cryptography\AutoEnrollment)
- MSCEP - Software\Microsoft\Cryptography\MSCEP
- Chain - Software\Microsoft\Cryptography\OID\EncodingType 0\CertDllCreateCertificateChainEngine\Config
- PolicyServers - Software\Microsoft\Cryptography\PolicyServers (Software\Policies\Microsoft\Cryptography\PolicyServers)
- Crypt32 - System\CurrentControlSet\Services\crypt32
- NGC - System\CurrentControlSet\Control\Cryptography\Ngc
- AutoUpdate - Software\Microsoft\SystemCertificates\AuthRoot\AutoUpdate
- Passport - Software\Policies\Microsoft\PassportForWork
- MDM - Software\Microsoft\Policies\PassportForWork

-setreg

Sets a registry value.

Windows Command Prompt

```
certutil [options] -setreg [{ca | restore | policy | exit | template | enroll | chain | PolicyServers}\[ProgId\]] RegistryValueName Value
```

Where:

- **ca** uses a Certificate Authority's registry key.
- **restore** uses Certificate Authority's restore registry key.
- **policy** uses the policy module's registry key.
- **exit** uses the first exit module's registry key.
- **template** uses the template registry key (use `-user` for user templates).
- **enroll** uses the enrollment registry key (use `-user` for user context).
- **chain** uses the chain configuration registry key.
- **PolicyServers** uses the Policy Servers registry key.
- **ProgId** uses the policy or exit module's ProgID (registry subkey name).
- **RegistryValueName** uses the registry value name (use `Name*` to prefix match).
- **Value** uses the new numeric, string, or date registry value or filename. If a numeric value starts with `+` or `-`, the bits specified in the new value are set or cleared in the existing registry value.

Options:

Windows Command Prompt

```
[-f] [-Enterprise] [-user] [-GroupPolicy] [-config Machine\CAName]
```

Remarks

- If a string value starts with `+` or `-`, and the existing value is a `REG_MULTI_SZ` value, the string is added to or removed from the existing registry value. To force creation of a `REG_MULTI_SZ` value, add `\n` to the end of the string value.
- If the value starts with `\@`, the rest of the value is the name of the file containing the hexadecimal text representation of a binary value.
- If it doesn't refer to a valid file, it's instead parsed as `[Date][+|-][dd:hh]` which is an optional date plus or minus optional days and hours.
- If both are specified, use a plus sign (+) or minus sign (-) separator. Use `now+dd:hh` for a date relative to the current time.
- Use `i64` as a suffix to create a REG_QWORD value.
- Use `chain\chaincacheresyncfiletime @now` to effectively flush cached CRLs.

-delreg

Deletes a registry value.

Windows Command Prompt

```
certutil [options] -delreg [{ca | restore | policy | exit | template | enroll
|chain | PolicyServers}\[ProgId\]][RegistryValueName]
```

Where:

- **ca** uses a Certificate Authority's registry key.
- **restore** uses Certificate Authority's restore registry key.
- **policy** uses the policy module's registry key.
- **exit** uses the first exit module's registry key.
- **template** uses the template registry key (use `-user` for user templates).
- **enroll** uses the enrollment registry key (use `-user` for user context).
- **chain** uses the chain configuration registry key.
- **PolicyServers** uses the Policy Servers registry key.
- **ProgId** uses the policy or exit module's ProgID (registry subkey name).
- **RegistryValueName** uses the registry value name (use `Name*` to prefix match).

- **Value** uses the new numeric, string or date registry value or filename. If a numeric value starts with `+` or `-`, the bits specified in the new value are set or cleared in the existing registry value.

Options:

Windows Command Prompt

```
[-f] [-Enterprise] [-user] [-GroupPolicy] [-config Machine\CAName]
```

Remarks

- If a string value starts with `+` or `-`, and the existing value is a `REG_MULTI_SZ` value, the string is added to or removed from the existing registry value. To force creation of a `REG_MULTI_SZ` value, add `\n` to the end of the string value.
- If the value starts with `\@`, the rest of the value is the name of the file containing the hexadecimal text representation of a binary value.
- If it doesn't refer to a valid file, it's instead parsed as `[Date][+|-][dd:hh]` which is an optional date plus or minus optional days and hours.
- If both are specified, use a plus sign (+) or minus sign (-) separator. Use `now+dd:hh` for a date relative to the current time.
- Use `i64` as a suffix to create a REG_QWORD value.
- Use `chain\chaincacheresyncfiletime @now` to effectively flush cached CRLs.
- Registry aliases:
  - Config
  - CA
  - Policy - PolicyModules
  - Exit - ExitModules
  - Restore - RestoreInProgress
  - Template - Software\Microsoft\Cryptography\CertificateTemplateCache
  - Enroll - Software\Microsoft\Cryptography\AutoEnrollment (Software\Policies\Microsoft\Cryptography\AutoEnrollment)
  - MSCEP - Software\Microsoft\Cryptography\MSCEP
  - Chain - Software\Microsoft\Cryptography\OID\EncodingType 0\CertDllCreateCertificateChainEngine\Config
  - PolicyServers - Software\Microsoft\Cryptography\PolicyServers (Software\Policies\Microsoft\Cryptography\PolicyServers)
  - Crypt32 - System\CurrentControlSet\Services\crypt32
  - NGC - System\CurrentControlSet\Control\Cryptography\Ngc
  - AutoUpdate - Software\Microsoft\SystemCertificates\AuthRoot\AutoUpdate

- Passport - Software\Policies\Microsoft\PassportForWork
- MDM - Software\Microsoft\Policies\PassportForWork

-importKMS

Imports user keys and certificates into the server database for key archival.

Windows Command Prompt

```
certutil [options] -importKMS UserKeyAndCertFile [CertId]
```

Where:

- **UserKeyAndCertFile** is a data file with user private keys and certificates that are to be archived. This file can be:
  - An Exchange Key Management Server (KMS) export file.
  - A PFX file.
- **CertId** is a KMS export file decryption certificate match token. For more information, see the `-store` parameter in this article.
- `-f` imports certificates not issued by the Certificate Authority.

Options:

Windows Command Prompt

```
[-f] [-Silent] [-split] [-config Machine\CAName] [-p Password] [-symkeyalg
SymmetricKeyAlgorithm[,KeyLength]]
```

-ImportCert

Imports a certificate file into the database.

Windows Command Prompt

```
certutil [options] -ImportCert Certfile [ExistingRow]
```

Where:

- **ExistingRow** imports the certificate in place of a pending request for the same key.
- `-f` imports certificates not issued by the Certificate Authority.

Options:

Windows Command Prompt

```
[-f] [-config Machine\CAName]
```

Remarks

The Certificate Authority may also need to be configured to support foreign certificates by running `certutil -setreg ca\KRAFlags +KRAF_ENABLEFOREIGN`.

-GetKey

Retrieves an archived private key recovery blob, generates a recovery script, or recovers archived keys.

Windows Command Prompt

```
certutil [options] -GetKey SearchToken [RecoveryBlobOutFile]
certutil [options] -GetKey SearchToken script OutputScriptFile
certutil [options] -GetKey SearchToken retrieve | recover OutputFileBaseName
```

Where:

- **script** generates a script to retrieve and recover keys (default behavior if multiple matching recovery candidates are found, or if the output file isn't specified).
- **retrieve** retrieves one or more Key Recovery Blobs (default behavior if exactly one matching recovery candidate is found, and if the output file is specified). Using this option truncates any extension and appends the certificate-specific string and the `.rec` extension for each key recovery blob. Each file contains a certificate chain and an associated private key, still encrypted to one or more Key Recovery Agent certificates.
- **recover** retrieves and recovers private keys in one step (requires Key Recovery Agent certificates and private keys). Using this option truncates any extension and appends the `.p12` extension. Each file contains the recovered certificate chains and associated private keys, stored as a PFX file.
- **SearchToken** selects the keys and certificates to be recovered, including:
  - Certificate Common Name
  - Certificate Serial Number
  - Certificate SHA-1 hash (thumbprint)
  - Certificate KeyId SHA-1 hash (Subject Key Identifier)
  - Requester Name (domain\user)
  - UPN (user@domain)

- **RecoveryBlobOutFile** outputs a file with a certificate chain and an associated private key, still encrypted to one or more Key Recovery Agent certificates.
- **OutputScriptFile** outputs a file with a batch script to retrieve and recover private keys.
- **OutputFileBaseName** outputs a file base name.

Options:

Windows Command Prompt

```
[-f] [-UnicodeText] [-Silent] [-config Machine\CAName] [-p Password] [-ProtectTo
SAMNameAndSIDList] [-csp Provider]
```

Remarks

- For **retrieve**, any extension is truncated and a certificate-specific string and the `.rec` extensions are appended for each key recovery blob. Each file contains a certificate chain and an associated private key, still encrypted to one or more Key Recovery Agent certificates.
- For **recover**, any extension is truncated and the `.p12` extension is appended. Contains the recovered certificate chains and associated private keys, stored as a PFX file.

-RecoverKey

Recovers an archived private key.

Windows Command Prompt

```
certutil [options] -RecoverKey RecoveryBlobInFile [PFXOutFile [RecipientIndex]]
```

Options:

Windows Command Prompt

```
[-f] [-user] [-Silent] [-split] [-p Password] [-ProtectTo SAMNameAndSIDList] [-csp
Provider] [-t Timeout]
```

-mergePFX

Merges PFX files.

Windows Command Prompt

```
certutil [options] -MergePFX PFXInFileList PFXOutFile [Modifiers]
```

Where:

- **PFXInFileList** is a comma-separated list of PFX input files.
- **PFXOutFile** is the name of the PFX output file.
- **Modifiers** are comma separated lists of one or more of the following:
  - **ExtendedProperties** includes any extended properties.
  - **NoEncryptCert** specifies to not encrypt the certificates.
  - **EncryptCert** specifies to encrypt the certificates.

Options:

Windows Command Prompt

```
[-f] [-user] [-split] [-p password] [-ProtectTo SAMNameAndSIDList] [-csp Provider]
```

Remarks

- The password specified on the command line must be a comma-separated password list.
- If more than one password is specified, the last password is used for the output file. If only one password is provided or if the last password is `*`, the user is prompted for the output file password.

-add-chain

Adds a certificate chain.

Windows Command Prompt

```
certutil [options] -add-chain LogId certificate OutFile
```

Options:

Windows Command Prompt

```
[-f]
```

-add-pre-chain

Adds a pre-certificate chain.

Windows Command Prompt

```
certutil [options] -add-pre-chain LogId pre-certificate OutFile
```

Options:

Windows Command Prompt

```
[-f]
```

-get-sth

Gets a signed tree head.

Windows Command Prompt

```
certutil [options] -get-sth [LogId]
```

Options:

Windows Command Prompt

```
[-f]
```

-get-sth-consistency

Gets signed tree head changes.

Windows Command Prompt

```
certutil [options] -get-sth-consistency LogId TreeSize1 TreeSize2
```

Options:

Windows Command Prompt

```
[-f]
```

-get-proof-by-hash

Gets proof of a hash from a timestamp server.

Windows Command Prompt

```
certutil [options] -get-proof-by-hash LogId Hash [TreeSize]
```

Options:

Windows Command Prompt

```
[-f]
```

-get-entries

Retrieves entries from an event log.

Windows Command Prompt

```
certutil [options] -get-entries LogId FirstIndex LastIndex
```

Options:

Windows Command Prompt

```
[-f]
```

-get-roots

Retrieves the root certificates from the certificate store.

Windows Command Prompt

```
certutil [options] -get-roots LogId
```

Options:

Windows Command Prompt

```
[-f]
```

-get-entry-and-proof

Retrieves an event log entry and its cryptographic proof.

Windows Command Prompt

```
certutil [options] -get-entry-and-proof LogId Index [TreeSize]
```

Options:

Windows Command Prompt

```
[-f]
```

-VerifyCT

Verifies a certificate against the Certificate Transparency log.

Windows Command Prompt

```
certutil [options] -VerifyCT Certificate SCT [precert]
```

Options:

Windows Command Prompt

```
[-f]
```

-?

Displays the list of parameters.

Windows Command Prompt

```
certutil -?
certutil <name_of_parameter> -?
certutil -? -v
```

Where:

- **-?** displays the list of parameters
- **-<name_of_parameter> -?** displays help content for the specified parameter.
- **-? -v** displays a verbose list of parameters and options.

Options

This section defines all of the options you're able to specify, based on the command. Each parameter includes information about which options are valid for use.

Expand table

Option	Description
-admin	Use ICertAdmin2 for CA properties.
-anonymous	Use anonymous SSL credentials.
-cert CertId	Signing certificate.
-clientcertificate clientCertId	Use X.509 Certificate SSL credentials. For selection UI, use `-clientcertificate`.
-config Machine\CAName	Certificate Authority and computer name string.
-csp provider	Provider: **KSP** - Microsoft Software Key Storage Provider **TPM** - Microsoft Platform Crypto Provider **NGC** - Microsoft Passport Key Storage Provider **SC** - Microsoft Smart Card Key Storage Provider
-dc DCName	Target a specific Domain Controller.
-enterprise	Use the local machine enterprise registry certificate store.
-f	Force overwrite.
-generateSSTFromWU SSTFile	Generate SST by using the automatic update mechanism.
-gmt	Display times using GMT.
-GroupPolicy	Use the group policy certificate store.
-idispatch	Use IDispatch instead of COM native methods.
-kerberos	Use Kerberos SSL credentials.
-location alternatestoragelocation	`(-loc)` AlternateStorageLocation.
-mt	Display machine templates.
-nocr	Encode text without CR characters.
-nocrlf	Encode text without CR-LF characters.
-nullsign	Use the hash of the data as a signature.
-oldpfx	Use old PFX encryption.

Option	Description
-out columnlist	Comma-separated column list.
-p password	Password
-pin PIN	Smart card PIN.
-policyserver URLorID	Policy Server URL or ID. For selection U/I, use `-policyserver`. For all Policy Servers, use `-policyserver *`
-privatekey	Display password and private key data.
-protect	Protect keys with password.
-protectto SAMnameandSIDlist	Comma-separated SAM name/SID list.
-restrict restrictionlist	Comma-separated Restriction List. Each restriction consists of a column name, a relational operator, and a constant integer, string, or date. One column name may be preceded by a plus or minus sign to indicate the sort order. For example: `requestID = 47`, `+requestername >= a, requestername`, or `-requestername > DOMAIN, Disposition = 21`.
-reverse	Reverse Log and Queue columns.
-seconds	Display times using seconds and milliseconds.
-service	Use service certificate store.
-sid	Numeric SID: **22** - Local System **23** - Local Service **24** - Network Service
-silent	Use the `silent` flag to acquire crypt context.
-split	Split embedded ASN.1 elements, and save to files.
-sslpolicy servername	SSL Policy matching ServerName.
-symkeyalg symmetrickeyalgorithm[,keylength]	Name of the Symmetric Key Algorithm with optional key length. For example: `AES,128` or `3DES`.
-syncWithWU DestinationDir	Sync with Windows Update.
-t timeout	URL fetch timeout in milliseconds.
-Unicode	Write redirected output in Unicode.
-UnicodeText	Write output file in Unicode.

Option	Description
-urlfetch	Retrieve and verify AIA Certs and CDP CRLs.
-user	Use the HKEY_CURRENT_USER keys or certificate store.
-username username	Use named account for SSL credentials. For selection UI, use `-username`.
-ut	Display user templates.
-v	Provide more detailed (verbose) information.
-v1	Use V1 interfaces.

Hash algorithms: MD2 MD4 MD5 SHA1 SHA256 SHA384 SHA512.

Related links

For more examples of how to use this command, see the following articles:

- Active Directory Certificate Services (AD CS)
- Certutil tasks for managing certificates
- Configure trusted roots and disallowed certificates in Windows

change

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Changes Remote Desktop Session Host server settings for logons, COM port mappings, and install mode.

> **Note**
>
> To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

Syntax

```
change logon
change port
change user
```

Parameters

Expand table

Parameter	Description
change logon command	Enables or disables logons from client sessions on an Remote Desktop Session Host server, or displays current logon status.
change port command	Lists or changes the COM port mappings to be compatible with MS-DOS applications.
change user command	Changes the install mode for the Remote Desktop Session Host server.

Related links

- Command-Line Syntax Key

- Remote Desktop Services (Terminal Services) Command Reference

Feedback

Was this page helpful? Yes No

change logon

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Enables or disables logons from client sessions, or displays current logon status. This utility is useful for system maintenance. You must be an administrator to run this command.

> **Note**
>
> To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

Syntax

```
change logon {/query | /enable | /disable | /drain | /drainuntilrestart}
```

Parameters

Expand table

Parameter	Description
/query	Displays the current logon status, whether enabled or disabled.
/enable	Enables logons from client sessions, but not from the console.
/disable	Disables subsequent logons from client sessions, but not from the console. Does not affect currently logged on users.
/drain	Disables logons from new client sessions, but allows reconnections to existing sessions.
/drainuntilrestart	Disables logons from new client sessions until the computer is restarted, but allows reconnections to existing sessions.
/?	Displays help at the command prompt.

Remarks

- Logons are re-enabled when you restart the system.
- If you're connected to the Remote Desktop Session Host server from a client session, and then you disable logons and log off before re-enabling logons, you won't be able to reconnect to your session. To re-enable logons from client sessions, log on at the console.

Examples

- To display the current logon status, type:

```
change logon /query
```

- To enable logons from client sessions, type:

```
change logon /enable
```

- To disable client logons, type:

```
change logon /disable
```

Related links

- Command-Line Syntax Key
- change command
- Remote Desktop Services (Terminal Services) Command Reference

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Feedback

Was this page helpful? Yes No

change port

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Lists or changes the COM port mappings to be compatible with MS-DOS applications.

> **Note**
>
> To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

Syntax

```
change port [<portX>=<portY>| /d <portX> | /query]
```

Parameters

Expand table

Parameter	Description
`<portX>` = `<portY>`	Maps COM `<portX>` to `<portY>`
/d `<portX>`	Deletes the mapping for COM `<portX>`
/query	Displays the current port mappings.
/?	Displays help at the command prompt.

Remarks

- Most MS-DOS applications support only COM1 through COM4 serial ports. The **change port** command maps a serial port to a different port number, allowing apps that don't support high-numbered COM ports to access the serial port. Remapping works only for the current session and is not retained if you log off from a session and then log on again.

- Use **change port** without any parameters to display the available COM ports and their current mappings.

Examples

- To map COM12 to COM1 for use by an MS-DOS-based application, type:

```
change port com12=com1
```

- To display the current port mappings, type:

```
change port /query
```

Related links

- Command-Line Syntax Key
- change command
- Remote Desktop Services (Terminal Services) Command Reference

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Feedback

Was this page helpful? Yes No

change user

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Changes the install mode for the Remote Desktop Session Host server.

> ⓘ Note
>
> To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

Syntax

```
change user {/execute | /install | /query}
```

Parameters

Expand table

Parameter	Description
/execute	Enables .ini file mapping to the home directory. This is the default setting.
/install	Disables .ini file mapping to the home directory. All .ini files are read and written to the system directory. You must disable .ini file mapping when installing applications on a Remote Desktop Session Host server.
/query	Displays the current setting for .ini file mapping.
/?	Displays help at the command prompt.

Remarks

- Use **change user /install** before installing an application to create .ini files for the application in the system directory. These files are used as the source when user-

specific .ini files are created. After installing the application, use **change user /execute** to revert to standard .ini file mapping.

- The first time you run the app, it searches the home directory for its .ini files. If the .ini files aren't found in the home directory, but are found in the system directory, Remote Desktop Services copies the .ini files to the home directory, ensuring that each user has a unique copy of the application .ini files. Any new .ini files are created in the home directory.

- Each user should have a unique copy of the .ini files for an application. This prevents instances where different users might have incompatible application configurations (for example, different default directories or screen resolutions).

- When the system is running **change user /install**, several things occur. All registry entries that are created are shadowed under **HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\Currentversion\Terminal Server\Install**, in either the **\SOFTWARE** subkey or the **\MACHINE** subkey. Subkeys added to **HKEY_CURRENT_USER** are copied under the **\SOFTWARE** subkey, and subkeys added to **HKEY_LOCAL_MACHINE** are copied under the **\MACHINE** subkey. If the application queries the Windows directory by using system calls, such as GetWindowsdirectory, the rd Session Host server returns the systemroot directory. If any .ini file entries are added by using system calls, such as WritePrivateProfileString, they are added to the .ini files under the systemroot directory.

- When the system returns to **change user /execute**, and the application tries to read a registry entry under **HKEY_CURRENT_USER** that does not exist, Remote Desktop Services checks to see whether a copy of the key exists under the **\Terminal Server\Install** subkey. If it does, the subkeys are copied to the appropriate location under **HKEY_CURRENT_USER**. If the application tries to read from an .ini file that does not exist, Remote Desktop Services searches for that .ini file under the system root. If the .ini file is in the system root, it is copied to the \Windows subdirectory of the user's home directory. If the application queries the Windows directory, the rd Session Host server returns the \Windows subdirectory of the user's home directory.

- When you log on, Remote Desktop Services checks whether its system .ini files are newer than the .ini files on your computer. If the system version is newer, your .ini file is either replaced or merged with the newer version. This depends on whether or not the INISYNC bit, 0x40, is set for this .ini file. Your previous version of the .ini file is renamed as Inifile.ctx. If the system registry values under the **\Terminal Server\Install** subkey are newer than your version under **HKEY_CURRENT_USER**,

your version of the subkeys is deleted and replaced with the new subkeys from **\Terminal Server\Install**.

Examples

- To disable .ini file mapping in the home directory, type:

```
change user /install
```

- To enable .ini file mapping in the home directory, type:

```
change user /execute
```

- To display the current setting for .ini file mapping, type:

```
change user /query
```

Related links

- Command-Line Syntax Key
- change command
- Remote Desktop Services (Terminal Services) Command Reference

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Feedback

Was this page helpful? Yes No

chcp

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Changes the active console code page. If used without parameters, **chcp** displays the number of the active console code page.

Syntax

```
chcp [<nnn>]
```

Parameters

Expand table

Parameter	Description
`<nnn>`	Specifies the code page.
/?	Displays help at the command prompt.

The following table lists each supported code page and its country/region or language:

Expand table

Code page	Country/region or language
437	United States
850	Multilingual (Latin I)
852	Slavic (Latin II)
855	Cyrillic (Russian)
857	Turkish
860	Portuguese

Code page	Country/region or language
861	Icelandic
863	Canadian-French
865	Nordic
866	Russian
869	Modern Greek
936	Chinese

Remarks

- Only the original equipment manufacturer (OEM) code page that is installed with Windows appears correctly in a Command Prompt window that uses Raster fonts. Other code pages appear correctly in full-screen mode or in Command Prompt windows that use TrueType fonts.
- You don't need to prepare code pages (as in MS-DOS).
- Programs that you start after you assign a new code page use the new code page. However, programs (except Cmd.exe) that you started before assigning the new code page will continue to use the original code page.

Examples

To view the active code page setting, type:

```
chcp
```

A message similar to the following appears: `Active code page: 437`

To change the active code page to 850 (Multilingual), type:

```
chcp 850
```

If the specified code page is invalid, the following error message appears: `Invalid code page`

Related links

- Command-Line Syntax Key
- Code Page BitFields
- Code Page Identifiers

Feedback

Was this page helpful? Yes No

chdir

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays the name of the current directory or changes the current directory. If used with only a drive letter (for example, `chdir C:`), **chdir** displays the names of the current directory in the specified drive. If used without parameters, **chdir** displays the current drive and directory.

> **Note**
>
> This command has been replaced by the **cd command**. For more information, including the syntax and parameter details, see cd command.

Related links

- Command-Line Syntax Key
- cd command

Feedback

Was this page helpful? Yes No

chglogon

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Enables or disables logons from client sessions on an Remote Desktop Session Host server, or displays current logon status.

> **Note**
>
> This command has been replaced by the **change log command**. For more information, including the syntax and parameter details, see change logon command.

Related links

- Command-Line Syntax Key
- change logon command
- Remote Desktop Services (Terminal Services) Command Reference

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Feedback

Was this page helpful? Yes No

chgport

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Lists or changes the COM port mappings to be compatible with MS-DOS applications.

> **Note**
>
> This command has been replaced by the **change port command**. For more information, including the syntax and parameter details, see change port command.

Related links

- Command-Line Syntax Key
- change port command
- Remote Desktop Services (Terminal Services) Command Reference

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Feedback

Was this page helpful? Yes No

chgusr

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Changes the install mode for the Remote Desktop Session Host server.

> **Note**
>
> This command has been replaced by the **change user command**. For more information, including the syntax and parameter details, see change user command.

Related links

- Command-Line Syntax Key
- change user command
- Remote Desktop Services (Terminal Services) Command Reference

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Feedback

Was this page helpful? Yes No

chkdsk

Article • 05/26/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Checks the file system and file system metadata of a volume for logical and physical errors. If used without parameters, **chkdsk** displays only the status of the volume and doesn't fix any errors. If used with the **/f**, **/r**, **/x**, or **/b** parameters, it fixes errors on the volume.

> **Important**
>
> Membership in the local **Administrators** group, or equivalent, is the minimum required to run **chkdsk**. To open a command prompt window as an administrator, right-click **Command prompt** in the **Start** menu, and then select **Run as administrator**.

> **Important**
>
> Interrupting **chkdsk** isn't recommended. However, canceling or interrupting **chkdsk** shouldn't leave the volume any more corrupt than it was before **chkdsk** was run. Running **chkdsk** again checks and should repair any remaining corruption on the volume.

> **Note**
>
> Chkdsk can be used only for local disks. The command can't be used with a local drive letter that has been redirected over the network.

Syntax

Windows Command Prompt

```
chkdsk [<volume>[[<path>]<filename>]] [/f] [/v] [/r] [/x] [/i] [/c] [/l[:<size>]]
[/b] [/scan] [/forceofflinefix] [/perf] [/spotfix] [/sdcleanup]
[/offlinescanandfix] [/freeorphanedchains] [/markclean] [/?]
```

Parameters

Expand table

Parameter	Description
`<volume>`	Specifies the drive letter (followed by a colon), mount point, or volume name.
`[<path>]<filename>`	Use with file allocation table (FAT) and FAT32 only. Specifies the location and name of a file or set of files that you want **chkdsk** to check for fragmentation. You can use the wildcard characters **?** and ***** to specify multiple files.
/f	Fixes errors on the disk. The disk must be locked. If **chkdsk** can't lock the drive, a message appears that asks you if you want to check the drive the next time you restart the computer.
/v	Displays the name of each file in every directory as the disk is checked.
/r	Locates bad sectors and recovers readable information. The disk must be locked. **/r** includes the functionality of **/f**, with the additional analysis of physical disk errors.
/x	Forces the volume to dismount first, if necessary. All open handles to the drive are invalidated. **/x** also includes the functionality of **/f**.
/i	Use with NTFS only. Performs a less vigorous check of index entries, which reduces the amount of time required to run **chkdsk**.
/c	Use with NTFS only. Doesn't check cycles within the folder structure, which reduces the amount of time required to run **chkdsk**.
/l[:`<size>`]	Use with NTFS only. Changes the log file size to the size you type. If you omit the size parameter, **/l** displays the current size.
/b	Use with NTFS only. Clears the list of bad clusters on the volume and rescans all allocated and free clusters for errors. **/b** includes the functionality of **/r**. Use this parameter after imaging a volume to a new hard disk drive.
/scan	Use with NTFS only. Runs an online scan on the volume.
/forceofflinefix	Use with NTFS only (must be used with **/scan**). Bypass all online repair; all defects found are queued for offline repair (for example, `chkdsk /spotfix`).
/perf	Use with NTFS only (must be used with **/scan**). Uses more system resources to complete a scan as fast as possible. This might have a negative performance impact on other tasks running on the system.
/spotfix	Use with NTFS only. Runs spot fixing on the volume.
/sdcleanup	Use with NTFS only. Garbage collect unneeded security descriptor data (implies **/f**).
/offlinescanandfix	Runs an offline scan and fix on the volume.
/freeorphanedchains	Use with FAT/FAT32/exFAT only. Frees any orphaned cluster chains instead of recovering their contents.

Parameter	Description
/markclean	Use with FAT/FAT32/exFAT only. Marks the volume clean if no corruption was detected, even if **/f** wasn't specified.
/?	Displays help at the command prompt.

Remarks

- The **/i** or **/c** switch reduces the amount of time required to run **chkdsk** by skipping certain volume checks.
- If you want **chkdsk** to correct disk errors, you can't have open files on the drive. If files are open, the following error message appears:

  error

  ```
  Chkdsk cannot run because the volume is in use by another process. Would you like to schedule this volume to be checked the next time the system restarts? (Y/N)
  ```

- If you choose to check the drive the next time you restart the computer, **chkdsk** checks the drive and corrects errors automatically when you restart the computer. If the drive partition is a boot partition, **chkdsk** automatically restarts the computer after it checks the drive.
- You can also use the `chkntfs /c` command to schedule the volume to be checked the next time the computer is restarted. Use the `fsutil dirty set` command to set the volume's dirty bit (indicating corruption), so that Windows runs **chkdsk** when the computer is restarted.
- You should use **chkdsk** occasionally on FAT and NTFS file systems to check for disk errors. **Chkdsk** examines disk space and disk use and provides a status report specific to each file system. The status report shows errors found in the file system. If you run **chkdsk** without the **/f** parameter on an active partition, it might report spurious errors because it can't lock the drive.
- **Chkdsk** corrects logical disk errors only if you specify the **/f** parameter. **Chkdsk** must be able to lock the drive to correct errors.

  Because repairs on FAT file systems usually change a disk's file allocation table and sometimes cause a loss of data, **chkdsk** might display a confirmation message similar to the following:

```
10 lost allocation units found in 3 chains.
Convert lost chains to files?
```

  - If you press **Y**, Windows saves each lost chain in the root directory as a file with a name in the format `File<nnnn>.chk`. When **chkdsk** finishes, you can check these files to see if they contain any data you need.
  - If you press **N**, Windows fixes the disk, but it doesn't save the contents of the lost allocation units.

- If you don't use the **/f** parameter, **chkdsk** displays a message that the file needs to be fixed, but it doesn't fix any errors.
- If you use `chkdsk /f*` on a large disk or a disk with a large number of files (for example, millions of files), `chkdsk /f` might take a long time to complete.
- Use the **/r** parameter to find physical disk errors in the file system and attempt to recover data from any affected disk sectors.
- If you specify the **/f** parameter, **chkdsk** displays an error message if there are open files on the disk. If you don't specify the **/f** parameter and open files exist, **chkdsk** might report lost allocation units on the disk. This could happen if open files haven't yet been recorded in the file allocation table. If **chkdsk** reports the loss of a large number of allocation units, consider repairing the disk.
- Because the Shadow Copies for Shared Folders source volume can't be locked while **Shadow Copies for Shared Folders** is enabled, running **chkdsk** against the source volume might report false errors or cause **chkdsk** to unexpectedly quit. You can, however, check shadow copies for errors by running **chkdsk** in Read-only mode (without parameters) to check the Shadow Copies for Shared Folders storage volume.
- The **chkdsk** command, with different parameters, is available from the Recovery Console.
- On servers that are infrequently restarted, you might want to use the **chkntfs** or the `fsutil dirty query` commands to determine whether the volume's dirty bit is already set before running chkdsk.

How chkdsk performs on different media

Hard Disk Drive (HDD)

- **Physical nature:** HDDs use spinning magnetic platters. Sectors and blocks are physically read from the platter.
- **When chkdsk runs:**
  - The physical head must move to each data location being checked, which is **slower** due to mechanical movement.
  - Checking for *bad sectors* (with `/r`) takes longer as every sector's physical integrity is checked and bad ones are replaced if possible.
  - High-hour runtimes aren't uncommon for large drives.

Normally, on large capacity HDDs, `/r` or `/b` would take a considerable amount of time as it reads every sector. Even without errors and with a mostly empty drive, the physical read speed bottlenecks the process.

There might be cases where on large capacity HDDs where `chkdsk` might complete its process too quickly. If this occurs, it might be that:

- The volume is mounted dirty or locked by the operating system (OS) or another process.
- The `chkdsk` operation didn't actually scan every sector on the drive.
- The HDD might have a failing read head or other hardware issue that causes `chkdsk` to behave unpredictably.
- The `chkdsk` operation only performed an online scan of the file system structure and wasn't explicitly made to run an offline scan before booting into the OS.

Viewing the `chkdsk` logs might point to a potential issue during scans. To learn more, see Viewing chkdsk logs.

Understanding exit codes

The following table lists the exit codes that **chkdsk** reports after it has finished.

Expand table

Exit code	Description
0	No errors were found.
1	Errors were found and fixed.

Exit code	Description
2	Performed disk cleanup (such as garbage collection) or didn't perform cleanup because **/f** was not specified.
3	Could not check the disk, errors could not be fixed, or errors were not fixed because **/f** was not specified.

Examples

To check the disk in drive D and have Windows fix errors, type:

Windows Command Prompt

```
chkdsk d: /f
```

If it encounters errors, **chkdsk** pauses and displays messages. **Chkdsk** finishes by displaying a report that lists the status of the disk. You can't open any files on the specified drive until `chkdsk` finishes.

To check all files on a FAT disk in the current directory for noncontiguous blocks, type:

Windows Command Prompt

```
chkdsk *.*
```

Chkdsk displays a status report, and then lists the files that match the file specifications that have noncontiguous blocks.

Viewing chkdsk logs

There are two methods that can be used to retrieve chkdsk log files in Windows, using the Event Viewer and PowerShell.

Event Viewer

To view logs with Event Viewer, perform the following actions:

1. Start > **Control Panel** > **Administrative Tools** > **Event Viewer**.

   Alternatively, press **Win + R** keys to bring up the run dialog box, type **eventvwr.msc**, and select **OK**.

2. Expand **Windows Logs** > right-click on **Application** > select **Filter Current Log**.
3. Within the **Filter Current Log** window, navigate to **Event sources** drop-down menu, select **Chkdsk** and **Wininit**.
4. Select **OK** to finish filtering for these two sources.

Related links

- Command-Line Syntax Key

chkntfs

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays or modifies automatic disk checking when the computer is started. If used without options, **chkntfs** displays the file system of the specified volume. If automatic file checking is scheduled to run, **chkntfs** displays whether the specified volume is dirty or is scheduled to be checked the next time the computer is started.

> **Note**
>
> To run **chkntfs**, you must be a member of the Administrators group.

Syntax

```
chkntfs <volume> [...]
chkntfs [/d]
chkntfs [/t[:<time>]]
chkntfs [/x <volume> [...]]
chkntfs [/c <volume> [...]]
```

Parameters

Expand table

Parameter	Description
`<volume>` [...]	Specifies one or more volumes to check when the computer starts. Valid volumes include drive letters (followed by a colon), mount points, or volume names.
/d	Restores all **chkntfs** default settings, except the countdown time for automatic file checking. By default, all volumes are checked when the computer is started, and **chkdsk** runs on those that are dirty.
/t [`:<time>`]	Changes the Autochk.exe initiation countdown time to the amount of time specified in seconds. If you do not enter a time, **/t** displays the current countdown time.

Parameter	Description
/x `<volume>` [...]	Specifies one or more volumes to exclude from checking when the computer is started, even if the volume is marked as requiring **chkdsk**.
/c `<volume>` [...]	Schedules one or more volumes to be checked when the computer is started, and runs **chkdsk** on those that are dirty.
/?	Displays help at the command prompt.

Examples

To display the type of file system for drive C, type:

```
chkntfs c:
```

> **Note**
>
> If automatic file checking is scheduled to run, additional output will display, indicating whether the drive is dirty or has been manually scheduled to be checked the next time the computer is started.

To display the Autochk.exe initiation countdown time, type:

```
chkntfs /t
```

To change the Autochk.exe initiation countdown time to 30 seconds, type:

```
chkntfs /t:30
```

> **Note**
>
> Although you can set the Autochk.exe initiation countdown time to zero, doing so will prevent you from canceling a potentially time-consuming automatic file check.

To exclude multiple volumes from being checked, you must list each of them in a single command. For example, to exclude both the D and E volumes, type:

```
chkntfs /x d: e:
```

> **Important**
>
> The **/x** command-line option isn't accumulative. If you type it more than once, the most recent entry overrides the previous entry.

To schedule automatic file checking on the D volume, but not the C or E volumes, type the following commands in order:

```
chkntfs /d
chkntfs /x c: d: e:
chkntfs /c d:
```

> **Important**
>
> The **/c** command-line option is accumulative. If you type **/c** more than once, each entry remains. To ensure that only a particular volume is checked, reset the defaults to clear all previous commands, exclude all volumes from being checked, and then schedule automatic file checking on the desired volume.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

choice

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Prompts the user to select one item from a list of single-character choices in a batch program, and then returns the index of the selected choice. If used without parameters, **choice** displays the default choices **Y** and **N**.

Syntax

```
choice [/c [<choice1><choice2><…>]] [/n] [/cs] [/t <timeout> /d <choice>]
[/m <text>]
```

Parameters

Expand table

Parameter	Description
/c `<choice1><choice2><…>`	Specifies the list of choices to be created. Valid choices include a-z, A-Z, 0-9, and extended ASCII characters (128-254). The default list is YN, which is displayed as `[Y,N]?`.
/n	Hides the list of choices, although the choices are still enabled and the message text (if specified by **/m**) is still displayed.
/cs	Specifies that the choices are case-sensitive. By default, the choices are not case-sensitive.
/t `<timeout>`	Specifies the number of seconds to pause before using the default choice specified by **/d**. Acceptable values are from **0** to **9999**. If **/t** is set to **0**, **choice** does not pause before returning the default choice.
/d `<choice>`	Specifies the default choice to use after waiting the number of seconds specified by **/t**. The default choice must be in the list of choices specified by **/c**.
/m `<text>`	Specifies a message to display before the list of choices. If **/m** is not specified, only the choice prompt is displayed.
/?	Displays help at the command prompt.

Remarks

- The **ERRORLEVEL** environment variable is set to the index of the key that the user selects from the list of choices. The first choice in the list returns a value of `1`, the second a value of `2`, and so on. If the user presses a key that is not a valid choice, **choice** sounds a warning beep.
- If **choice** detects an error condition, it returns an **ERRORLEVEL** value of `255`. If the user presses CTRL+BREAK or CTRL+C, **choice** returns an **ERRORLEVEL** value of `0`.

> ⓘ Note
>
> When you use **ERRORLEVEL** values in a batch program, you must list them in decreasing order.

Examples

To present the choices **Y**, **N**, and **C**, type the following line in a batch file:

```
choice /c ync
```

The following prompt appears when the batch file runs the **choice** command:

```
[Y,N,C]?
```

To hide the choices **Y**, **N**, and **C**, but display the text **Yes**, **No**, or **Continue**, type the following line in a batch file:

```
choice /c ync /n /m "Yes, No, or Continue?"
```

> ⓘ Note
>
> If you use the **/n** parameter, but do not use **/m**, the user is not prompted when **choice** is waiting for input.

To show both the text and the options used in the previous examples, type the following line in a batch file:

```
choice /c ync /m "Yes, No, or Continue"
```

To set a time limit of five seconds and specify **N** as the default value, type the following line in a batch file:

```
choice /c ync /t 5 /d n
```

> **Note**
>
> In this example, if the user doesn't press a key within five seconds, **choice** selects **N** by default and returns an error value of `2`. Otherwise, **choice** returns the value corresponding to the user's choice.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

cipher

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays or alters the encryption of directories and files on NTFS volumes. If used without parameters, **cipher** displays the encryption state of the current directory and any files it contains.

Syntax

```
cipher [/e | /d | /c] [/s:<directory>] [/b] [/h] [pathname [...]]
cipher /k
cipher /r:<filename> [/smartcard]
cipher /u [/n]
cipher /w:<directory>
cipher /x[:efsfile] [filename]
cipher /y
cipher /adduser [/certhash:<hash> | /certfile:<filename>] [/s:directory]
[/b] [/h] [pathname [...]]
cipher /removeuser /certhash:<hash> [/s:<directory>] [/b] [/h] [<pathname>
[...]]
cipher /rekey [pathname [...]]
```

Parameters

Expand table

Parameters	Description
/b	Aborts if an error is encountered. By default, **cipher** continues to run even if errors are encountered.
/c	Displays information on the encrypted file.
/d	Decrypts the specified files or directories.
/e	Encrypts the specified files or directories. Directories are marked so that files that are added afterward will be encrypted.
/h	Displays files with hidden or system attributes. By default, these files are not

Parameters	Description
	encrypted or decrypted.
/k	Creates a new certificate and key for use with Encrypting File System (EFS) files. If the **/k** parameter is specified, all other parameters are ignored.
/r:`<filename>` [/smartcard]	Generates an EFS recovery agent key and certificate, then writes them to a .pfx file (containing certificate and private key) and a .cer file (containing only the certificate). If **/smartcard** is specified, it writes the recovery key and certificate to a smart card, and no .pfx file is generated.
/s:`<directory>`	Performs the specified operation on all subdirectories in the specified *directory*.
/u [/n]	Finds all encrypted files on the local drive(s). If used with the **/n** parameter, no updates are made. If used without **/n**, **/u** compares the user's file encryption key or the recovery agent's key to the current ones, and updates them if they have changed. This parameter works only with **/n**.
/w:`<directory>`	Removes data from available unused disk space on the entire volume. If you use the **/w** parameter, all other parameters are ignored. The directory specified can be located anywhere in a local volume. If it is a mount point or points to a directory in another volume, the data on that volume is removed.
/x[:efsfile] [`<FileName>`]	Backs up the EFS certificate and keys to the specified file name. If used with **:efsfile**, **/x** backs up the user's certificate(s) that were used to encrypt the file. Otherwise, the user's current EFS certificate and keys are backed up.
/y	Displays your current EFS certificate thumbnail on the local computer.
/adduser [/certhash: `<hash>`	/certfile:`<filename>`]
/rekey	Updates the specified encrypted file(s) to use the currently configured EFS key.
/removeuser /certhash: `<hash>`	Removes a user from the specified file(s). The *Hash* provided for **/certhash** must be the SHA1 hash of the certificate to remove.
/?	Displays help at the command prompt.

Remarks

- If the parent directory is not encrypted, an encrypted file could become decrypted when it is modified. Therefore, when you encrypt a file, you should also encrypt the parent directory.

- An administrator can add the contents of a .cer file to the EFS recovery policy to create the recovery agent for users, and then import the .pfx file to recover individual files.
- You can use multiple directory names and wildcards.
- You must put spaces between multiple parameters.

Examples

To display the encryption status of each of the files and subdirectories in the current directory, type:

```
cipher
```

Encrypted files and directories are marked with an **E**. Unencrypted files and directories are marked with a **U**. For example, the following output indicates that the current directory and all its contents are currently unencrypted:

```
Listing C:\Users\MainUser\Documents\
New files added to this directory will not be encrypted.
U Private
U hello.doc
U hello.txt
```

To enable encryption on the Private directory used in the previous example, type:

```
cipher /e private
```

The following output displays:

```
Encrypting files in C:\Users\MainUser\Documents\
Private             [OK]
1 file(s) [or directorie(s)] within 1 directorie(s) were encrypted.
```

The **cipher** command displays the following output:

```
Listing C:\Users\MainUser\Documents\
New files added to this directory will not be encrypted.
E Private
U hello.doc
U hello.txt
```

Where the **Private** directory is now marked as encrypted.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

cleanmgr

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Clears unnecessary files from your computer's hard disk. You can use command-line options to specify that **Cleanmgr** cleans up Temp files, Internet files, downloaded files, and Recycle Bin files. You can then schedule the task to run at a specific time by using the **Scheduled Tasks** tool.

Syntax

```
cleanmgr [/d <driveletter>] [/sageset:n]  [/sagerun:n] [/TUNEUP:n]
[/LOWDISK] [/VERYLOWDISK]
```

Parameters

Expand table

Parameter	Description
/d `<driveletter>`	Specifies the drive that you want Disk Cleanup to clean. **NOTE:** The **/d** option isn't utilized with `/sagerun:n`.
/sageset:n	Displays the **Disk Cleanup Settings** dialog box and also creates a registry key to store the settings that you select. The `n` value, which is stored in the registry, allows you to specify tasks for Disk Cleanup to run. The `n` value can be any integer value from 0 to 9999. For more information, see Registry key information.
/sagerun:n	Runs the specified tasks that are assigned to the n value if you use the **/sageset** option. All drives on the computer are enumerated and the selected profile runs against each drive.
/tuneup:n	Run **/sageset** and **/sagerun** for the same `n`.
/lowdisk	Run with the default settings.
/verylowdisk	Run with the default settings, no user prompts.

Parameter	Description
/autoclean	Automatically deletes the files that are left behind after you upgrade Windows.
/?	Displays help at the command prompt.

Options

The options for the files that you can specify for Disk Cleanup by using **/sageset** and **/sagerun** include:

- **Temporary Setup Files** - These are files that were created by a Setup program that is no longer running.
- **Downloaded Program Files** - Downloaded program files are ActiveX controls and Java programs that are downloaded automatically from the Internet when you view certain pages. These files are temporarily stored in the Downloaded Program Files folder on the hard disk. This option includes a View Files button so that you can see the files before Disk Cleanup removes them. The button opens the C:\Winnt\Downloaded Program Files folder.
- **Temporary Internet Files** - The Temporary Internet Files folder contains Web pages that are stored on your hard disk for quick viewing. Disk Cleanup removes these pages but leaves your personalized settings for Web pages intact. This option also includes a View Files button, which opens the C:\Documents and Settings\Username\Local Settings\Temporary Internet Files\Content.IE5 folder.
- **Old Chkdsk Files** - When Chkdsk checks a disk for errors, Chkdsk might save lost file fragments as files in the root folder on the disk. These files are unnecessary.
- **Recycle Bin** - The Recycle Bin contains files that you have deleted from the computer. These files aren't permanently removed until you empty the Recycle Bin. This option includes a View Files button that opens the Recycle Bin.

  Note: A Recycle Bin may appear in more than one drive, for example, not just in %SystemRoot%.
- **Temporary Files** - Programs sometimes store temporary information in a Temp folder. Before a program quits, the program usually deletes this information. You can safely delete temporary files that haven't been modified within the last week.
- **Temporary Offline Files** - Temporary offline files are local copies of recently used network files. These files are automatically cached so that you can use them after

you disconnect from the network. A **View Files** button opens the Offline Files folder.

- **Offline Files** - Offline files are local copies of network files that you specifically want to have available offline so that you can use them after you disconnect from the network. A **View Files** button opens the Offline Files folder.

- **Compress Old Files** - Windows can compress files that you haven't used recently. Compressing files saves disk space, but you can still use the files. No files are deleted. Because files are compressed at different rates, the displayed amount of disk space that you'll gain is approximate. An Options button permits you to specify the number of days to wait before Disk Cleanup compresses an unused file.

- **Catalog Files for the Content Indexer** - The Indexing service speeds up and improves file searches by maintaining an index of the files that are on the disk. These Catalog files remain from a previous indexing operation and can be deleted safely.

  Note: Catalog File may appear in more than one drive, for example, not just in `%SystemRoot%`.

> ⓘ **Note**
>
> If you specify cleaning up the drive that contains the Windows installation, all of these options are available on the **Disk Cleanup** tab. If you specify any other drive, only the Recycle Bin and the Catalog files for content index options are available on the **Disk Cleanup** tab.

Examples

To run the Disk Cleanup app so that you can use its dialog box to specify options for use later, saving the settings to the set **1**, type the following:

```
cleanmgr /sageset:1
```

To run Disk Cleanup and include the options that you specified with the cleanmgr /sageset:1 command, type:

```
```

```
cleanmgr /sagerun:1
```

To run `cleanmgr /sageset:1` and `cleanmgr /sagerun:1` together, type:

```
cleanmgr /tuneup:1
```

Related links

- Free up drive space in Windows 10
- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

clip

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Redirects the command output from the command line to the Windows clipboard. You can use this command to copy data directly into any application that can receive text from the Clipboard. You can also paste this text output into other programs.

Syntax

```
<command> | clip
clip < <filename>
```

Parameters

Expand table

Parameter	Description
`<command>`	Specifies a command whose output you want to send to the Windows clipboard.
`<filename>`	Specifies a file whose contents you want to send to the Windows clipboard.
/?	Displays help at the command prompt.

Examples

To copy the current directory listing to the Windows clipboard, type:

```
dir | clip
```

To copy the output of a program called *generic.awk* to the Windows clipboard, type:

```
awk -f generic.awk input.txt | clip
```

To copy the contents of a file called *readme.txt* to the Windows clipboard, type:

```
clip < readme.txt
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

cls

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Clears the Command Prompt window.

Syntax

```
cls
```

Parameters

Expand table

Parameter	Description
/?	Displays help at the command prompt.

Examples

To clear all information that appears in the Command Prompt window and return to a blank window, type:

```
cls
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

cmd

Article • 05/23/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Starts a new instance of the command interpreter, cmd.exe. If used without parameters, `cmd` displays the version and copyright information of the operating system.

> **Note**
>
> Users seeking more advanced capabilities are encouraged to explore PowerShell for enhanced scripting and automation.

Syntax

```
cmd [/c|/k] [/s] [/q] [/d] [/a|/u] [/t:{<b><f> | <f>}] [/e:{on | off}] [/f:{on | off}] [/v:{on | off}] [<string>]
```

Parameters

Expand table

Parameter	Description
/c	Carries out the command specified by `<string>` and then exits the command processor.
/k	Carries out the command specified by `<string>` and keeps the command processor running.
/s	When used with `/c` or `/k`, triggers special non-parsing rules that strip the first and last quotes (`"`) around the `<string>` but leaves the rest of the command unchanged.
/q	Turns echo off.
/d	Disables execution of AutoRun commands.
/a	Formats command output as American National Standards Institute (ANSI).
/u	Formats command output as Unicode.
/t:{`<b><f>` |	Sets the background (*b*) and foreground (*f*) colors.

Parameter	Description
`<f>`}	
/e:on	Enables command extensions.
/e:off	Disables commands extensions.
/f:on	Enables file and directory name completion.
/f:off	Disables file and directory name completion.
/v:on	Enables delayed environment variable expansion.
/v:off	Disables delayed environment variable expansion.
`<string>`	Specifies the command you want to carry out.
/?	Displays help at the command prompt.

The following table lists valid hexadecimal digits that you can use as the values for `<b>` and `<f>`:

Expand table

Value	Color
0	Black
1	Blue
2	Green
3	Aqua
4	Red
5	Purple
6	Yellow
7	White
8	Gray
9	Light blue
a	Light green
b	Light aqua
c	Light red

Value	Color
d	Light purple
e	Light yellow
f	Bright white

Remarks

- To redirect command output to the input of another command, use the pipe (`|`) operator. For example:

```
<command1> | <command2>
```

- The double pipe (`||`) operator is used to execute the next command only if the previous command fails. In the given scenario, `command2` is executed only if `command1` fails. For example:

```
<command1> || <command2>
```

- To redirect command output to a file, use the greater-than angle bracket `>` character. For example:

```
<command1> > <file1.txt>
```

- To group or nest multiple commands together, use `&`. For example:

```
<command1> & <command2>
```

- To use multiple commands for `<string>`, separate them by the command separator `&&`. For example:

```
```

```
<command1> && <command2> && <command3>
```

> **Note**
>
> When you use `&&`, the command following `&&` runs only if the preceding command completes successfully. `Cmd` executes the first command, and proceeds to the next only if the previous command was successful. Otherwise, if the previous command fails, the subsequent commands won't be executed.

- The ampersand `&`, pipe `|`, and parentheses `( )` are special characters that must be preceded by the escape character `^` or quotation marks when you pass them as arguments.
- If a command completes an operation successfully, it returns an exit code of zero (0) or no exit code.
- If the directory path, files, or any information you supply contains spaces, you must use double quotation marks (`" "`) around the text, such as `"Computer Name"`. For example:

```
mkdir Test && mkdir "Test 2" && move "Test 2" Test
```

- You must use quotation marks around the following special characters: & < > [ ] | { } ^ = ; ! ' + , ` ~ [white space].
- If you specify `/c` or `/k`, the `cmd` processes, the remainder of `<string>`, and the quotation marks are preserved only if all of the following conditions are met:
  - You don't also use `/s`.
  - You use exactly one set of quotation marks.
  - You don't use any special characters within the quotation marks, for example: & < > ( ) @ ^ |.
  - You use one or more white-space characters within the quotation marks.
  - The `<string>` within quotation marks is the name of an executable file.

  If the previous conditions aren't met, `<string>` is processed by examining the first character to verify whether it's an opening quotation mark. If the first character is an opening quotation mark, it's stripped along with the closing quotation mark. Any text following the closing quotation marks is preserved.
- If you don't specify `/d`, `cmd` looks for the following registry subkeys:

- **HKEY_LOCAL_MACHINE\Software\Microsoft\Command Processor\AutoRun\REG_SZ**
- **HKEY_CURRENT_USER\Software\Microsoft\Command Processor\AutoRun\REG_EXPAND_SZ**

If one or both registry subkeys are present, they're executed before all other variables.

- You can disable command extensions for a particular process by using `/e:off`. You can enable or disable extensions for all `cmd` command-line options on a computer or user session by setting the following **REG_DWORD** values:
  - **HKEY_LOCAL_MACHINE\Software\Microsoft\Command Processor\EnableExtensions\REG_DWORD**
  - **HKEY_CURRENT_USER\Software\Microsoft\Command Processor\EnableExtensions\REG_DWORD**

  Set the **REG_DWORD** value to either **0×1** (enabled) or **0×0** (disabled) in the registry by using Regedit.exe. User-specified settings take precedence over computer settings, and command-line options take precedence over registry settings.

  > **⊗ Caution**
  >
  > Incorrectly editing the registry might severely damage your system. Before making changes to the registry, you should back up any valued data on the computer.

- When you enable command extensions, the following commands are affected:
  - `assoc`
  - `call`
  - `chdir` (`cd`)
  - `color`
  - `del` (`erase`)
  - `endlocal`
  - `for`
  - `ftype`
  - `goto`
  - `if`
  - `mkdir` (`md`)
  - `popd`
  - `prompt`
  - `pushd`

    - `set`
    - `setlocal`
    - `shift`
    - `start` (also includes changes to external command processes)

- If you enable delayed environment variable expansion, you can use the exclamation point character (`!`) to substitute the value of an environment variable at run time.

- File and directory name completion isn't enabled by default. You can enable or disable file name completion for a particular process of the `cmd` command with `/f:{on | off}`.

- You can enable or disable file and directory name completion for all processes of the `cmd` command on a computer or for a user logon session by setting the following **REG_DWORD** values:

    - **HKEY_LOCAL_MACHINE\Software\Microsoft\Command Processor\CompletionChar\REG_DWORD**

    - **HKEY_LOCAL_MACHINE\Software\Microsoft\Command Processor\PathCompletionChar\REG_DWORD**

    - **HKEY_CURRENT_USER\Software\Microsoft\Command Processor\CompletionChar\REG_DWORD**

    - **HKEY_CURRENT_USER\Software\Microsoft\Command Processor\PathCompletionChar\REG_DWORD**

    To set the **REG_DWORD** value, run Regedit.exe and use the hexadecimal value of a control character for a particular function (for example, **0×9** is TAB and **0×08** is BACKSPACE). Use the [**0×20**] value for white space, because it isn't a valid control character. User-specified settings take precedence over computer settings, and command-line options take precedence over registry settings.

    > **⊗ Caution**
    >
    > Incorrectly editing the registry might severely damage your system. Before making changes to the registry, you should back up any valued data on the computer.

- If you enable file and directory name completion by using `/f:on`, use **CTRL+D** for directory name completion and **CTRL+F** for file name completion. The only difference between the key combinations **CTRL+D** and **CTRL+F** is that **CTRL+D** only matches directory names and **CTRL+F** matches both file and directory names. If you use file and

directory name completion on the built-in directory commands `cd`, `md`, or `rd`, directory completion is assumed.

- Pressing **CTRL+D** or **CTRL+F** processes the file and directory name completion. These key combination functions append a wildcard character to `<string>` if one isn't present, build a list of paths that match, and then display the first matching path. If none of the paths match, the file and directory name completion function beeps and doesn't change the display.
  - To move through the list of matching paths, press **CTRL+D** or **CTRL+F** repeatedly.
  - To move through the list backwards, press the **SHIFT** key and **CTRL+D** or **CTRL+F** simultaneously.
  - To discard the saved list of matching paths and generate a new list, edit `<string>` and press **CTRL+D** or **CTRL+F**.
  - If you switch between **CTRL+D** and **CTRL+F**, the saved list of matching paths is discarded, and a new list is generated.

- File and directory name completion correctly processes file names that contain white space or special characters if you place quotation marks around the matching path.

- If you process file and directory name completion from within `<string>`, at the point in `<string>` where the completion was processed, any part of the *path* to the right of the cursor is discarded.

Using environment variables

The `cmd` command-shell environment is defined by variables that determine the behavior of the command shell and the operating system. You can define the behavior of the command-shell environment or the entire operating system environment by using two types of environment variables, system and local. System environment variables define the behavior of the global operating system environment. Local environment variables define the behavior of the environment of the current instance of `cmd`.

Only users with administrative privileges can change system variables. These variables are most commonly used in logon scripts. Local environment variables are only available when the user for whom they were created is logged on to the computer. Local variables set in the **HKEY_CURRENT_USER** hive are valid only for the current user, but define the behavior of the global operating system environment.

The following list describes the various types of variables in descending order of precedence:

1. Built-in system variables.
2. System variables found in the **HKEY_LOCAL_MACHINE** hive.

3. Local variables found in the **HKEY_CURRENT_USER** hive.
4. All environment variables and paths set in the Autoexec.bat file.
5. All environment variables and paths set in a logon script (if present).
6. Variables used interactively in a script or batch file.

In the command shell, each instance of `cmd` inherits the environment of its parent application. Therefore, you can change the variables in the new `cmd` environment without affecting the environment of the parent application.

You can view the environment variables in your environment by running one of the following commands:

Windows Command Prompt

```
set
```

PowerShell

```
Get-ChildItem Env:
```

Setting environment variables

To view a variable, type:

Windows Command Prompt

```
set <VariableName>
```

To add a variable, type:

Windows Command Prompt

```
set <VariableName>=<value>
```

To delete a variable, type:

Windows Command Prompt

```
set <VariableName>=
```

You can use most characters as variable values, including white space. If you use the special characters <, >, |, &, or ^, you must precede them with the escape character `^` *or* double

quotation marks. If you use double quotation marks, they're included as part of the value because everything following the equal sign is taken as the value. Consider the following examples:

To create the variable value `New&Name`, type:

Windows Command Prompt

```
set <VariableName>=New^&Name
```

To create the variable value `"New&Name"`, type:

Windows Command Prompt

```
set <VariableName>="New&Name"
```

If you type `set <VariableName>=New&Name` (without the carat `^` or double quotes) at the command prompt, the following error message might display:

error

```
"'Name' is not recognized as an internal or external command, operable program or
batch file."
```

Variable names aren't case-sensitive. However, `set` displays the variable exactly as you typed it. You can combine uppercase and lowercase letters in your variable names to make your code more readable, for example, *UserName*.

> **Note**
>
> - The maximum individual environment variable size is 8,192 bytes.
> - The maximum total size for all environment variables, including names, values, and equal signs, is 65,536 characters for a process.

Substituting environment variable values

To substitute variable values in the command line or scripts, enclose the variable name in percent signs (%VariableName%). This allows `cmd` to reference the variable's value instead of treating it as literal text. When a script is run, `cmd` replaces instances of the variable with its

value. For example, using %USERNAME% in a script automatically substitutes in the actual username. Variable substitution isn't recursive; `cmd` checks variables only once.

Related links

- Command-Line Syntax Key

cmdkey

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Creates, lists, and deletes stored user names and passwords or credentials.

Syntax

```
cmdkey [{/add:<targetname>|/generic:<targetname>}] {/smartcard | /user:
<username> [/pass:<password>]} [/delete{:<targetname> | /ras}] /list:
<targetname>
```

Parameters

Expand table

Parameters	Description
/add:`<targetname>`	Adds a user name and password to the list. Requires the parameter of `<targetname>` which identifies the computer or domain name that this entry will be associated with.
/generic:`<targetname>`	Adds generic credentials to the list. Requires the parameter of `<targetname>` which identifies the computer or domain name that this entry will be associated with.
/smartcard	Retrieves the credential from a smart card. If more than one smart card is found on the system when this option is used, **cmdkey** displays information about all available smart cards, and then prompts the user to specify which one to use.
/user:`<username>`	Specifies the user or account name to store with this entry. If `<username>` isn't supplied, it will be requested.
/pass:`<password>`	Specifies the password to store with this entry. If `<password>` isn't supplied, it will be requested. Passwords are not displayed after they're stored.
/delete:`{<targetname> | /ras}`	Deletes a user name and password from the list. If `<targetname>` is specified, that entry is deleted. If `/ras` is specified, the stored remote access entry is deleted.

Parameters	Description
/list:`<targetname>`	Displays the list of stored user names and credentials. If `<targetname>` isn't specified, all stored user names and credentials are listed.
/?	Displays help at the command prompt.

Examples

To display a list of all user names and credentials that are stored, type:

```
cmdkey /list
```

To add a user name and password for user *Mikedan* to access computer *Server01* with the password *Kleo*, type:

```
cmdkey /add:server01 /user:mikedan /pass:Kleo
```

To add a user name and password for user *Mikedan* to access computer *Server01* and prompt for the password whenever Server01 is accessed, type:

```
cmdkey /add:server01 /user:mikedan
```

To delete a credential stored by remote access, type:

```
cmdkey /delete /ras
```

To delete a credential stored for *Server01*, type:

```
cmdkey /delete:server01
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

cmstp

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Installs or removes a Connection Manager service profile. Used without optional parameters, **cmstp** installs a service profile with default settings appropriate to the operating system and to the user's permissions.

Syntax

Syntax 1 - This is the typical syntax used in a custom installation application. To use this syntax, you must run **cmstp** from the directory that contains the `<serviceprofilefilename>.exe` file.

```
<serviceprofilefilename>.exe /q:a /c:cmstp.exe <serviceprofilefilename>.inf
[/nf] [/s] [/u]
```

Syntax 2

```
cmstp.exe [/nf] [/s] [/u] [drive:][path]serviceprofilefilename.inf
```

Parameters

Expand table

Parameter	Description
`<serviceprofilefilename>.exe`	Specifies, by name, the installation package that contains the profile that you want to install. Required for Syntax 1, but not valid for Syntax 2.
/q:a	Specifies that the profile should be installed without prompting the user. The verification message that the installation has succeeded will still appear. Required for Syntax 1, but not valid for Syntax 2.

Parameter	Description
[drive:][path] `<serviceprofilefilename>.inf`	Required. Specifies, by name, the configuration file that determines how the profile should be installed. The [drive:][path] parameter isn't valid for Syntax 1.
/nf	Specifies that the support files should not be installed.
/s	Specifies that the service profile should be installed or uninstalled silently (without prompting for user response or displaying verification message). This is the only parameter that you can use in combination with **/u**.
/u	Specifies that the service profile should be uninstalled.
/?	Displays help at the command prompt.

Examples

To install the *fiction* service profile without any support files, type:

```
fiction.exe /c:cmstp.exe fiction.inf /nf
```

To silently install the *fiction* service profile for a single user, type:

```
fiction.exe /c:cmstp.exe fiction.inf /s /su
```

To silently uninstall the *fiction* service profile, type:

```
fiction.exe /c:cmstp.exe fiction.inf /s /u
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

color

Article • 09/20/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Changes the foreground and background colors in the Command Prompt window for the current session. If used without parameters, **color** restores the default Command Prompt window foreground and background colors.

Syntax

Windows Command Prompt

```
color [attr]
```

Parameters

Expand table

Parameter	Description
`attr`	Specifies color attribute of console output.
/?	Displays help at the command prompt.

The following table lists valid hexadecimal digits that you can use as the values for `attr`:

Expand table

Value	Color
0	Black
1	Blue
2	Green
3	Aqua
4	Red
5	Purple

Value	Color
6	Yellow
7	White
8	Gray
9	Light blue
a	Light green
b	Light aqua
c	Light red
d	Light purple
e	Light yellow
f	Bright white

Remarks

- You can specify one or two hexadecimal digits. The first is used as the foreground color and the second is used as the background color. If you specify two hexadecimal digits, don't use space characters between them.
- If you specify only one hexadecimal digit, the corresponding color is used as the foreground color and the background color is set to the default color.
- To set the default Command Prompt window color, select the upper-left corner of the **Command Prompt** window, select **Defaults**, select the **Colors** tab, and then select the colors that you want to use for the **Screen Text** and **Screen Background**.
- If you specify the same value for two hexadecimal digits, the ERRORLEVEL is set to `1` and no change is made to either the foreground or the background color.

Examples

To change the Command Prompt window background color to gray and the foreground color to red, type:

Windows Command Prompt

```
color 84
```

To change the Command Prompt window foreground color to light yellow, type:

Windows Command Prompt

```
color e
```

> **Note**
>
> In this example, the background is set to the default color because only one hexadecimal digit is specified.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

comp

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Compares the contents of two files or sets of files byte-by-byte. These files can be stored on the same drive or on different drives, and in the same directory or in different directories. When this command compares files, it displays their location and file names. If used without parameters, **comp** prompts you to enter the files to compare.

Syntax

```
comp [<data1>] [<data2>] [/d] [/a] [/l] [/n=<number>] [/c]
```

Parameters

Expand table

Parameter	Description
`<data1>`	Specifies the location and name of the first file or set of files that you want to compare. You can use wildcard characters (* and ?) to specify multiple files.
`<data2>`	Specifies the location and name of the second file or set of files that you want to compare. You can use wildcard characters (* and ?) to specify multiple files.
/d	Displays differences in decimal format. (The default format is hexadecimal.)
/a	Displays differences as characters.
/l	Displays the number of the line where a difference occurs, instead of displaying the byte offset.
/n= `<number>`	Compares only the number of lines that are specified for each file, even if the files are different sizes.
/c	Performs a comparison that is not case-sensitive.
/off[line]	Processes files with the offline attribute set.
/?	Displays Help at the command prompt.

Remarks

- During the comparison, **comp** displays messages that identify the locations of unequal information between the files. Each message indicates the offset memory address of the unequal bytes and the contents of the bytes (in hexadecimal notation unless the **/a** or **/d** command-line parameter is specified). Messages appear in the following format:

  ```
  Compare error at OFFSET xxxxxxxx
  file1 = xx
  file2 = xx
  ```

  After ten unequal comparisons, **comp** stops comparing the files and displays the following message:

  `10 Mismatches - ending compare`

- If you omit necessary components of either *data1* or *data2*, or if you omit *data2* entirely, this command prompts you for the missing information.

- If *data1* contains only a drive letter or a directory name with no file name, this command compares all of the files in the specified directory to the file specified in *data1*.

- If *data2* contains only a drive letter or a directory name, the default file name for *data2* becomes the same name as for *data1*.

- If the **comp** command can't find the specified files, it will prompt you with a message about whether you want to compare additional files.

- The files that you compare can have the same file name, provided they're in different directories or on different drives. You can use wildcard characters (***** and **?**) to specify file names.

- You must specify **/n** to compare files of different sizes. If the file sizes are different and **/n** isn't specified, the following message is displayed:

  ```
  Files are different sizes
  Compare more files (Y/N)?
  ```

To compare these files anyway, press **N** to stop the command. Then, run the **comp** command again, using the **/n** option to compare only the first portion of each file.

- If you use wildcard characters (* and ?) to specify multiple files, **comp** finds the first file that matches *data1* and compares it with the corresponding file in *data2*, if it exists. The **comp** command reports the results of the comparison for each file matching *data1*. When finished, **comp** displays the following message:

  `Compare more files (Y/N)?`

  To compare more files, press **Y**. The **comp** command prompts you for the locations and names of the new files. To stop the comparisons, press **N**. When you press **Y**, you're prompted for which command-line options to use. If you don't specify any command-line options, **comp** uses the ones you specified before.

Examples

To compare the contents of the directory *c:\reports* with the backup directory `\\sales\backup\april`, type:

```
comp c:\reports \\sales\backup\april
```

To compare the first ten lines of the text files in the *\invoice* directory and display the result in decimal format, type:

```
comp \invoice\*.txt \invoice\backup\*.txt /n=10 /d
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

compact

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays or alters the compression of files or directories on NTFS partitions. If used without parameters, **compact** displays the compression state of the current directory and any files it contains.

Syntax

```
compact [/C | /U] [/S[:dir]] [/A] [/I] [/F] [/Q] [/EXE[:algorithm]]
[/CompactOs[:option] [/windir:dir]] [filename [...]]
```

Parameters

Expand table

Parameter	Description
/c	Compresses the specified directory or file. Directories are marked so any files added afterwards are compressed, unless the /EXE parameter is specified.
/u	Uncompresses the specified directory or file. Directories are marked so any files added afterwards aren't compressed. If the /EXE parameter is specified, only files compressed as executables are uncompressed; if you don't specify the /EXE parameter, only NTFS compressed files are uncompressed.
/s`[:<dir>]`	Performs the chosen operation on files in the specified directory and all subdirectories. By default, the current directory is used as the `<dir>` value.
/a	Displays hidden or system files. By default, these files aren't included.
/i	Continues performing the specified operation, ignoring errors. By default, this command stops when an error is encountered.
/f	Forces compression or uncompression of the specified directory or file. Already-compressed files are skipped by default. The **/f** parameter is used in the case of a file that was partly compressed when the operation was interrupted by a system crash. To force the file to be compressed in its entirety, use the **/c** and **/f** parameters and specify the partially compressed file.

Parameter	Description
/q	Reports only the most essential information.
/EXE	Uses compression optimized for executable files that are read frequently, but not modified. Supported algorithms are: • **XPRESS4K** (fastest and default value) • **XPRESS8K** • **XPRESS16K** • **LZX** (most compact)
/CompactOs	Sets or queries the system's compression state. Supported options are: • **query** - Queries the system's **Compact** state. • **always** - Compresses all operating system binaries and sets the system state to Compact, which remains unless administrator changes it. • **never** - Uncompresses all operating system binaries and sets the system state to non-Compact, which remains unless administrator changes it.
/windir	Used with the **/CompactOs:query** parameter, when querying the offline operating system. Specifies the directory where Windows is installed.
`<filename>`	Specifies a pattern, file, or directory. You can use multiple file names, and the * and **?** wildcard characters.
/?	Displays help at the command prompt.

Remarks

- This command is the command-line version of the NTFS file system compression feature. The compression state of a directory indicates whether files are automatically compressed when they are added to the directory. Setting the compression state of a directory does not necessarily change the compression state of files that are already in the directory.
- You can't use this command to read, write, or mount volumes compressed using DriveSpace or DoubleSpace. You also can't use this command to compress file allocation table (FAT) or FAT32 partitions.

Examples

To set the compression state of the current directory, its subdirectories, and existing files, type:

```
```

```
compact /c /s
```

To set the compression state of files and subdirectories within the current directory, without altering the compression state of the current directory itself, type:

```
compact /c /s *.*
```

To compress a volume, from the root directory of the volume, type:

```
compact /c /i /s:\
```

> **Note**
>
> This example sets the compression state of all directories (including the root directory on the volume) and compresses every file on the volume. The **/i** parameter prevents error messages from interrupting the compression process.

To compress all files with the .bmp file name extension in the \tmp directory and all subdirectories of \tmp, without modifying the compressed attribute of the directories, type:

```
compact /c /s:\tmp *.bmp
```

To force complete compression of the file *zebra.bmp*, which was partially compressed during a system crash, type:

```
compact /c /f zebra.bmp
```

To remove the compressed attribute from the directory c:\tmp, without changing the compression state of any files in that directory, type:

```
```

```
compact /u c:\tmp
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

copy

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Copies one or more files from one location to another.

> **Note**
>
> You can also use the **copy** command, with different parameters, from the Recovery Console. For more information about the recovery console, see Windows Recovery Environment (Windows RE).

Syntax

```
copy [/d] [/v] [/n] [/y | /-y] [/z] [/a | /b] <source> [/a | /b] [+<source> [/a | /b] [+ ...]] [<destination> [/a | /b]]
```

Parameters

Expand table

Parameter	Description
/d	Allows the encrypted files being copied to be saved as decrypted files at the destination.
/v	Verifies that new files are written correctly.
/n	Uses a short file name, if available, when copying a file with a name longer than eight characters, or with a file name extension longer than three characters.
/y	Suppresses prompting to confirm that you want to overwrite an existing destination file.
/-y	Prompts you to confirm that you want to overwrite an existing destination file.
/z	Copies networked files in restartable mode.

Parameter	Description
/a	Indicates an ASCII text file.
/b	Indicates a binary file.
`<source>`	Required. Specifies the location from which you want to copy a file or set of files. *Source* can consist of a drive letter and colon, a directory name, a file name, or a combination of these.
`<destination>`	Required. Specifies the location to which you want to copy a file or set of files. *Destination* can consist of a drive letter and colon, a directory name, a file name, or a combination of these.
/?	Displays help at the command prompt.

Remarks

- You can copy an ASCII text file that uses an end-of-file character (CTRL+Z) to indicate the end of the file.
- If **/a** precedes or follows a list of files on the command line, it applies to all files listed until **copy** encounters **/b**. In this case, **/b** applies to the file preceding **/b**.

  The effect of **/a** depends on its position in the command-line string: - If **/a** follows *source*, the **copy** command treats the file as an ASCII file and copies data that precedes the first end-of-file character (CTRL+Z). - If **/a** follows *destination*, the **copy** command adds an end-of-file character (CTRL+Z) as the last character of the file.
- If **/b** directs the command interpreter to read the number of bytes specified by the file size in the directory. **/b** is the default value for **copy**, unless **copy** combines files.
- If **/b** precedes or follows a list of files on the command line, it applies to all listed files until **copy** encounters **/a**. In this case, **/a** applies to the file preceding **/a**.

  The effect of **/b** depends on its position in the command–line string: - If **/b** follows *source*, the **copy** command copies the entire file, including any end-of-file character (CTRL+Z). - If **/b** follows *destination*, the **copy** command doesn't add an end-of-file character (CTRL+Z).
- If a write operation cannot be verified, an error message appears. Although recording errors rarely occur with the **copy** command , you can use **/v** to verify that critical data has been correctly recorded. The **/v** command-line option also slows

down the **copy** command, because each sector recorded on the disk must be checked.

- If **/y** is preset in the **COPYCMD** environment variable, you can override this setting by using **/-y** at the command line. By default, you are prompted when you replace this setting, unless the **copy** command is executed in a batch script.
- To append files, specify a single file for *destination*, but multiple files for *source* (use wildcard characters or *file1*+*file2*+*file3* format).
- If the connection is lost during the copy phase (for example, if the server going offline breaks the connection), you can use **copy /z** to resume after the connection is re-established. The **/z** option also displays the percentage of the copy operation that is completed for each file.
- You can substitute a device name for one or more occurrences of *source* or *destination*.
- If *destination* is a device (for example, Com1 or Lpt1), the **/b** option copies data to the device in binary mode. In binary mode, **copy /b** copies all characters (including special characters such as CTRL+C, CTRL+S, CTRL+Z, and ENTER) to the device, as data. However, if you omit **/b**, the data is copied to the device in ASCII mode. In ASCII mode, special characters might cause files to combine during the copying process.
- If you don't specify a destination file, a copy is created with the same name, modified date, and modified time as the original file. The new copy is stored in the current directory on the current drive. If the source file is on the current drive and in the current directory and you do not specify a different drive or directory for the destination file, the **copy** command stops and displays the following error message:

```
File cannot be copied onto itself
0 File(s) copied
```

- If you specify more than one file in *source*, the **copy** command combines them all into a single file using the file name specified in *destination*. The **copy** command assumes the combined files are ASCII files unless you use the **/b** option.
- To copy files that are 0 bytes long, or to copy all of a directory's files and subdirectories, use the xcopy command.

- To assign the current time and date to a file without modifying the file, use the following syntax:

  ```
  copy /b <source> +,,
  ```

  Where the commas indicate that the *destination* parameter has been intentionally left out.

Examples

To copy a file called *memo.doc* to *letter.doc* in the current drive and ensure that an end-of-file character (CTRL+Z) is at the end of the copied file, type:

```
copy memo.doc letter.doc /a
```

To copy a file named *robin.typ* from the current drive and directory to an existing directory named *Birds* that is located on drive C, type:

```
copy robin.typ c:\birds
```

> ⓘ **Note**
>
> If the *Birds* directory doesn't exist, the file *robin.typ* is copied into a file named *Birds* that is located in the root directory on the disk in drive C.

To combine *Mar89.rpt*, *Apr89.rpt*, and *May89.rpt*, which are located in the current directory, and place them in a file named *Report* (also in the current directory), type:

```
copy mar89.rpt + apr89.rpt + may89.rpt Report
```

> ⓘ **Note**

If you combine files, the **copy** command marks the destination file with the current date and time. If you omit *destination*, the files are combined and stored under the name of the first file in the list.

To combine all files in *Report*, when a file named *Report* already exists, type:

```
copy report + mar89.rpt + apr89.rpt + may89.rpt
```

To combine all files in the current directory that have the .txt file name extension into a single file named *Combined.doc*, type:

```
copy *.txt Combined.doc
```

To combine several binary files into one file by using wildcard characters, include **/b**. This prevents Windows from treating CTRL+Z as an end-of-file character. For example, type:

```
copy /b *.exe Combined.exe
```

⊗ Caution

If you combine binary files, the resulting file might be unusable due to internal formatting.

- Combining each file that has a .txt extension with its corresponding .ref file creates a file with the same file name, but with a .doc extension. The **Copy** command combines *file1.txt* with *file1.ref* to form *file1.doc*, and then the command combines *file2.txt* with *file2.ref* to form *file2.doc*, and so on. For example, type:

```
copy *.txt + *.ref *.doc
```

To combine all files with the .txt extension, and then to combine all files with the .ref extension into one file named *Combined.doc*, type:

```
copy *.txt + *.ref Combined.doc
```

Related links

- Command-Line Syntax Key
- xcopy command

Feedback

Was this page helpful? Yes No

cscript

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Starts a script to run in a command-line environment.

> **ⓘ Important**
>
> Performing this task does not require you to have administrative credentials. Therefore, as a security best practice, consider performing this task as a user without administrative credentials.

Syntax

```
cscript <scriptname.extension> [/b] [/d] [/e:<engine>] [{/h:cscript | /h:wscript}] [/i] [/job:<identifier>] [{/logo | /nologo}] [/s] [/t:<seconds>] [x] [/u] [/?] [<scriptarguments>]
```

Parameters

Expand table

Parameter	Description
scriptname.extension	Specifies the path and file name of the script file with optional file name extension.
/b	Specifies batch mode, which does not display alerts, scripting errors, or input prompts.
/d	Starts the debugger.
/e:`<engine>`	Specifies the engine that is used to run the script.
/h:cscript	Registers cscript.exe as the default script host for running scripts.
/h:wscript	Registers wscript.exe as the default script host for running scripts. The default.

Parameter	Description
/i	Specifies interactive mode, which displays alerts, scripting errors, and input prompts. The default, and the opposite of `/b`.
/job:`<identifier>`	Runs the job identified by *identifier* in a .wsf script file.
/logo	Specifies that the Windows Script Host banner is displayed in the console before the script runs. The default, and the opposite of `/nologo`.
/nologo	Specifies that the Windows Script Host banner is not displayed before the script runs.
/s	Saves the current command-prompt options for the current user.
/t:`<seconds>`	Specifies the maximum time the script can run (in seconds). You can specify up to 32,767 seconds. The default is no time limit.
/u	Specifies Unicode for input and output that is redirected from the console.
/x	Starts the script in the debugger.
/?	Displays available command parameters and provides help for using them. The same as typing **cscript.exe** with no parameters and no script.
scriptarguments	Specifies the arguments passed to the script. Each script argument must be preceded by a slash (**/**).

Remarks

- Each parameter is optional; however, you can't specify script arguments without specifying a script. If you don't specify a script or any script arguments, cscript.exe displays the cscript.exe syntax and the valid host options.
- The **/t** parameter prevents excessive running of scripts by setting a timer. When the run time exceeds the specified value, cscript interrupts the script engine and ends the process.
- Windows script files usually have one of the following file name extensions: .wsf, .vbs, .js. Windows Script Host can use .wsf script files. Each .wsf file can use multiple scripting engines and perform multiple jobs.
- if you double-click a script file with an extension that has no association, the **Open With** dialog box appears. Select wscript or cscript, and then select **Always use this program to open this file type**. This registers wscript.exe or cscript as the default script host for files of this file type.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

date

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays or sets the system date. If used without parameters, **date** displays the current system date setting and prompts you to enter a new date.

> **ⓘ Important**
>
> You must be an administrator to use this command.

Syntax

```
date [/t | <month-day-year>]
```

Parameters

Expand table

Parameter	Description
`<month-day-year>`	Sets the date specified, where *month* is the month (one or two digits, including values 1 through 12), *day* is the day (one or two digits, including values 1 through 31), and *year* is the year (two or four digits, including the values 00 through 99 or 1980 through 2099). You must separate values for *month*, *day*, and *year* with periods (.), hyphens (-), or slash marks (/). **Note:** Be mindful if you use 2 digits to represent the year, the values 80-99 correspond to 1980 through 1999.
/t	Displays the current date without prompting you for a new date.
/?	Displays help at the command prompt.

Examples

If command extensions are enabled, to display the current system date, type:

```
date /t
```

To change the current system date to August 3, 2007, you can type any of the following:

```
date 08.03.2007
date 08-03-07
date 8/3/07
```

To display the current system date, followed by a prompt to enter a new date, type:

```
date
```

To keep the current date and return to the command prompt, press **ENTER**. To change the current date, type the new date based on your current date configuration, as seen in the second example above, and then press **ENTER**.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

DCDiag

Article • 11/15/2023

> Applies To: Windows Server 2022, Windows Server 2019, Windows Server 2016

`DCDiag.exe` analyzes the state of domain controllers (DC) in a forest or enterprise and reports any problems to help in troubleshooting. As an end-user reporting program, DCDiag is a command-line tool that encapsulates detailed knowledge of how to identify abnormal behavior in the system.

By default, DCDiag is readily available if logged into a DC. An alternative method to access DCDiag is to install the Remote Server Administration Tools (RSAT) on your device. DCDiag must be ran with administrative rights from an elevated command prompt (CMD) or PowerShell.

DCDiag consists of a framework for executing tests and a series of tests to verify different functional areas of the system. This framework selects which DC is tested according to scope directives from the user, such as enterprise, site, or single server. Testing the overall connectivity and responsiveness of the DC includes verifying:

- The DC can be located in DNS
- The DC responds to Internet Control Message Protocol (ICMP) pings
- The DC allows Lightweight Directory Access Protocol (LDAP) connectivity by binding to the instance
- The DC allows binding to the AD RPC interface using the DsBindWithCred function.

> **Note**
>
> Blocking ICMP prevents DCDiag from functioning as intended. While blocking ICMP is recommended at the Internet-edge of your network, internally blocking ICMP traffic leads to administrative issues that break legacy group policies, black hole router detection, or inefficient MTU sizes due to lack of a discovery option. Troubleshooting tools such as `ping.exe` or `tracert.exe` are also affected.

DCDiag syntax

```
dcdiag [/s:<DomainController>] [/n:<NamingContext>] [/u:<Domain>\<UserName> /p:{* | <Password> | ""}] [{/a | /e}] [{/q | /v}] [/i] [/f:<LogFile>] [/c [/skip:<Test>]] [/test:<Test>] [/fix] [{/h | /?}] [/ReplSource:<SourceDomainController>]
```

DCDiag uses the following parameters:

Parameter	Description
/s:`<DomainController>`	Specifies the name of the server to run the command against. If this parameter isn't specified, the tests are run against the local domain controller. This parameter is ignored for DcPromo and RegisterInDns tests, which can only be run locally.
/n:`<NamingContext>`	Uses NamingContext as the naming context to test. You can specify domains in NetBIOS, Domain Name System (DNS), or distinguished name format.
/u:`<Domain>`\`<UserName>` /p:{`<Password>` | ""}	Uses Domain\UserName. DCDiag uses the current credentials of the user (or process) that is logged on. If alternate credentials are needed, use the following options to provide those credentials for binding with Password as the password: Use quotation marks ("") for an empty or null password. Use the wildcard character (*) to prompt for the password.
/a	Tests all the servers on this AD DS site.
/e	Tests all the servers in the enterprise. This overrides `/a`.
/q	Quiet. Prints only error messages.
/v	Verbose. Prints extended information.
/fix	Affects the MachineAccount test only. This parameter causes the test to fix the Service Principal Names (SPNs) on the Machine Account object of the domain controller.
/f:`<LogFile>`	Redirects all output to a log file.
/c	Comprehensive. Runs all tests except DCPromo and RegisterInDNS, including non-default tests. Optionally, you can use this parameter with the `/skip` parameter to skip specified tests. The following tests aren't run by default: • Topology • CutoffServers • OutboundSecureChannels.
/h or /?	Displays help at the command prompt.
/test:`<Test>`	Runs this test only. The connectivity test can't be skipped with the `/skip` parameter.
/ReplSource:`<SourceDomainController>`	Tests the connection between the domain controller on which you run the command and the source domain controller. (This parameter is used for the CheckSecurityError test.) SourceDomainController is the DNS name, NetBIOS name, or distinguished name of a real or potential server that will be the source domain controller for replication, as represented by a real or potential connection object.

DCDiag known tests

The following table displays known tests that run by default unless specified otherwise.

Test	Description
Advertising	Checks whether each domain controller advertises itself in the roles that it should be capable of performing. This test validates that the public DsGetDcName function used by computers to locate domain controllers will correctly locate any DCs. This test fails if the Netlogon Service has stopped or failed to start. If the Key Distribution Key (KDC) service is stopped, the Advertising test fails since the flag returned from DsGetDcName won't include KDC. If port 88 over TCP and UDP is blocked on a firewall, the Advertising test will pass despite the KDC not being able to answer Kerberos tickets requests.
CheckSDRefDom	Checks that all application directory partitions have appropriate security descriptor reference domains. This test uses LDAP and validates cross reference objects located in **cn=partitions,cn=configuration,dc=<forest root domain>** contain the correct domain names in their msDS-SDReferenceDomain attributes.
CheckSecurityError	Test is **not** run by default. Performs various security checks for errors related to the security components of the DC, such as issues with the security policy or the security database using LDAP, RPC, RPC over SMB, and ICMP. It checks: • That at least one KDC is online for each domain and is reachable. • That DCs computer object has replicated to other DCs. • If packet fragmentation of Kerberos over UDP might be an issue based on current MTU size by sending non-fragmenting ICMP packets. • That there aren't any replication or Knowledge Consistency Checker (KCC) connection issues for connected partners by querying the function DsReplicaGetInfo to get any security-related errors. • If the DCs computer account exists in Active Directory (AD) as part of the default "Domain Controllers" OU along with the correct UserAccountControl flags for DCs, the correct ServerReference attributes are set, and has the minimum required Service Principal Names (SPN) configured. When the **/ReplSource** parameter is added, the partner also checks: • The time skew calculated between the servers to verify it's less than 300 seconds (5 minutes) for Kerberos. It **doesn't** check the Kerberos policy to see if the allowed skew has been modified. • Permissions on all the naming contexts (such as Schema, Configuration, etc.) on the source DC that validates replication and connectivity functions between DCs. • Connectivity to validate that the user running DCDiag can connect to and read the SYSVOL and NETLOGON shares without any security errors. • The **Access this computer from the network** privilege on the DC is checked to verify it's held by the *Administrators*, *Authenticated Users*, and *Everyone* groups. • The DC's computer object is checked to ensure it's the latest version on the DCs. This is done to prove replication convergence along with checking versions, USNs, originating servers, and timestamps.

Test	Description
Connectivity	Verifies that the DSA and DNS are registered and reachable using LDAP and RPC.
CrossRefValidation	Retrieves a list of naming contexts located in *cn=partitions,cn=configuration,dc=<forest root domain>* with their cross references and then validates them similar to the **CheckSDRefDom** test using LDAP. This test looks at the nCName, dnsRoot, nETBIOSName, and systemFlags attributes to: • Ensure DNs names aren't invalid or null. • Confirm DNs haven't been altered by CNF or 0ADEL. • Ensure systemFlags are correct for that object. • Call out empty (orphaned) replica sets.
CutoffServers	Tests AD replication to ensure there aren't DCs without working connection objects between partners. Any servers that can't replicate inbound or outbound from any DCs are considered "cut off" using the DsReplicaSyncAll function, which triggers replication on the DCs. Use the `/e` parameter with caution if there are poorly implemented WAN links that are kept clean using schedules. If a server can't be contacted or is unavailable to LDAP on the network, then it provides no error or test results, even if the `/v` parameter is specified. This test uses RPC.
DcPromo	Tests against the server specified in the client DNS settings if the infrastructure meets the necessary requirements to promote your device to a DC. This test uses DNS on the network and checks: • If at least one network adapter has a primary DNS server set. • If there's a disjointed namespace based on the DNS suffix. • That proposed authoritative DNS zone can be contacted. • If dynamic DNS updates are possible for the server's A record. It checks both the setting on the authoritative DNS zone and client registry configuration for DnsUpdateOnAllAdapters and DisableDynamicUpdate. • If an LDAP DClocator record, like **_ldap._tcp.dc._msdcs.<domain>**, is returned when querying for existing forests. The following arguments are required: • `/DnsDomain`:<Active_Directory_Domain_DNS_Name> ◦ Using this parameter requires one of the following arguments: `/ChildDomain`, `/NewForest`, `/NewTree`, or `/ReplicaDC` • If `/NewTree` is specified, the following argument is required: ◦ `/ForestRoot`:<Forest_Root_Domain_DNS_Name>
DFSREvent	This test validates the Distributed File System Replication (DFSR) service's health by checking the DFSR event log warning and error entries from the past 24 hours. This test uses RPC and EventLog Remoting Protocol.
DNS	Tests enterprise-wide DNS health checks using DNS, RPC, and WMI protocols. Not ran by default and must be explicitly requested. See DNS syntax.
FrsEvent	Checks to see if there are errors in the File Replication Service (FRS) event log from the past 24 hours as failing replication of the SysVol share can cause policy problems. This

Test	Description
	test uses RPC and EventLog Remoting Protocol.
Intersite	Checks for failures that would prevent or temporarily hold up intersite replication and predicts how long it would take for the KCC to recover. This test uses DRS functions to check for conditions that would prevent intersite AD replication within a specific site or all sites by: • Locating and connecting to the Intersite Topology Generators (ISTG). • Locating and connecting to the bridgehead servers. • Reporting back any replication failures after triggering a replication. • Validating all DCs within sites with inbound connections to this site are available. • Checking the KCC values for **IntersiteFailuresAllowed** and **MaxFailureTimeForIntersiteLink** overrides within the registry key: **KEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\NTDS\Parameters**. The `/a` or `/e` parameter must be used as not providing a site would allow the test to run but skips actual testing. This test uses RPC over the network to test the replication aspects and asks registry connections to check for NTDS override entries. LDAP is also used to locate connection info.
KccEvent	This test queries the KCC on the DC for errors and warnings generated in the Directory Services event log during the last 15 minutes. The 15-minute threshold is irrespective of the Repl topology update period (secs) registry value on the DC. If firewall rules are causing this test to fail, see KB2512643 that covers enabling those rules to allow the test to succeed. This test uses RPC along with the EventLog Remoting Protocol.
KnowsOfRoleHolders	This test returns the DCs knowledge of the five Flexible Single Master Operation (FSMO) roles but doesn't check all DCs knowledge for consistency. Using the `/e` parameter provides data for comparison. This test uses RPC to return DSListRoles within the Directory Replication Service (DRS) functions.
MachineAccount	Checks whether the machine account has properly registered and that the services are advertised using LDAP and RPC over SMB, including checking: • The DC's computer account exists in AD. • It's within the Domain Controllers OU. • It has the correct UserAccountControl flags for DCs. • The correct ServerReference attributes are set. • The minimum Service Principal Names (SPN) are set. This test is identical to **CheckSecurityError** as they use the same internal test. This test also has two repair options: • `/RecreateMachineAccount` - Recreates a missing DC computer object. This is **not** a recommended fix as it doesn't recreate any child objects of a DC, such as FRS and DFSR subscriptions. The best practice is to use a valid SystemState backup to authoritatively restore the DC's deleted object and child objects. If you use this option, the DC needs to be gracefully demoted and promoted to repair all the missing relationships.

Test	Description
	• `/FixMachineAccount` - Adds the UserAccountControl flags **TRUSTED_FOR_DELEGATION** and **SERVER_TRUST_ACCOUNT** to a DCs computer object. Using this repair option is preferred over trying to set these flags yourself through **ADSIEDIT** or other LDAP editors.
NCSecDesc	Checks permissions on all the naming contexts (such as Schema, Configuration, etc.) on the source DC to validate that replication and connectivity works between DCs. It makes sure that *Enterprise Domain Controllers* and *Administrators* groups have the correct minimum permissions, which is the same test performed within **CheckSecurityError**. This test uses LDAP.
NetLogons	Validates that the user running DCDiag can connect to and read the SYSVOL and NETLOGON shares without any security errors. It also verifies that the *Administrators*, *Authenticated Users*, and *Everyone* group have the **access this computer from the network** privilege on the DC.
ObjectsReplicated	Checks that the Machine Account and Directory System Agent (DSA) objects have replicated. Two objects are validated by default and that they exist in each DC and are up to date on all other DCs: • **CN=NTDS Settings** • **CN=<DC name>** You can use the `/objectdn:dn` parameter with the `/n:nc` parameter to specify an additional object to check. This test is done using RPC with DRS functions.
OutboundSecureChannels	This test doesn't run by default. It checks that secure channels exist from all of the domain controllers in the domain to the domains specified by the `/testdomain` parameter. The `/nositerestriction` parameter prevents DCDiag from limiting the test to the domain controllers in the site.
RegisterInDNS	Tests whether the directory server can register the directory Server Locator DNS records. These records must be present in DNS in order for other computers to locate this directory server for the **<Active_Directory_Domain_DNS_Name>** domain. This also reports if any modifications to the existing DNS infrastructure are required. The parameter `/DnsDomain`:<Active_Directory_Domain_DNS_Name> must be used. This test checks: • The authoritative DNS zone can be contacted. • If at least one network adapter has a primary DNS server set. • If you would have a disjoint namespace based on the DNS suffix. • The proposed authoritative DNS zone can be contacted. • If dynamic DNS updates are possible for the server's A record. It checks settings on the authoritative DNS zone and the client registry configuration of **DnsUpdateOnAllAdapters** and **DisableDynamicUpdate**. • If an LDAP DClocator record, like **_ldap._tcp.dc._msdcs.<domain>**, is returned when querying for existing forests.
Replications	This test checks all AD replication connection objects for all naming contexts on specified DC(s) if:

Test	Description
	• The last replication attempted was successful or returns an error. • That replication is disabled. • Replication latency is more than 12 hours.
RidManager	Checks whether the relative identifier (RID) master is accessible and if: • It contains the proper information. • Can be located and contacted through a DsBind. • Has valid RID pool values. The role holder must be online and accessible for DCs to be able to create security principals (users, computers, and groups) as well as for further DCs to be promoted within a domain. This test uses LDAP and RPC.
Services	This test validates that various AD-dependent services are running, accessible, and set to specific start types. These services start automatically and run in a shared process unless specified otherwise: • DFSR (runs in own process) • DNSCACHE • EVENTSYSTEM • IISADMIN (If using SMTP-based AD replication) • ISMSERV • KDC • NETLOGON • NTDS • NTFRS (runs in own process) • RPCSS • SAMSS • SERVER • SMTPSVC (If using SMTP-based AD replication) • W32TIME (starts automatically or manually) • WORKSTATION These service names are listed in the registry path **HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services**. This test uses RPC and the Service Control Manager remote protocol.
SysVolCheck	This test reads the DCs Netlogon SysVolReady registry key to validate that SYSVOL is ready. The value name has to exist with a value of 1 to pass this test and works with either FRS or DFSR replicated SYSVOLs. It doesn't check if the SYSVOL and NELOGON shares are accessible as this is performed by **CheckSecurityError**. This test uses RPC over SMB.
SystemLog	Validates the System Event Log's health by reading and writing entries from the past 60 minutes for any errors and warnings. This test uses RPC and the Service Control Manager remote protocol.
Topology	Checks that the generated AD replication topology is fully connected for all DSAs. This test isn't performed by default and must be ran explicitly. It checks:

Test	Description
	• If automatic intra-site topology generation is disabled. • If automatic inter-site topology generation is disabled. • For disconnected topologies (missing connection objects), both upstream and downstream from each reference DC. This test uses RPC, LDAP, and DsReplicaSyncAll with the flag **DS_REPSYNCALL_DO_NOT_SYNC**, meaning that it analyzes and validates replication topology without actually replicating changes. This test doesn't validate the availability of replication partners. Having a partner offline won't cause failures in this test. It also doesn't test if the schedule is closed, preventing replication. To see those active replication results, use tests **Replications** or **CutoffServers**.
VerifyEnterpriseReferences	Checks that specified system references are intact for the FRS and replication infrastructure across all objects in the enterprise on each domain controller. This includes the following DC site attributes and objects: • **frsComputerReference**: cn=domain system volume (sysvol share),cn=ntfrs subscriptions,cn=<DC Name>,ou=domain controllers,DC=<domain> • **frsComputerReferenceBL**: cn=<DC Name>,cn=domain system volume (sysvol share),cn=file replication service,cn=system,dc=<domain> • **hasMasterNCs**: cn=ntds settings,cn=<DC Name>,cn=<site>,cn=sites,cn=configuration,dc=<domain> • **msDFSR-ComputerReference**: cn=<DC Name>,cn=topology,cn=domain system volume,cn=dfsr-blobalsettings,cn=system,dc=<domain> • **msDFSR-ComputerReferenceBL**: cn=<DC Name>,ou=domain controllers,dc=<domain> • **nCName**: cn=<partition name>,cn=partitions,cn=configuration,dc=<domain> • **ServerReference**: cn=<DC name>,cn=<site>,cn=sites,cn=configuration,dc=<domain> • **ServerReferenceBL**: cn=<DC Name>,ou=domain controllers,dc=<domain> The two **DFSR** tests are only performed if domain functional level is Windows Server 2008 or higher. This means there will be an expected failure if **DFSR** hasn't been migrated to SYSVOL. This test uses LDAP and only the specified DCs are contacted.
VerifyReferences	Checks that certain system references are intact for the FRS and replication infrastructure. This test verifies computer reference attributes for a single DC, including the following DC site attributes and objects: • **frsComputerReference**: cn=domain system volume (sysvol share),cn=ntfrs subscriptions,cn=<DC Name>,ou=domain controllers,DC=<domain> • **frsComputerReferenceBL**: cn=<DC Name>,cn=domain system volume (sysvol share),cn=file replication service,cn=system,dc=<domain> • **msDFSR-ComputerReference**: cn=<DC Name>,cn=topology,cn=domain system volume,cn=dfsr-blobalsettings,cn=system,dc=<domain> • **msDFSR-ComputerReferenceBL**: cn=<DC Name>,ou=domain controllers,dc=<domain> • **ServerReference**: cn=<DC name>,cn=<site>,cn=sites,cn=configuration,dc=<domain> • **ServerReferenceBL**: cn=<DC Name>,ou=domain controllers,dc=<domain>

Test	Description
	This test uses LDAP and is similar to the **VerifyEnterpriseRefrences** test except that it doesn't check partition cross references or all other DC objects.
VerifyReplicas	Checks that all application directory partitions are fully instantiated on all replica servers. It verifies that the specified server hosts the application partitions specified by its crossref attributes in the partitions container. It operates like **CheckSDRefDom** except that it doesn't show output data and validates hosting. This test uses LDAP.

> ⓘ Note
>
> Connectivity checks on domain controllers that are registered on the network and connect with other resources like DNS, LDAP, and RPC **can't** be skipped.

DNS syntax

```
dcdiag /test:DNS [/DnsBasic | /DnsForwarders | /DnsDelegation | /DnsDynamicUpdate |
/DnsRecordRegistration | /DnsResolveExtName [/DnsInternetName:<InternetName>] |
/DnsAll] [/f:<LogFile>] [/x:<XMLLog.xml>] [/xsl:<XSLFile.xsl> or <XSLTFile.xslt>]
[/s:<DomainController>] [/e] [/v]
```

DNS test uses the following parameters:

Parameter	Description
/test:DNS	Performs the specified DNS test. If no test is specified, defaults to `/DnsAll`.
/DnsBasic	Performs basic DNS tests, including network connectivity, DNS client configuration, service availability, and zone existence.
/DnsForwarders	Performs the `/DnsBasic` tests and also checks the configuration of forwarders.
/DnsDelegation	Performs the `/DnsBasic` tests and also checks for proper delegations.
/DnsDynamicUpdate	Performs `/DnsBasic` tests and also determines if dynamic update is enabled in the Active Directory zone.
/DnsRecordRegistration	Performs the `/DnsBasic` tests, and also checks if the address (A), canonical name (CNAME) and well-known service (SRV) resource records are registered. In addition, creates an inventory report based on the test results.
/DnsResolveExtName [/DnsInternetName:\ `<InternetName>`]	Performs the `/DnsBasic` tests, and also attempts to resolve InternetName. If `/DnsInternetName` isn't specified, attempts to resolve the name

Parameter	Description
	<www.microsoft.com>. If `/DnsInternetName` is specified, attempts to resolve the Internet name supplied by the user.
/DnsAll	Performs all tests, except for the `/DnsResolveExtName` test, and generates a report.
/f:`<LogFile>`	Redirects all output to a log file.
/s:`<DomainController>`	Runs the tests against the domain controller. If this parameter isn't specified, the tests are run against the local domain controller.
/e	Runs all tests specified by `/test:DNS` against all domain controllers in the Active Directory forest.
/v	Verbose. Presents extended information about successful test results in addition to information about errors and warnings. When the `/v` parameter isn't used, provides only error and warning information. Use the `/v` switch when errors or warnings are reported in the summary table.
/x:`<XMLLog.xml>`	Redirects all output to xmllog.xml. This parameter works only with the `/test:DNS` option.
/xsl:`<XSLFile.xsl>` or /xsl:`<XSLTFile.xslt>`	Adds the processing instructions that reference the specified sheet. This parameter only works with the `/test:DNS /x:<XMLLog.xml` option.

> **Note**
>
> Run times for DNS tests can be significant in large enterprises when the `/e` parameter is used. Domain controllers and DNS servers that are offline will increase run times due to long time-out periods for RPC and other protocols.

Examples

Connectivity test

To run a series of connectivity tests on the local domain, run the following command:

PowerShell

```
dcdiag
```

A successful connectivity test output:

Output

```
Directory Server Diagnosis
```

```
Performing initial setup:

   Trying to find home server...

   Home Server = MapleWaffle-WS22

   * Identified AD Forest.
   Done gathering initial info.


Doing initial required tests


   Testing server: Default-First-Site-Name\MAPLEWAFFLE-WS2

      Starting test: Connectivity

         ..... MAPLEWAFFLE-WS2 passed test Connectivity


Doing primary tests

      Testing server: Default-First-Site-Name\MAPLEWAFFLE-WS2

      Starting test: Advertising
         ..... MAPLEWAFFLE-WS2 passed test Advertising

      Starting test: FrsEvent
         ..... MAPLEWAFFLE-WS2 passed test FrsEvent

      Starting test: DFSREvent
         ..... MAPLEWAFFLE-WS2 passed test DFSREvent

      Starting test: SysVolCheck
         ..... MAPLEWAFFLE-WS2 passed test SysVolCheck

      Starting test: KccEvent
         ..... MAPLEWAFFLE-WS2 passed test KccEvent

      Starting test: KnowsOfRoleHolders
         ..... MAPLEWAFFLE-WS2 passed test KnowsOfRoleHolders

      Starting test: MachineAccount
         ..... MAPLEWAFFLE-WS2 passed test MachineAccount

      Starting test: NCSecDesc
         ..... MAPLEWAFFLE-WS2 passed test NCSecDesc

      Starting test: NetLogons
         ..... MAPLEWAFFLE-WS2 passed test NetLogons

      Starting test: ObjectsReplicated
         ..... MAPLEWAFFLE-WS2 passed test ObjectsReplicated

      Starting test: Replications
         ..... MAPLEWAFFLE-WS2 passed test Replications

      Starting test: RidManager
         ..... MAPLEWAFFLE-WS2 passed test RidManager
```

```
      Starting test: Services
         ..... MAPLEWAFFLE-WS2 passed test Services

      Starting test: SystemLog
         ..... MAPLEWAFFLE-WS2 passed test SystemLog

      Starting test: VerifyReferences
         ..... MAPLEWAFFLE-WS2 passed test VerifyReferences

   Running partition tests on : ForestDnsZones

      Starting test: CheckSDRefDom
         ..... ForestDnsZones passed test CheckSDRefDom

      Starting test: CrossRefValidation
         ..... ForestDnsZones passed test CrossRefValidation

   Running partition tests on : DomainDnsZones

      Starting test: CheckSDRefDom
         ..... DomainDnsZones passed test CheckSDRefDom

      Starting test: CrossRefValidation
         ..... DomainDnsZones passed test CrossRefValidation

   Running partition tests on : Schema

      Starting test: CheckSDRefDom
         ..... Schema passed test CheckSDRefDom

      Starting test: CrossRefValidation
         ..... Schema passed test CrossRefValidation

   Running partition tests on : Configuration

      Starting test: CheckSDRefDom
         ..... Configuration passed test
         CheckSDRefDom

      Starting test: CrossRefValidation
         ..... Configuration passed test
         CrossRefValidation

   Running partition tests on : corp

      Starting test: CheckSDRefDom
         ..... corp passed test CheckSDRefDom

      Starting test: CrossRefValidation
         ..... corp passed test CrossRefValidation

   Running enterprise tests on : corp.contoso.com

      Starting test: LocatorCheck
         ..... corp.contoso.com passed test
         LocatorCheck

      Starting test: Intersite
```

```
         ..... corp.contoso.com passed test
         Intersite
```

To run a series of connectivity tests on a specific domain controller, run the following command:

```PowerShell
dcdiag /s:<DomainControllerName>
```

It should generate similar results as the local test if no issues are encountered.

Output to a log file

DCDiag is able to save the output results to a text file by running:

```PowerShell
dcdiag /s:<DomainControllerName> /f:<FileName.txt>
```

If no `<FilePath>` is specified, results are saved to `C:\Users\<UserName>\<FileName.txt>` by default.

To save to a specific location, run:

```PowerShell
dcdiag /s:<DomainControllerName> /f:<DriveLetter>\<FilePath>\<FileName.txt>
```

See also

Command-Line Syntax Key

dcgpofix

Article • 04/22/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Recreates the default Group Policy Objects (GPOs) for a domain. To get to the Group Policy Management Console (GPMC), you must install Group Policy Management as a feature through Server Manager. `Dcgpofix.exe` is included with Windows Server and is located in the `C:\Windows\system32\` folder.

`Dcgpofix.exe` restores only the policy settings that are contained in the Default Domain Policy GPO and Default Domain Controller GPO. `Dcgpofix.exe` doesn't restore other GPOs that administrators create, it's only intended for disaster recovery of the default GPOs.

> **ⓘ Important**
>
> As a best practice, you should configure the Default Domain Policy GPO only to manage the default **Account Policies** settings, Password Policy, Account Lockout Policy, and Kerberos Policy. Additionally, you should configure the Default Domain Controllers Policy GPO only to set user rights and audit policies.

Syntax

```
dcgpofix [/ignoreschema] [/target: {domain | dc | both}] [/?]
```

Parameters

Expand table

Parameter	Description
/ignoreschema	Ignores the version of the Active Directory schema when you run this command. Otherwise, the command only works on the same schema version as the Windows version in which the command was shipped.

Parameter	Description
`/target {domain | dc | both`	Specifies whether to target the Default Domain policy, the Default Domain Controllers policy, or both types of policies.
/?	Displays Help at the command prompt.

Examples

To manage the default **Account Policies** settings, Password Policy, Account Lockout Policy, and Kerberos Policy, while ignoring the Active Directory schema version, type:

```
dcgpofix /ignoreschema /target:domain
```

To configure the Default Domain Controllers Policy GPO only to set user rights and audit policies, while ignoring the Active Directory schema version, type:

```
dcgpofix /ignoreschema /target:dc
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

dcpromo

10/22/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Applies To: Windows Server (All supported versions)

The dcpromo command installs, promotes, demotes, and removes Active Directory Domain Services (AD DS) from a Windows Server.

For examples of how to use **dcpromo**, see Examples.

Syntax

```
dcpromo [/answer[:\<filename>] | /unattend[:\<filename>] | /unattend | /adv]
/uninstallBinaries [/CreateDCAccount | /UseExistingAccount:Attach] /? /?[:{Promotion |
CreateDCAccount | UseExistingAccount | Demotion}]
```

Dcpromo.exe parameters

Expand table

Parameter	Description
/answer[:<filename>]	Specifies an answer file that contains installation parameters and values.
/unattend[:<filename>]	Specifies an answer file that contains installation parameters and values. This command provides the same function as /answer[:<filename>].
/unattend	Specifies an unattended installation in which you provide installation parameters and values at the command line.
/adv	Performs an install from media (IFM) operation.
/UninstallBinaries	Uninstalls AD DS binaries.
/CreateDCAccount	Creates a read-only domain controller (RODC) account. Only a member of the Domain Admins group or the Enterprise Admins group can run this command.
/UseExistingAccount:Attach	Attaches a server to an existing RODC account. A member of the Domain Admins group or a delegated user can run this

Parameter	Description
	command.
/?	Displays Help for Dcpromo parameters.
/?[:{Promotion | CreateDCAccount | UseExistingAccount | Demotion}]	Displays parameters that apply to the dcpromo operation. For example, `dcpromo /?:Promotion displays` all of the parameters that you can use for a promotion operation.

dcpromo promotion operation parameters

The following table shows the parameters available when running `dcpromo.exe` from a command prompt for the unattended installation of a domain controller that runs Windows Server.

For more information about creating a new forest, see Installing a New Windows Server 2008 Forest.

For more information about creating a new domain, see Installing a New Windows Server 2008 Child Domain.

For more information about creating a new domain tree, see Installing a New Windows Server 2008 Domain Tree.

For more information about creating another domain controller for a domain, see Installing an Additional Windows Server 2008 Domain Controller.

Expand table

Parameter and value pair	Description and default
AllowDomainControllerReinstall:{Yes | <No> | NoAndNoPromptEither}	Specifies whether to continue installing this domain controller, even when another domain controller account with the same name is detected. Use Yes only if you're sure that the account isn't currently used by another domain controller. The default is **No**.
AllowDomainReinstall:{Yes | <No> | NoAndNoPromptEither}	Specifies whether an existing domain is recreated. The default is **No**.
ApplicationPartitionsToReplicate:""	Specifies the application directory partitions that dcpromo will replicate. Use the following format:

Parameter and value pair	Description and default
	"**partition1**" "**partition2**" "**partitionN**" Use * to replicate all application directory partitions.
AutoConfigDNS:{Yes | No} This parameter has been renamed to **InstallDNS**.	Specifies whether the DNS Server service should be installed. The default is automatically computed based on the environment.
ChildName:"child_domain_name"	Specifies the single-label Domain Name System (DNS) name of the child domain.
ConfirmGc:{Yes | No}	Specifies whether you want the domain controller to be a global catalog server.
CreateDNSDelegation: { Yes | No}	Indicates whether to create a DNS delegation that references the new DNS server that you're installing along with the domain controller. Valid for Active Directory-integrated DNS only. The default is computed automatically based on the environment.
CriticalReplicationOnly:{Yes | <No>}	Specifies whether the AD DS installation operation performs only critical replication before reboot and then continues, skipping the noncritical (and potentially lengthy) portion of replication. The noncritical replication happens after the installation finishes and the computer reboots. The default is **No**.
DatabasePath:"path_to_database_files"	Specifies the fully qualified, non-Universal Naming Convention (UNC) path to a directory on a fixed disk of the local computer that contains the domain database, for example, **C:\Windows\NTDS**. The default is **%SYSTEMROOT%\NTDS**.
DelegatedAdmin:"name of user or group"	Specifies the name of the user or group that will install and administer the RODC.
DNSDelegationPassword:"password"	Specifies the password for the user name (account credentials) for creating DNS delegation.
DNSDelegationUserName:"user_name"	Specifies the user name (account credentials) for creating DNS delegation.
DNSOnNetwork:{<Yes> | No}	Specifies whether DNS service is available on the network. This parameter is used only when the IP setting of the network adapter for this computer isn't configured with the name of a DNS server for

Parameter and value pair	Description and default
	name resolution. No indicates that a DNS server will be installed on this computer for name resolution. Otherwise, the IP settings of the network adapter must be configured with a DNS server name first. The default is **Yes**.
DomainLevel:{0 | 2 | 3 | 4 | 5 | 6 | 7 | 10}	This entry specifies the domain functional level. This entry is based on the levels that exist in the forest when a new domain is created in an existing forest. Value descriptions are as follows: - 0 = Windows 2000 Server native mode - 2 = Windows Server 2003 - 3 = Windows Server 2008 - 4 = Windows Server 2008 R2 - 5 = Windows Server 2012 - 6 = Windows Server 2012 R2 - 7 = Windows Server 2016 - 10 = Windows Server 2025 The domain functional level can't be lower than the forest functional level, but it can be higher. The default is automatically computed and set to the existing forest functional level or the value that is set for **/ForestLevel**.
DomainNetBiosName:"domain_NetBIOS_name"	Assigns a NetBIOS name to the new domain.
ForestLevel:{ 0 | 2 | 3 | 4 | 5 | 6 | 7 | 10}	This entry specifies the forest functional level when a new domain is created in a new forest as follows: - 0 = Windows 2000 Server native mode - 2 = Windows Server 2003 - 3 = Windows Server 2008 - 4 = Windows Server 2008 R2 - 5 = Windows Server 2012 - 6 = Windows Server 2012 R2 - 7 = Windows Server 2016 - 10 = Windows Server 2025 The default forest functional level in Windows Server 2008 when you create a new forest is Windows 2000 (0). Don't use this parameter when you install a domain controller in an existing forest.
InstallDNS:{Yes | No}	Specifies whether the DNS Server service should be installed. The default is automatically computed

Parameter and value pair	Description and default
	based on the environment. This parameter replaces **AutoConfigDNS**.
LogPath:"path_to_log_files"	Specifies the fully qualified, non-UNC path to a directory on a fixed disk of the local computer that contains the domain log files, for example, **C:\Windows\Logs**. The default is **%SYSTEMROOT%\NTDS**.
NewDomain:{Tree | Child | <Forest>}	Indicates the type of domain that you want to create: a new domain tree in an existing forest, a child of an existing domain, or a new forest. The default is new forest.
NewDomainDNSName:"DNS_name_of_domain"	Specifies the fully qualified domain name (FQDN) for the new domain.
ParentDomainDNSName:"DNS_name_of_domain"	Specifies the FQDN of an existing parent domain. You use this parameter when you install a child domain.
Password:"password"	Specifies the password that corresponds to the user name (account credentials) that is used to install the domain controller. Use this parameter with the **UserName** parameter. Use * to prompt the user to supply a password.
PasswordReplicationAllowed:{"security_principal" | None}	Specifies the names of user accounts, group accounts, and computer accounts whose passwords can be replicated to this RODC. Use None if you want to keep the value empty. By default, only the Allowed RODC Password Replication Group is allowed, and it's originally created empty.
PasswordReplicationDenied:{"security_principal" | None}	Specifies the names of user accounts, group accounts, and computer accounts whose passwords aren't to be replicated to this RODC. Use None if you don't want to deny the replication of credentials of any users or computers. By default, Administrators, Server Operators, Backup Operators, Account Operators, and the Denied RODC Password Replication Group are denied. By default, the Denied RODC Password Replication Group includes: • Cert Publishers • Domain Admins • Enterprise Admins • Enterprise Domain Controllers

Parameter and value pair	Description and default
	• Enterprise Read-Only Domain Controllers • Group Policy Creator Owners • The krbtgt account • Schema Admins
RebootOnCompletion:{<Yes> | No}	Specifies whether to restart the computer upon completion of the command, regardless of success. The default is **Yes**.
RebootOnSuccess:{<Yes> | No | NoAndNoPromptEither}	Specifies whether to restart the computer upon successful completion of the command. The default is **Yes**.
ReplicaDomainDNSName:"DNS_name_of_domain"	Specifies the FQDN of the domain in which you want to install another domain controller.
ReplicaOrNewDomain:{<Replica> | ReadOnlyReplica | Domain}	Specifies whether to install another domain controller (a writable domain controller or an RODC) or to create a new domain. The default is to install another writable domain controller.
ReplicationSourceDC:"DNS_name_of_DC"	Indicates the FQDN of the partner domain controller from which you replicate the domain information.
ReplicationSourcePath:"replication_source_path"	Indicates the location of the installation media that will be used to install a new domain controller.
SafeModeAdminPassword:"password"	Supplies the password for the administrator account when the computer is started in Safe Mode or a variant of Safe Mode, such as Directory Services Restore Mode. The default is an empty password. You must supply a password.
SiteName:"site_name"	Specifies the name of an existing site where you can place the new domain controller. The default value depends on the type of installation. For a new forest, the default is Default-First-Site-Name. For all other installations, the default is the site that is associated with the subnet that includes the IP address of this server. If no such

Parameter and value pair	Description and default
	site exists, the default is the site of the replication source domain controller.
SkipAutoConfigDns	Skips automatic configuration of DNS client settings, forwarders, and root hints. This parameter is in effect only if the DNS Server service is already installed.
Syskey:{<none> | system key}	Specifies the system key for the media from which you replicate the data. The default is **none**.
SysVolPath:"path_to_database_file"	Specifies the fully qualified, non-UNC path to a directory on a fixed disk of the local computer, for example, **C:\Windows\SYSVOL**. The default is **%SYSTEMROOT%\SYSVOL**.
TransferIMRoleIfNecessary:{Yes | <No>}	Specifies whether to transfer the infrastructure master operations master role (also known as flexible single master operations or FSMO) to the domain controller that you're creating. Perform the transfer in case it's currently hosted on a global catalog server and you don't plan to make the domain controller that you're creating a global catalog server. Use **Yes** to transfer the infrastructure master role to the domain controller that you're creating in case the transfer is needed; in this case, make sure to use `/ConfirmGC:No`. Use **No** if you want the infrastructure master role to remain where it currently is. The default is **No**.
UserDomain:"domain_name"	Specifies the domain name for the user name (account credentials) for installing a domain controller. Use this parameter with the **UserName** parameter.
UserName:"user_name"	Specifies the user name (account credentials) for the operation. If no value is specified, the credentials of the current user are used for the operation.

dcpromo /CreateDCAccount operation parameters

The following table shows the parameters that you can use when you create an RODC account.

For more information about creating an RODC account, see Performing a Staged RODC Installation.

Expand table

Parameter and value pair	Description and default
PAutoConfigDNS:{Yes | No} This parameter has been renamed to InstallDNS.	Specifies whether the DNS Server service should be installed. The default is computed automatically based on the environment.
ConfirmGc:{Yes | No}	Specifies whether the domain controller will be a global catalog server.
DCAccountName:"name of the domain controller to create"	Specifies the name of the RODC account that you're creating.
DelegatedAdmin:"name of user or group"	Specifies the name of the user or group that will install and administer the RODC.
InstallDNS:{Yes | No}	Specifies whether the DNS Server service should be installed. The default is computed automatically based on the environment. This parameter replaces **/AutoConfigDNS**.
password:"password"	Specifies the password that corresponds to the user name (account credentials) that is used to install the domain controller. Use this parameter with the **UserName** parameter. Specify * to prompt the user to supply a password.
PasswordReplicationAllowed:{"security_principal" | None}	Specifies the names of user accounts, group accounts, and computer accounts whose passwords can be replicated to this RODC. Use None if you want to keep this value empty. By default, only the Allowed RODC Password Replication Group is allowed, and it's originally created empty.
PasswordReplicationDenied:{"security_principal" | None}	Specifies the names of user accounts, group accounts, and computer accounts whose passwords aren't to be replicated to this RODC. Use None if you don't want to deny the replication of credentials of any users or computers. By default, Administrators, Server Operators, Backup Operators, Account Operators, and the Denied RODC Password Replication Group are denied. By default, the Denied RODC Password Replication Group includes:

Parameter and value pair	Description and default
	• Cert Publishers • Domain Admins • Enterprise Admins • Enterprise Domain Controllers • Enterprise Read-Only Domain Controllers • Group Policy Creator Owners • The krbtgt account • Schema Admins
ReplicaDomainDNSName:"DNS_name_of_domain"	Specifies the FQDN of the domain in which you want to install another domain controller.
ReplicationSourceDC:"DNS_name_of_DC"	Indicates the FQDN of the partner domain controller from which you replicate the domain information.
SiteName:"site_name"	Specifies the name of an existing site where you can place the new domain controller. The default value depends on the type of installation. For a new forest, the default is Default-First-Site-Name. For all other installations, the default is the site that is associated with the subnet that includes the IP address of this server. If no such site exists, the default is the site of the replication source domain controller.
UserDomain:"domain_name"	Specifies the domain name for the user name (account credentials) for the operation. This parameter also helps to specify the forest where you plan to install the domain controller or create an RODC account. If no value is specified, the domain of the computer is used.
UserName:"user_name"	Specifies the user name (account credentials) for the operation. If no value is specified, the credentials of the current user are used for the operation.

dcpromo /UseExistingAccount operation parameters

You can use parameters in the following list when you attach a server to an RODC account.

For more information about attaching a server to an RODC account, see Performing a Staged RODC Installation.

Expand table

Parameter and value pair	Description and default
ApplicationPartitionsToReplicate:""	Specifies the application directory partitions that dcpromo will replicate. Use the following format: "**partition1**" "**partition2**" "**partitionN**" Use * to replicate all application directory partitions.
CriticalReplicationOnly:{Yes | <No>}	Specifies whether the installation performs only critical replication before reboot and then continues, skipping the noncritical (and potentially lengthy) portion of replication. The noncritical replication happens after the role installation finishes and the computer reboots. The default is No.
DatabasePath:"path_to_database_files"	Specifies the fully qualified, non-UNC path to a directory on a fixed disk of the local computer that contains the domain database, for example, **C:\Windows\NTDS**. The default is **%SYSTEMROOT%\NTDS**.
DNSOnNetwork:{<Yes> | No}	Specifies whether the DNS Server service is available on the network. This parameter is used only when the IP setting of the network adapter for this computer isn't configured with the name of a DNS server for name resolution. No indicates that DNS server will be installed on this computer for name resolution. Otherwise, the IP settings of network adapter must be configured with a DNS server name first. The default is **Yes**.
LogPath:"path_to_log_files"	Specifies the fully qualified, non-UNC path to a directory on a fixed disk of the local computer that contains the domain log files, for example, C:\Windows\Logs. The default is **%SYSTEMROOT%\NTDS**.
Password:"password"	Specifies the password that corresponds to the user name (account credentials) that is used to install the domain controller. Use this parameter with the **UserName** parameter. Use * to prompt the user to supply a password.

Parameter and value pair	Description and default
RebootOnCompletion:{<Yes> | No}	Specifies whether to restart the computer upon completion, regardless of success. The default is **Yes**.
RebootOnSuccess:{<Yes> | No | NoAndNoPromptEither}	Specifies whether to restart the computer upon successful completion. The default is **Yes**.
ReplicaDomainDNSName:"DNS_name_of_domain"	Specifies the FQDN of the domain in which you want to install another domain controller.
ReplicationSourceDC:"DNS_name_of_DC"	Indicates the FQDN of the partner domain controller from which you replicate the domain information.
ReplicationSourcePath:"replication_source_path"	Indicates the location of the installation media that will be used to install a new domain controller.
SafeModeAdminPassword:"password"	Supplies the password for the administrator account when the computer is started in Safe Mode or a variant of Safe Mode, such as Directory Service Restore Mode. The default is an empty password. You must supply a password.
SkipAutoConfigDns	Skips automatic configuration of DNS client settings, forwarders, and root hints. This parameter is in effect only if the DNS Server service is already installed.
Syskey:{<none> | system key}	Specifies the system key for the media from which you replicate the data. The default is **none**.
SysVolPath:"path_to_database_file"	Specifies the fully qualified, non-UNC path to a directory on a fixed disk of the local computer, for example, **C:\Windows\SYSVOL**. The default is **%SYSTEMROOT%\SYSVOL**.
UserDomain:"domain_name"	Specifies the domain name for the user name (account credentials) for the operation. This parameter also helps to specify the forest where you plan to install the domain controller or create an RODC account. If no value is specified, the domain of the computer will be used.

Parameter and value pair	Description and default
UserName:"user_name"	Specifies the user name (account credentials) for the operation. If no value is specified, the credentials of the current user are used for the operation.

dcpromo Demotion operation parameters

You can use parameters in the following list when you remove AD DS from a domain controller that runs Windows Server.

For more information about removing a domain controller from a domain, see Removing a Windows Server 2008 Domain Controller from a Domain.

For more information about removing the last domain controller in a domain, see Removing the Last Windows Server 2008 Domain Controller in a Domain.

For more information about removing the last domain controller in a forest, see Removing the Last Windows Server 2008 Domain Controller in a Forest.

For more information about forcing the removal of a domain controller, see Forcing the Removal of a Windows Server 2008 Domain Controller.

Expand table

Parameter and value pair	Description and default
AdministratorPassword:"administrator password"	Specifies a local administrator account password when AD DS is removed from a domain controller. The default is an empty password.
DemoteFSMO:{Yes | <No>}	Indicates that (forced) demotion should continue, even if an operations master role is discovered on domain controller from which AD DS is being removed. The default is **No**.
DNSDelegationPassword {Password | *}	Specifies the password to use for the user name (the account credentials) when you create or remove the DNS delegation. Specify `*` to prompt the user to enter credentials.
DNSDelegationUserName: "user_name"	Specifies the user name to use when you create or remove the DNS delegation. If you don't specify a value, then the account credentials that you specify for the AD DS installation or removal are used to for the DNS delegation.

Parameter and value pair	Description and default
IgnoreIsLastDcInDomainMismatch: {Yes | <No>}	Used with `/IsLastDCInDomain`. This parameter specifies whether Dcpromo.exe ignores any inconsistency that it detects with the value that you specify for `/IsLastDCInDomain`. For example, if you specify `/IsLastDCInDomain:Yes` but dcpromo detects that there's actually another active domain controller in the domain, you can specify `/IgnoreIsLastDcInDomainMismatch:Yes` to have dcpromo continue the removal of AD DS from the domain controller despite the inconsistency that it has detected. Similarly, if you specify `/IsLastDCInDomain:No` but dcpromo can't detect that another domain controller is in the domain, you can specify `/IgnoreIsLastDcInDomainMismatch:Yes` to have dcpromo continue to remove AD DS from the domain controller. The default is **No**. The default causes the wizard to prompt the user to continue and causes the command-line tool to exit with an error.
IgnoreIsLastDNSServerForZone:{Yes | <No>}	Specifies whether to continue the removal of AD DS, even when the domain controller is the last DNS server for one or more of the Active Directory-integrated DNS zones that it hosts. The default is **No**.
IsLastDCInDomain:{Yes | <No>}	Specifies whether the computer from which AD DS is being removed is the last domain controller in the domain. The default is **No**.
Password:"password"	Specifies the password that corresponds to the user name (account credentials) that is used to install the domain controller. Use this parameter with the **UserName** parameter. Specify `*` to prompt the user to supply a password.
RebootOnCompletion:{<Yes> | No}	Specifies whether to restart the computer upon completion, regardless of success. The default is **Yes**.
RebootOnSuccess:{<Yes> | No | NoAndNoPromptEither}	Specifies whether to restart the computer upon successful completion. The default is **Yes**.
RemoveApplicationPartitions:{Yes | <No>}	Specifies whether to remove application partitions during the removal of AD DS from a domain controller. The default is **No**.

Parameter and value pair	Description and default
RemoveDNSDelegation:{<Yes> | No}	Specifies whether to remove DNS delegations that point to this DNS server from the parent DNS zone. The default is **Yes**.
RetainDCMetadata:{Yes | <No>}	Retains domain controller metadata in the domain after AD DS removal to allow a delegated administrator to remove AD DS from an RODC. **The default is No**.
UserDomain:"domain_name"	Specifies the domain name for the user name (account credentials) for the operation. This parameter also helps to specify the forest where you plan to install the domain controller or create an RODC account. If no value is specified, the domain of the computer will be used.
UserName:"user_name"	Specifies the user name (account credentials) for the operation. If no value is specified, the credentials of the current user are used for the operation.

Examples

The following example supplies an answer file named **NewForestInstallation**:

cli

```
dcpromo /answer:NewForestInstallation
```

The following example creates the first domain controller in a new child domain where you expect to install at least some Windows Server 2025 domain controllers:

cli

```
dcpromo /unattend /InstallDns:yes /ParentDomainDNSName:contoso.com
/replicaOrNewDomain:domain /newDomain:child /newDomainDnsName:east.contoso.com
/childName:east /DomainNetbiosName:east /databasePath:"e:\ntds"
/logPath:"e:\ntdslogs" /sysvolpath:"g:\sysvol" /safeModeAdminPassword:FH#3573.cK
/forestLevel:10 /domainLevel:10 /rebootOnCompletion:yes
```

The following example creates another domain controller with the global catalog, and it installs and configures the DNS Server service:

cli

```
dcpromo /unattend /InstallDns:yes /confirmGC:yes /replicaOrNewDomain:replica
/databasePath:"e:\ntds" /logPath:"e:\ntdslogs" /sysvolpath:"g:\sysvol"
/safeModeAdminPassword:M6$,U8Gvx4 /rebootOnCompletion:yes
```

Change History

Expand table

Date	Revision
10/04/2010	Removed the DNSDelegation, DNSDelegationUserName, DNSDelegationPassword, and TransferIMRoleIfNecessary parameters from the /UseExistingAccount operation.

defrag

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Locates and consolidates fragmented files on local volumes to improve system performance.

Membership in the local **Administrators** group, or equivalent, is the minimum required to run this command.

Syntax

```
defrag <volumes> | /c | /e <volumes>    [/h] [/m [n]| [/u] [v]]
defrag <volumes> | /c | /e <volumes> /a [/h] [/m [n]| [/u] [v]]
defrag <volumes> | /c | /e <volumes> /x [/h] [/m [n]| [/u] [v]]
defrag <volume> [<parameters>]
```

Parameters

Expand table

Parameter	Description
`<volume>`	Specifies the drive letter or mount point path of the volume to be defragmented or analyzed.
/a	Performs an analysis on the specified volumes.
/b	Performs boot optimization to increase boot performance.
/c	Performs the operation on all volumes.
/d	Performs traditional defrag (this is the default). On a tiered volume though, traditional defrag is performed only on the Capacity tier.
/e	Performs the operation on all volumes except those specified.
/g	Optimizes the storage tiers on the specified volumes.
/h	Runs the operation at normal priority (default is low).

Parameter	Description
/i [n]	Tier optimization would run for at most n seconds on each volume.
/k	Performs slab consolidation on the specified volumes.
/l	Performs retrim on the specified volumes.
/m [n]	Runs the operation on each volume in parallel in the background. At most, *n* threads optimize the storage tiers in parallel.
/o	Performs the proper optimization for each media type.
/t	Tracks an operation already in progress on the specified volume.
/u	Prints the progress of the operation on the screen.
/v	Prints verbose output containing the fragmentation statistics.
/x	Performs free space consolidation on the specified volumes.
/?	Displays the help information.

Remarks

- You can't defragment specific file system volumes or drives, including:
  - Volumes locked by the file system.
  - Volumes the file system marked as dirty, indicating possible corruption. You must run `chkdsk` before you can defragment this volume or drive. You can determine if a volume is dirty by using the `fsutil dirty` command.
  - Network drives.
  - CD-ROMs.
  - File system volumes that aren't **NTFS**, **ReFS**, **Fat** or **Fat32**.
- To perform this procedure, you must be a member of the Administrators group on the local computer, or you must have been delegated the appropriate authority. If the computer is joined to a domain, members of the Domain Admins group might be able to perform this procedure. As a security best practice, consider using **Run As** to perform this procedure.
- A volume must have at least 15% free space for **defrag** to completely and adequately defragment it. **defrag** uses this space as a sorting area for file fragments. If a volume has less than 15% free space, **defrag** will only partially

defragment it. To increase the free space on a volume, delete unneeded files or move them to another disk.

- While **defrag** is analyzing and defragmenting a volume, it displays a blinking cursor. When **defrag** is finished analyzing and defragmenting the volume, it displays the analysis report, the defragmentation report, or both reports, and then exits to the command prompt.
- By default, **defrag** displays a summary of both the analysis and defragmentation reports if you don't specify the **/a** or **/v** parameters.
- You can send the reports to a text file by typing *>FileName.txt*, where *FileName.txt* is a file name you specify. For example: `defrag volume /v > FileName.txt`
- To interrupt the defragmentation process, at the command line, press **CTRL+C**.
- Running the **defrag** command and Disk defragmenter are mutually exclusive. If you're using Disk defragmenter to defragment a volume and you run the **defrag** command at a command-line, the **defrag** command fails. Conversely, if you run the **defrag** command and open Disk defragmenter, the defragmentation options in Disk defragmenter are unavailable.

Examples

To defragment the volume on drive C while providing progress and verbose output, type:

```
defrag c: /u /v
```

To defragment the volumes on drives C and D in parallel in the background, type:

```
defrag c: d: /m
```

To perform a fragmentation analysis of a volume mounted on drive C and provide progress, type:

```
defrag c: mountpoint /a /u
```

To defragment all volumes with normal priority and provide verbose output, type:

```
defrag /c /h /v
```

Scheduled task

The defragmentation process runs scheduled task as a maintenance task, which typically runs every week. As an Administrator, you can change how often the task runs by using the **Optimize Drives** app.

- When run from the scheduled task, **defrag** uses the below policy guidelines for SSDs:
    - **Traditional optimization processes**. Includes **traditional defragmentation**, for example moving files to make them reasonably contiguous and **retrim**. This is done once per month. However, if both **traditional defragmentation** and **retrim** are skipped, then **analysis** isn't run. Changing the frequency of the scheduled task doesn't affect the once per month cadence for the SSDs.
    - If you manually run **traditional defragmentation** on an SSD, between your normally scheduled runs, the next scheduled task run performs **analysis** and **retrim**, but skips **traditional defragmentation** on that SSD.
    - If you skip **analysis**, you won't see an updated **Last run** time in the **Optimize Drives** app. Because of that, the **Last run** time can be up to a month old.
    - You might find that scheduled task hasn't defragmented all volumes. This is typically because:
        - The process won't wake the computer to run.
        - The computer isn't plugged in. The process won't run if the computer is running on battery power.
        - The computer is resumed from being idle.

Related links

- Command-Line Syntax Key
- chkdsk

- fsutil
- fsutil dirty
- Optimize-Volume PowerShell

Feedback

Was this page helpful? Yes No

del

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Deletes one or more files. This command performs the same actions as the **erase** command.

The **del** command can also run from the Windows Recovery Console, using different parameters. For more information, see Windows Recovery Environment (WinRE).

> ⚠ **Warning**
>
> If you use **del** to delete a file from your disk, you can't retrieve it.

Syntax

```
del [/p] [/f] [/s] [/q] [/a[:]<attributes>] <names>
erase [/p] [/f] [/s] [/q] [/a[:]<attributes>] <names>
```

Parameters

Expand table

Parameter	Description
`<names>`	Specifies a list of one or more files or directories. Wildcards may be used to delete multiple files. If a directory is specified, all files within the directory will be deleted.
/p	Prompts for confirmation before deleting the specified file.
/f	Forces deletion of read-only files.
/s	Deletes specified files from the current directory and all subdirectories. Displays the names of the files as they are being deleted.
/q	Specifies quiet mode. You are not prompted for delete confirmation.

Parameter	Description
/a[:] `<attributes>`	Deletes files based on the following file attributes: • **r** Read-only files • **h** Hidden files • **i** Not content indexed files • **s** System files • **a** Files ready for archiving • **l** Reparse points • **-** Used as a prefix meaning 'not' .
/?	Displays help at the command prompt.

Remarks

- If you use the `del /p` command, you'll see the following message:

  `FileName, Delete (Y/N)?`

  To confirm the deletion, press **Y**. To cancel the deletion and to display the next file name (if you specified a group of files), press **N**. To stop the **del** command, press CTRL+C.

- If you disable command extension, the **/s** parameter will display the names of any files that weren't found ,instead of displaying the names of files that are being deleted.

- If you specify specific folders in the `<names>` parameter, all of the included files will also be deleted. For example, if you want to delete all of the files in the *\work* folder, type:

  ```
  del \work
  ```

- You can use wildcards (***** and **?**) to delete more than one file at a time. However, to avoid deleting files unintentionally, you should use wildcards cautiously. For example, if you type the following command:

  ```
  del *.*
  ```

The **del** command displays the following prompt:

```
Are you sure (Y/N)?
```

To delete all of the files in the current directory, press **Y** and then press ENTER. To cancel the deletion, press **N** and then press ENTER.

> **Note**
>
> Before you use wildcard characters with the **del** command, use the same wildcard characters with the **dir** command to list all the files that will be deleted.

Examples

To delete all the files in a folder named Test on drive C, type either of the following:

```
del c:\test
del c:\test\*.*
```

To delete all the files in a folder where the folder has a space in its name, the full path needs to be wrapped in double quotes. Type either of the following:

```
del "c:\test folder\"
del "c:\test folder\*.*"
```

To delete all files with the .bat file name extension from the current directory, type:

```
del *.bat
```

To delete all read-only files in the current directory, type:

```
del /a:r *.*
```

Related links

- Command-Line Syntax Key
- Windows Recovery Environment (WinRE)

Feedback

Was this page helpful? Yes No

dfsdiag

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Provides diagnostic information for DFS Namespaces.

Syntax

```
dfsdiag /testdcs [/domain:<domain name>]
dfsdiag /testsites </machine:<server name>| /DFSPath:<namespace root or DFS
folder> [/recurse]> [/full]
dfsdiag /testdfsconfig /DFSRoot:<namespace>
dfsdiag /testdfsintegrity /DFSRoot:<DFS root path> [/recurse] [/full]
dfsdiag /testreferral /DFSpath:<DFS path to get referrals> [/full]
```

Parameters

Expand table

Parameter	Description
dfsdiag testdcs	Checks domain controller configuration.
dfsdiag testsites	Checks site associations.
dfsdiag testdfsconfig	Checks DFS Namespace configuration.
dfsdiag testdfsintegrity	Checks DFS Namespace integrity.
dfsdiag testreferral	Checks referral responses.
/?	Displays help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

dfsdiag testdcs

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Checks the configuration of domain controllers by performing the following tests on each domain controller in the specified domain:

- Verifies that the Distributed File System (DFS) Namespace service is running and that its startup type is set to **Automatic**.
- Checks for the support of site-costed referrals for NETLOGON and SYSvol.
- Verifies the consistency of the site association by hostname and IP address.

Syntax

```
dfsdiag /testdcs [/domain:<domain_name>]
```

Parameters

Expand table

Parameter	Description
/domain: `<domain_name>`	Name of the domain to check. This parameter is optional. The default value is the local domain to which the local host is joined.

Examples

To verify the configuration of domain controllers in the *contoso.com* domain, type:

```
dfsdiag /testdcs /domain:contoso.com
```

Related links

- Command-Line Syntax Key
- dfsdiag command

Feedback

Was this page helpful? Yes No

dfsdiag testdfsconfig

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Checks the configuration of a Distributed File System (DFS) namespace by performing the following actions:

- Verifies that the DFS Namespace service is running and that its startup type is set to **Automatic** on all namespace servers.
- Verifies that the DFS registry configuration is consistent among namespace servers.
- Validates the following dependencies on clustered namespace servers:
  - Namespace root resource dependency on network name resource.
  - Network name resource dependency on IP address resource.
  - Namespace root resource dependency on physical disk resource.

Syntax

```
dfsdiag /testdfsconfig /DFSroot:<namespace>
```

Parameters

Expand table

Parameter	Description
/DFSroot:`<namespace>`	The namespace (DFS root) to diagnose.

Examples

To verify the configuration of Distributed File System (DFS) namespaces in *contoso.com\MyNamespace*, type:

```
dfsdiag /testdfsconfig /DFSroot:\\contoso.com\MyNamespace
```

Related links

- Command-Line Syntax Key
- dfsdiag command

Feedback

Was this page helpful? Yes No

dfsdiag testdfsintegrity

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Checks the integrity of the Distributed File System (DFS) namespace by performing the following tests:

- Checks for DFS metadata corruption or inconsistencies between domain controllers.
- Validates the configuration of access-based enumeration to ensure that it is consistent between DFS metadata and the namespace server share.
- Detects overlapping DFS folders (links), duplicate folders, and folders with overlapping folder targets.

Syntax

```
dfsdiag /testdfsintegrity /DFSroot: <DFS root path> [/recurse] [/full]
```

Parameters

Expand table

Parameter	Description
/DFSroot: `<DFS root path>`	The DFS namespace to diagnose.
/recurse	Performs the testing, including any namespace interlinks.
/full	Verifies the consistency of the share and NTFS ACLs, along with the client side configuration on all folder targets. It also verifies that the online property is set.

Examples

To verify the integrity and consistency of the Distributed File System (DFS) namespaces in *contoso.com\MyNamespace*, including any interlinks, type:

```
dfsdiag /testdfsintegrity /DFSRoot:\contoso.com\MyNamespace /recurse /full
```

Related links

- Command-Line Syntax Key
- dfsdiag command

Feedback

Was this page helpful? Yes No

dfsdiag testreferral

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Checks Distributed File System (DFS) referrals by performing the following tests:

- If you use the **DFSpath*** parameter without arguments, the command validates that the referral list includes all trusted domains.
- If you specify a domain, the command performs a health check of domain controllers (`dfsdiag /testdcs`) and tests the site associations and domain cache of the local host.
- If you specify a domain and \SYSvol or \NETLOGON, the command performs the same domain controller health checks, along with checking that the time **To Live (TTL)** of SYSvol or NETLOGON referrals matches the default value of 900 seconds.
- If you specify a namespace root, the command performs the same domain controller health checks, along with performing a DFS configuration check (`dfsdiag /testdfsconfig`) and a namespace integrity check (`dfsdiag /testdfsintegrity`).
- If you specify a DFS folder (link), the command performs the same namespace root health checks, along with validating the site configuration for folder targets (dfsdiag /testsites) and validating the site association of the local host.

Syntax

```
dfsdiag /testreferral /DFSpath:<DFS path to get referrals> [/full]
```

Parameters

Expand table

Parameter	Description
/DFSpath:`<path to get referrals>`	Can be one of the following: • **Blank:** Tests only trusted domains.

Parameter	Description
	• `\\Domain:` Tests only domain controller referrals. • `\\Domain\SYSvol:` Tests only SYSvol referrals. • `\\Domain\NETLOGON:` Tests only NETLOGON referrals. • `\\<domain or server>\<namespace root>:` Tests only namespace root referrals. • `\\<domain or server>\<namespace root>\<DFS folder>:` Tests only the DFS folder (link) referrals.
/full	Applies only to Domain and Root referrals. Verifies the consistency of site association information between the registry and active directory Domain Services (AD DS).

Examples

To check the Distributed File System (DFS) referrals in *contoso.com\MyNamespace*, type:

```
dfsdiag /testreferral /DFSpath:\\contoso.com\MyNamespace
```

To check the Distributed File System (DFS) referrals in all trusted domains, type:

```
dfsdiag /testreferral /DFSpath:
```

Related links

- Command-Line Syntax Key
- dfsdiag command

Feedback

Was this page helpful? Yes No

dfsdiag testsites

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Checks the configuration of active directory Domain Services (AD DS) sites by verifying that servers that act as namespace servers or folder (link) targets have the same site associations on all domain controllers.

Syntax

```
dfsdiag /testsites </machine:<server name>| /DFSpath:<namespace root or DFS folder> [/recurse]> [/full]
```

Parameters

Expand table

Parameter	Description
`/machine:<server name>`	The name of the server on which to verify the site association.
`/DFSpath:<namespace root or DFS folder>`	The namespace root or Distributed File System (DFS) folder (link) with targets for which to verify the site association.
/recurse	Enumerates and verifies the site associations for all folder targets under the specified namespace root.
/full	Verifies that AD DS and the registry of the server contain the same site association information.

Examples

To check the site associations on *machine\MyServer*, type:

```
dfsdiag /testsites /machine:MyServer
```

To check a Distributed File System (DFS) folder to verify the site association, along with verifying that AD DS and the registry of the server contain the same site association information, type:

```
dfsdiag /TestSites /DFSpath:\\contoso.com\namespace1\folder1 /full
```

To check a namespace root to verify the site association, along with enumerating and verifying the site associations for all folder targets under the specified namespace root, and verifying that AD DS and the registry of the server contain the same site association information, type:

```
dfsdiag /testsites /DFSpath:\\contoso.com\namespace2 /recurse /full
```

Related links

- Command-Line Syntax Key
- dfsdiag command

Feedback

Was this page helpful? Yes No

dfsrmig

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

The migration tool for the DFS Replication service, dfsrmig.exe, is installed with the DFS Replication service. This tool migrates SYSvol replication from File Replication Service (FRS) to Distributed File System (DFS) Replication. It also provides information about the progress of the migration and modifies Active Directory Domain Services (AD DS) objects to support the migration.

Syntax

```
dfsrmig [/setglobalstate <state> | /getglobalstate | /getmigrationstate |
/createglobalobjects |
/deleterontfrsmember [<read_only_domain_controller_name>] |
/deleterodfsrmember [<read_only_domain_controller_name>] | /?]
```

Parameters

Expand table

Parameter	Description
`/setglobalstate <state>`	Sets the domain's global migration state to one that corresponds to the value specified by *state*. You can only set the global migration state to a stable state. The *state* values include: • **0** - Start state • **1** - Prepared state • **2** - Redirected state • **3** - Eliminated state
/getglobalstate	Retrieves the current global migration state for the domain from the local copy of the AD DS database, when run on the PDC emulator. Use this option to confirm that you set the correct global migration state.

Parameter	Description
	Important: You should only run this command on the PDC emulator.
/getmigrationstate	Retrieves the current local migration state for all domain controllers in the domain and determines whether those local states match the current global migration state. Use this option to determine if all domain controllers have reached the global migration state.
/createglobalobjects	Creates the global objects and settings in AD DS used by DFS Replication uses. The only situations where you should use this option to manually create objects and settings, are: • **A new read-only domain controller is promoted during migration**. If a new read-only domain controller is promoted in the domain after moving into the **Prepared** state, but before migration to the **Eliminated** state, then the objects that correspond to the new domain controller aren't created, causing replication and the migration to fail. • **Global settings for the DFS Replication service are missing or were deleted**. If these settings are missing for a domain controller, migration from the **Start** state to the **Prepared** state will stall at the **Preparing** transition state. **Note:** Because the global AD DS settings for the DFS Replication service for a read-only domain controller are created on the PDC emulator, these settings need to replicate to the read-only domain controller from the PDC emulator before the DFS Replication service on the read-only domain controller can use these settings. Because of Active Directory replication latencies, this replication can take some time to occur.
`/deleterontfrsmember [<read_only_domain_controller_name>]`	Deletes the global AD DS settings for FRS replication that correspond to the specified read-only domain controller, or deletes the global AD DS settings for FRS replication for all read-only domain controllers if no value is specified for `<read_only_domain_controller_name>`. You shouldn't need to use this option during a normal migration process, because the DFS Replication service automatically deletes these AD DS settings during the migration from the **Redirected** state to the **Eliminated** state. Use this option to manually delete the AD DS

Parameter	Description
	settings only when the automatic deletion fails on a read-only domain controller and stalls the read-only domain controller for a long time during the migration from the **Redirected** state to the **Eliminated** state.
`/deleterodfsrmember [<read_only_domain_controller_name>]`	Deletes the global AD DS settings for DFS Replication that correspond to the specified read-only domain controller, or deletes the global AD DS settings for DFS Replication for all read-only domain controllers if no value is specified for `<read_only_domain_controller_name>`. Use this option to manually delete the AD DS settings only when the automatic deletion fails on a read-only domain controller and stalls the read-only domain controller for a long time when rolling back the migration from the Prepared state to the start state.
/?	Displays help at the command prompt.

Remarks

- Use the `/setglobalstate <state>` command to set the global migration state in AD DS on the PDC emulator to initiate and control the migration process. If the PDC emulator isn't available, this command fails.
- Migration to the **Eliminated** state is irreversible and rollback isn't possible, so use a value of **3** for *state* only when you are fully committed to using DFS Replication for SYSvol replication.
- Global migration states must be a stable migration state.
- Active Directory replication replicates the global state to other domain controllers in the domain, but because of replication latencies, you can get inconsistencies if you run `dfsrmig /getglobalstate` on a domain controller other than the PDC emulator.
- The output of `dfsrmig /getmigrationstate` indicates whether migration to the current global state is complete, listing the local migration state for any domain controllers that haven't yet reached the current global migration state. The local migration state for domain controllers can also include transition states for domain controllers that have not reached the current global migration state.

- Read-only domain controllers can't delete settings from AD DS, the PDC emulator performs this operation, and the changes eventually replicate to the read-only domain controllers after the applicable latencies for active directory replication.
- The **dfsrmig** command is supported only on domain controllers that run at the Windows Server domain functional level, because SYSvol migration from FRS to DFS Replication is only possible on domain controllers that operate at that level.
- You can run the **dfsrmig** command on any domain controller, but operations that create or manipulate AD DS objects are only allowed on read-write capable domain controllers (not on read-only domain controllers).

Examples

To set the global migration state to Prepared (**1**) and to initiate migration or to rollback from the Prepared state, type:

```
dfsrmig /setglobalstate 1
```

To set the global migration state to Start (**0**) and to initiate rollback to the Start state, type:

```
dfsrmig /setglobalstate 0
```

To display the global migration state, type:

```
dfsrmig /getglobalstate
```

Output from the `dfsrmig /getglobalstate` command:

```
Current DFSR global state: Prepared
Succeeded.
```

To display information about whether the local migration states on all the domain controllers match the global migration state and if there are any local migration states

where the local state doesn't match the global state, type:

```
dfsrmig /GetMigrationState
```

Output from the `dfsrmig /getmigrationstate` command when the local migration states on all of the domain controllers match the global migration state:

```
All Domain Controllers have migrated successfully to Global state
(Prepared).
Migration has reached a consistent state on all Domain Controllers.
Succeeded.
```

Output from the `dfsrmig /getmigrationstate` command when the local migration states on some domain controllers don't match the global migration state.

```
The following Domain Controllers are not in sync with Global state
(Prepared):
Domain Controller (Local Migration State) DC type
=========
CONTOSO-DC2 (start) ReadOnly DC
CONTOSO-DC3 (Preparing) Writable DC
Migration has not yet reached a consistent state on all domain controllers
State information might be stale due to AD latency.
```

To create the global objects and settings that DFS Replication uses in AD DS on domain controllers where those settings were not created automatically during migration or where those settings are missing, type:

```
dfsrmig /createglobalobjects
```

To delete the global AD DS settings for FRS replication for a read-only domain controller named contoso-dc2 if those settings were not deleted automatically by the migration process, type:

```
dfsrmig /deleterontfrsmember contoso-dc2
```

To delete the global AD DS settings for FRS replication for all read-only domain controllers if those settings were not deleted automatically by the migration process, type:

```
dfsrmig /deleterontfrsmember
```

To delete the global AD DS settings for DFS Replication for a read-only domain controller named contoso-dc2 if those settings were not deleted automatically by the migration process, type:

```
dfsrmig /deleterodfsrmember contoso-dc2
```

To delete the global AD DS settings for DFS Replication for all read-only domain controllers if those settings were not deleted automatically by the migration process, type:

```
dfsrmig /deleterodfsrmember
```

To display help at the command prompt:

```
dfsrmig
```

```
dfsrmig /?
```

Related links

- Command-Line Syntax Key
- SYSvol Migration Series: Part 2 dfsrmig.exe: The SYSvol Migration Tool
- Active Directory Domain Services

Feedback

Was this page helpful? Yes No

dfsutil

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The dfsutil command manages DFS Namespaces, servers, and clients.

Functionality available in PowerShell

The DFSN PowerShell module provides equivalent functionality to the following dfsutil parameters.

Expand table

Parameter	Description
root	Displays, creates, removes, imports, exports namespace roots.
link	Displays, creates, removes, or moves folders (links).
target	Displays, create, remove folder target or namespace server.
property	Displays or modifies a folder target or namespace server.
server	Displays or modifies namespace configuration.
domain	Displays all domain-based namespaces in a domain.

Functionality available only in dfsutil

The following functionality is available only as dfsutil parameters:

Expand table

Parameter	Description
client	Displays or modifies client information or registry keys.
diag	Perform diagnostics or view dfsdirs/dfspath.
cache	Displays or flushes the client cache.

For more info about each of these commands, open a command prompt on a server with the DFS Namespaces management tools installed, and then type `dfsutil client /?`, `dfsutil diag /?`, or `dfsutil cache /?`.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

diantz

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Package existing files into a cabinet (.cab) file. This command performs the same actions as the updated makecab command.

Syntax

```
diantz [/v[n]] [/d var=<value> ...] [/l <dir>] <source> [<destination>]
diantz [/v[<n>]] [/d var=<value> ...] /f <directives_file> [...]
```

Parameters

Expand table

Parameter	Description
`<source>`	File to compress.
`<destination>`	File name to give compressed file. If omitted, the last character of the source file name is replaced with an underscore (_) and used as the destination.
/f `<directives_file>`	A file with **diantz** directives (may be repeated).
/d var=`<value>`	Defines variable with specified value.
/l `<dir>`	Location to place destination (default is current directory).
/v[`<n>`]	Set debugging verbosity level (0=none,...,3=full).
/?	Displays help at the command prompt.

Related links

- Command-Line Syntax Key

- Microsoft Cabinet format

Feedback

Was this page helpful? Yes No

dir

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays a list of a directory's files and subdirectories. If used without parameters, this command displays the disk's volume label and serial number, followed by a list of directories and files on the disk (including their names and the date and time each was last modified). For files, this command displays the name extension and the size in bytes. This command also displays the total number of files and directories listed, their cumulative size, and the free space (in bytes) remaining on the disk.

The **dir** command can also run from the Windows Recovery Console, using different parameters. For more information, see Windows Recovery Environment (WinRE).

Syntax

```
dir [<drive>:][<path>][<filename>] [...] [/p] [/q] [/w] [/d] [/a[[:]
<attributes>]][/o[[:]<sortorder>]] [/t[[:]<timefield>]] [/s] [/b] [/l] [/n]
[/x] [/c] [/4] [/r]
```

Parameters

Expand table

Parameter	Description
`[<drive>:][<path>]`	Specifies the drive and directory for which you want to see a listing.
`[<filename>]`	Specifies a particular file or group of files for which you want to see a listing.
/p	Displays one screen of the listing at a time. To see the next screen, press any key.
/q	Displays file ownership information.
/w	Displays the listing in wide format, with as many as five file names or directory names on each line.

Parameter	Description
/d	Displays the listing in the same format as **/w**, but the files are sorted by column.
/a[[:]`<attributes>`]	Displays only the names of those directories and files with your specified attributes. If you don't use this parameter, the command displays the names of all files except hidden and system files. If you use this parameter without specifying any *attributes*, the command displays the names of all files, including hidden and system files. The list of possible *attributes* values are: • **d** - Directories • **h** - Hidden files • **s** - System files • **l** - Reparse points • **r** - Read-only files • **a** - Files ready for archiving • **i** - Not content indexed files You can use any combination of these values, but don't separate your values using spaces. Optionally you can use a colon (:) separator, or you can use a hyphen (-) as a prefix to mean, "not". For example, using the **-s** attribute won't show the system files.
/o[[:]`<sortorder>`]	Sorts the output according to *sortorder*, which can be any combination of the following values: • **n** - Alphabetically by name • **e** - Alphabetically by extension • **g** - Group directories first • **s** - By size, smallest first • **d** - By date/time, oldest first • Use the **-** prefix to reverse the sort order Multiple values are processed in the order in which you list them. Don't separate multiple values with spaces, but you can optionally use a colon (:). If *sortorder* isn't specified, **dir /o** lists the directories alphabetically, followed by the files, which are also sorted alphabetically.
/t[[:]`<timefield>`]	Specifies which time field to display or to use for sorting. The available *timefield* values are: • **c** - Creation • **a** - Last accessed • **w** - Last written
/s	Lists every occurrence of the specified file name within the specified directory and all subdirectories.
/b	Displays a bare list of directories and files, with no additional information. The

Parameter	Description
	/b parameter overrides **/w**.
/l	Displays unsorted directory names and file names, using lowercase.
/n	Displays a long list format with file names on the far right of the screen.
/x	Displays the short names generated for non-8dot3 file names. The display is the same as the display for **/n**, but the short name is inserted before the long name.
/c	Displays the thousand separator in file sizes. This is the default behavior. Use **/-c** to hide separators.
/4	Displays years in four-digit format.
/r	Display alternate data streams of the file.
/?	Displays help at the command prompt.

Remarks

- To use multiple *filename* parameters, separate each file name with a space, comma, or semicolon.
- You can use wildcard characters (***** or **?**), to represent one or more characters of a file name and to display a subset of files or subdirectories.
- You can use the wildcard character, *****, to substitute for any string of characters, for example:
  - `dir *.txt` lists all files in the current directory with extensions that begin with .txt, such as .txt, .txt1, .txt_old.
  - `dir read *.txt` lists all files in the current directory that begin with read and with extensions that begin with .txt, such as .txt, .txt1, or .txt_old.
  - `dir read *.*` lists all files in the current directory that begin with read with any extension.

  The asterisk wildcard always uses short file name mapping, so you might get unexpected results. For example, the following directory contains two files (t.txt2 and t97.txt):

```
C:\test>dir /x
Volume in drive C has no label.
```

```
 Volume Serial Number is B86A-EF32

 Directory of C:\test

 11/30/2004  01:40 PM <DIR>  .
 11/30/2004  01:40 PM <DIR> ..
 11/30/2004  11:05 AM 0 T97B4~1.TXT t.txt2
 11/30/2004  01:16 PM 0 t97.txt
```

You might expect that typing `dir t97\*` would return the file t97.txt. However, typing `dir t97\*` returns both files, because the asterisk wildcard matches the file t.txt2 to t97.txt by using its short name map *T97B4~1.TXT*. Similarly, typing `del t97\*` would delete both files.

- You can use the question mark (?) as a substitute for a single character in a name. For example, typing `dir read???.txt` lists any files in the current directory with the .txt extension that begin with read and are followed by up to three characters. This includes Read.txt, Read1.txt, Read12.txt, Read123.txt, and Readme1.txt, but not Readme12.txt.

- If you use **/a** with more than one value in *attributes*, this command displays the names of only those files with all the specified attributes. For example, if you use **/a** with **r** and **-h** as attributes (by using either `/a:r-h` or `/ar-h`), this command will only display the names of the read-only files that aren't hidden.

- If you specify more than one *sortorder* value, this command sorts the file names by the first criterion, then by the second criterion, and so on. For example, if you use **/o** with the **e** and **-s** parameters for *sortorder* (by using either `/o:e-s` or `/oe-s`), this command sorts the names of directories and files by extension, with the largest first, and then displays the final result. The alphabetic sorting by extension causes file names with no extensions to appear first, then directory names, and then file names with extensions.

- If you use the redirection symbol (`>`) to send this command's output to a file, or if you use a pipe (`|`) to send this command's output to another command, you must use `/a:-d` and **/b** to only list the file names. You can use *filename* with **/b** and **/s** to specify that this command is to search the current directory and its subdirectories for all file names that match *filename*. This command lists only the drive letter, directory name, file name, and file name extension (one path per line), for each file name it finds. Before you use a pipe to send this command's output to another command, you should set the *TEMP* environment variable in your Autoexec.nt file.

Examples

To display all directories one after the other, in alphabetical order, in wide format, and pausing after each screen, make sure that the root directory is the current directory, and then type:

```
dir /s/w/o/p
```

The output lists the root directory, the subdirectories, and the files in the root directory, including extensions. This command also lists the subdirectory names and the file names in each subdirectory in the tree.

To alter the preceding example so that **dir** displays the file names and extensions, but omits the directory names, type:

```
dir /s/w/o/p/a:-d
```

To print a directory listing, type:

```
dir > prn
```

When you specify **prn**, the directory list is sent to the printer that is attached to the LPT1 port. If your printer is attached to a different port, you must replace **prn** with the name of the correct port.

You can also redirect output of the **dir** command to a file by replacing **prn** with a file name. You can also type a path. For example, to direct **dir** output to the file dir.doc in the Records directory, type:

```
dir > \records\dir.doc
```

If dir.doc does not exist, **dir** creates it, unless the **Records** directory does not exist. In that case, the following message appears:

```
```

```
File creation error
```

To display a list of all the file names with the .txt extension in all directories on drive C, type:

```
dir c:\*.txt /w/o/s/p
```

The **dir** command displays, in wide format, an alphabetized list of the matching file names in each directory, and it pauses each time the screen fills until you press any key to continue.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

diskcomp

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Compares the contents of two floppy disks. If used without parameters, **diskcomp** uses the current drive to compare both disks.

Syntax

```
diskcomp [<drive1>: [<drive2>:]]
```

Parameters

Expand table

Parameter	Description
`<drive1>`	Specifies the drive containing one of the floppy disks.
/?	Displays help at the command prompt.

Remarks

- The **diskcomp** command works only with floppy disks. You cannot use **diskcomp** with a hard disk. If you specify a hard disk drive for *drive1* or *drive2*, **diskcomp** displays the following error message:

  ```
  Invalid drive specification
  Specified drive does not exist
  or is nonremovable
  ```

- If all tracks on the two disks being compared are the same (it ignores a disk's volume number), **diskcomp** displays the following message:

```
Compare OK
```

If the tracks aren't the same, **diskcomp** displays a message similar to the following:

```

Compare error on
side 1, track 2
```

When **diskcomp** completes the comparison, it displays the following message:

```

Compare another diskette (Y/N)?
```

If you press **Y**, **diskcomp** prompts you to insert the disk for the next comparison. If you press **N**, **diskcomp** stops the comparison.

- If you omit the *drive2* parameter, **diskcomp** uses the current drive for *drive2*. If you omit both drive parameters, **diskcomp** uses the current drive for both. If the current drive is the same as *drive1*, **diskcomp** prompts you to swap disks as necessary.

- If you specify the same floppy disk drive for *drive1* and *drive2*, **diskcomp** compares them by using one drive and prompts you to insert the disks as necessary. You might have to swap the disks more than once, depending on the capacity of the disks and the amount of available memory.

- **Diskcomp** can't compare a single-sided disk with a double-sided disk, nor a high-density disk with a double-density disk. If the disk in *drive1* isn't of the same type as the disk in *drive2*, **diskcomp** displays the following message:

  ```

  Drive types or diskette types not compatible
  ```

- **Diskcomp** doesn't work on a network drive or on a drive created by the **subst** command. If you attempt to use **diskcomp** with a drive of any of these types, **diskcomp** displays the following error message:

  ```
  ```

```
Invalid drive specification
```

- If you use **diskcomp** with a disk that you made by using **copy**, **diskcomp** might display a message similar to the following:

  ```
  Compare error on
  side 0, track 0
  ```

  This type of error can occur even if the files on the disks are identical. Although **copy** duplicates information, it doesn't necessarily place it in the same location on the destination disk.

- **diskcomp** exit codes:

  Expand table

Exit code	Description
0	Disks are the same
1	Differences were found
3	Hard error occurred
4	Initialization error occurred

  To process exit codes that are returned by **diskcomp**, you can use the *ERRORLEVEL* environment variable on the **if** command line in a batch program.

Examples

If your computer has only one floppy disk drive (for example, drive A), and you want to compare two disks, type:

```
diskcomp a: a:
```

Diskcomp prompts you to insert each disk, as needed.

To illustrates how to process a **diskcomp** exit code in a batch program that uses the *ERRORLEVEL* environment variable on the **if** command line:

```
rem Checkout.bat compares the disks in drive A and B
echo off
diskcomp a: b:
if errorlevel 4 goto ini_error
if errorlevel 3 goto hard_error
if errorlevel 1 goto no_compare
if errorlevel 0 goto compare_ok
:ini_error
echo ERROR: Insufficient memory or command invalid
goto exit
:hard_error
echo ERROR: An irrecoverable error occurred
goto exit
:break
echo You just pressed CTRL+C to stop the comparison
goto exit
:no_compare
echo Disks are not the same
goto exit
:compare_ok
echo The comparison was successful; the disks are the same
goto exit
:exit
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

diskcopy

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Copies the contents of the floppy disk in the source drive to a formatted or unformatted floppy disk in the destination drive. If used without parameters, **diskcopy** uses the current drive for the source disk and the destination disk.

Syntax

```
diskcopy [<drive1>: [<drive2>:]] [/v]
```

Parameters

Expand table

Parameter	Description
`<drive1>`	Specifies the drive that contains the source disk.
/v	Verifies that the information is copied correctly. This option slows down the copying process.
/?	Displays help at the command prompt.

Remarks

- **Diskcopy** works only with removable disks such as floppy disks, which must be the same type. You cannot use **diskcopy** with a hard disk. If you specify a hard disk drive for *drive1* or *drive2*, **diskcopy** displays the following error message:

```
Invalid drive specification
Specified drive does not exist or is nonremovable
```

The **diskcopy** command prompts you to insert the source and destination disks and waits for you to press any key on the keyboard before continuing.

After it copies the disk, **diskcopy** displays the following message:

```
Copy another diskette (Y/N)?
```

If you press **Y**, **diskcopy** prompts you to insert source and destination disks for the next copy operation. To stop the **diskcopy** process, press **N**.

If you're copying to an unformatted floppy disk in *drive2*, **diskcopy** formats the disk with the same number of sides and sectors per track as are on the disk in *drive1*. **Diskcopy** displays the following message while it formats the disk and copies the files:

```
Formatting while copying
```

- If the source disk has a volume serial number, **diskcopy** creates a new volume serial number for the destination disk and displays the number when the copy operation is complete.
- If you omit the *drive2* parameter, **diskcopy** uses the current drive as the destination drive. If you omit both drive parameters, **diskcopy** uses the current drive for both. If the current drive is the same as *drive1*, **diskcopy** prompts you to swap disks as necessary.
- Run **diskcopy** from a drive other than the floppy disk drive, for example the C drive. If floppy disk *drive1* and floppy disk *drive2* are the same, **diskcopy** prompts you to switch disks. If the disks contain more information than the available memory can hold, **diskcopy** cannot read all of the information at once. **Diskcopy** reads from the source disk, writes to the destination disk, and prompts you to insert the source disk again. This process continues until you have copied the entire disk.
- Fragmentation is the presence of small areas of unused disk space between existing files on a disk. A fragmented source disk can slow down the process of finding, reading, or writing files.

  Because **diskcopy** makes an exact copy of the source disk on the destination disk, any fragmentation on the source disk is transferred to the destination disk. To

avoid transferring fragmentation from one disk to another, use the copy command or the xcopy command to copy your disk. Because **copy** and **xcopy** copy files sequentially, the new disk is not fragmented.

> **Note**
>
> You cannot use **xcopy** to copy a startup disk.

- **diskcopy** exit codes:

Expand table

Exit code	Description
0	Copy operation was successful
1	Nonfatal Read/Write error occurred
3	Fatal hard error occurred
4	Initialization error occurred

To process the exit codes that are returned by **diskcomp**, you can use the *ERRORLEVEL* environment variable on the **if** command line in a batch program.

Examples

To copy the disk in drive B to the disk in drive A, type:

```
diskcopy b: a:
```

To use floppy disk drive A to copy one floppy disk to another, first switch to the C drive and then type:

```
diskcopy a: a:
```

Related links

- Command-Line Syntax Key
- xcopy command
- copy command

Feedback

Was this page helpful? Yes No

diskpart

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

> Applies to: Windows Server 2022, Windows 10, Windows 8.1, Windows 8, Windows 7, Windows Server 2019, Windows Server 2016, Windows Server 2012 R2, Windows Server 2012, and Windows Server 2008 R2, Windows Server 2008

The diskpart command interpreter helps you manage your computer's drives (disks, partitions, volumes, or virtual hard disks).

Before you can use **diskpart** commands, you must first list, and then select an object to give it focus. After an object has focus, any diskpart commands that you type will act on that object.

Determine focus

When you select an object, the focus remains on that object until you select a different object. For example, if the focus is set on disk 0 and you select volume 8 on disk 2, the focus shifts from disk 0 to disk 2, volume 8.

Some commands automatically change the focus. For example, when you create a new partition, the focus automatically switches to the new partition.

You can only give focus to a partition on the selected disk. After a partition has focus, the related volume (if any) also has focus. After a volume has focus, the related disk and partition also have focus if the volume maps to a single specific partition. If this isn't the case, focus on the disk and partition are lost.

Syntax

To start the diskpart command interpreter, at the command prompt type:

Windows Command Prompt

```
diskpart <parameter>
```

> **Important**
>
> You must be in your local **Administrators** group, or a group with similar permissions, to run diskpart.

Parameters

You can run the following commands from the Diskpart command interpreter:

Expand table

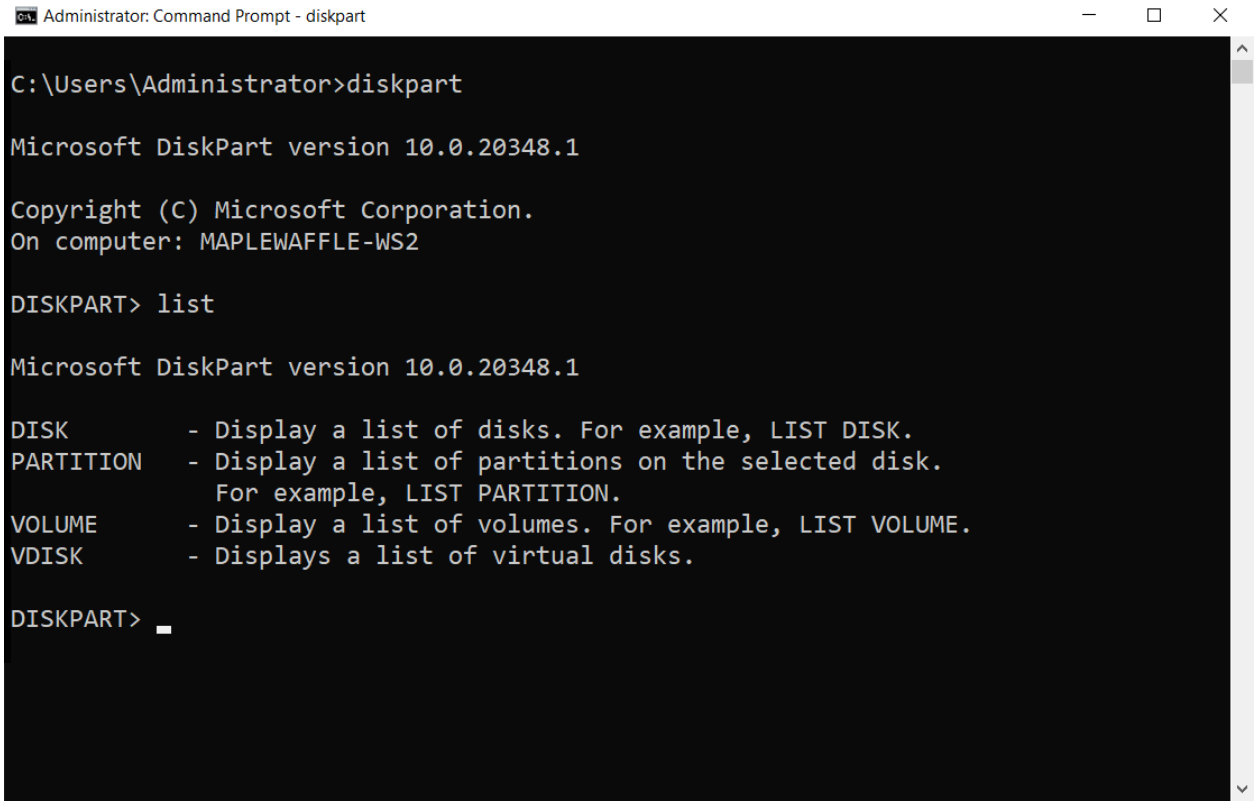
Command	Description
active	Marks the disk's partition with focus, as active.
add	Mirrors the simple volume with focus to the specified disk.
assign	Assigns a drive letter or mount point to the volume with focus.
attach vdisk	Attaches (sometimes called mounts or surfaces) a virtual hard disk (VHD) so that it appears on the host computer as a local hard disk drive.
attributes	Displays, sets, or clears the attributes of a disk or volume.
automount	Enables or disables the automount feature.
break	Breaks the mirrored volume with focus into two simple volumes.
clean	Removes any and all partition or volume formatting from the disk with focus.
compact vdisk	Reduces the physical size of a dynamically expanding virtual hard disk (VHD) file.
convert	Converts file allocation table (FAT) and FAT32 volumes to the NTFS file system, leaving existing files and directories intact.
create	Creates a partition on a disk, a volume on one or more disks, or a virtual hard disk (VHD).
delete	Deletes a partition or a volume.
detach vdisk	Stops the selected virtual hard disk (VHD) from appearing as a local hard disk drive on the host computer.
detail	Displays information about the selected disk, partition, volume, or virtual hard disk (VHD).
exit	Exits the diskpart command interpreter.

Command	Description
expand vdisk	Expands a virtual hard disk (VHD) to the size that you specify.
extend	Extends the volume or partition with focus, along with its file system, into free (unallocated) space on a disk.
filesystems	Displays information about the current file system of the volume with focus and lists the file systems that are supported for formatting the volume.
format	Formats a disk to accept files.
gpt	Assigns the gpt attribute(s) to the partition with focus on basic GUID partition table (gpt) disks.
help	Displays a list of the available commands or detailed help information on a specified command.
import	Imports a foreign disk group into the disk group of the local computer.
inactive	Marks the system partition or boot partition with focus as inactive on basic master boot record (MBR) disks.
list	Displays a list of disks, of partitions in a disk, of volumes in a disk, or of virtual hard disks (VHDs).
merge vdisk	Merges a differencing virtual hard disk (VHD) with its corresponding parent VHD.
offline	Takes an online disk or volume to the offline state.
online	Takes an offline disk or volume to the online state.
recover	Refreshes the state of all disks in a disk group, attempt to recover disks in an invalid disk group, and resynchronizes mirrored volumes and RAID-5 volumes that have stale data.
rem	Provides a way to add comments to a script.
remove	Removes a drive letter or mount point from a volume.
repair	Repairs the RAID-5 volume with focus by replacing the failed disk region with the specified dynamic disk.
rescan	Locates new disks that may have been added to the computer.
retain	Prepares an existing dynamic simple volume to be used as a boot or system volume.
san	Displays or sets the storage area network (san) policy for the operating system.
select	Shifts the focus to a disk, partition, volume, or virtual hard disk (VHD).

Command	Description
set id	Changes the partition type field for the partition with focus.
shrink	Reduces the size of the selected volume by the amount you specify.
uniqueid	Displays or sets the GUID partition table (GPT) identifier or master boot record (MBR) signature for the disk with focus.

Listing available objects

You can view a list of options associated to each command by running the main command followed by what is available to that specific command. Running **list** by itself will display the four parameters below:



> **Note**
>
> After you run the **list** command, an asterisk (*) appears next to the object of focus.

Examples

To see available disk(s), run **list disk**:

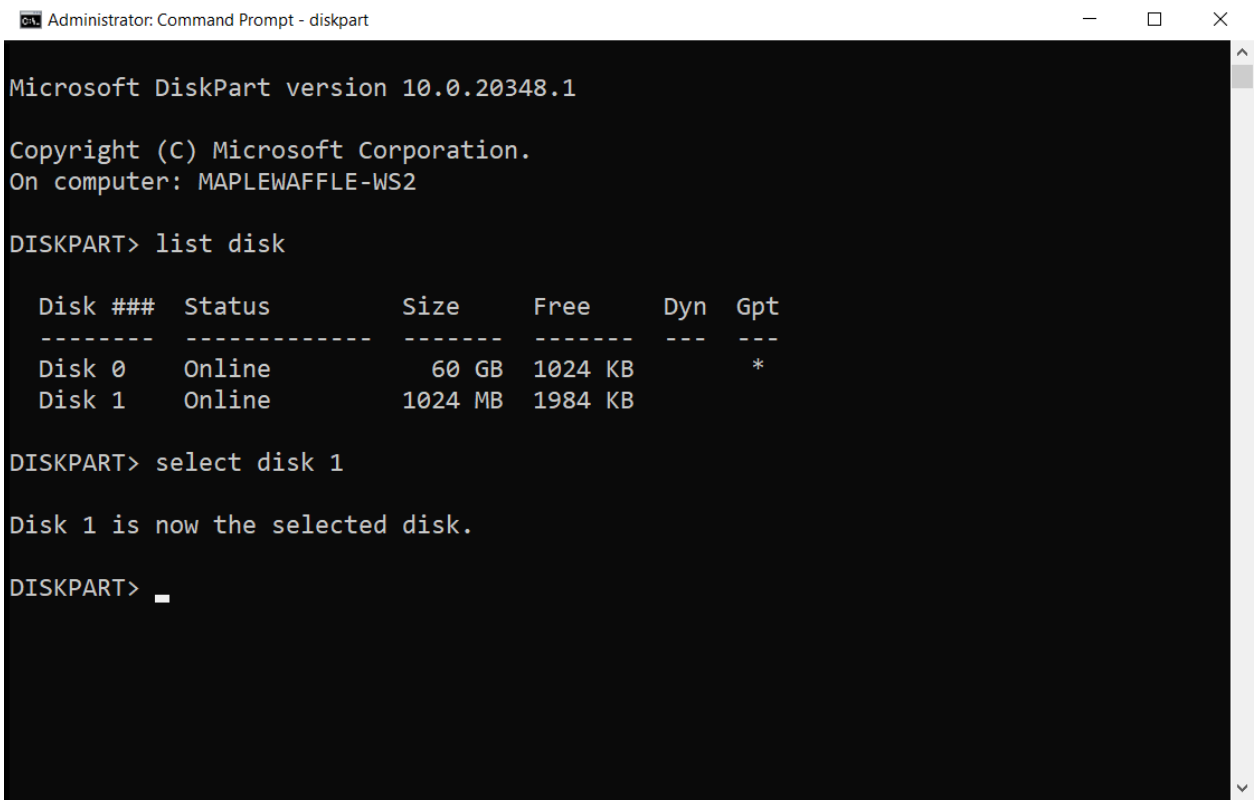
```
Windows Command Prompt
```

```
list disk
```

To select a disk, run **select disk** followed by the disk number. For example:

Windows Command Prompt

```
select disk 1
```



Before disk 1 can be utilized, a partition will need to be created by running **create partition primary**:

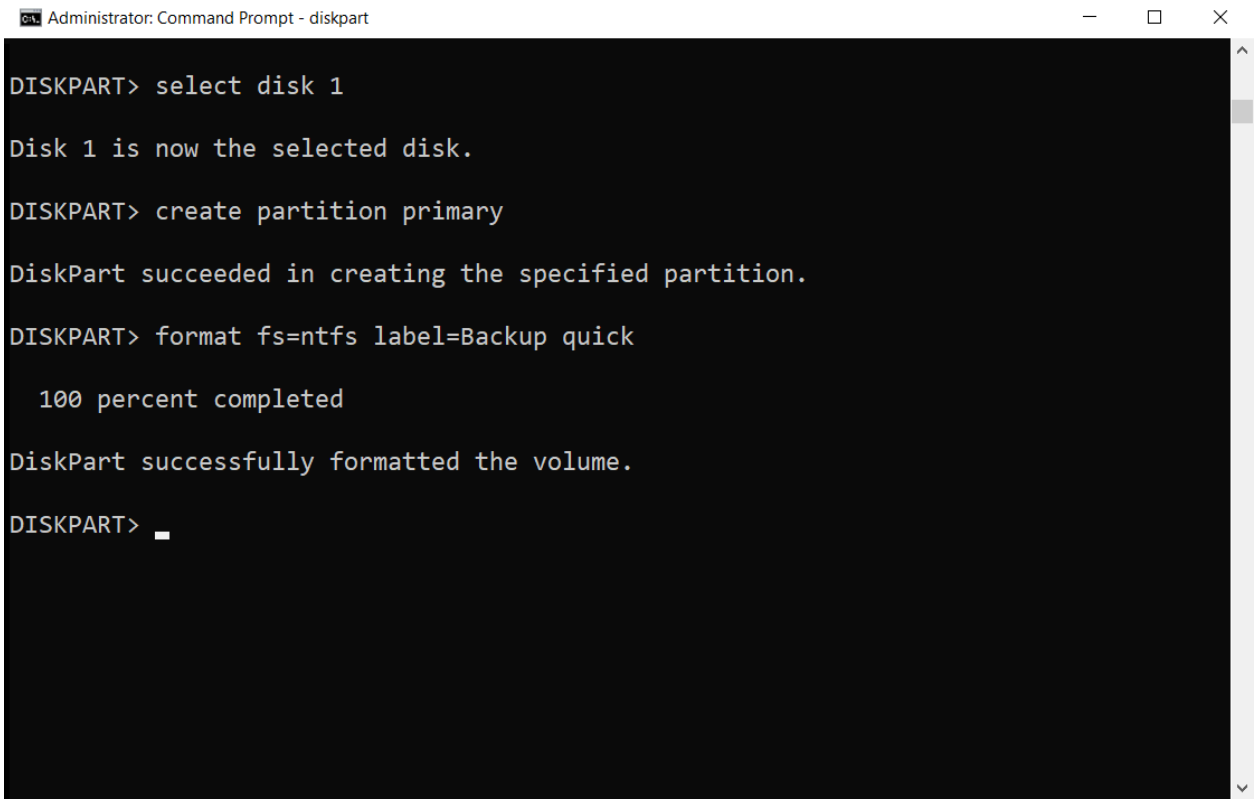
Windows Command Prompt

```
create partition primary
```

Lastly, we can perform a quick format of disk 1 to NTFS with the label "Backup" by running **format fs=ntfs label=Backup quick** as seen below:

Windows Command Prompt

```
format fs=ntfs label=Backup quick
```



Related links

- Command-Line Syntax Key
- Disk management overview
- Storage Cmdlets in Windows PowerShell

Feedback

Was this page helpful? Yes No

diskpart scripts and examples

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Use `diskpart /s` to run scripts that automate disk-related tasks, such as creating volumes or converting disks to dynamic disks. Scripting these tasks is useful if you deploy Windows by using unattended Setup or the Sysprep tool, which do not support creating volumes other than the boot volume.

To create a diskpart script, create a text file that contains the Diskpart commands that you want to run, with one command per line, and no empty lines. You can start a line with `rem` to make the line a comment. For example, here's a script that wipes a disk and then creates a 300 MB partition for the Windows Recovery Environment:

```
select disk 0
clean
convert gpt
create partition primary size=300
format quick fs=ntfs label=Windows RE tools
assign letter=T
```

Examples

- To run a diskpart script, at the command prompt, type the following command, where *scriptname* is the name of the text file that contains your script:

```
diskpart /s scriptname.txt
```

- To redirect diskpart's scripting output to a file, type the following command, where *logfile* is the name of the text file where diskpart writes its output:

```
diskpart /s scriptname.txt > logfile.txt
```

Remarks

- When using the **diskpart** command as a part of a script, we recommend that you complete all of the diskpart operations together as part of a single diskpart script. You can run consecutive diskpart scripts, but you must allow at least 15 seconds between each script for a complete shutdown of the previous execution before running the **diskpart** command again in successive scripts. Otherwise, the successive scripts might fail. You can add a pause between consecutive diskpart scripts by adding the `timeout /t 15` command to your batch file along with your diskpart scripts.
- When diskpart starts, the diskpart version and computer name display at the command prompt. By default, if diskpart encounters an error while attempting to perform a scripted task, diskpart stops processing the script and displays an error code (unless you specified the **noerr** parameter). However, diskpart always returns errors when it encounters syntax errors, regardless of whether you used the **noerr** parameter. The **noerr** parameter enables you to perform useful tasks such as using a single script to delete all partitions on all disks regardless of the total number of disks.

Related links

- Command-Line Syntax Key
- Sample: Configure UEFI/GPT-Based Hard Drive Partitions by Using Windows PE and DiskPart
- Sample: Configure BIOS/MBR-Based Hard Disk Partitions by Using Windows PE and DiskPart
- Storage Cmdlets in Windows PowerShell

Feedback

Was this page helpful? Yes No

active

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

On basic disks, marks the partition with focus as active. Only partitions can be marked as active. A partition must be selected for this operation to succeed. Use the **select partition** command to select a partition and shift the focus to it.

> ⊗ **Caution**
>
> DiskPart only informs the basic input/output system (BIOS) or Extensible Firmware Interface (EFI) that the partition or volume is a valid system partition or system volume, and is capable of containing the operating system startup files. DiskPart does not check the contents of the partition. If you mistakenly mark a partition as active and it does not contain the operating system startup files, your computer might not start.

Syntax

```
active
```

Examples

To mark the partition with focus as the active partition, type:

```
active
```

Related links

- Command-Line Syntax Key
- select partition command

Feedback

Was this page helpful? Yes No

add

Article • 10/16/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Mirrors the simple volume with focus to the specified active disk.

> **Note**
>
> This DiskPart command isn't available in any edition of Windows Vista.

Syntax

```
add disk=<n> [align=<n>] [wait] [noerr]
```

Parameters

Expand table

Parameter	Description
disk=`<n>`	Specifies a disk, other than the one containing the existing simple volume, to contain the mirror. You can mirror only simple volumes. The specified disk must have unallocated space at least as large as the size of the simple volume you want to mirror.
align=`<n>`	Typically used with hardware RAID Logical Unit Number (LUN) arrays to improve performance. Aligns all volume or partition extents to the closest alignment boundary. `n` is the number of kilobytes (KB) from the beginning of the disk to the closest alignment boundary.
wait	Waits for the volume to finish synchronizing with the added disk before returning. Without this parameter, DiskPart returns after the mirrored volume is created and doesn't wait for the synchronization to complete.
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error didn't occur. Without this parameter, an error causes DiskPart to exit with an error code.

Remarks

- A volume must be selected for this operation to succeed. Use the select volume command to select a volume and shift the focus to it.

Examples

To create a mirror of the volume with focus on disk 2, type:

```
add disk=2
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

assign

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Assigns a drive letter or mount point to the volume with focus. You can also use this command to change the drive letter associated with a removable drive. If no drive letter or mount point is specified, the next available drive letter is assigned. If the drive letter or mount point is already in use, an error is generated.

A volume must be selected for this operation to succeed. Use the **select volume** command to select a volume and shift the focus to it.

> **Important**
>
> You can't assign drive letters to system volumes, boot volumes, or volumes that contain the paging file. In addition, you cannot assign a drive letter to an Original Equipment Manufacturer (OEM) partition or any GUID Partition Table (gpt) partition other than a basic data partition.

Syntax

```
assign [{letter=<d> | mount=<path>}] [noerr]
```

Parameters

Expand table

Parameter	Description
`letter=<d>`	The drive letter you want to assign to the volume.
`mount=<path>`	The mount point path you want to assign to the volume. For instructions about how to use this command, see Assign a mount point folder path to a drive.
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes

Parameter	Description
	DiskPart to exit with an error code.

Examples

To assign the letter E to the volume in focus, type:

```
assign letter=e
```

Related links

- Command-Line Syntax Key
- select volume command

Feedback

Was this page helpful? Yes No

attach vdisk

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Attaches (sometimes called mounts or surfaces) a virtual hard disk (VHD) so that it appears on the host computer as a local hard disk drive. If the VHD already has a disk partition and file system volume when you attach it, the volume inside the VHD is assigned a drive letter.

> **ⓘ Important**
>
> You must choose and detach a VHD for this operation to succeed. Use the **select vdisk** command to select a VHD and shift the focus to it.

Syntax

```
attach vdisk [readonly] { [sd=<SDDL>] | [usefilesd] } [noerr]
```

Parameters

Expand table

Parameter	Description
readonly	Attaches the VHD as read-only. Any write operation returns an error.
`sd=<SDDL string>`	Sets the user filter on the VHD. The filter string must be in the Security Descriptor Definition Language (SDDL) format. By default the user filter allows access like on a physical disk. SDDL strings can be complex, but in its simplest form, a security descriptor that protects access is known as a discretionary access control list (DACL). It uses the form: `D:<dacl_flags><string_ace1><string_ace2>...<string_acen>` Common DACL flags are: • **A**. Allow access • **D**. Deny access

Parameter	Description
	Common rights are: • **GA**. All access • **GR**. Read access • **GW**. Write access Common user accounts are: • **BA**. Built in administrators • **AU**. Authenticated users • **CO**. Creator owner • **WD**. Everyone Examples: • **D:P:(A;;GR;;;AU**. Gives read-access to all authenticated users. • **D:P:(A;;GA;;;WD**. Gives everyone full access.
usefilesd	Specifies that the security descriptor on the .vhd file should be used on the VHD. If the **Usefilesd** parameter is not specified, the VHD will not have an explicit security descriptor unless it is specified with the **Sd** parameter.
noerr	Used for scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Examples

To attach the selected VHD as read-only, type:

```
attach vdisk readonly
```

Related links

- Command-Line Syntax Key
- select vdisk
- compact vdisk
- detail vdisk
- detach vdisk

- expand vdisk
- merge vdisk
- list

Feedback

Was this page helpful? Yes No

attributes

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays, sets, or clears the attributes of a disk or volume.

Syntax

```
attributes disk
attributes volume
```

Parameters

Expand table

Parameter	Description
attributes disk	Displays, sets, or clears the attributes of a disk.
attributes volume	Displays, sets, or clears the attributes of a volume.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

attributes disk

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays, sets, or clears the attributes of a disk. When this command is used to display the current attributes of a disk, the startup disk attribute denotes the disk used to start the computer. For a dynamic mirror, it displays the disk that contains the boot plex of the boot volume.

> **ⓘ Important**
>
> A disk must be selected for the **attributes disk** command to succeed. Use the **select disk** command to select a disk and shift the focus to it.

Syntax

```
attributes disk [{set | clear}] [readonly] [noerr]
```

Parameters

Expand table

Parameter	Description
set	Sets the specified attribute of the disk with focus.
clear	Clears the specified attribute of the disk with focus.
readonly	Specifies that the disk is read-only.
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Examples

To view the attributes of the selected disk, type:

```
attributes disk
```

To set the selected disk as read-only, type:

```
attributes disk set readonly
```

Related links

- Command-Line Syntax Key
- select disk command

Feedback

Was this page helpful? Yes No

attributes volume

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays, sets, or clears the attributes of a volume.

Syntax

```
attributes volume [{set | clear}] [{hidden | readonly | nodefaultdriveletter | shadowcopy}] [noerr]
```

Parameters

Expand table

Parameter	Description
set	Sets the specified attribute of the volume with focus.
clear	Clears the specified attribute of the volume with focus.
readonly	Specifies that the volume is read-only.
hidden	Specifies that the volume is hidden.
nodefaultdriveletter	Specifies that the volume does not receive a drive letter by default.
shadowcopy	Specifies that the volume is a shadow copy volume.
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Remarks

- On basic master boot record (MBR) disks, the **hidden**, **readonly**, and **nodefaultdriveletter** parameters apply to all volumes on the disk.

- On basic GUID partition table (GPT) disks, and on dynamic MBR and gpt disks, the **hidden**, **readonly**, and **nodefaultdriveletter** parameters apply only to the selected volume.
- A volume must be selected for the **attributes volume** command to succeed. Use the **select volume** command to select a volume and shift the focus to it.

Examples

To display the current attributes on the selected volume, type:

```
attributes volume
```

To set the selected volume as hidden and read-only, type:

```
attributes volume set hidden readonly
```

To remove the hidden and read-only attributes on the selected volume, type:

```
attributes volume clear hidden readonly
```

Related links

- Command-Line Syntax Key
- select volume command

Feedback

Was this page helpful? Yes No

automount

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

- Command-Line Syntax Key

> **ⓘ Important**
>
> In storage area network (SAN) configurations, disabling automount prevents Windows from automatically mounting or assigning drive letters to any new basic volumes that are visible to the system.

Syntax

automount [{ enable | disable | scrub }] [noerr]

Parameters

Expand table

Parameter	Description
enable	Enables Windows to automatically mount new basic and dynamic volumes that are added to the system and to assign them drive letters.
disable	Prevents Windows from automatically mounting any new basic and dynamic volumes that are added to the system. **Note**: Disabling automount can cause failover clusters to fail the storage portion of the Validate a Configuration Wizard.
scrub	Removes volume mount point directories and registry settings for volumes that are no longer in the system. This prevents volumes that were previously in the system from being automatically mounted and given their former volume mount point(s) when they are added back to the system.
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Examples

To see if the automount feature is enabled, type the following commands from within the diskpart command:

```
automount
```

To enable the automount feature, type:

```
automount enable
```

To disable the automount feature, type:

```
automount disable
```

Related links

- Command-Line Syntax Key
- diskpart commands

Feedback

Was this page helpful? Yes No

break

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

> **ⓘ Important**
>
> This command is no longer in use. It is included only to preserve compatibility with existing MS-DOS files, but it has no effect at the command line because the functionality is automatic.

Sets or clears extended CTRL+C checking on MS-DOS systems. If used without parameters, **break** displays the existing setting value.

If command extensions are enabled and running on the Windows platform, inserting the **break** command into a batch file enters a hard-coded breakpoint if being debugged by a debugger.

Syntax

```
break=[on|off]
```

> **ⓘ Note**
>
> Because the break command has no effect, it is often used to create empty files or delete the content of an existing file. For example:

```
rem -- cleans the content of the file --
break>log
```

Related links

- Command-Line Syntax Key

- break command

Feedback

Was this page helpful? Yes No

clean

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Removes all partitions or volume formatting from the disk with focus.

> **Note**
>
> For a PowerShell version of this command, see clear-disk command.

Syntax

```
clean [all]
```

Parameters

Expand table

Parameter	Description
all	Specifies that each and every sector on the disk is set to zero, which completely deletes all data contained on the disk.

Remarks

- On master boot record (MBR) disks, only the MBR partitioning information and hidden sector information is overwritten.
- On GUID Partition Table (gpt) disks, the gpt partitioning information, including the Protective MBR, is overwritten. There is no hidden sector information.
- A disk must be selected for this operation to succeed. Use the **select disk** command to select a disk and shift the focus to it.

Examples

To remove all formatting from the selected disk, type:

```
clean
```

Related links

- clear-disk command
- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

compact vdisk

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Reduces the physical size of a dynamically expanding virtual hard disk (VHD) file. This parameter is useful because dynamically expanding VHDs increase in size as you add files, but they do not automatically reduce in size when you delete files.

Syntax

```
compact vdisk
```

Remarks

- A dynamically expanding VHD must be selected for this operation to succeed. Use the select vdisk command to select a VHD and shift the focus to it.
- You can only use compact dynamically expanding VHDs that are detached or attached as read-only.

Related links

- Command-Line Syntax Key
- attach vdisk command
- detail vdisk command
- Detach vdisk command
- expand vdisk command
- Merge vdisk command
- select vdisk command
- list command

Feedback

Was this page helpful? Yes No

convert

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Converts a disk from one disk type to another.

Syntax

```
convert basic
convert dynamic
convert gpt
convert mbr
```

Parameters

Expand table

Parameter	Description
convert basic command	Converts an empty dynamic disk into a basic disk.
convert dynamic command	Converts a basic disk into a dynamic disk.
convert gpt command	Converts an empty basic disk with the master boot record (MBR) partition style into a basic disk with the GUID partition table (GPT) partition style.
convert mbr command	Converts an empty basic disk with the GUID Partition Table (GPT) partition style into a basic disk with the master boot record (MBR) partition style.

Related links

- Command-Line Syntax Key

Feedback